

Installation of 19c RAC on Linux7 in Virtual Box VMs

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Pre-Requisites

Oracle Virtual Box Installed

(Preferably configured with a default location for VMs)

Some familiarity with Oracle Virtual Box

Vagrant Installed

Oracle 19c Grid Infrastructure and Database Installer files

(download from <https://www.oracle.com/database/technologies/oracle19c-linux-downloads.html>)

X-Server for Display (eg mobaxterm)

puTTY software for terminal sessions

Awareness of Oracle GI, RAC and Database Architecture

Familiarity with Linux

PC as Underlying Host

Recent Intel/AMD CPU

8 Cores

32GB RAM

Setup the srv1 node

Create the vm

Create srv1 using Oracle Linux 7 from Oracle's github page for vagrant projects (<https://github.com/oracle/vagrant-projects>)

```
git clone https://github.com/oracle/vagrant-projects
```

Edit "vagrantfile" under C:\vagrant-projects\OracleLinux\7 (i.e. the downloaded folder) to set the hostname (srv1)

Run "vagrant up" to create srv1 as a Virtual Box VM

Vagrant will refresh all the packages when it creates the VM

```
C:\Virtual_Box_VMs\vagrant-projects\OracleLinux\7>vagrant up
==> vagrant: Getting Proxy Configuration from Host...
==> vagrant: A new version of Vagrant is available: 2.4.1 (installed
version: 2.3.2)!
==> vagrant: To upgrade visit: https://www.vagrantup.com/downloads.html

Bringing machine 'srv1' up with 'virtualbox' provider...
==> srv1: Importing base box 'oraclelinux/7'...
==> srv1: Matching MAC address for NAT networking...
==> srv1: Checking if box 'oraclelinux/7' version '7.9.517' is up to
date...
==> srv1: Setting the name of the VM: srv1
==> srv1: Clearing any previously set network interfaces...
==> srv1: Preparing network interfaces based on configuration...
    srv1: Adapter 1: nat
==> srv1: Forwarding ports...
    srv1: 22 (guest) => 2222 (host) (adapter 1)
==> srv1: Running 'pre-boot' VM customizations...
==> srv1: Booting VM...
==> srv1: Waiting for machine to boot. This may take a few minutes...
    srv1: SSH address: 127.0.0.1:2222
    srv1: SSH username: vagrant
    srv1: SSH auth method: private key
    srv1:
    srv1: Vagrant insecure key detected. Vagrant will automatically replace
    srv1: this with a newly generated keypair for better security.
    srv1:
    srv1: Inserting generated public key within guest...
    srv1: Removing insecure key from the guest if it's present...
    srv1: Key inserted! Disconnecting and reconnecting using new SSH key...
==> srv1: Machine booted and ready!
==> srv1: Checking for guest additions in VM...
==> srv1: Setting hostname...
==> srv1: Configuring proxy environment variables...
==> srv1: Configuring proxy for Yum...
==> srv1: Mounting shared folders...
```

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```
srv1: /vagrant => C:/Virtual_Box_VMs/vagrant-projects/OracleLinux/7
==> srv1: Running provisioner: shell...
srv1: Running: C:/Users/HEMANT~1/AppData/Local/Temp/vagrant-shell120240319-4376-rcrrbg.sh
srv1: INSTALLER: Started up
srv1: Resolving Dependencies
srv1: --> Running transaction check
srv1: ---> Package curl.x86_64 0:7.29.0-59.0.3.el7_9.1 will be updated
srv1: ---> Package curl.x86_64 0:7.29.0-59.0.3.el7_9.2 will be an
update
srv1: ---> Package dracut.x86_64 0:033-572.0.9.el7 will be updated
srv1: ---> Package dracut.x86_64 0:033-572.0.13.el7 will be an update
srv1: ---> Package kernel-headers.x86_64 0:3.10.0-1160.105.1.0.1.el7
will be updated
srv1: ---> Package kernel-headers.x86_64 0:3.10.0-1160.108.1.0.1.el7
will be an update
srv1: ---> Package kernel-tools.x86_64 0:3.10.0-1160.105.1.0.1.el7 will
be updated
srv1: ---> Package kernel-tools.x86_64 0:3.10.0-1160.108.1.0.1.el7 will
be an update
srv1: ---> Package kernel-tools-libs.x86_64 0:3.10.0-1160.105.1.0.1.el7
will be updated
srv1: ---> Package kernel-tools-libs.x86_64 0:3.10.0-1160.108.1.0.1.el7
will be an update
srv1: ---> Package kernel-uek.x86_64 0:5.4.17-2136.329.3.1.el7uek will
be installed
srv1: ---> Package kernel-uek-devel.x86_64 0:5.4.17-2136.329.3.1.el7uek
will be installed
srv1: ---> Package libcurl.x86_64 0:7.29.0-59.0.3.el7_9.1 will be
updated
srv1: ---> Package libcurl.x86_64 0:7.29.0-59.0.3.el7_9.2 will be an
update
srv1: ---> Package libzstd.x86_64 0:1.4.4-1.el7 will be updated
srv1: ---> Package libzstd.x86_64 0:1.5.5-1.el7 will be an update
srv1: ---> Package linux-firmware.noarch 999:20231102-
999.29.git2b304bfe.el7 will be updated
srv1: ---> Package linux-firmware.noarch 999:20240122-
999.31.gitbf0987d3.el7 will be an update
srv1: ---> Package openssh.x86_64 0:7.4p1-23.0.1.el7_9 will be updated
srv1: ---> Package openssh.x86_64 0:7.4p1-23.0.3.el7_9 will be an
update
srv1: ---> Package openssh-clients.x86_64 0:7.4p1-23.0.1.el7_9 will be
updated
srv1: ---> Package openssh-clients.x86_64 0:7.4p1-23.0.3.el7_9 will be
an update
srv1: ---> Package openssh-server.x86_64 0:7.4p1-23.0.1.el7_9 will be
updated
srv1: ---> Package openssh-server.x86_64 0:7.4p1-23.0.3.el7_9 will be
an update
srv1: ---> Package python-chardet.noarch 0:2.2.1-3.el7 will be
obsoleted
srv1: ---> Package python2-chardet.noarch 0:3.0.4-3.el7 will be
obsoleting
srv1: ---> Package systemd.x86_64 0:219-78.0.10.el7_9.9 will be updated
srv1: ---> Package systemd.x86_64 0:219-78.0.13.el7_9.9 will be an
update
srv1: ---> Package systemd-libs.x86_64 0:219-78.0.10.el7_9.9 will be
updated
srv1: ---> Package systemd-libs.x86_64 0:219-78.0.13.el7_9.9 will be an
update
```

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```
srv1: ---> Package systemd-sysv.x86_64 0:219-78.0.10.el7_9.9 will be
updated
srv1: ---> Package systemd-sysv.x86_64 0:219-78.0.13.el7_9.9 will be an
update
srv1: ---> Package tzdata.noarch 0:2023c-1.el7 will be updated
srv1: ---> Package tzdata.noarch 0:2024a-1.el7 will be an update
srv1: --> Finished Dependency Resolution
srv1:
srv1: Dependencies Resolved
srv1:
srv1:
=====
=====
srv1: Package Arch Version Repository
Size
srv1:
=====
=====
srv1: Installing:
srv1: kernel-uek x86_64 5.4.17-2136.329.3.1.el7uek ol7_UEKR6
108 M
srv1: kernel-uek-devel x86_64 5.4.17-2136.329.3.1.el7uek ol7_UEKR6
19 M
srv1: python2-chardet noarch 3.0.4-3.el7
ol7_developer 185 k
srv1: replacing python-chardet.noarch 2.2.1-3.el7
srv1: Updating:
srv1: curl x86_64 7.29.0-59.0.3.el7_9.2 ol7_latest
272 k
srv1: dracut x86_64 033-572.0.13.el7 ol7_latest
331 k
srv1: kernel-headers x86_64 3.10.0-1160.108.1.0.1.el7 ol7_latest
9.1 M
srv1: kernel-tools x86_64 3.10.0-1160.108.1.0.1.el7 ol7_latest
8.2 M
srv1: kernel-tools-libs x86_64 3.10.0-1160.108.1.0.1.el7 ol7_latest
8.1 M
srv1: libcurl x86_64 7.29.0-59.0.3.el7_9.2 ol7_latest
224 k
srv1: libzstd x86_64 1.5.5-1.el7
ol7_developer_EPEL 292 k
srv1: linux-firmware noarch 999:20240122-999.31.gitbf0987d3.el7
srv1: ol7_latest
353 M
srv1: openssh x86_64 7.4p1-23.0.3.el7_9 ol7_latest
485 k
srv1: openssh-clients x86_64 7.4p1-23.0.3.el7_9 ol7_latest
655 k
srv1: openssh-server x86_64 7.4p1-23.0.3.el7_9 ol7_latest
460 k
srv1: systemd x86_64 219-78.0.13.el7_9.9 ol7_latest
5.1 M
srv1: systemd-libs x86_64 219-78.0.13.el7_9.9 ol7_latest
421 k
srv1: systemd-sysv x86_64 219-78.0.13.el7_9.9 ol7_latest
99 k
srv1: tzdata noarch 2024a-1.el7 ol7_latest
497 k
srv1:
srv1: Transaction Summary
```

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```
srv1:
=====
=====
srv1: Install    3 Packages
srv1: Upgrade   15 Packages
srv1:
srv1: Total download size: 515 M
srv1: Downloading packages:
....
....
....
srv1: Complete!
srv1: INSTALLER: System updated
srv1: INSTALLER: Locale set
==> srv1: Running provisioner: shell...
srv1: Running: inline script
srv1: Reloading your VM to activate the latest kernel
==> srv1: Running provisioner: reload...
==> srv1: Attempting graceful shutdown of VM...
==> srv1: Checking if box 'oraclelinux/7' version '7.9.517' is up to
date...
==> srv1: Clearing any previously set forwarded ports...
==> srv1: Clearing any previously set network interfaces...
==> srv1: Preparing network interfaces based on configuration...
srv1: Adapter 1: nat
==> srv1: Forwarding ports...
srv1: 22 (guest) => 2222 (host) (adapter 1)
==> srv1: Running 'pre-boot' VM customizations...
==> srv1: Booting VM...
==> srv1: Waiting for machine to boot. This may take a few minutes...
==> srv1: Machine booted and ready!
==> srv1: Checking for guest additions in VM...
==> srv1: Setting hostname...
==> srv1: Configuring proxy environment variables...
==> srv1: Configuring proxy for Yum...
==> srv1: Mounting shared folders...
srv1: /vagrant => C:/Virtual_Box_VMs/vagrant-projects/OracleLinux/7
==> srv1: Machine already provisioned. Run `vagrant provision` or use the
`--provision`
==> srv1: flag to force provisioning. Provisioners marked to run always
will still run.
==> srv1: Running provisioner: shell...
srv1: Running: inline script
srv1: INSTALLER: Installation complete, Oracle Linux 7 ready to use!

C:\Virtual_Box_VMs\vagrant-projects\OracleLinux\7>
```

The actual VM files would be under “srv1” folder under default folder for Virtual Box (thus, in my case, at C:\Virtual_Box_VMs\srv1)

Set the root password

From the CMD window at C:\vagrant-projects\OracleLinux\7
run “vagrant ssh” to create an SSH session into “srv1”

From the opened session (which logs in as “vagrant”)

run "sudo su -"

and then “passwd” at the “#” prompt for root to change the root password

```
C:\Virtual_Box_VMs\vagrant-projects\OracleLinux\7>vagrant ssh
==> vagrant: Getting Proxy Configuration from Host...
```

```
Welcome to Oracle Linux Server release 7.9 (GNU/Linux 5.4.17-
2136.325.5.el7uek.x86_64)
```

The Oracle Linux End-User License Agreement can be viewed here:

```
* /usr/share/eula/eula.en_US
```

For additional packages, updates, documentation and community help, see:

```
* https://yum.oracle.com/
```

```
[vagrant@srv1 ~]$ sudo su -
[root@srv1 ~]# passwd
Changing password for user root.
New password:
BAD PASSWORD: The password fails the dictionary check - it is based on a
dictionary word
Retype new password:
Sorry, passwords do not match.
New password:
BAD PASSWORD: The password fails the dictionary check - it is based on a
dictionary word
Retype new password:
passwd: all authentication tokens updated successfully.
[root@srv1 ~]#
```

Edit /etc/ssh/sshd_config

```
[root@srv1 ~]# vi /etc/ssh/sshd_config
```

and set :

```
PasswordAuthentication yes
```

To allow remote login from a puTty session

Configure the eth0 network interface

At /etc/sysconfig/network-scripts , edit the file “ifcfg-eth0”

```
root@srv1 network-scripts]# vi ifcfg-eth0
```

and set the parameters in the file as :

```
DEVICE="eth0"
BOOTPROTO="static"
IPADDR="192.168.56.90"
NETMASK="255.255.255.0"
GATEWAY="0.0.0.0"
ONBOOT="yes"
TYPE="Ethernet"
USERCTL="yes"
PEERDNS="yes"
IPV6INIT="no"
```

```
NM_CONTROLLED="no"
```

Update resolv.conf for NameServer

Update /etc/resolv.conf :

```
[root@srv1 network-scripts]# vi /etc/resolv.conf  
and set :  
nameserver 192.168.1.254
```

Edit the hosts file

```
[root@srv1 network-scripts]# vi /etc/hosts
```

And set the entries as :

```
#127.0.1.1 srv1 srv1  
  
#Host IPs  
192.168.56.90  srv1 srv1.localdomain  
192.168.56.91  srv1-vip srv1-vip.localdomain  
192.168.10.8   srv1-priv srv1-priv  
  
192.168.56.92  srv2 srv2.localdomain  
192.168.56.93  srv2-vip srv2-vip.localdomain  
192.168.10.10  srv2-priv srv2-priv  
  
192.168.56.80  dnssrvr dnssrvr.localdomain  
#SCAN IPs (DNS Server now configured to dnssrvr  
#192.168.56.94 ora19cscan ora19cscan.localdomain  
#192.168.56.95 ora19cscan ora19cscan.localdomain  
#192.168.56.96 ora19cscan ora19cscan.localdomain  
  
192.168.56.99  stdby stdby.localdomain
```

Add Other Network Interfaces

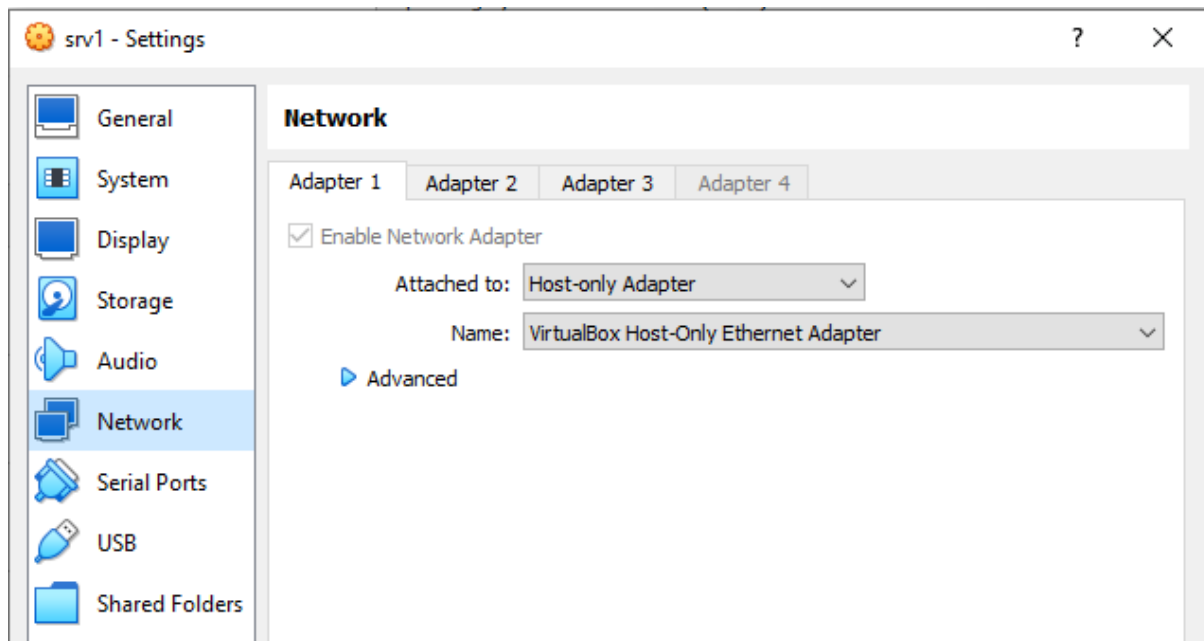
Shutdown the vm with

```
[root@srv1 network-scripts]# shutdown -h now
```

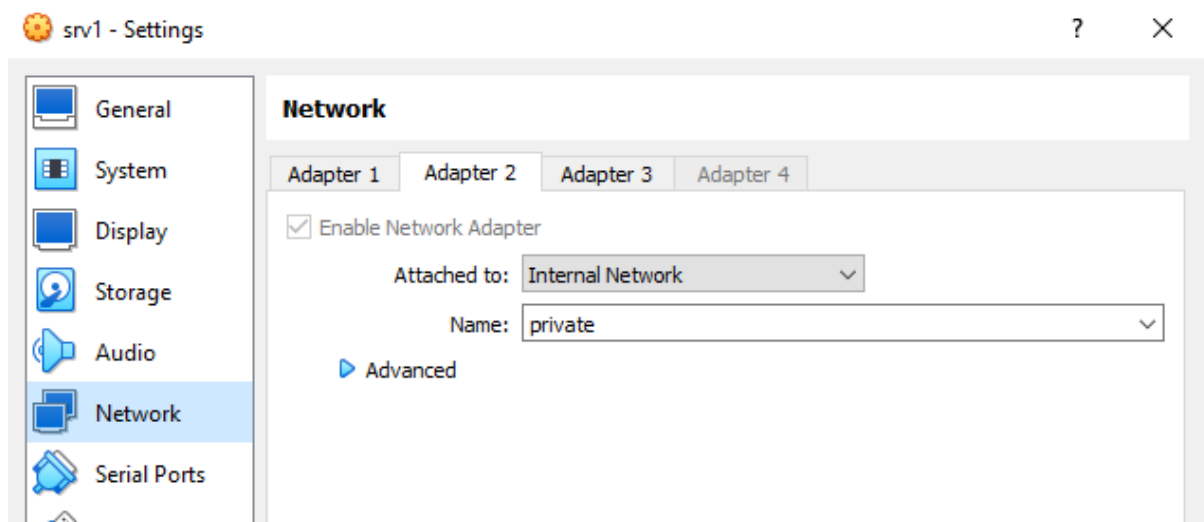
(Henceforth all startups will be from Virtual Box)

From the Virtual Box, Settings → Network Page, configure

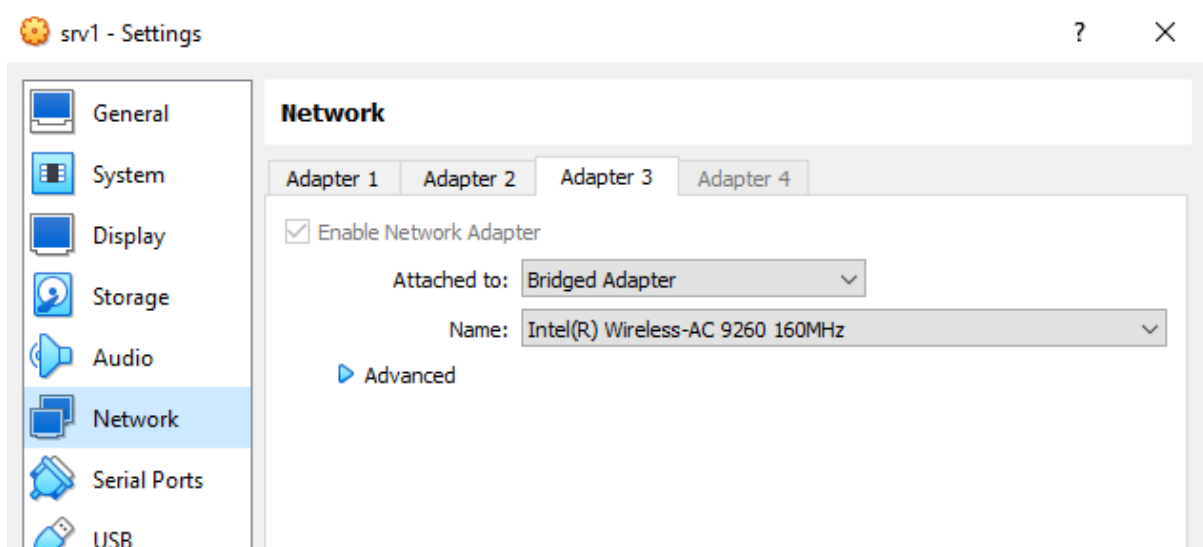
Adapter 1 : Change Attached To : “Host-Only Adapter”



Adapter 2 : Attached To : “Internal Network” and name the network “private”



Adapter 3 : Attached to “Bridged Adapter” (it should automatically show the Host’s Network Interface that connects to the Internet)



Start the VM to configure the Interfaces

Login as root from puTTY

At `/etc/sysconfig/network-scripts` , create the file “`ifcfg-eth1`” as : [this will be the Private Network for the Cluster]

```
DEVICE="eth1"
BOOTPROTO="static"
IPADDR="192.168.10.8"
NETMASK="255.255.255.0"
GATEWAY="0.0.0.0"
ONBOOT="yes"
TYPE="Ethernet"
USERCTL="yes"
PEERDNS="yes"
IPV6INIT="no"
NM_CONTROLLED="no"
```

And “`ifcfg-eth2`” as : [this will be using the Bridged-Adapter]

```
DEVICE="eth2"
BOOTPROTO="dhcp"
ONBOOT="yes"
TYPE="Ethernet"
USERCTL="yes"
PEERDNS="yes"
IPV6INIT="no"
PERSISTENT_DHCLIENT="1"
NM_CONTROLLED="no"
```

Reboot the VM to Test Network connectivity

Do a puTty login to the VM

Test “`ping www.google.com`”

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```
[root@srv1 ~]# ping www.google.com
PING www.google.com (142.251.10.99) 56(84) bytes of data.
64 bytes from sd-in-f99.1e100.net (142.251.10.99): icmp_seq=1 ttl=54
time=8.28 ms
64 bytes from sd-in-f99.1e100.net (142.251.10.99): icmp_seq=2 ttl=54
time=8.23 ms
^C
--- www.google.com ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 8.236/8.259/8.283/0.093 ms
```

Verify all 3 Network Interfaces

```
[root@srv1 ~]# ifconfig -a
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.90 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::a00:27ff:fec5:de6f prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:c5:de:6f txqueuelen 1000 (Ethernet)
    RX packets 115 bytes 13462 (13.1 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 101 bytes 16860 (16.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.10.8 netmask 255.255.255.0 broadcast 192.168.10.255
    inet6 fe80::a00:27ff:feb9:86d7 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:b9:86:d7 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 15 bytes 1086 (1.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.16 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::a00:27ff:fea4:ea6f prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:a4:ea:6f txqueuelen 1000 (Ethernet)
    RX packets 107 bytes 11365 (11.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 43 bytes 4078 (3.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

[root@srv1 ~]#
```

(eth2 would have a dynamically assigned IP address from DHCP)

Pre-Clone Configuration

Install the Oracle Pre-Installation RPM

The RPM is : oracle-database-preinstall-19c

```
[root@srv1 ~]# yum install oracle-database-preinstall-19c
...
...
... Long list of messages about packages

Installed:
  oracle-database-preinstall-19c.x86_64 0:1.0-3.el7

Dependency Installed:
...
... List of dependencies

Complete!
[root@srv1 ~]#
```

Install Name Service Caching Daemon (optional)

```
[root@srv1 ~]# yum install nscd
```

Create Groups and Users

```
[root@srv1 ~]# groupadd grid
[root@srv1 ~]# groupadd dba
[root@srv1 ~]# useradd grid -g grid ## Preferable : -g oinstall
[root@srv1 ~]# useradd oracle -G dba,grid ## Preferable : -g oinstall
[root@srv1 /]# usermod -G grid,oinstall grid
[root@srv1 /]# usermod -G grid,dba,oper,oinstall oracle
```

Note : Preferably specify “oinstall” as the Primary Group for both grid and oracle. Else, if the grid Primary group is grid, the DBCA may fail if the Primary Group doesn’t match. Similarly, when configuring oracleasmilb, the group is specified as oinstall and when installing grid and oracle, the group is oinstall.

(Note : groups “oinstall” and “dba” and user “oracle” would have already been created by the oracle-database-preinstall-19c RPM installation)

Optionally, run

“usermod -G grid,dba,oper,oinstall,backupdba,dgdba,kmdba,racdba oracle”
so that the additional groups can also be configured for the corresponding privileges when installing the RDBMS software.

Edit and Verify the groups file /etc/group

```
oinstall:x:54321:oracle,grid
dba:x:54322:oracle
oper:x:54323:
backupdba:x:54324:
```

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```
dgdba:x:54325:
kmdba:x:54326:
racdba:x:54330:
grid:x:54331:grid,oracle
```

Set Passwords

```
[root@srv1 ~]# passwd grid
Changing password for user grid.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@srv1 ~]# passwd oracle
Changing password for user oracle.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@srv1 ~]#
```

Run oracle-database-preinstall-19c-verify

```
[root@srv1 ~]# ls -l /usr/bin/ora*
lrwxrwxrwx. 1 root root 83 Mar 19 08:31 /usr/bin/oracle-database-
preinstall-19c-verify -> /etc/sysconfig/oracle-database-preinstall-
19c/oracle-database-preinstall-19c-verify
[root@srv1 ~]# /usr/bin/oracle-database-preinstall-19c-verify
Getting all the available kernels on the system

index=0:kernel=/boot/vmlinuz-3.10.0-1160.108.1.0.1.el7.x86_64:args="ro
console=tty0 crashkernel=auto rd.lvm.lv=vg_main/lv_root
rd.lvm.lv=vg_main/lv_swap numa=off transparent_hugepage=never
LANG=en_US.UTF-8":root=/dev/mapper/vg_main-lv_root:initrd=/boot/initramfs-
3.10.0-1160.108.1.0.1.el7.x86_64.img:title=Oracle Linux Server 7.9, with
Linux 3.10.0-1160.108.1.0.1.el7.x86_64:
index=1:kernel=/boot/vmlinuz-5.4.17-2136.329.3.1.el7uek.x86_64:args="ro
console=tty0 crashkernel=auto rd.lvm.lv=vg_main/lv_root
rd.lvm.lv=vg_main/lv_swap numa=off
transparent_hugepage=never":root=/dev/mapper/vg_main-
lv_root:initrd=/boot/initramfs-5.4.17-
2136.329.3.1.el7uek.x86_64.img:title=Oracle Linux Server 7.9, with
Unbreakable Enterprise Kernel 5.4.17-2136.329.3.1.el7uek.x86_64:
index=2:kernel=/boot/vmlinuz-5.4.17-2136.325.5.el7uek.x86_64:args="ro
console=tty0 crashkernel=auto rd.lvm.lv=vg_main/lv_root
rd.lvm.lv=vg_main/lv_swap numa=off
transparent_hugepage=never":root=/dev/mapper/vg_main-
lv_root:initrd=/boot/initramfs-5.4.17-
2136.325.5.el7uek.x86_64.img:title=Oracle Linux Server 7.9, with
Unbreakable Enterprise Kernel 5.4.17-2136.325.5.el7uek.x86_64:
index=3:kernel=/boot/vmlinuz-0-rescue-
633e848f812a5c408945853266376911:args="ro console=tty0 crashkernel=auto
rd.lvm.lv=vg_main/lv_root rd.lvm.lv=vg_main/lv_swap numa=off
transparent_hugepage=never":root=/dev/mapper/vg_main-
lv_root:initrd=/boot/initramfs-0-rescue-
633e848f812a5c408945853266376911.img:title=Oracle Linux Server 7.9, with
Linux 0-rescue-633e848f812a5c408945853266376911:
```

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```
index=4:non linux entry:Default has been set to - /boot/vmlinuz-5.4.17-  
2136.329.3.1.el7uek.x86_64  
[root@srv1 ~]#
```

Configure the limits file for the grid user

The Oracle Pre-Installation RPM would have configured the limits file

/etc/security/limits.d/oracle-database-preinstall-19c.conf for the default “oracle” user. Edit this file to set the limits for “grid” as well.

Thus, the entries would be :

```
[root@srv1 limits.d]# pwd  
/etc/security/limits.d  
[root@srv1 limits.d]#  
[root@srv1 limits.d]# cat oracle-database-preinstall-19c.conf  
  
# oracle-database-preinstall-19c setting for nofile soft limit is 1024  
oracle soft nofile 1024  
grid soft nofile 1024  
  
# oracle-database-preinstall-19c setting for nofile hard limit is 65536  
oracle hard nofile 65536  
grid hard nofile 65536  
  
# oracle-database-preinstall-19c setting for nproc soft limit is 16384  
# refer orabug15971421 for more info.  
oracle soft nproc 16384  
grid soft nproc 16384  
  
# oracle-database-preinstall-19c setting for nproc hard limit is 16384  
oracle hard nproc 16384  
grid hard nproc 16384  
  
# oracle-database-preinstall-19c setting for stack soft limit is 10240KB  
oracle soft stack 10240  
grid soft stack 10240  
  
# oracle-database-preinstall-19c setting for stack hard limit is 32768KB  
oracle hard stack 32768  
grid hard stack 32768  
  
# oracle-database-preinstall-19c setting for memlock hard limit is maximum  
of 128GB on x86_64 or 3GB on x86 OR 90 % of RAM  
oracle hard memlock 134217728  
grid hard memlock 134217728  
  
# oracle-database-preinstall-19c setting for memlock soft limit is maximum  
of 128GB on x86_64 or 3GB on x86 OR 90% of RAM  
oracle soft memlock 134217728  
grid soft memlock 134217728  
  
# oracle-database-preinstall-19c setting for data soft limit is 'unlimited'  
oracle soft data unlimited  
grid soft data unlimited  
  
# oracle-database-preinstall-19c setting for data hard limit is 'unlimited'
```

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```
oracle hard data unlimited
grid hard data unlimited
[root@srv1 limits.d]#
```

Also, verify and, if necessary, increase the process count limit for users other than root. The file name would be “xx-nproc.conf” where “xx” is a number. On my installed servers it is 20-nproc.conf on both servers. The limit should be 4096

```
[root@srv1 limits.d]# cat 20-nproc.conf
# Default limit for number of user's processes to prevent
# accidental fork bombs.
# See rhbz #432903 for reasoning.

*            soft    nproc      4096
root         soft    nproc      unlimited
[root@srv1 limits.d]#
```

Install and Configure ASMLib

Installation

The installation consists of

1. oracleasm-support
2. kmod-oracleasm

```
[root@srv1 ~]# yum install oracleasm-support
Resolving Dependencies
--> Running transaction check
---> Package oracleasm-support.x86_64 0:2.1.11-2.el7 will be installed
--> Finished Dependency Resolution
```

Dependencies Resolved

```
=====
... Installation Messages
```

```
Installed:
  oracleasm-support.x86_64 0:2.1.11-2.el7
```

```
Complete!
[root@srv1 ~]#
```

```
[root@srv1 ~]# yum install kmod-oracleasm
Resolving Dependencies
--> Running transaction check
---> Package kmod-oracleasm.x86_64 0:2.0.8-28.0.1.el7 will be installed
--> Processing Dependency: kernel(add_timer) = 0xbe2c0274 for package:
kmod-oracleasm-2.0.8-28.0.1.el7.x86_64
...
... Long list of messages about packages
```

```
Installed:
  kmod-oracleasm.x86_64 0:2.0.8-28.0.1.el7
```

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Dependency Installed:

```
kernel.x86_64 0:3.10.0-1160.108.1.0.1.el7
```

Complete!

```
[root@srv1 ~]#
```

Configuration

```
[root@srv1 ~]# oracleasm configure -i
Configuring the Oracle ASM library driver.
```

This will configure the on-boot properties of the Oracle ASM library driver. The following questions will determine whether the driver is loaded on boot and what permissions it will have. The current values will be shown in brackets ('[]'). Hitting <ENTER> without typing an answer will keep that current value. Ctrl-C will abort.

```
Default user to own the driver interface []: grid
Default group to own the driver interface []: oinstall
Start Oracle ASM library driver on boot (y/n) [n]: y
Scan for Oracle ASM disks on boot (y/n) [y]: y
Writing Oracle ASM library driver configuration: done
[root@srv1 ~]#
```

Initialisation

```
[root@srv1 ~]# /usr/sbin/oracleasm init
Creating /dev/oracleasm mount point: /dev/oracleasm
Loading module "oracleasm": oracleasm
Configuring "oracleasm" to use device physical block size
Mounting ASMLib driver filesystem: /dev/oracleasm
[root@srv1 ~]#
```

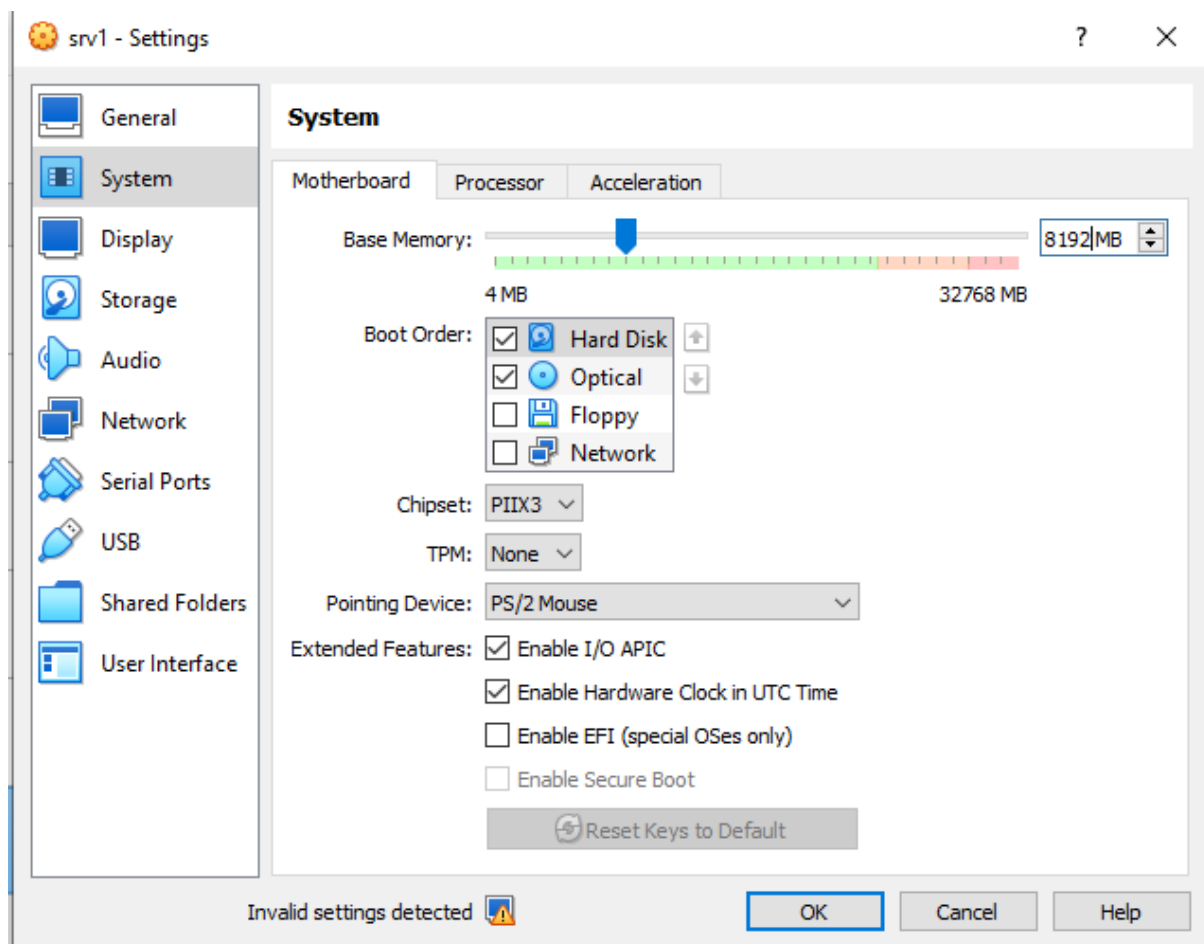
Clone srv1 to srv2

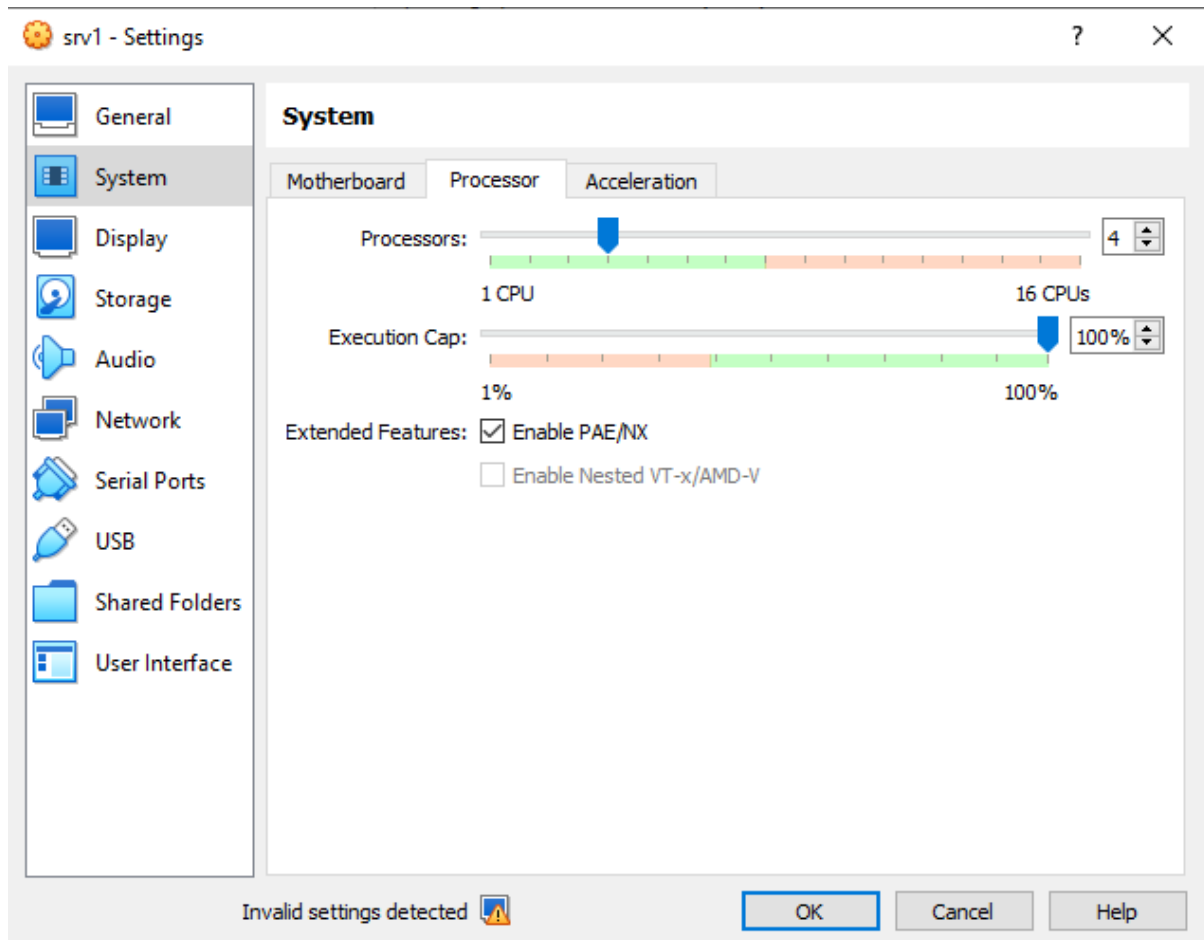
Shutdown srv1

```
#shutdown -h now
```

Change the Allocated Memory and Processor Count

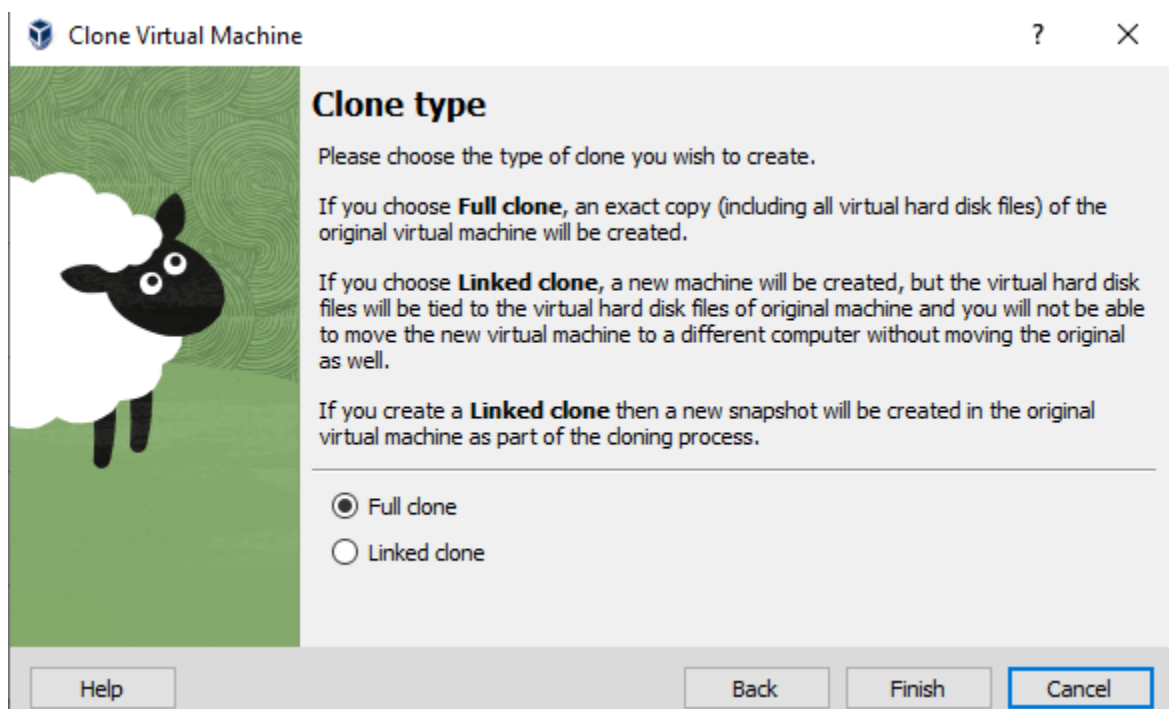
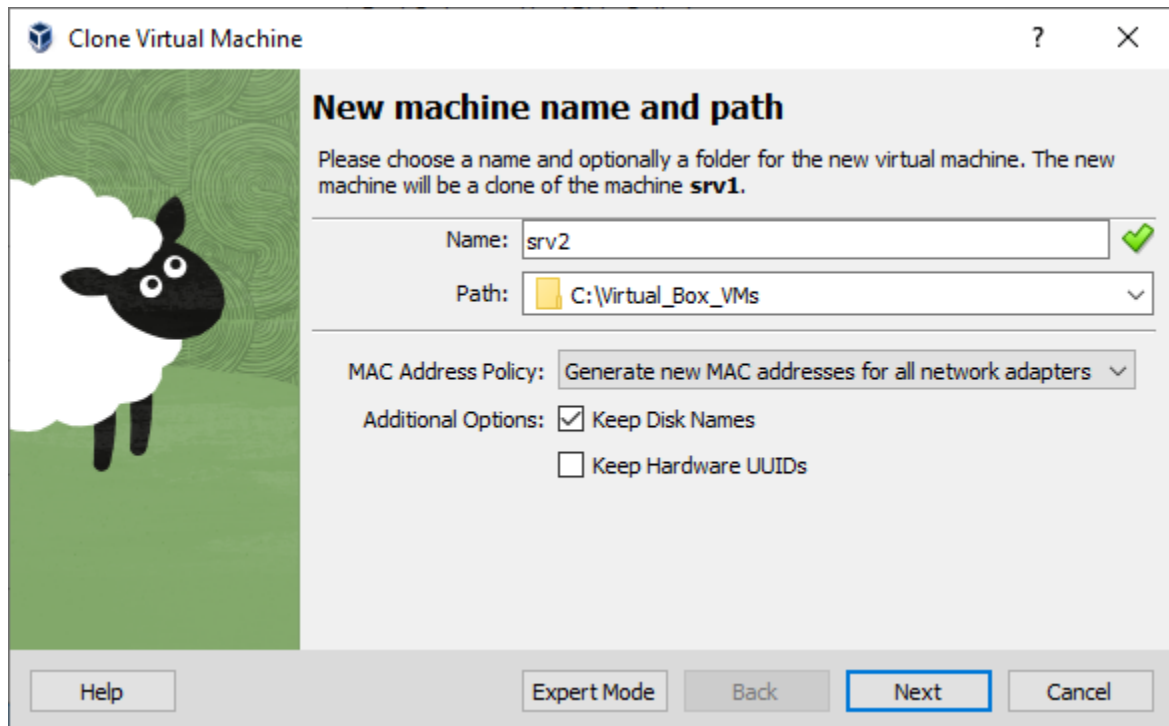
Recommended : 8GB Memory and 4 Processors





Use The Clone VM facility in Virtual Box

Right click on “srv1” and select “Clone” to clone srv1 to srv2 as a “Full Clone” (Ensure that “Generate new MAC addresses” is selected)



Update the Hostname and Network Configuration

Login to srv2 from the default console (in Virtual Box) and

```
[root@srv1 ~]# hostname  
srv1  
[root@srv1 ~]# hostname srv2
```

Edit /etc/hostname using “vi”

```
[root@srv1 ~]# cat /etc/hostname  
srv2  
[root@srv1 ~]#
```

(ignore the “@srv1” because the server hasn’t inherited the updated /etc/hostname yet)

Edit ifcfg-eth0 and ifcfg-eth1 under /etc/sysconfig/network-scripts to set the IP Addresses (all other values do not need to be updated as the server has been cloned from srv1)

```
[root@srv1 network-scripts]# cat ifcfg-eth0  
DEVICE="eth0"  
BOOTPROTO="static"  
IPADDR="192.168.56.92"  
NETMASK="255.255.255.0"  
GATEWAY="0.0.0.0"  
ONBOOT="yes"  
TYPE="Ethernet"  
USERCTL="yes"  
PEERDNS="yes"  
IPV6INIT="no"  
NM_CONTROLLED="no"  
[root@srv1 network-scripts]# cat ifcfg-eth1  
DEVICE="eth1"  
BOOTPROTO="static"  
IPADDR="192.168.10.10"  
NETMASK="255.255.255.0"  
GATEWAY="0.0.0.0"  
ONBOOT="yes"  
TYPE="Ethernet"  
USERCTL="yes"  
PEERDNS="yes"  
IPV6INIT="no"  
NM_CONTROLLED="no"  
[root@srv1 network-scripts]# _
```

Reboot the server and login from puTty

(the root password is the same as on srv1 as srv2 has been cloned from srv1)

Setup the DNS Server

Create the vm

Create srv1 using Oracle Linux 7 from Oracle's github page

Git clone Oracle's vagrant-projects

Edit "vagrantfile" under \vagrant-projects\OracleLinux\7 to set the hostname (srv1)

Run "vagrant up" to create srv1 as a Virtual Box VM

Vagrant will refresh all the packages

The actual VM files would be under "srv1" folder under default folder for Virtual Box (thus, in my case, at C:\Virtual_Box_VMs\srv1)

Set the root password

From window run "vagrant ssh" at \vagrant-projects\OracleLinux\7

From the opened session (which logs in as "vagrant")

run "sudo su -"

and then "passwd" at the "#" prompt for root

Edit /etc/ssh/sshd_config and set :

PasswordAuthentication yes

Configure the eth0 network interface

At /etc/sysconfig/network-scripts , edit the file "ifcfg-eth0" as :

```
DEVICE="eth0"
BOOTPROTO="static"
IPADDR="192.168.56.80"
NETMASK="255.255.255.0"
GATEWAY="0.0.0.0"
ONBOOT="yes"
TYPE="Ethernet"
USERCTL="yes"
PEERDNS="yes"
IPV6INIT="no"
NM_CONTROLLED="no"
```

Update resolv.conf for NameServer

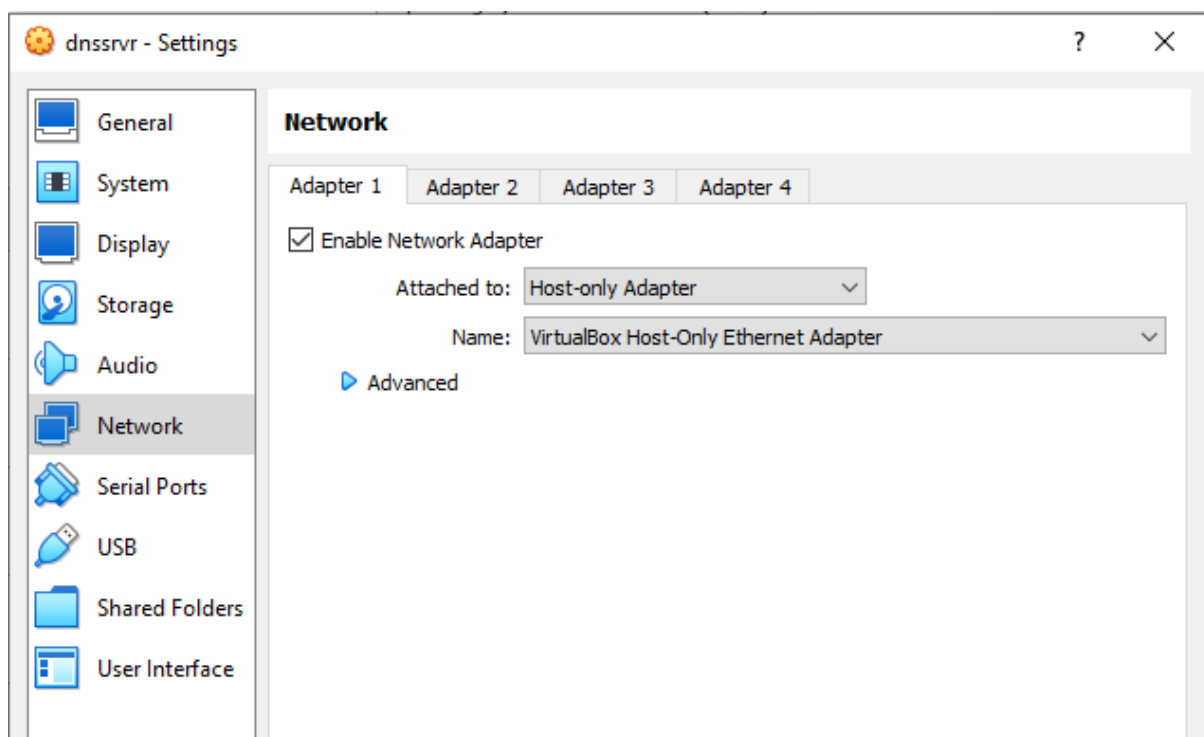
Update /etc/resolv.conf :
nameserver 192.168.1.254

Add The Other Network Interfaces

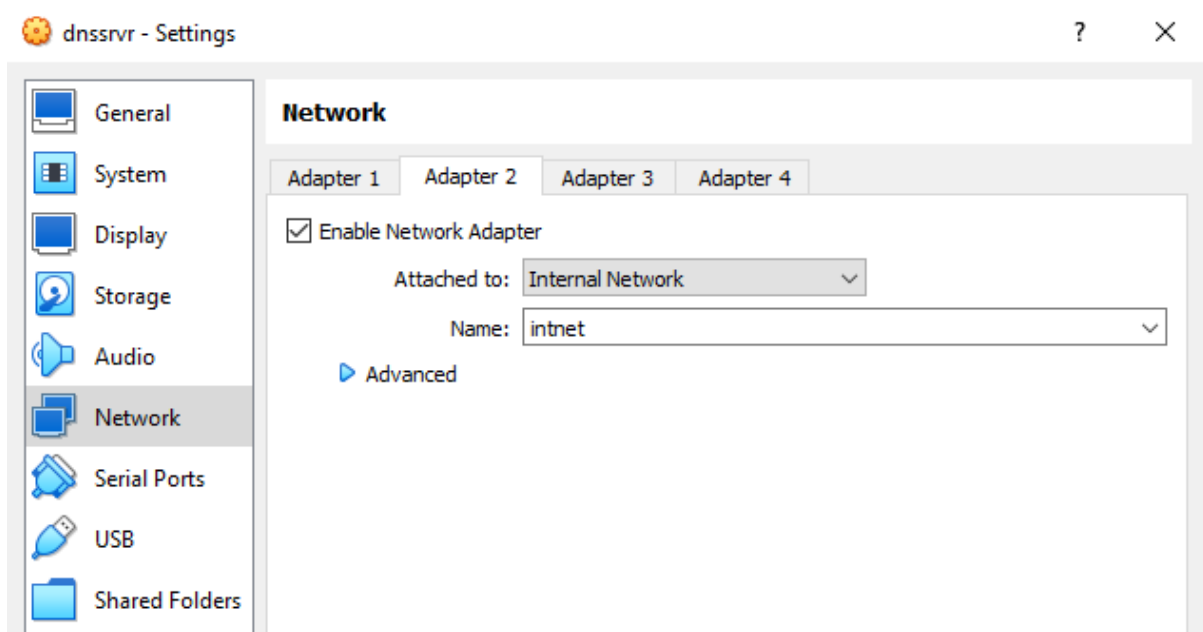
Shutdown the vm with “shutdown -h now”

From the Virtual Box, Settings → Network Page, configure

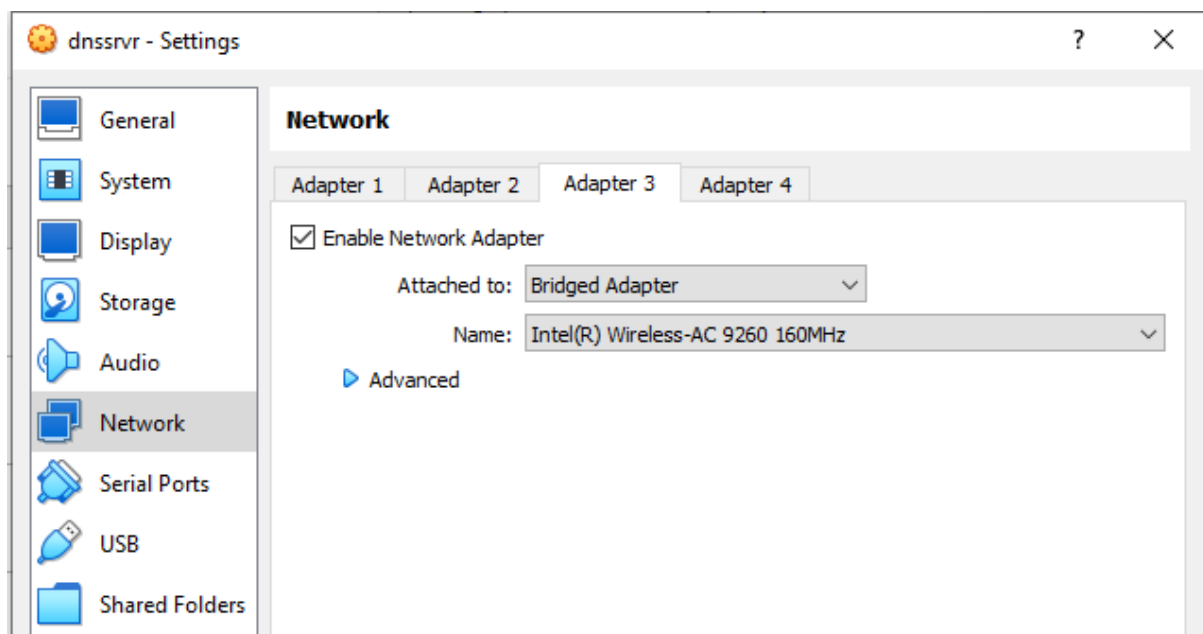
Adapter 1 : Change Attached To : “Host-Only Adapter”



Adapter 2 : Attached to “Internal Network”



Adapter 3 : Adapter 3 : Attached to “Bridged Adapter” (it should automatically show the Host’s Network Interface that connects to the Internet)



Start the VM to configure the Interfaces

Login

At `/etc/sysconfig/network-scripts` , create

The file “`ifcfg-eth1`” as :

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```
DEVICE="eth1"
BOOTPROTO="static"
IPADDR="192.168.10.6"
NETMASK="255.255.255.0"
GATEWAY="0.0.0.0"
ONBOOT="yes"
TYPE="Ethernet"
USERCTL="yes"
PEERDNS="yes"
IPV6INIT="no"
NM_CONTROLLED="no"
```

the file "ifcfg-eth2" as :

```
DEVICE="eth2"
BOOTPROTO="dhcp"
ONBOOT="yes"
TYPE="Ethernet"
USERCTL="yes"
PEERDNS="yes"
IPV6INIT="no"
PERSISTENT_DHCLIENT="1"
NM_CONTROLLED="no"
```

Reboot the VM to Test Network connectivity

Do a puTty login to the VM

Test "ping www.google.com"

Verify all 3 Network Interfaces

```
[root@dnssrvr ~]# ping www.google.com
PING www.google.com (64.233.170.104) 56(84) bytes of data.
64 bytes from sg-in-f104.1e100.net (64.233.170.104): icmp_seq=1 ttl=100
time=3.79 ms
64 bytes from sg-in-f104.1e100.net (64.233.170.104): icmp_seq=2 ttl=100
time=4.40 ms
^C
--- www.google.com ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 3.791/4.098/4.405/0.307 ms
[root@dnssrvr ~]# ifconfig -a
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.80 netmask 255.255.255.0 broadcast 192.168.56.255
    inet6 fe80::a00:27ff:fec5:de6f prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:c5:de:6f txqueuelen 1000 (Ethernet)
    RX packets 159 bytes 14834 (14.4 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 116 bytes 16869 (16.4 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.10.6 netmask 255.255.255.0 broadcast 192.168.10.255
    inet6 fe80::a00:27ff:fe35:84c5 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:35:84:c5 txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
```

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```

    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 15  bytes 1086 (1.0 KiB)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

eth2: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.1.14  netmask 255.255.255.0  broadcast 192.168.1.255
    inet6 fe80::a00:27ff:fe1c:292d  prefixlen 64  scopeid 0x20<link>
    ether 08:00:27:1c:29:2d  txqueuelen 1000  (Ethernet)
    RX packets 85  bytes 9583 (9.3 KiB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 48  bytes 4526 (4.4 KiB)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 128  scopeid 0x10<host>
    loop txqueuelen 1000  (Local Loopback)
    RX packets 0  bytes 0 (0.0 B)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 0  bytes 0 (0.0 B)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

[root@dnssrvr ~]#
```

Install and Configure dnsmasq

Install dnsmasq

```
[root@dnssrvr ~]# yum install dnsmasq
ol7_UCKR6
| 3.0 kB  00:00:00
ol7_addons
| 3.0 kB  00:00:00
ol7_developer
| 3.0 kB  00:00:00
ol7_developer_EPEL
| 3.6 kB  00:00:00
ol7_latest
| 3.6 kB  00:00:00
ol7_optional_latest
| 3.0 kB  00:00:00
ol7_software_collections
| 3.0 kB  00:00:00
Resolving Dependencies
--> Running transaction check
---> Package dnsmasq.x86_64 0:2.76-17.0.3.el7_9.3 will be installed
--> Processing Dependency: libnettle.so.4()(64bit) for package: dnsmasq-
2.76-17.0.3.el7_9.3.x86_64
--> Running transaction check
---> Package nettle.x86_64 0:2.7.1-9.el7_9 will be installed
--> Finished Dependency Resolution
```

Dependencies Resolved

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```
=====
=====
Package              Arch              Version
Repository           Size
=====
=====
Installing:
  dnsmasq             x86_64           2.76-17.0.3.el7_9.3
  ol7_latest           281 k
Installing for dependencies:
  nettle              x86_64           2.7.1-9.el7_9
  ol7_latest           327 k

Transaction Summary
=====
=====
Install 1 Package (+1 Dependent package)

Total download size: 608 k
Installed size: 1.3 M
Is this ok [y/d/N]: y
Downloading packages:
(1/2): dnsmasq-2.76-17.0.3.el7_9.3.x86_64.rpm
| 281 kB  00:00:00
(2/2): nettle-2.7.1-9.el7_9.x86_64.rpm
| 327 kB  00:00:00
-----
-----
Total
1.9 MB/s | 608 kB  00:00:00
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
  Installing : nettle-2.7.1-9.el7_9.x86_64
1/2
  Installing : dnsmasq-2.76-17.0.3.el7_9.3.x86_64
2/2
  Verifying  : dnsmasq-2.76-17.0.3.el7_9.3.x86_64
1/2
  Verifying  : nettle-2.7.1-9.el7_9.x86_64
2/2

Installed:
  dnsmasq.x86_64 0:2.76-17.0.3.el7_9.3

Dependency Installed:
  nettle.x86_64 0:2.7.1-9.el7_9

Complete!
[root@dnssrvr ~]#
```

Configure dnsmasq

```
[root@dnssrvr ~]# systemctl enable dnsmasq.service
```

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```
Created symlink from /etc/systemd/system/multi-  
user.target.wants/dnsmasq.service to  
/usr/lib/systemd/system/dnsmasq.service.  
[root@dnssrvr ~]#
```

Edit the /etc/hosts file to add the RAC servers and SCAN

```
192.168.56.80 dnssrvr dnssrvr.localdomain  
  
#My 19c RAC Host IPs  
192.168.56.90 srv1 srv1.localdomain  
192.168.56.91 srv1-vip srv1-vip.localdomain  
192.168.10.8 srv1-priv srv1-priv  
  
192.168.56.92 srv2 srv2.localdomain  
192.168.56.93 srv2-vip srv2-vip.localdomain  
192.168.10.10 srv2-priv srv2-priv  
  
#SCAN IPs  
192.168.56.94 ora19cscan ora19cscan.localdomain  
192.168.56.95 ora19cscan ora19cscan.localdomain  
192.168.56.96 ora19cscan ora19cscan.localdomain  
  
192.168.56.99 stdby stdby.localdomain
```

Start dnsmasq

```
[root@dnssrvr etc]# systemctl start dnsmasq.service  
  
[root@dnssrvr etc]# systemctl status dnsmasq.service  
● dnsmasq.service - DNS caching server.  
   Loaded: loaded (/usr/lib/systemd/system/dnsmasq.service; enabled; vendor  
  preset: disabled)  
   Active: active (running) since Fri 2024-03-15 02:57:02 UTC; 1min 35s ago  
 Main PID: 1326 (dnsmasq)  
    CGroup: /system.slice/dnsmasq.service  
            └─1326 /usr/sbin/dnsmasq -k  
  
Mar 15 02:57:02 dnssrvr systemd[1]: Started DNS caching server..  
Mar 15 02:57:02 dnssrvr dnsmasq[1326]: started, version 2.76 cachesize 150  
Mar 15 02:57:02 dnssrvr dnsmasq[1326]: compile time options: IPv6 GNU-  
getopt DBus no-ill8n IDN DHCP DHCPv6 no-Lua TFTP...notify  
Mar 15 02:57:02 dnssrvr dnsmasq[1326]: reading /etc/resolv.conf  
Mar 15 02:57:02 dnssrvr dnsmasq[1326]: using nameserver 192.168.1.254#53  
Mar 15 02:57:02 dnssrvr dnsmasq[1326]: read /etc/hosts - 12 addresses  
Hint: Some lines were ellipsized, use -l to show in full.  
[root@dnssrvr etc]#
```

Note : The configuration file does not need to be updated for this RAC Cluster IP addresses as it will read them from the “local” /etc/hosts file

Test DNS Round Robin

On srv1 and srv2, edit /etc/hosts

Comment out the entries for the SCAN

```
192.168.56.80 dnssrvr dnssrvr.localdomain
#SCAN IPs (DNS Server now configured to dnssrvr
#192.168.56.94 ora19cscan
#192.168.56.95 ora19cscan
#192.168.56.96 ora19cscan
```

On srv1 and srv2, edit /etc/resolv.conf

(This change will have to be done on srv2 as well)

```
; generated by /usr/sbin/dhclient-script
; nameserver 192.168.1.254
nameserver 192.168.56.80
```

On srv1 and srv2, edit /etc/nsswitch.conf

Edit the entry for hosts to specify “dns” before “file”

```
#hosts:      db files nisplus nis dns
#hosts:      files dns myhostname
#edited to specify dns before files
hosts:      dns files myhostname
```

On srv1 and srv2, set /etc/resolv.conf to be immutable

(The “chattr +i” command makes the file immutable – else, every reboot will cause DHCP to update and reset the file and remove the 192.168.56.80 nameserver entry)

```
[root@srv1 ~]# vi /etc/hosts
[root@srv1 ~]# vi /etc/resolv.conf
[root@srv1 ~]# vi /etc/nsswitch.conf
[root@srv1 ~]# chattr +i /etc/resolv.conf
```

```
[root@srv2 ~]# vi /etc/hosts
[root@srv2 ~]# vi /etc/resolv.conf
[root@srv2 ~]# vi /etc/nsswitch.conf
[root@srv2 ~]# chattr +i /etc/resolv.conf
```

On srv1 and srv2, run nslookup for the SCAN name

It should show a different sequence of IP addresses, using Round-Robin

```
[root@srv1 ~]# nslookup ora19cscan
Server:      192.168.56.80
Address:     192.168.56.80#53
```

```
Name:   ora19cscan
Address: 192.168.56.96
Name:   ora19cscan
Address: 192.168.56.94
Name:   ora19cscan
Address: 192.168.56.95
```

```
[root@srv1 ~]# ora19cscan
-bash: ora19cscan: command not found
[root@srv1 ~]# nslookup ora19cscan
Server:      192.168.56.80
```

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Address: 192.168.56.80#53

Name: ora19cscan
Address: 192.168.56.94
Name: ora19cscan
Address: 192.168.56.95
Name: ora19cscan
Address: 192.168.56.96

[root@srv1 ~]#

[root@srv2 ~]# nslookup ora19cscan
Server: 192.168.56.80
Address: 192.168.56.80#53

Name: ora19cscan
Address: 192.168.56.95
Name: ora19cscan
Address: 192.168.56.96
Name: ora19cscan
Address: 192.168.56.94

[root@srv2 ~]# nslookup ora19cscan
Server: 192.168.56.80
Address: 192.168.56.80#53

Name: ora19cscan
Address: 192.168.56.96
Name: ora19cscan
Address: 192.168.56.94
Name: ora19cscan
Address: 192.168.56.95

[root@srv2 ~]#

Test Other addresses

[root@srv1 etc]# nslookup www.google.com
Server: 192.168.56.80
Address: 192.168.56.80#53

Non-authoritative answer:

Name: www.google.com
Address: 74.125.24.99
Name: www.google.com
Address: 74.125.24.104
Name: www.google.com
Address: 74.125.24.106
Name: www.google.com
Address: 74.125.24.103
Name: www.google.com
Address: 74.125.24.105
Name: www.google.com
Address: 74.125.24.147
Name: www.google.com
Address: 2404:6800:4003:c01::67
Name: www.google.com
Address: 2404:6800:4003:c01::6a
Name: www.google.com
Address: 2404:6800:4003:c01::69

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Name: www.google.com
Address: 2404:6800:4003:c01::68

```
[root@srv1 etc]#
```

Reboot all three servers and test DNS Lookups again

Note : dnssrvr must reboot first so that DNS lookups from srvr and srv2 are successful

After the reboot, check that /etc/resolv.conf did not get auto-updated by DHCP

Note how nslookup for the SCAN name returns the IP addresses in different orders (round-robin) each time

```
[root@srv1 ~]# cat /etc/resolv.conf
; generated by /usr/sbin/dhclient-script
; nameserver 192.168.1.254
nameserver 192.168.56.80
[root@srv1 ~]# nslookup ora19cscan
Server:          192.168.56.80
Address:         192.168.56.80#53
```

```
Name:   ora19cscan
Address: 192.168.56.96
Name:   ora19cscan
Address: 192.168.56.94
Name:   ora19cscan
Address: 192.168.56.95
```

```
[root@srv1 ~]# nslookup ora19cscan
Server:          192.168.56.80
Address:         192.168.56.80#53
```

```
Name:   ora19cscan
Address: 192.168.56.94
Name:   ora19cscan
Address: 192.168.56.95
Name:   ora19cscan
Address: 192.168.56.96
```

```
[root@srv1 ~]# ping www.oracle.com
PING e2581.dscx.akamaiedge.net (23.209.64.118) 56(84) bytes of data.
64 bytes from a23-209-64-118.deploy.static.akamaitechnologies.com
(23.209.64.118): icmp_seq=1 ttl=42 time=198 ms
64 bytes from a23-209-64-118.deploy.static.akamaitechnologies.com
(23.209.64.118): icmp_seq=2 ttl=42 time=194 ms
^C
--- e2581.dscx.akamaiedge.net ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1000ms
rtt min/avg/max/mdev = 194.557/196.663/198.769/2.106 ms
[root@srv1 ~]#
```

```
[root@srv2 ~]# cat /etc/resolv.conf
; generated by /usr/sbin/dhclient-script
; nameserver 192.168.1.254
nameserver 192.168.56.80
[root@srv2 ~]# nslookup ora19cscan
Server:          192.168.56.80
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

Address: 192.168.56.80#53

Name: ora19cscan
Address: 192.168.56.95
Name: ora19cscan
Address: 192.168.56.96
Name: ora19cscan
Address: 192.168.56.94

```
[root@srv2 ~]# nslookup ora19cscan
Server:      192.168.56.80
Address:     192.168.56.80#53
```

Name: ora19cscan
Address: 192.168.56.96
Name: ora19cscan
Address: 192.168.56.94
Name: ora19cscan
Address: 192.168.56.95

```
[root@srv2 ~]# ping www.google.com
PING www.google.com (74.125.130.105) 56(84) bytes of data.
64 bytes from sb-in-f105.1e100.net (74.125.130.105): icmp_seq=1 ttl=100
time=6.46 ms
64 bytes from sb-in-f105.1e100.net (74.125.130.105): icmp_seq=2 ttl=100
time=8.60 ms
^C
--- www.google.com ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 6.464/7.534/8.605/1.074 ms
[root@srv2 ~]#
```

Create the ASM “Disks”

Note : Shutdown both srv1 and srv2 before creating and attaching the ASM disks

Note : Although we could create 1 single Disk and DiskGroup for each of “OCR+VOTE”, “DATA” and “FRA”, I plan to create HighRedundancy DiskGroup (**5 Disks**) for “OCR+VOTE” and NormalRedundancy (**2 Disks**) for each of “DATA” and “FRA”. So I need 5 Disks of 2GB each for “OCR+VOTE” and 4 Disks of 15GB each for “DATA” and “FRA”

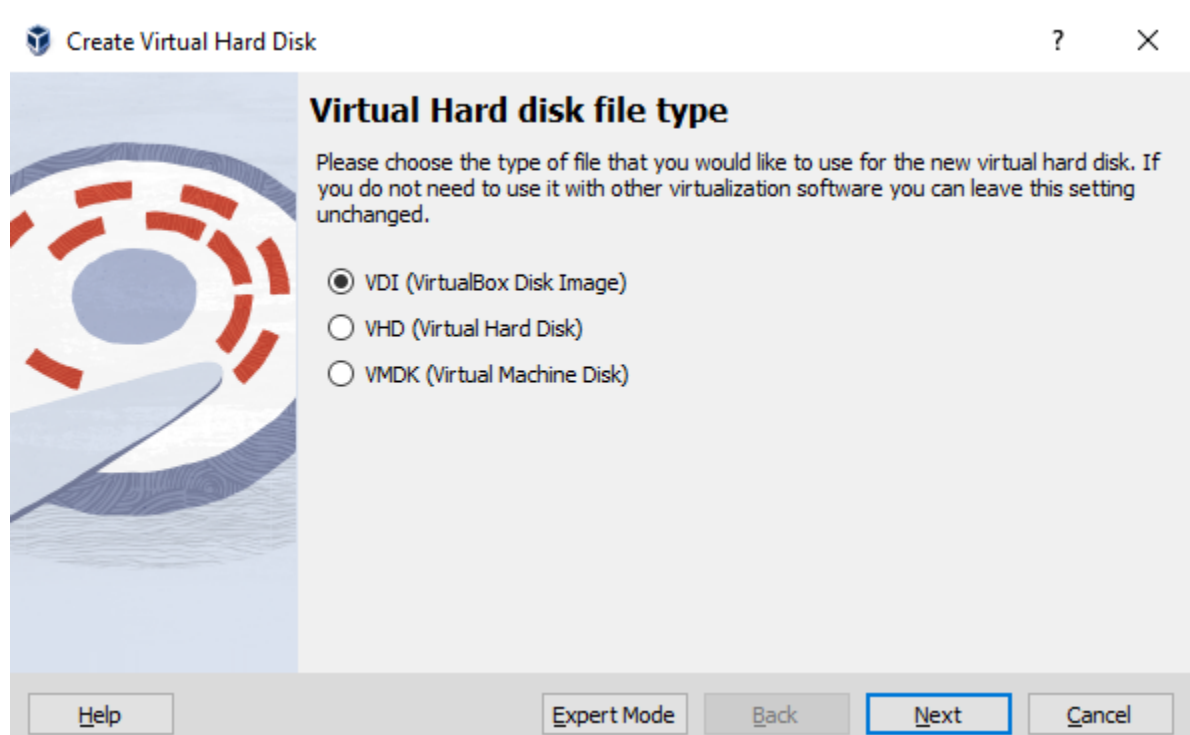
As GIMR (Grid Infrastructure Management Repository) is not mandatory in 19c, I am not installing it. Else, I would need 100GB

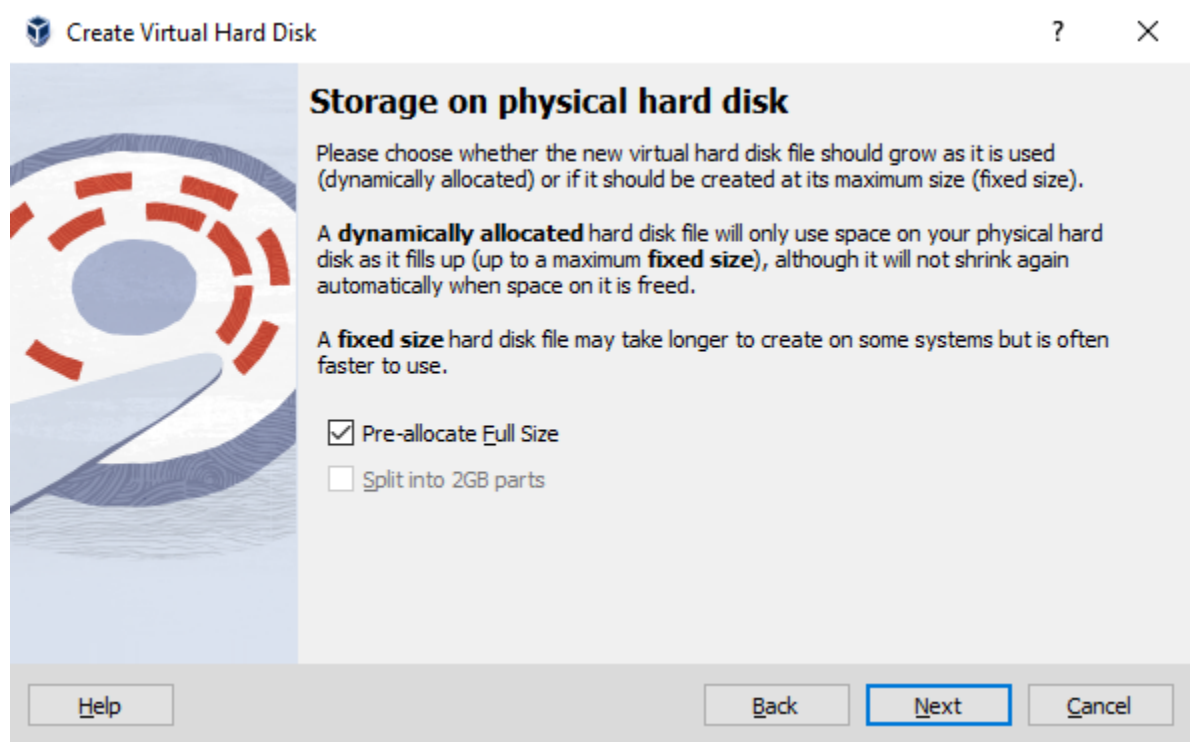
However, with multiple disks of different sizes, you may see that that two servers do not show the same disk names with the same sizes at each reboot. The consistency will be implemented when the disks are labelled by ASM during the Grid Infrastructure installation. See the note under “[Create Partitions from srv1](#)”

Create a separate folder for the ASM Disks on the Host PC.

From Virtual Box select File→Tools→Virtual Media Manager

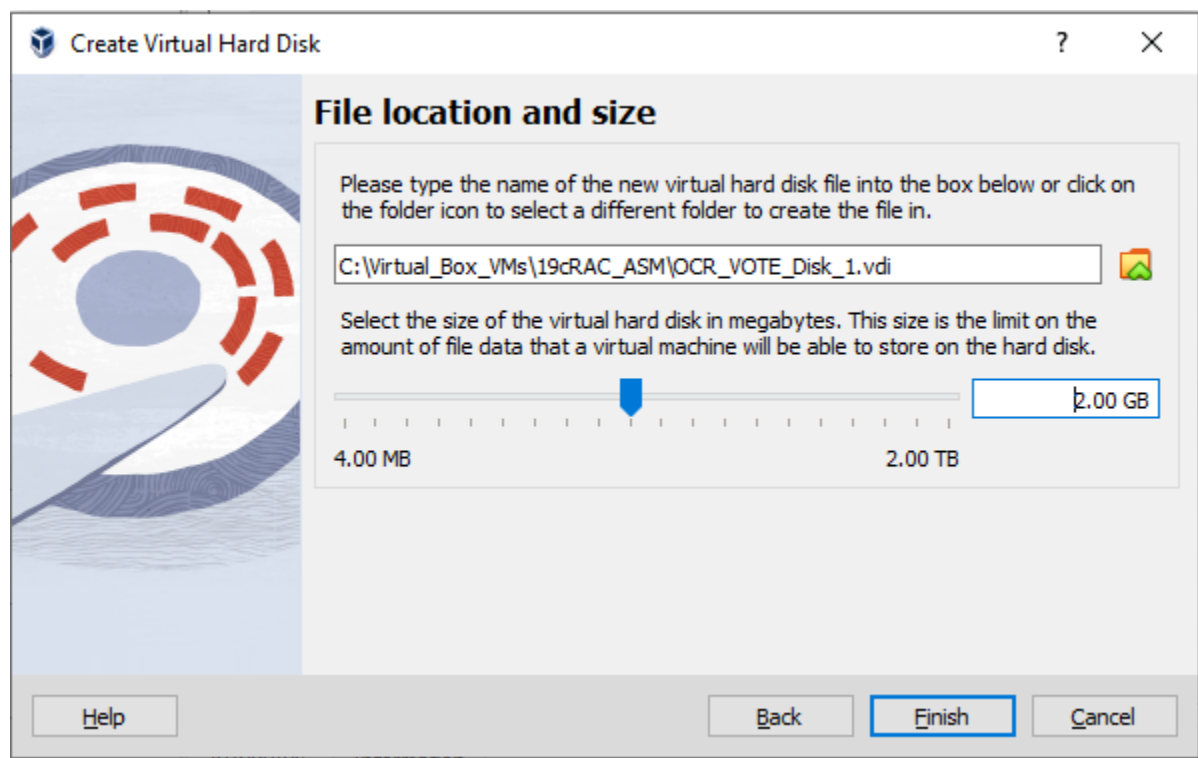
Select “Create Disk Image” and “Virtual Disk Image” for each of these 9 Disks (the first 5 are of 2GB each, the next 4 are of 15GB each)





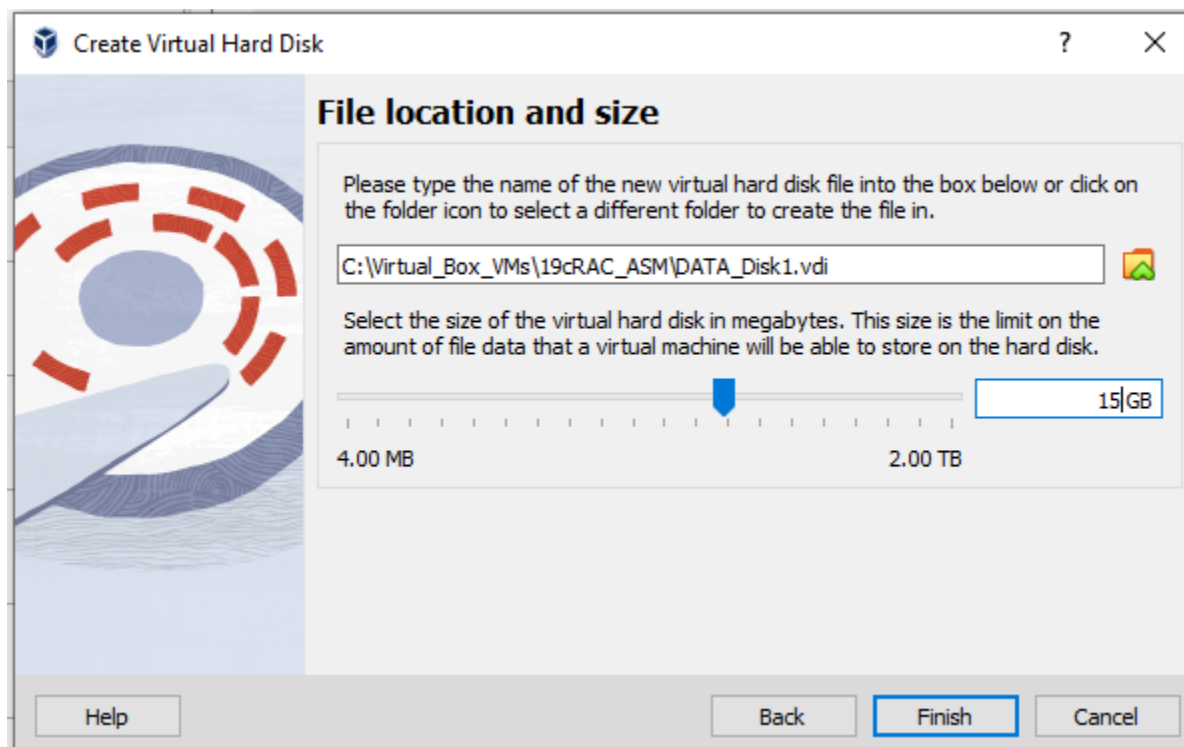
OCR+VOTE

For OCR+VOTE create 5 disks of 2GB each (Disk1 shown below)



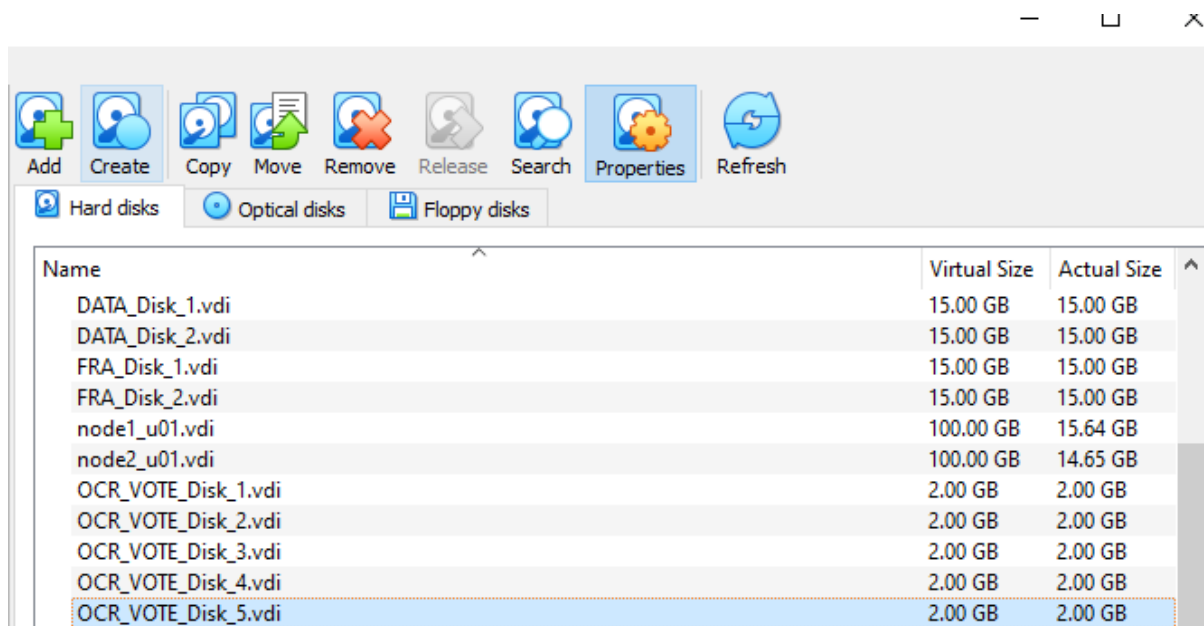
DATA and FRA

For each of DATA and FRA create 2 Disks of 15GB each (DATA_Disk1 shown below)

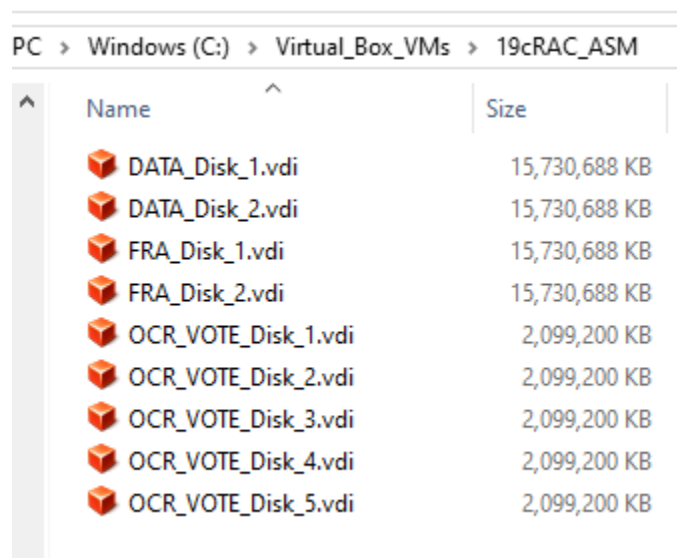


List of Disks

(Ignore “node1_u01” and “node2_u01” -- they are of two other VMs. The “srv1” and “srv2” disks are not seen as I have shown only some of the Virtual Box disks in the screenshot above)



You would have 9 “Disks” in the Folder :



Make the Disks Sharable between srv1 and srv2

Still in the Virtual Box Virtual Media Manager, right-click on each Disk, select “Properties” and set the Type to “Shareable” and click on “Apply” for all 9 Disks

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OCR_VOTE_Disk_1.vdi	2.00 GB	2.00 GB
OCR_VOTE_Disk_2.vdi	2.00 GB	2.00 GB
OCR_VOTE_Disk_3.vdi	2.00 GB	2.00 GB
OCR_VOTE_Disk_4.vdi	2.00 GB	2.00 GB
OCR_VOTE_Disk_5.vdi	2.00 GB	2.00 GB

Attributes Information

Type: Shareable

Location: C:\Virtual_Box_VMs\19cRAC_ASM\OCR_VOTE_Disk_1.vdi

Description:

Size: 4.00 MB 2.00 TB

Apply Reset

Each will currently appear to be “Not Attached”

OCR_VOTE_Disk_1.vdi	2.00 GB	2.00 GB
OCR_VOTE_Disk_2.vdi	2.00 GB	2.00 GB
OCR_VOTE_Disk_3.vdi	2.00 GB	2.00 GB
OCR_VOTE_Disk_4.vdi	2.00 GB	2.00 GB
OCR_VOTE_Disk_5.vdi	2.00 GB	2.00 GB

Attributes Information

Format: VDI

Storage details: Fixed size storage

Attached to: Not Attached

Encrypted with key: Not Encrypted

UUID: {dd3f3e0d-40aa-4ef2-a345-504253144585}

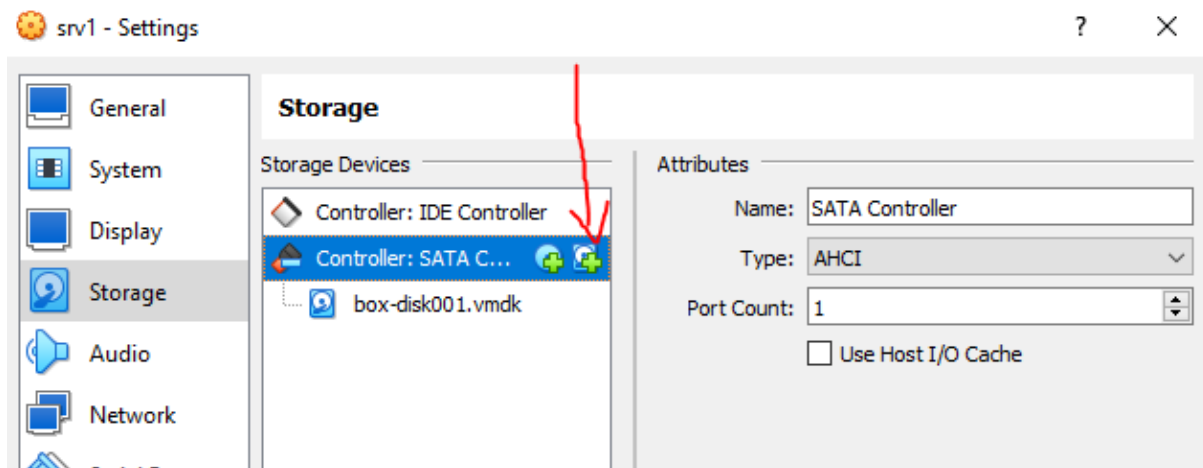
Attach the Disks to srv1 and srv2

(The Disks do not need to be added to the DNS Server dnssrvr)

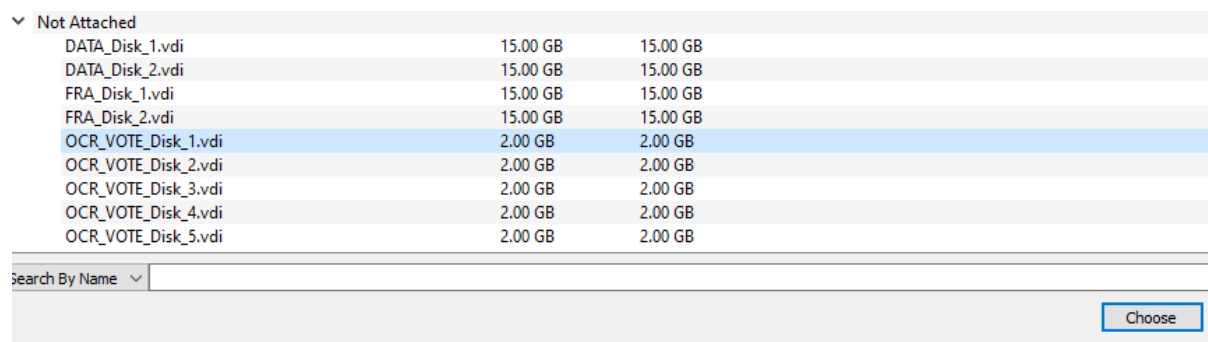
In Virtual Box, on srv1, select Settings → Storage and Add Hard Disk to the existing *SATA* Controller (not the IDE Controller) for all 9 disks

Remember to add each of the Disks in the same sequence that they were created (OCR_VOTE_Disk_1..Disk_5, DATA_Disk_1..Disk_2, FRA_Disk_1..Disk_2)

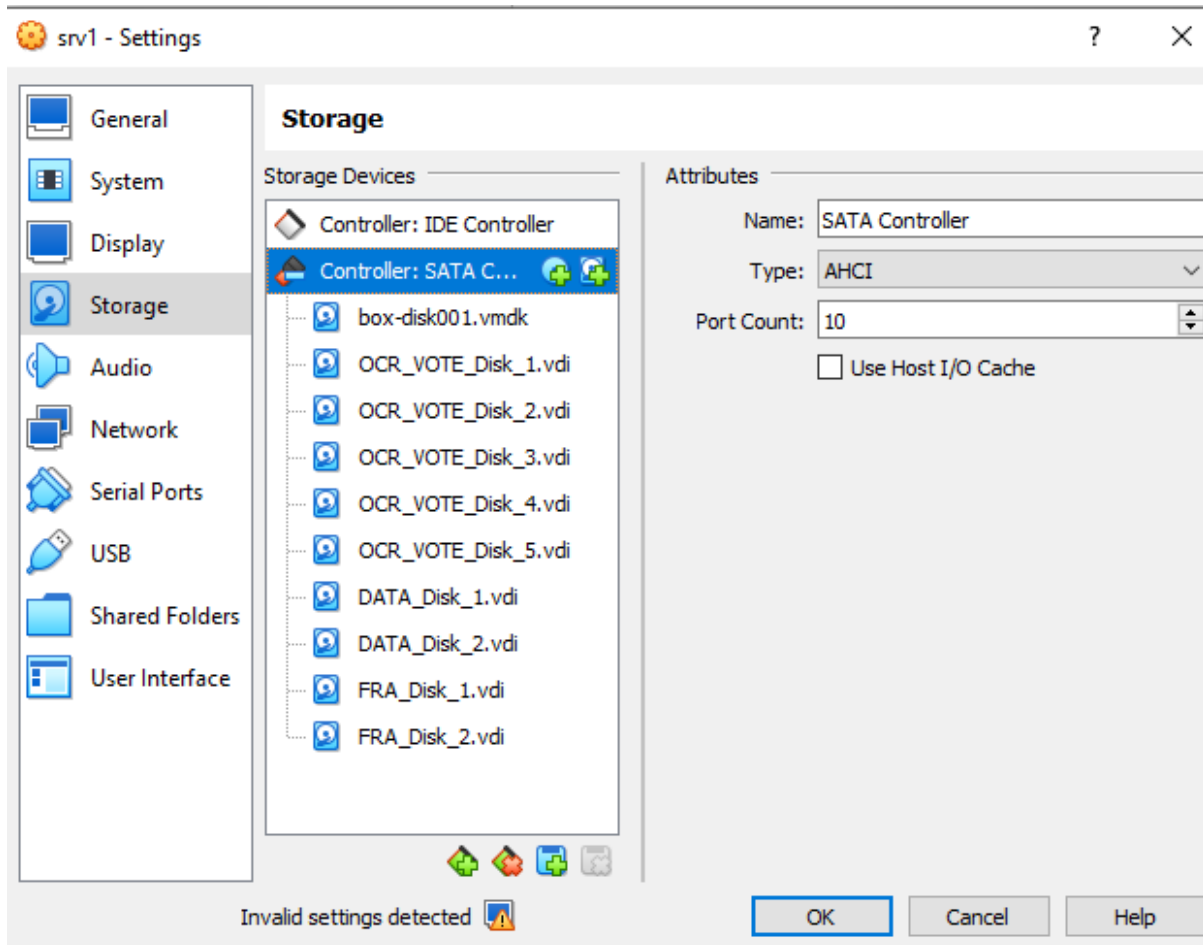
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Select the Disk and click on “Choose”

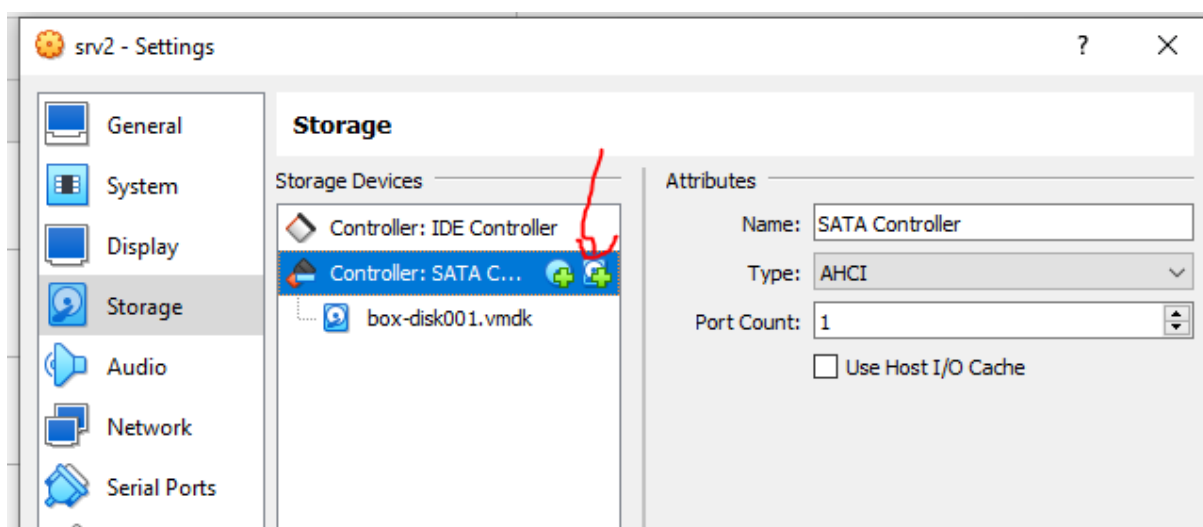


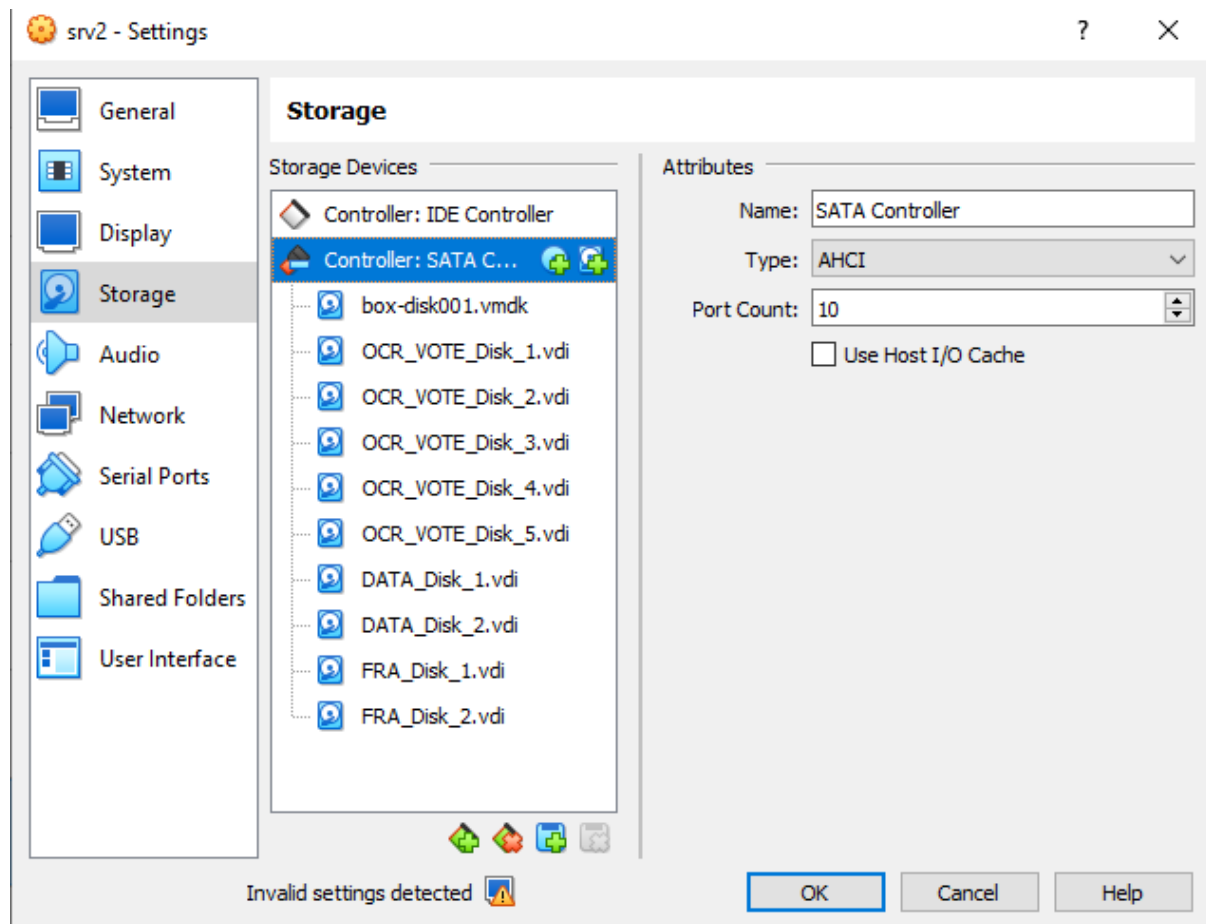
After adding each Disk, you should see them all as Attached :



Repeat the Steps on srv2

(when adding them on srv2, they will appear in the “Attached” list of Virtual Media, but since they have been set to “Shareable” they can be attached to srv2)





Start srv1 (only) and Verify the Disks

They Disks will appear as /dev/sdb to /dev/sdj

```
[root@srv1 ~]# cd /dev
[root@srv1 dev]# ls -l sd*
brw-rw----. 1 root disk 8,  0 Mar 21 07:50 sda
brw-rw----. 1 root disk 8,  1 Mar 21 07:50 sda1
brw-rw----. 1 root disk 8,  2 Mar 21 07:50 sda2
brw-rw----. 1 root disk 8, 16 Mar 21 07:50 sdb
brw-rw----. 1 root disk 8, 32 Mar 21 07:50 sdc
brw-rw----. 1 root disk 8, 48 Mar 21 07:50 sdd
brw-rw----. 1 root disk 8, 64 Mar 21 07:50 sde
brw-rw----. 1 root disk 8, 80 Mar 21 07:50 sdf
brw-rw----. 1 root disk 8, 96 Mar 21 07:50 sdg
brw-rw----. 1 root disk 8, 112 Mar 21 07:50 sdh
brw-rw----. 1 root disk 8, 128 Mar 21 07:50 sdi
brw-rw----. 1 root disk 8, 144 Mar 21 07:50 sdj
[root@srv1 dev]#
```

Check the Disk Sizes

This shows that /dev/sdb is 2GB in size ::

```
[root@srv1 dev]# fdisk /dev/sdb
Welcome to fdisk (util-linux 2.23.2).
```

Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table
Building a new DOS disklabel with disk identifier 0x71c5c030.

Command (m for help): m

Command action

- a toggle a bootable flag
- b edit bsd disklabel
- c toggle the dos compatibility flag
- d delete a partition
- g create a new empty GPT partition table
- G create an IRIX (SGI) partition table
- l list known partition types
- m print this menu
- n add a new partition
- o create a new empty DOS partition table
- p print the partition table
- q quit without saving changes
- s create a new empty Sun disklabel
- t change a partition's system id
- u change display/entry units
- v verify the partition table
- w write table to disk and exit
- x extra functionality (experts only)

Command (m for help): p

```
Disk /dev/sdb: 2147 MB, 2147483648 bytes, 4194304 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk label type: dos
Disk identifier: 0x71c5c030
```

Device	Boot	Start	End	Blocks	Id	System
--------	------	-------	-----	--------	----	--------

Command (m for help): q

```
[root@srv1 dev]#
```

Check ALL 9 DISKS

This shows all the disks from srv1 (the 2GB disks will appear as “2147 MB” and the 15GB disks will appear “as 16.1 GB)

NOTE : If you reboot the server before you execute the steps to label the disks with ASM, the sequence may change at each reboot. (e.g /dev/sdd may appear as 16.1 GB and some other disk lower may appear as 2147 MB) This is because drive names are not persistent unless “udev rules” are configured for persistency.

```
[root@srv1 dev]# fdisk -l /dev/sd[b-j]
```

```
Disk /dev/sdb: 2147 MB, 2147483648 bytes, 4194304 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disk /dev/sdc: 2147 MB, 2147483648 bytes, 4194304 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disk /dev/sdd: 2147 MB, 2147483648 bytes, 4194304 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disk /dev/sde: 2147 MB, 2147483648 bytes, 4194304 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disk /dev/sdf: 2147 MB, 2147483648 bytes, 4194304 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disk /dev/sdg: 16.1 GB, 16106127360 bytes, 31457280 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disk /dev/sdh: 16.1 GB, 16106127360 bytes, 31457280 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disk /dev/sdi: 16.1 GB, 16106127360 bytes, 31457280 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disk /dev/sdj: 16.1 GB, 16106127360 bytes, 31457280 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
[root@srv1 dev]#
```

Create partitions from srv1

Partitions on the new Disks need to be created only from srv1. They can then be verified from both servers.

For each of the 9 disks, create a single “Primary” Partition as large as the disk.
For example this sequence of commands creates the 2GB Partition from the first sector (2048) to the last sector (4194303) on /dev/sdb so it is then listed as /dev/sdb1

Execute similar commands for all 9 disks always specifying the default “First sector” and “Last sector”

NOTE : If you reboot the server before you execute the steps to label the disks with ASM, the sequence may change at each reboot. (e.g /dev/sdd may appear as 16.1 GB and some other disk lower may appear as 2147 MB) This is because drive names are not persistent unless “udev rules” are configured for persistency.

“m” shows the menu of commands

“n” is to create a new partition

“p” (after “n”) is to create a primary partition

“1” to create 1 partition only

select the default “first sector” and “last sector” to specify the whole disk

“p” (at the “Command” menu) to print the partition table

“w” to write the partition table to disk

```
[root@srv1 dev]# fdisk /dev/sdb
Welcome to fdisk (util-linux 2.23.2).
```

Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table
Building a new DOS disklabel with disk identifier 0x4fe81e49.

Command (m for help): m

Command action

- a toggle a bootable flag
- b edit bsd disklabel
- c toggle the dos compatibility flag
- d delete a partition
- g create a new empty GPT partition table
- G create an IRIX (SGI) partition table
- l list known partition types
- m print this menu
- n add a new partition
- o create a new empty DOS partition table
- p print the partition table

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```
q    quit without saving changes
s    create a new empty Sun disklabel
t    change a partition's system id
u    change display/entry units
v    verify the partition table
w    write table to disk and exit
x    extra functionality (experts only)
```

Command (m for help): n

Partition type:

```
p    primary (0 primary, 0 extended, 4 free)
e    extended
```

Select (default p): p

Partition number (1-4, default 1): 1

First sector (2048-4194303, default 2048):

Using default value 2048

Last sector, +sectors or +size{K,M,G} (2048-4194303, default 4194303):

Using default value 4194303

Partition 1 of type Linux and of size 2 GiB is set

Command (m for help): p

Disk /dev/sdb: 2147 MB, 2147483648 bytes, 4194304 sectors

Units = sectors of 1 * 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk label type: dos

Disk identifier: 0x4fe81e49

Device	Boot	Start	End	Blocks	Id	System
/dev/sdb1		2048	4194303	2096128	83	Linux

Command (m for help): w

The partition table has been altered!

Calling ioctl() to re-read partition table.

Syncing disks.

[root@srv1 dev]#

Verify that it shows that Partition 1 has been created

[root@srv1 dev]# ls -l /dev/sdb*

brw-rw----. 1 root disk 8, 16 Mar 21 08:00 /dev/sdb

brw-rw----. 1 root disk 8, 17 Mar 21 08:00 /dev/sdb1

[root@srv1 dev]# fdisk /dev/sdb

Welcome to fdisk (util-linux 2.23.2).

Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Command (m for help): p

Disk /dev/sdb: 2147 MB, 2147483648 bytes, 4194304 sectors

Units = sectors of 1 * 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk label type: dos

Disk identifier: 0x4fe81e49

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Device	Boot	Start	End	Blocks	Id	System
/dev/sdb1		2048	4194303	2096128	83	Linux

Command (m for help): q

[root@srv1 dev]#

Run this sequence for the 5 disks of 2GB and the 4 disks of 15GB

This shows a partition being created on /dev/sdg (15GB) as “16.1 GB) :

```
[root@srv1 dev]# fdisk /dev/sdg
Welcome to fdisk (util-linux 2.23.2).
```

Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table
Building a new DOS disklabel with disk identifier 0x015e8b2d.

Command (m for help): n

Partition type:

 p primary (0 primary, 0 extended, 4 free)
 e extended

Select (default p): p

Partition number (1-4, default 1): 1

First sector (2048-31457279, default 2048):

Using default value 2048

Last sector, +sectors or +size{K,M,G} (2048-31457279, default 31457279):

Using default value 31457279

Partition 1 of type Linux and of size 15 GiB is set

Command (m for help): p

Disk /dev/sdg: 16.1 GB, 16106127360 bytes, 31457280 sectors

Units = sectors of 1 * 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk label type: dos

Disk identifier: 0x015e8b2d

Device	Boot	Start	End	Blocks	Id	System
/dev/sdg1		2048	31457279	15727616	83	Linux

Command (m for help): w

The partition table has been altered!

Calling ioctl() to re-read partition table.

Syncing disks.

[root@srv1 dev]#

Check all the disk partitions

Here /dev/sdb, /dev/sdc, /dev/sdd, /dev/sde and /dev/sdf are the three 2GB disks, each with 1 Partition of the full size of the disk. (Thus, the partition /dev/sdb1 is 2146 MB, i.e. 1MB smaller than the disk /dev/sdb of 2147 MB).

/dev/sdg, /dev/sdh, /dev/sdi and /dev/sdj are the four 15GB (shown as 16.1B) disks

```
[root@srv1 dev]# ls -l /dev/sd[b-j]*
brw-rw----. 1 root disk 8, 16 Mar 21 08:03 /dev/sdb
brw-rw----. 1 root disk 8, 17 Mar 21 08:03 /dev/sdb1
brw-rw----. 1 root disk 8, 32 Mar 21 08:05 /dev/sdc
brw-rw----. 1 root disk 8, 33 Mar 21 08:05 /dev/sdc1
brw-rw----. 1 root disk 8, 48 Mar 21 08:06 /dev/sdd
brw-rw----. 1 root disk 8, 49 Mar 21 08:06 /dev/sdd1
brw-rw----. 1 root disk 8, 64 Mar 21 08:06 /dev/sde
brw-rw----. 1 root disk 8, 65 Mar 21 08:06 /dev/sde1
brw-rw----. 1 root disk 8, 80 Mar 21 08:06 /dev/sdf
brw-rw----. 1 root disk 8, 81 Mar 21 08:06 /dev/sdf1
brw-rw----. 1 root disk 8, 96 Mar 21 08:09 /dev/sdg
brw-rw----. 1 root disk 8, 97 Mar 21 08:09 /dev/sdg1
brw-rw----. 1 root disk 8, 112 Mar 21 08:07 /dev/sdh
brw-rw----. 1 root disk 8, 113 Mar 21 08:07 /dev/sdh1
brw-rw----. 1 root disk 8, 128 Mar 21 08:09 /dev/sdi
brw-rw----. 1 root disk 8, 129 Mar 21 08:09 /dev/sdi1
brw-rw----. 1 root disk 8, 144 Mar 21 08:10 /dev/sdj
brw-rw----. 1 root disk 8, 145 Mar 21 08:10 /dev/sdj1
[root@srv1 dev]# fdisk -l /dev/sd[b-j]*
```

Disk /dev/sdb: 2147 MB, 2147483648 bytes, 4194304 sectors
 Units = sectors of 1 * 512 = 512 bytes
 Sector size (logical/physical): 512 bytes / 512 bytes
 I/O size (minimum/optimal): 512 bytes / 512 bytes
 Disk label type: dos
 Disk identifier: 0x4fe81e49

Device	Boot	Start	End	Blocks	Id	System
/dev/sdb1		2048	4194303	2096128	83	Linux

Disk /dev/sdb1: 2146 MB, 2146435072 bytes, 4192256 sectors
 Units = sectors of 1 * 512 = 512 bytes
 Sector size (logical/physical): 512 bytes / 512 bytes
 I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/sdc: 2147 MB, 2147483648 bytes, 4194304 sectors
 Units = sectors of 1 * 512 = 512 bytes
 Sector size (logical/physical): 512 bytes / 512 bytes
 I/O size (minimum/optimal): 512 bytes / 512 bytes
 Disk label type: dos
 Disk identifier: 0x5c215198

Device	Boot	Start	End	Blocks	Id	System
/dev/sdc1		2048	4194303	2096128	83	Linux

Disk /dev/sdc1: 2146 MB, 2146435072 bytes, 4192256 sectors
 Units = sectors of 1 * 512 = 512 bytes
 Sector size (logical/physical): 512 bytes / 512 bytes
 I/O size (minimum/optimal): 512 bytes / 512 bytes

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Disk /dev/sdd: 2147 MB, 2147483648 bytes, 4194304 sectors

Units = sectors of 1 * 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk label type: dos

Disk identifier: 0x41e7f408

Device	Boot	Start	End	Blocks	Id	System
/dev/sdd1		2048	4194303	2096128	83	Linux

Disk /dev/sdd1: 2146 MB, 2146435072 bytes, 4192256 sectors

Units = sectors of 1 * 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/sde: 2147 MB, 2147483648 bytes, 4194304 sectors

Units = sectors of 1 * 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk label type: dos

Disk identifier: 0x9caace14

Device	Boot	Start	End	Blocks	Id	System
/dev/sde1		2048	4194303	2096128	83	Linux

Disk /dev/sde1: 2146 MB, 2146435072 bytes, 4192256 sectors

Units = sectors of 1 * 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/sdf: 2147 MB, 2147483648 bytes, 4194304 sectors

Units = sectors of 1 * 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk label type: dos

Disk identifier: 0x82074365

Device	Boot	Start	End	Blocks	Id	System
/dev/sdf1		2048	4194303	2096128	83	Linux

Disk /dev/sdf1: 2146 MB, 2146435072 bytes, 4192256 sectors

Units = sectors of 1 * 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/sdg: 16.1 GB, 16106127360 bytes, 31457280 sectors

Units = sectors of 1 * 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk label type: dos

Disk identifier: 0x015e8b2d

Device	Boot	Start	End	Blocks	Id	System
/dev/sdg1		2048	31457279	15727616	83	Linux

Disk /dev/sdg1: 16.1 GB, 16105078784 bytes, 31455232 sectors

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Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/sdh: 16.1 GB, 16106127360 bytes, 31457280 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk label type: dos
Disk identifier: 0x271c3701

Device	Boot	Start	End	Blocks	Id	System
/dev/sdh1		2048	31457279	15727616	83	Linux

Disk /dev/sdh1: 16.1 GB, 16105078784 bytes, 31455232 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/sdi: 16.1 GB, 16106127360 bytes, 31457280 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk label type: dos
Disk identifier: 0x2b8a4b66

Device	Boot	Start	End	Blocks	Id	System
/dev/sdi1		2048	31457279	15727616	83	Linux

Disk /dev/sdi1: 16.1 GB, 16105078784 bytes, 31455232 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/sdj: 16.1 GB, 16106127360 bytes, 31457280 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disk label type: dos
Disk identifier: 0xf374b074

Device	Boot	Start	End	Blocks	Id	System
/dev/sdj1		2048	31457279	15727616	83	Linux

Disk /dev/sdj1: 16.1 GB, 16105078784 bytes, 31455232 sectors
Units = sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

[root@srv1 dev]#

Label the Disks using ASMLib

The command is “oracleasm”

```
[root@srv1 dev]# cd
[root@srv1 ~]#
[root@srv1 ~]# oracleasm
Usage: oracleasm [--exec-path=<exec_path>] <command> [ <args> ]
        oracleasm --exec-path
        oracleasm -h
        oracleasm -V
```

The basic oracleasm commands are:

configure	Configure the Oracle Linux ASMLib driver
init	Load and initialize the ASMLib driver
exit	Stop the ASMLib driver
scandisks	Scan the system for Oracle ASMLib disks
status	Display the status of the Oracle ASMLib driver
listdisks	List known Oracle ASMLib disks
listiids	List the iid files
deleteiids	Delete the unused iid files
querydisk	Determine if a disk belongs to Oracle ASMLib
createdisk	Allocate a device for Oracle ASMLib use
deletedisk	Return a device to the operating system
renamedisk	Change the label of an Oracle ASMLib disk
update-driver	Download the latest ASMLib driver

Create ASM Disks

Use “oracleasm createdisk” to label all the 9 Disk.

Thus, the first 2GB disk (for OCR_VOTE) is labelled as “OCR_VOTE_DISK_1”
(The labels do not have to match the names of the disks created using Virtual Box Virtual Media Manager and visible in Windows Explorer, but it is sensible to provide the same names as labels).

```
[root@srv1 ~]# oracleasm createdisk OCR_VOTE_DISK_1 /dev/sdb1
Writing disk header: done
Instantiating disk: done
[root@srv1 ~]#
```

```
[root@srv1 ~]# oracleasm createdisk OCR_VOTE_DISK_2 /dev/sdc1
Writing disk header: done
Instantiating disk: done
[root@srv1 ~]#
[root@srv1 ~]# oracleasm createdisk OCR_VOTE_DISK_3 /dev/sdd1
Writing disk header: done
Instantiating disk: done
[root@srv1 ~]# oracleasm createdisk OCR_VOTE_DISK_4 /dev/sde1
Writing disk header: done
Instantiating disk: done
[root@srv1 ~]# oracleasm createdisk OCR_VOTE_DISK_5 /dev/sdf1
Writing disk header: done
Instantiating disk: done
[root@srv1 ~]#
```

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```
[root@srv1 ~]# oracleasm createdisk DATA_DISK_1 /dev/sdg1
Writing disk header: done
Instantiating disk: done
[root@srv1 ~]# oracleasm createdisk DATA_DISK_2 /dev/sdh1
Writing disk header: done
Instantiating disk: done
[root@srv1 ~]#

[root@srv1 ~]# oracleasm createdisk FRA_DISK_1 /dev/sdi1
Writing disk header: done
Instantiating disk: done
[root@srv1 ~]# oracleasm createdisk FRA_DISK_2 /dev/sdj1
Writing disk header: done
Instantiating disk: done
[root@srv1 ~]#
```

Scan the ASM Disks

This will refresh the status of the ASM Disks

```
[root@srv1 ~]# oracleasm scandisks
Reloading disk partitions: done
Cleaning any stale ASM disks...
Scanning system for ASM disks...
[root@srv1 ~]#
```

List the ASM Disks

This will verify that the ASM Disks are visible

```
[root@srv1 ~]# oracleasm listdisks
DATA_DISK_1
DATA_DISK_2
FRA_DISK_1
FRA_DISK_2
OCR_VOTE_DISK_1
OCR_VOTE_DISK_2
OCR_VOTE_DISK_3
OCR_VOTE_DISK_4
OCR_VOTE_DISK_5
[root@srv1 ~]#
```

Reboot both Nodes and Verify ASM listing

Reboot both nodes and run “oracleasm listdisks”

Note : Reboot and Verify on srv1 first

After reboot

```
[root@srv1 ~]# oracleasm listdisks
DATA_DISK_1
DATA_DISK_2
```

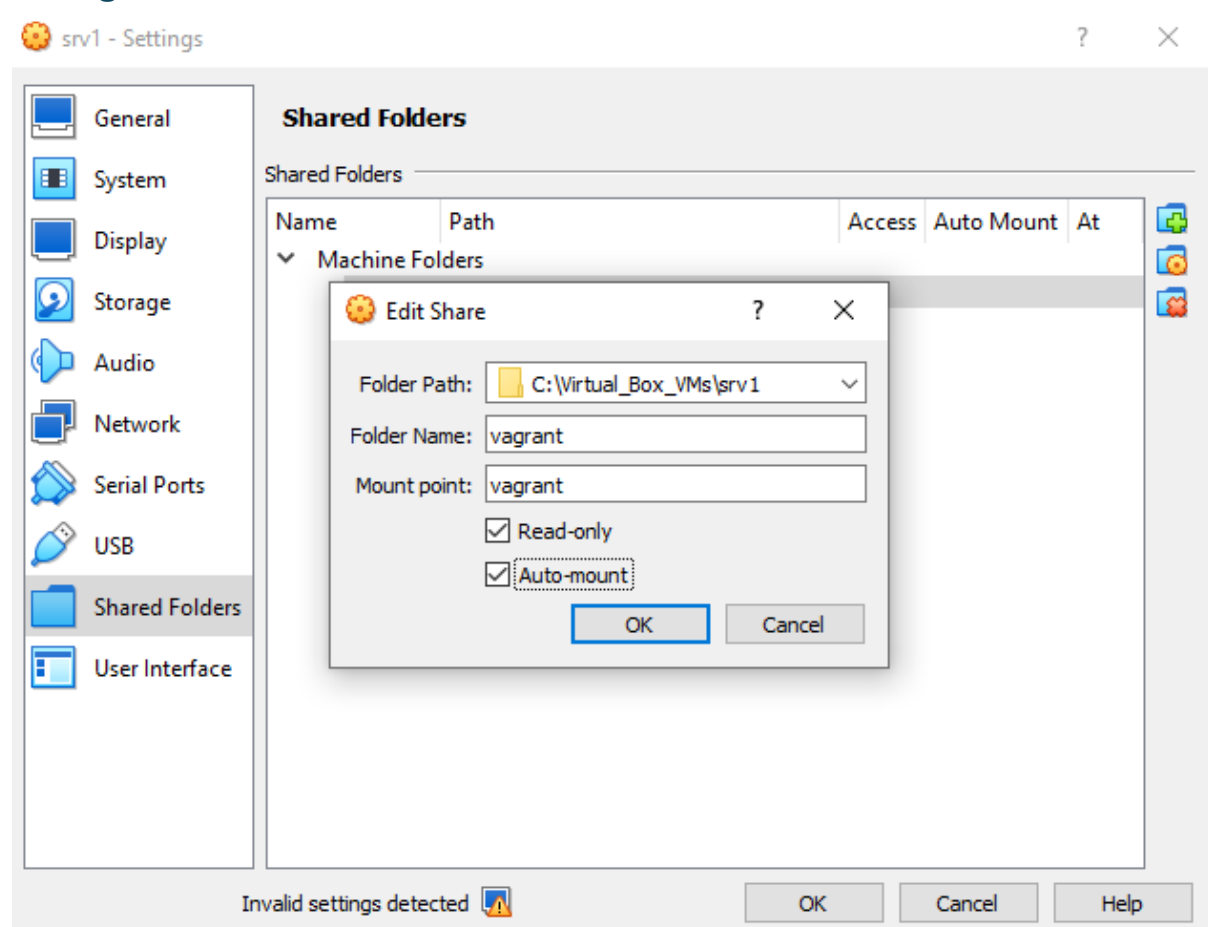
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```
FRA_DISK_1  
FRA_DISK_2  
OCR_VOTE_DISK_1  
OCR_VOTE_DISK_2  
OCR_VOTE_DISK_3  
OCR_VOTE_DISK_4  
OCR_VOTE_DISK_5  
[root@srv1 ~]#
```

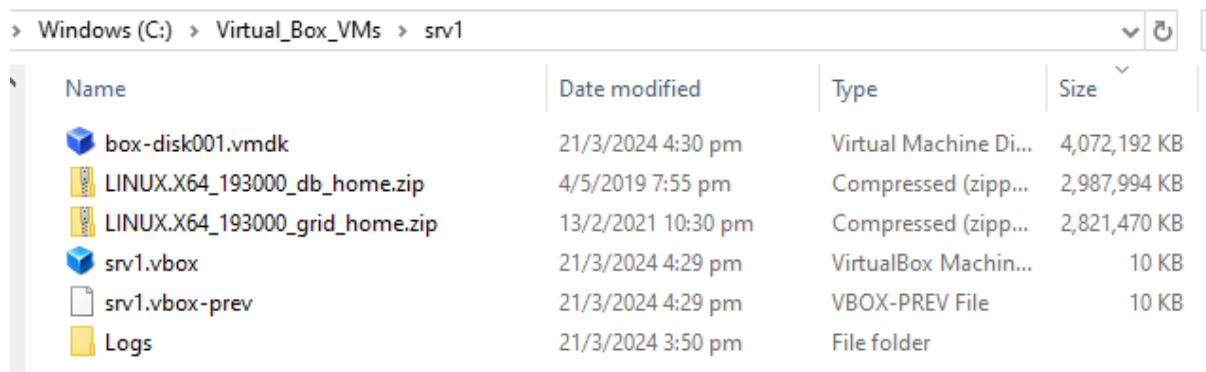
```
[root@srv2 ~]# oracledisk listdisks  
DATA_DISK_1  
DATA_DISK_2  
FRA_DISK_1  
FRA_DISK_2  
OCR_VOTE_DISK_1  
OCR_VOTE_DISK_2  
OCR_VOTE_DISK_3  
OCR_VOTE_DISK_4  
OCR_VOTE_DISK_5  
[root@srv2 ~]#
```

Copy the 19c Installers to srv1

Configure the Shared Folder for the VM



Copy the Installer files to the folder



Name	Date modified	Type	Size
box-disk001.vmdk	21/3/2024 4:30 pm	Virtual Machine Di...	4,072,192 KB
LINUX.X64_193000_db_home.zip	4/5/2019 7:55 pm	Compressed (zipp...	2,987,994 KB
LINUX.X64_193000_grid_home.zip	13/2/2021 10:30 pm	Compressed (zipp...	2,821,470 KB
srv1.vbox	21/3/2024 4:29 pm	VirtualBox Machin...	10 KB
srv1.vbox-prev	21/3/2024 4:29 pm	VBOX-PREV File	10 KB
Logs	21/3/2024 3:50 pm	File folder	

Startup srv1 and verify that the files are available

```
[root@srv1 ~]# df -h
Filesystem                Size      Used Avail Use% Mounted on
devtmpfs                   3.8G         0   3.8G   0% /dev
tmpfs                      3.8G         0   3.8G   0% /dev/shm
tmpfs                      3.8G   8.5M   3.8G   1% /run
tmpfs                      3.8G         0   3.8G   0% /sys/fs/cgroup
/dev/mapper/vg_main-lv_root 33G   3.3G   30G   11% /
/dev/sda1                  497M   163M   335M   33% /boot
vagrant                   954G   792G   162G   84% /media/sf_vagrant
tmpfs                     766M         0   766M   0% /run/user/0

[root@srv1 ~]# cd /media/sf_vagrant
[root@srv1 sf_vagrant]# ls -l *zip
-rwxrwx---. 1 root vboxsf 3059705302 May  4  2019
LINUX.X64_193000_db_home.zip
-rwxrwx---. 1 root vboxsf 2889184573 Feb 13  2021
LINUX.X64_193000_grid_home.zip
[root@srv1 sf_vagrant]#
```

Prepare for Installation of Grid Infrastructure

(Before beginning installation ensure that dnssrvr, srv1 and srv2 are all started up and can “ping” each other by hostname and IP address and test “nslookup ora19cscan” from both srv1 and srv2)

Note : The installation must be initiated only from the first node of the cluster (srv1) – all the files will be automatically copied and configured on the second node (srv2) as part of the execution of the installation.

Create the folders on both nodes

Note : The ORACLE_HOME for the Grid Infrastructure installation must ***NOT*** be a sub-folder of the ORACLE_BASE (as we normally do for the RDBMS installation). The oraInventory folder is separate from the ORACLE_BASE and ORACLE_HOME.

```
[root@srv1 ~]# mkdir -p /u01/app/oracle/product/19.3
[root@srv1 ~]# chown -R oracle:oinstall /u01/app/oracle
[root@srv1 ~]# mkdir -p /u01/app/grid_base
[root@srv1 ~]# mkdir -p /u01/app/grid/product/19.3
[root@srv1 ~]# chown -R grid:oinstall /u01/app/grid_base
[root@srv1 ~]# chown -R grid:oinstall /u01/app/grid/product/19.3
[root@srv1 ~]# mkdir /u01/app/oraInventory
[root@srv1 ~]# chown grid:oinstall /u01/app/oraInventory
[root@srv1 ~]# chmod 775 /u01/app/oraInventory

[root@srv2 dev]# mkdir -p /u01/app/oracle/product/19.3
[root@srv2 dev]# chown -R oracle:oinstall /u01/app/oracle
[root@srv2 dev]# mkdir -p /u01/app/grid_base
[root@srv2 dev]# mkdir -p /u01/app/grid/product/19.3
[root@srv2 dev]# chown -R grid:oinstall /u01/app/grid_base
[root@srv2 dev]# chown -R grid:oinstall /u01/app/grid/product/19.3
[root@srv2 ~]# mkdir /u01/app/oraInventory
[root@srv2 ~]# chown grid:oinstall /u01/app/oraInventory
[root@srv2 ~]# chmod 775 /u01/app/oraInventory
```

Extract the Grid Infrastructure Installer on srv1 (only)

The unzip command unzips the file from the media folder (i.e. mapped drive from Host) to the target location for Grid Infrastructure installation.

The chown command changes ownership to grid.

The chown command must also be run for the hidden folder .patch_storage

```
[root@srv1 19.3]# cd /media/sf_vagrant
[root@srv1 sf_vagrant]# ls *grid*.zip
LINUX.X64_193000_grid_home.zip
[root@srv1 sf_vagrant]# cd /u01/app/grid/product/19.3
[root@srv1 19.3]# unzip /media/sf_vagrant/LINUX.X64_193000_grid_home.zip -d
.
...list of folders and files being extracted appears
...list of folders and files being extracted appears
...listing of symbolic links created by the unzip appears

[root@srv1 19.3]# pwd
/u01/app/grid/product/19.3
[root@srv1 19.3]# ls -l |head -5
total 132
drwxr-xr-x.  2 root root   102 Apr 18  2019 addnode
drwxr-xr-x. 10 root root  4096 Apr 17  2019 assistants
drwxr-xr-x.  2 root root  8192 Apr 18  2019 bin
```

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```
drwxr-xr-x. 3 root root 19 Apr 17 2019 cha
[root@srv1 19.3]#

[root@srv1 19.0]# chown -R grid:oinstall *
[root@srv1 19.0]#
[root@srv1 19.3]# cd .patch_storage
[root@srv1 .patch_storage]# ls -l |head -5
total 24
drwxr-xr-x. 3 root root 74 Apr 18 2019 29401763_Apr_11_2019_22_26_25
drwxr-xr-x. 3 root root 74 Apr 18 2019 29517242_Apr_17_2019_23_27_10
drwxr-xr-x. 3 root root 74 Apr 18 2019 29517247_Apr_1_2019_15_08_20
drwxr-xr-x. 3 root root 74 Apr 18 2019 29585399_Apr_9_2019_19_12_47
[root@srv1 .patch_storage]# chown -R grid:oinstall .
[root@srv1 .patch_storage]# ls -l |head -5
total 24
drwxr-xr-x. 3 grid oinstall 74 Apr 18 2019
29401763_Apr_11_2019_22_26_25
drwxr-xr-x. 3 grid oinstall 74 Apr 18 2019
29517242_Apr_17_2019_23_27_10
drwxr-xr-x. 3 grid oinstall 74 Apr 18 2019 29517247_Apr_1_2019_15_08_20
drwxr-xr-x. 3 grid oinstall 74 Apr 18 2019 29585399_Apr_9_2019_19_12_47
[root@srv1 .patch_storage]# cd ..
[root@srv1 19.3]# pwd
/u01/app/grid/product/19.3
[root@srv1 19.3]# ls -l |head -5
total 132
drwxr-xr-x. 2 grid oinstall 102 Apr 18 2019 addnode
drwxr-xr-x. 10 grid oinstall 4096 Apr 17 2019 assistants
drwxr-xr-x. 2 grid oinstall 8192 Apr 18 2019 bin
drwxr-xr-x. 3 grid oinstall 19 Apr 17 2019 cha
[root@srv1 19.3]#
[root@srv1 19.3]# ls -la .patch_storage
total 32
drwxr-xr-x. 7 grid oinstall 4096 Apr 18 2019 .
drwxr-xr-x. 65 grid oinstall 4096 Mar 21 08:45 ..
drwxr-xr-x. 3 grid oinstall 74 Apr 18 2019
29401763_Apr_11_2019_22_26_25
drwxr-xr-x. 3 grid oinstall 74 Apr 18 2019
29517242_Apr_17_2019_23_27_10
drwxr-xr-x. 3 grid oinstall 74 Apr 18 2019 29517247_Apr_1_2019_15_08_20
drwxr-xr-x. 3 grid oinstall 74 Apr 18 2019 29585399_Apr_9_2019_19_12_47
-rw-r--r--. 1 grid oinstall 6460 Apr 18 2019 interim_inventory.txt
-rw-r--r--. 1 grid oinstall 92 Apr 18 2019
LatestOPatchSession.properties
drwxr-xr-x. 10 grid oinstall 4096 Apr 18 2019 NApply
-rw-r--r--. 1 grid oinstall 5128 Apr 18 2019 record_inventory.txt
[root@srv1 19.3]# ls -ld .patch_storage
drwxr-xr-x. 7 grid oinstall 4096 Apr 18 2019 .patch_storage
[root@srv1 19.3]#
```

Install the cvuqdisk rpm that is in the unzipped installer

```
[root@srv1 19.3]# cd /u01/app/grid/product/19.3/cv/rpm
[root@srv1 rpm]# ls -l cvuq*
-rw-r--r--. 1 grid oinstall 11412 Mar 13 2019 cvuqdisk-1.0.10-1.rpm
[root@srv1 rpm]# rpm -iv cvuqdisk-1.0.10-1.rpm
Preparing packages...
```

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```
Using default group oinstall to install package
cvuqdisk-1.0.10-1.x86_64
[root@srv1 rpm]#
```

Copy this rpm to srv2 and install it there as well

```
[root@srv2 ~]# cd /tmp
[root@srv2 tmp]# scp root@srv1:/u01/app/grid/product/19.3/cv/rpm/cvuqdisk-1.0.10-1.rpm .
root@srv1's password:
cvuqdisk-1.0.10-1.rpm
100% 11KB 6.4MB/s 00:00
[root@srv2 tmp]# rpm -iv cvuqdisk-1.0.10-1.rpm
Preparing packages...
Using default group oinstall to install package
cvuqdisk-1.0.10-1.x86_64
[root@srv2 tmp]#
```

Configure grid user Environment

Edit the .bash_profile for grid to set the environment parameters for ORACLE_BASE, ORACLE_HOME, ORACLE_SID and PATH and set umask

```
[root@srv1 ~]# su - grid
Last login: Thu Mar 14 10:21:04 UTC 2024 on pts/0
[grid@srv1 ~]$ echo $SHELL
/bin/bash
[grid@srv1 ~]$ vi .bash_profile
[grid@srv1 ~]$ cat .bash_profile
# .bash_profile

# Get the aliases and functions
if [ -f ~/.bashrc ]; then
    . ~/.bashrc
fi

# User specific environment and startup programs

PATH=$PATH:$HOME/.local/bin:$HOME/bin

export PATH

ORACLE_BASE=/u01/app/grid_base ; export ORACLE_BASE
ORACLE_HOME=/u01/app/grid/product/19.3 ; export ORACLE_HOME
ORACLE_SID=+ASM1 ; export ORACLE_SID
PATH=$PATH:$ORACLE_HOME/bin ; export PATH

umask 0022

[grid@srv1 ~]$
```

On srv2, set the same variables, **except that ORACLE_SID would be "+ASM2"**

```
[root@srv2 ~]# su - grid
[grid@srv2 ~]$ vi .bash_profile
```

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```
[grid@srv2 ~]$ cat .bash_profile
# .bash_profile

# Get the aliases and functions
if [ -f ~/.bashrc ]; then
    . ~/.bashrc
fi

# User specific environment and startup programs

PATH=$PATH:$HOME/.local/bin:$HOME/bin

export PATH

ORACLE_BASE=/u01/app/grid_base ; export ORACLE_BASE
ORACLE_HOME=/u01/app/grid/product/19.3 ; export ORACLE_HOME
ORACLE_SID=+ASM2 ; export ORACLE_SID
PATH=$PATH:$ORACLE_HOME/bin ; export PATH

umask 0022

[grid@srv2 ~]$
```

Verify that the variables are set :

```
[grid@srv1 ~]$ exit
logout
[root@srv1 ~]# su - grid
Last login: Thu Mar 21 08:52:16 UTC 2024 on pts/0
[grid@srv1 ~]$ env |grep ORA
ORACLE_SID=+ASM1
ORACLE_BASE=/u01/app/grid_base
ORACLE_HOME=/u01/app/grid/product/19.3
[grid@srv1 ~]$ umask
0022
[grid@srv1 ~]$
```

```
[grid@srv2 ~]$ env |grep ORA
ORACLE_SID=+ASM2
ORACLE_BASE=/u01/app/grid_base
ORACLE_HOME=/u01/app/grid/product/19.3
[grid@srv2 ~]$ umask
0022
[grid@srv2 ~]$
```

Install Grid Infrastructure / ClusterWare

Pre-Checks

nslookup

Note : Recheck nslookup to confirm that the ora19cscan entry is being returned by the DNS server in a round-robin manner at both servers srv1 and srv2

```
[root@srv1 ~]# nslookup ora19cscan
Server:          192.168.56.80
Address:         192.168.56.80#53
```

```
Name:   ora19cscan
Address: 192.168.56.96
Name:   ora19cscan
Address: 192.168.56.94
Name:   ora19cscan
Address: 192.168.56.95
```

```
[root@srv1 ~]# nslookup ora19cscan
Server:          192.168.56.80
Address:         192.168.56.80#53
```

```
Name:   ora19cscan
Address: 192.168.56.94
Name:   ora19cscan
Address: 192.168.56.95
Name:   ora19cscan
Address: 192.168.56.96
```

```
[root@srv1 ~]# nslookup srv2
Server:          192.168.56.80
Address:         192.168.56.80#53
```

```
Name:   srv2
Address: 192.168.56.92
```

```
[root@srv1 ~]#
```

```
[root@srv2 ~]# nslookup ora19cscan
Server:          192.168.56.80
Address:         192.168.56.80#53
```

```
Name:   ora19cscan
Address: 192.168.56.95
Name:   ora19cscan
Address: 192.168.56.96
Name:   ora19cscan
Address: 192.168.56.94
```

```
[root@srv2 ~]# nslookup ora19cscan
Server:          192.168.56.80
Address:         192.168.56.80#53
```

```
Name:   ora19cscan
Address: 192.168.56.96
Name:   ora19cscan
Address: 192.168.56.94
```

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Name: ora19cscan
Address: 192.168.56.95

```
[root@srv2 ~]# nslookup srv1
Server:      192.168.56.80
Address:     192.168.56.80#53
```

Name: srv1
Address: 192.168.56.90

```
[root@srv2 ~]#
```

oracleasm listdisks

```
[root@srv1 ~]# oracleasm listdisks
DATA_DISK_1
DATA_DISK_2
FRA_DISK_1
FRA_DISK_2
OCR_VOTE_DISK_1
OCR_VOTE_DISK_2
OCR_VOTE_DISK_3
OCR_VOTE_DISK_4
OCR_VOTE_DISK_5
[root@srv1 ~]#
[root@srv1 ~]# ls -l /dev/oracleasm/disks
total 0
brw-rw----. 1 grid oinstall 8,  97 Mar 21 08:34 DATA_DISK_1
brw-rw----. 1 grid oinstall 8, 113 Mar 21 08:34 DATA_DISK_2
brw-rw----. 1 grid oinstall 8, 129 Mar 21 08:34 FRA_DISK_1
brw-rw----. 1 grid oinstall 8, 145 Mar 21 08:34 FRA_DISK_2
brw-rw----. 1 grid oinstall 8,  17 Mar 21 08:34 OCR_VOTE_DISK_1
brw-rw----. 1 grid oinstall 8,  33 Mar 21 08:34 OCR_VOTE_DISK_2
brw-rw----. 1 grid oinstall 8,  65 Mar 21 08:34 OCR_VOTE_DISK_3
brw-rw----. 1 grid oinstall 8,  49 Mar 21 08:34 OCR_VOTE_DISK_4
brw-rw----. 1 grid oinstall 8,  81 Mar 21 08:34 OCR_VOTE_DISK_5
[root@srv1 ~]#
```

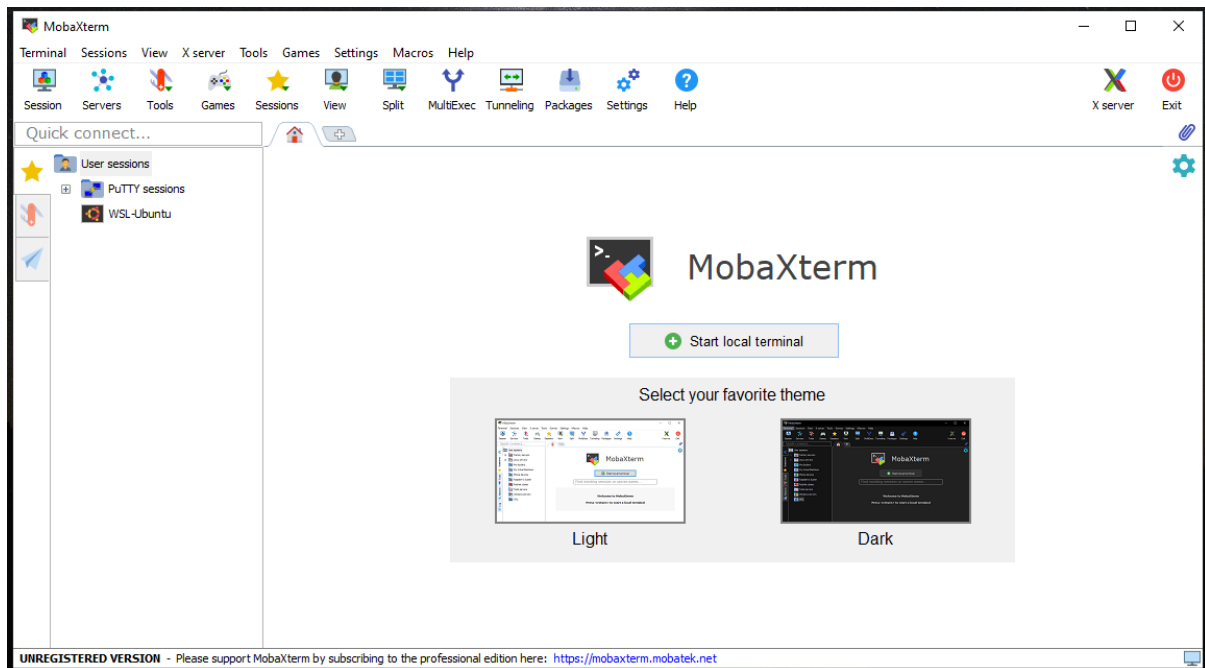
```
[root@srv2 ~]# oracleasm listdisks
DATA_DISK_1
DATA_DISK_2
FRA_DISK_1
FRA_DISK_2
OCR_VOTE_DISK_1
OCR_VOTE_DISK_2
OCR_VOTE_DISK_3
OCR_VOTE_DISK_4
OCR_VOTE_DISK_5
[root@srv2 ~]#
[root@srv2 ~]# ls -l /dev/oracleasm/disks
total 0
brw-rw----. 1 grid oinstall 8,  81 Mar 21 08:31 DATA_DISK_1
brw-rw----. 1 grid oinstall 8, 113 Mar 21 08:31 DATA_DISK_2
brw-rw----. 1 grid oinstall 8, 129 Mar 21 08:31 FRA_DISK_1
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
brw-rw----. 1 grid oinstall 8, 145 Mar 21 08:31 FRA_DISK_2
brw-rw----. 1 grid oinstall 8,  17 Mar 21 08:31 OCR_VOTE_DISK_1
brw-rw----. 1 grid oinstall 8,  33 Mar 21 08:31 OCR_VOTE_DISK_2
brw-rw----. 1 grid oinstall 8,  49 Mar 21 08:31 OCR_VOTE_DISK_3
brw-rw----. 1 grid oinstall 8,  65 Mar 21 08:31 OCR_VOTE_DISK_4
brw-rw----. 1 grid oinstall 8,  97 Mar 21 08:31 OCR_VOTE_DISK_5
[root@srv2 ~]#
```

Start X-Server

Start the installed X-Server (MobaXterm on my Desktop PC)

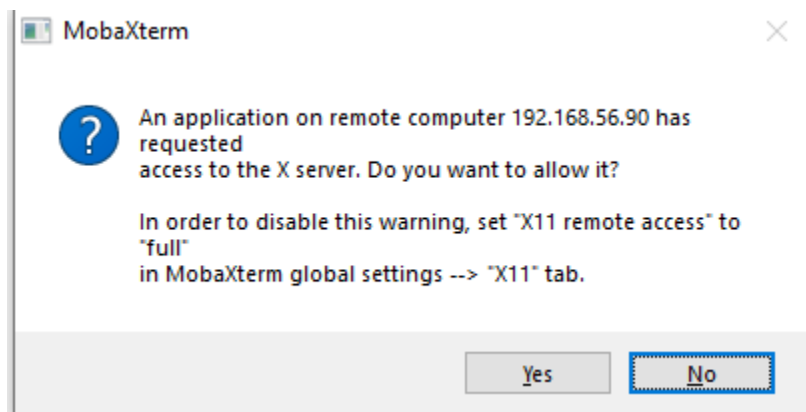


Grid Infrastructure Installation

Note : Set “DISPLAY” to the gateway IP address that is hosted by Virtual Box on your PC

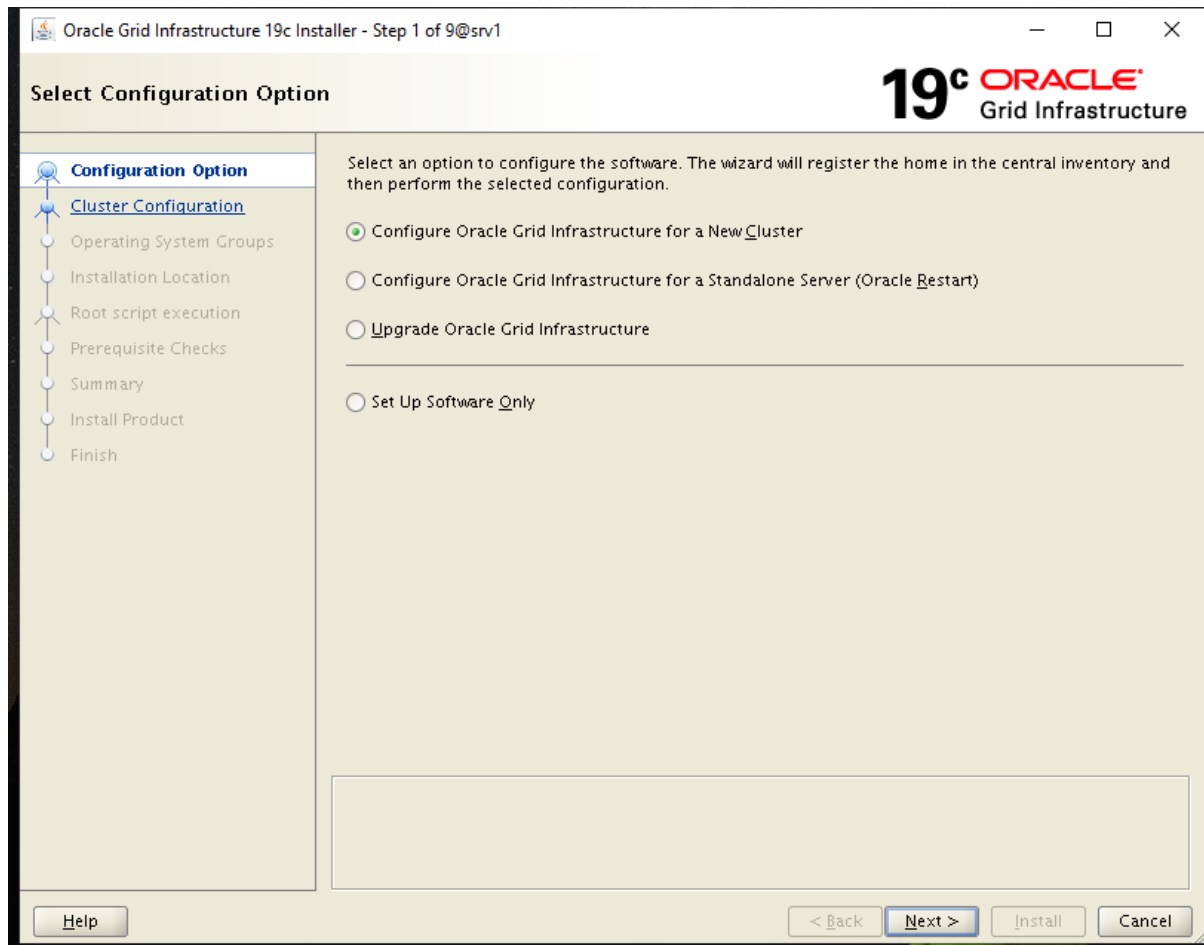
```
[root@srv1 ~]# su - grid
[grid@srv1 ~]$ env |grep ORA
ORACLE_SID=+ASM1
ORACLE_BASE=/u01/app/grid_base
ORACLE_HOME=/u01/app/grid/product/19.3
[grid@srv1 ~]$ umask
0022
[grid@srv1 ~]$ export DISPLAY=192.168.56.1:0.0
[grid@srv1 ~]$ cd $ORACLE_HOME
[grid@srv1 19.3]$ ls -l grid*
-rwxr-x---. 1 grid oinstall 3294 Mar  8 2017 gridSetup.sh
[grid@srv1 19.3]$ ./gridSetup.sh
```

Allow the gridSetup connection to X-Server with “Yes” at : (You will see this prompt twice)



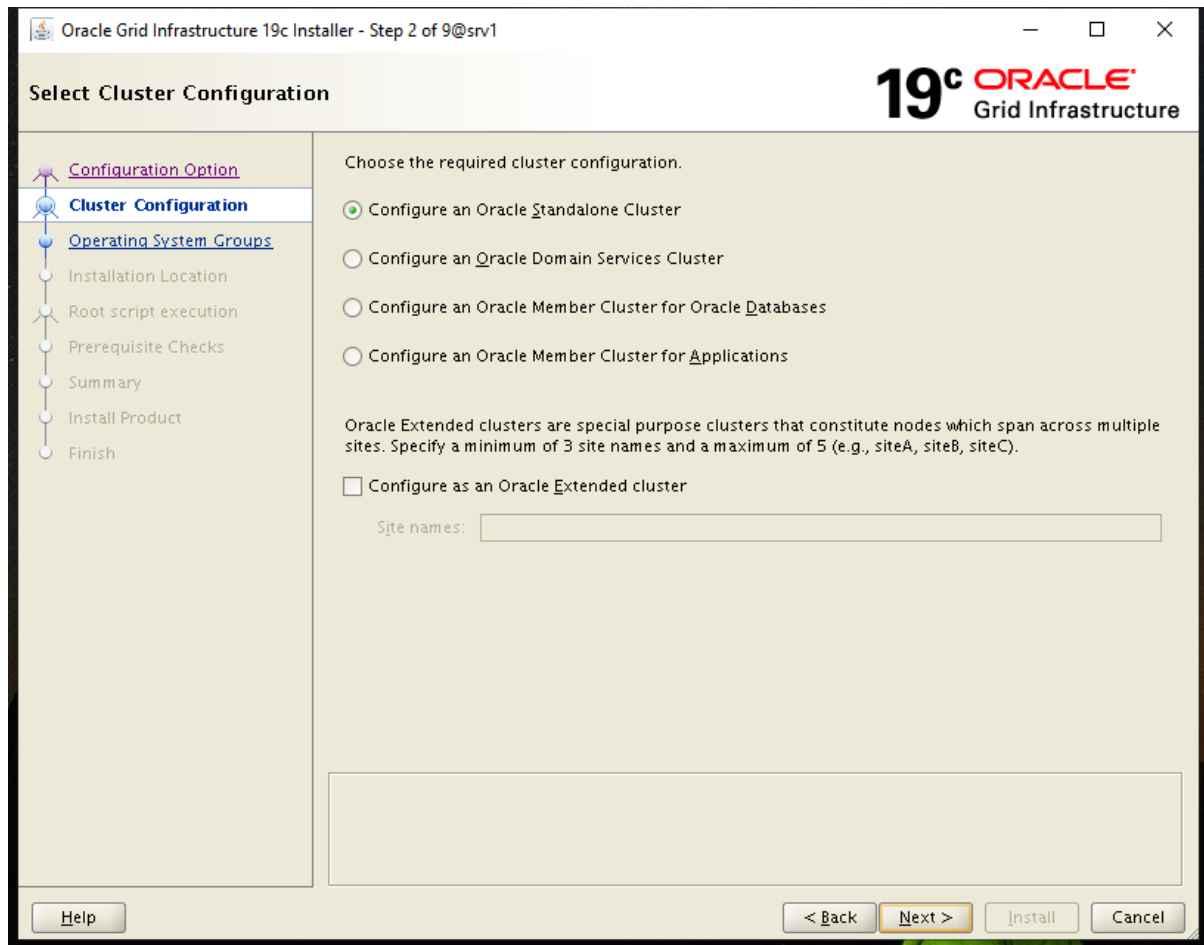
Select “Configure Oracle Grid Infrastructure for a New Cluster”

Installation of 19c RAC on Linux7 in Virtual Box VMs



Select “Configure as an Oracle Standalone Cluster”

Installation of 19c RAC on Linux7 in Virtual Box VMs



Specify a Cluster Name and provide the SCAN name that is configured in DNS :

Oracle Grid Infrastructure 19c Installer - Step 3 of 17@srv1

Grid Plug and Play Information

Single Client Access Name (SCAN) allows clients to use one name in connection strings to connect to the cluster as a whole. Client connect requests to the SCAN name can be handled by any cluster node.

☒ Create Local SCAN

Cluster Name:

SCAN Name:

SCAN Port:

☐ Use Shared SCAN

SCAN Client Data:

☐ Configure GNS

☐ Configure nodes Virtual IPs as assigned by the Dynamic Networks

☒ Create a new GNS

GNS VIP Address:

GNS Sub Domain:

☐ Use Shared GNS

GNS Client Data:

The installer will validate the SCAN name (it does nslookup to validate that IP addresses are assigned to the name)

Note : If the SCAN name is specified as “ora19cscan” (without a domain), the cluster verification utility for the SCAN will raise errors. [These are fixed in the section [Correct the SCAN to include domain](#) later]. It is preferable to specify “ora19cscan.localdomain” in this screen itself.

Installation of 19c RAC on Linux7 in Virtual Box VMs

Oracle Grid Infrastructure 19c Installer - Step 3 of 17@srv1

Grid Plug and Play Information

19c ORACLE Grid Infrastructure

Single Client Access Name (SCAN) allows clients to use one name in connection strings to connect to the cluster as a whole. Client connect requests to the SCAN name can be handled by any cluster node.

☒ Create Local SCAN

Cluster Name: ora19c-cluster
SCAN Name: ora19cscan
SCAN Port: 1521

☐ Use Shared SCAN

SCAN Client Data:

☐  Validating scan name

GNS VIP Address:
GNS Sub Domain:

☐ Use Shared GNS

Add the second node (ie. srv2) to the new Cluster (using the “Add” button) :

Oracle Grid Infrastructure 19c Installer - Step 4 of 17@srv1

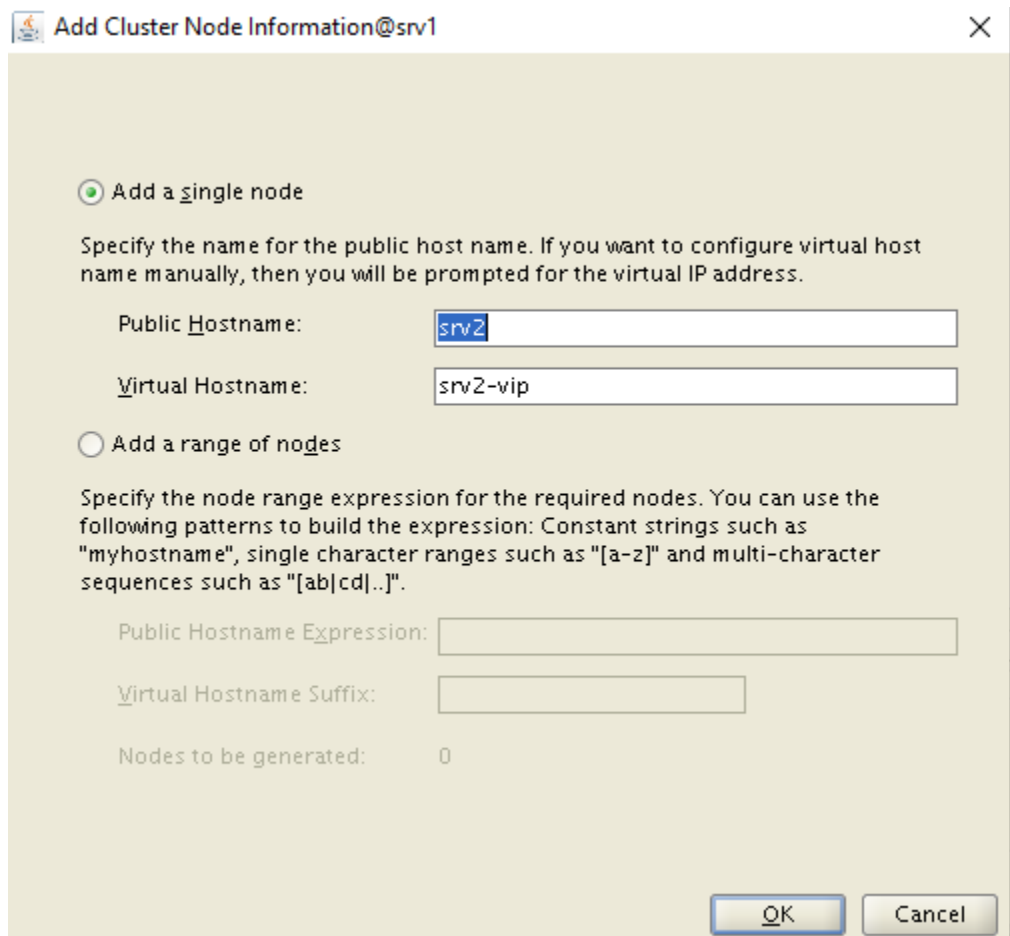
Cluster Node Information

19c ORACLE Grid Infrastructure

Provide the list of nodes to be managed by Oracle Grid Infrastructure with their Public Hostname and Virtual Hostname.

Public Hostname	Virtual Hostname
srv1	srv1-vip





The image shows a Windows-style dialog box titled "Add Cluster Node Information@srv1". It has a close button (X) in the top right corner. The dialog is divided into two main sections. The first section, "Add a single node", is selected with a radio button. It contains a text box for "Public Hostname" with the value "srv2" and a text box for "Virtual Hostname" with the value "srv2-vip". The second section, "Add a range of nodes", is unselected. It contains text boxes for "Public Hostname Expression" and "Virtual Hostname Suffix", both of which are empty. Below these is a label "Nodes to be generated:" followed by the value "0". At the bottom right are "OK" and "Cancel" buttons.

Add Cluster Node Information@srv1

☒ Add a single node

Specify the name for the public host name. If you want to configure virtual host name manually, then you will be prompted for the virtual IP address.

Public Hostname:

Virtual Hostname:

☐ Add a range of nodes

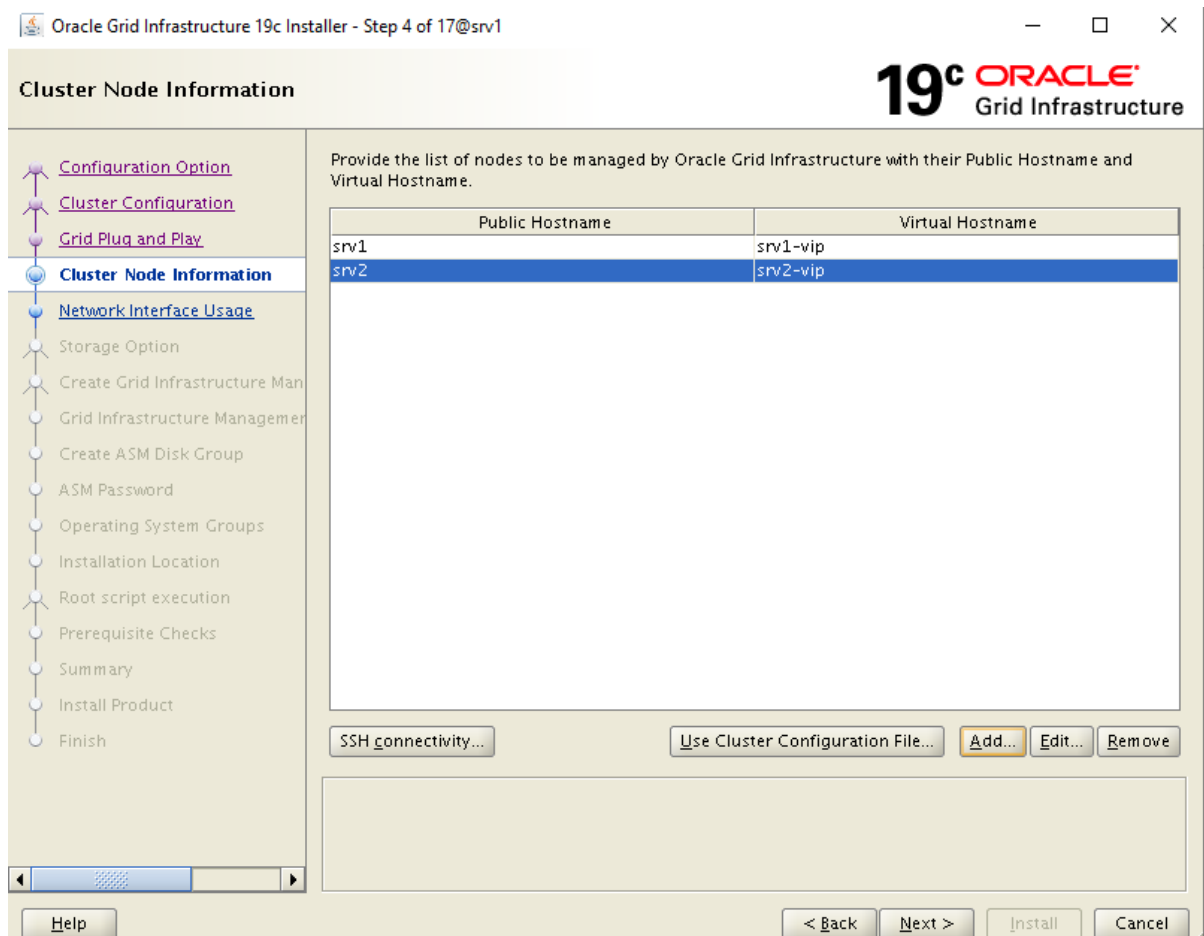
Specify the node range expression for the required nodes. You can use the following patterns to build the expression: Constant strings such as "myhostname", single character ranges such as "[a-z]" and multi-character sequences such as "[ab|cd|..]".

Public Hostname Expression:

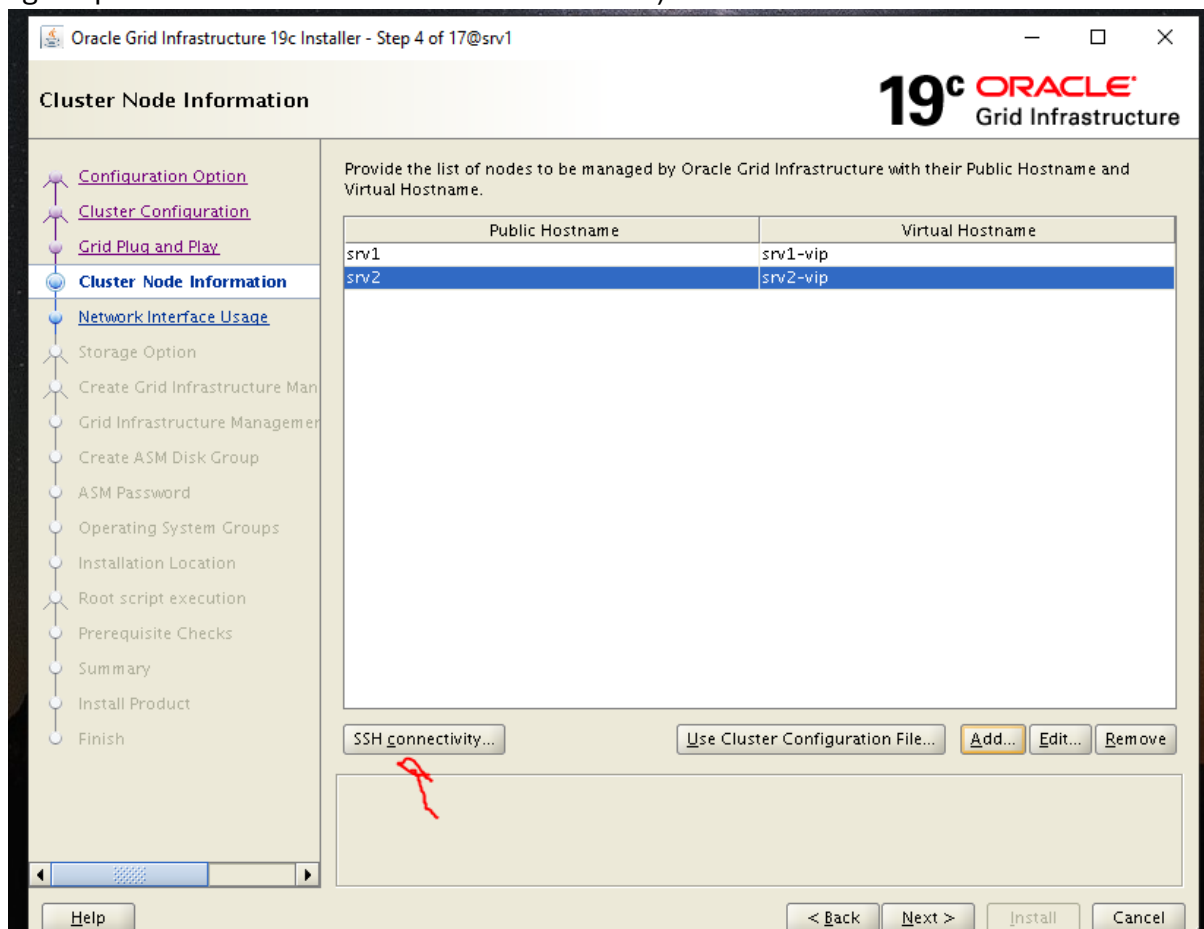
Virtual Hostname Suffix:

Nodes to be generated: 0

Installation of 19c RAC on Linux7 in Virtual Box VMs



Configure SSH Connectivity with the “SSH connectivity” button : (You will need the “grid” password to be the same on both servers)



Installation of 19c RAC on Linux7 in Virtual Box VMs

Oracle Grid Infrastructure 19c Installer - Step 4 of 17@srv1

Cluster Node Information

Provide the list of nodes to be managed by Oracle Grid Infrastructure with their Public Hostname and Virtual Hostname.

Public Hostname	Virtual Hostname
srv1	srv1-vip
srv2	srv2-vip

SSH connectivity... Use Cluster Configuration File... Add... Edit... Remove

OS Username: grid OS Password: [masked]

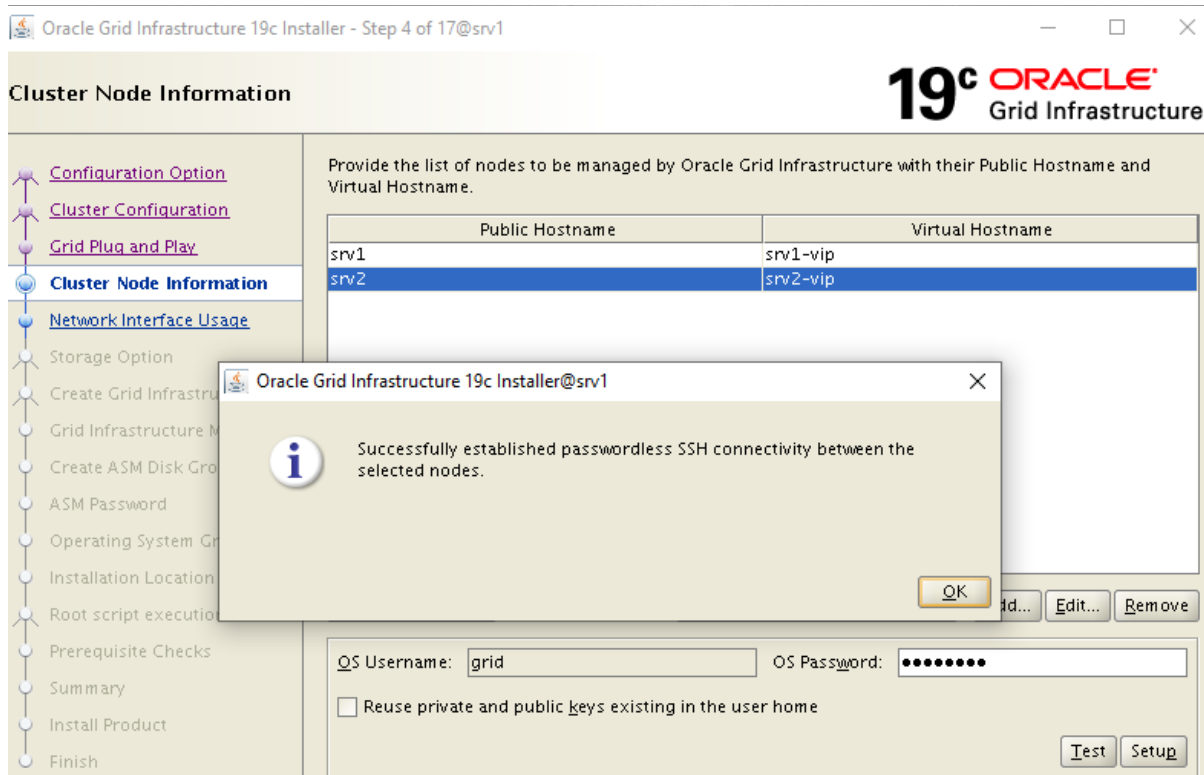
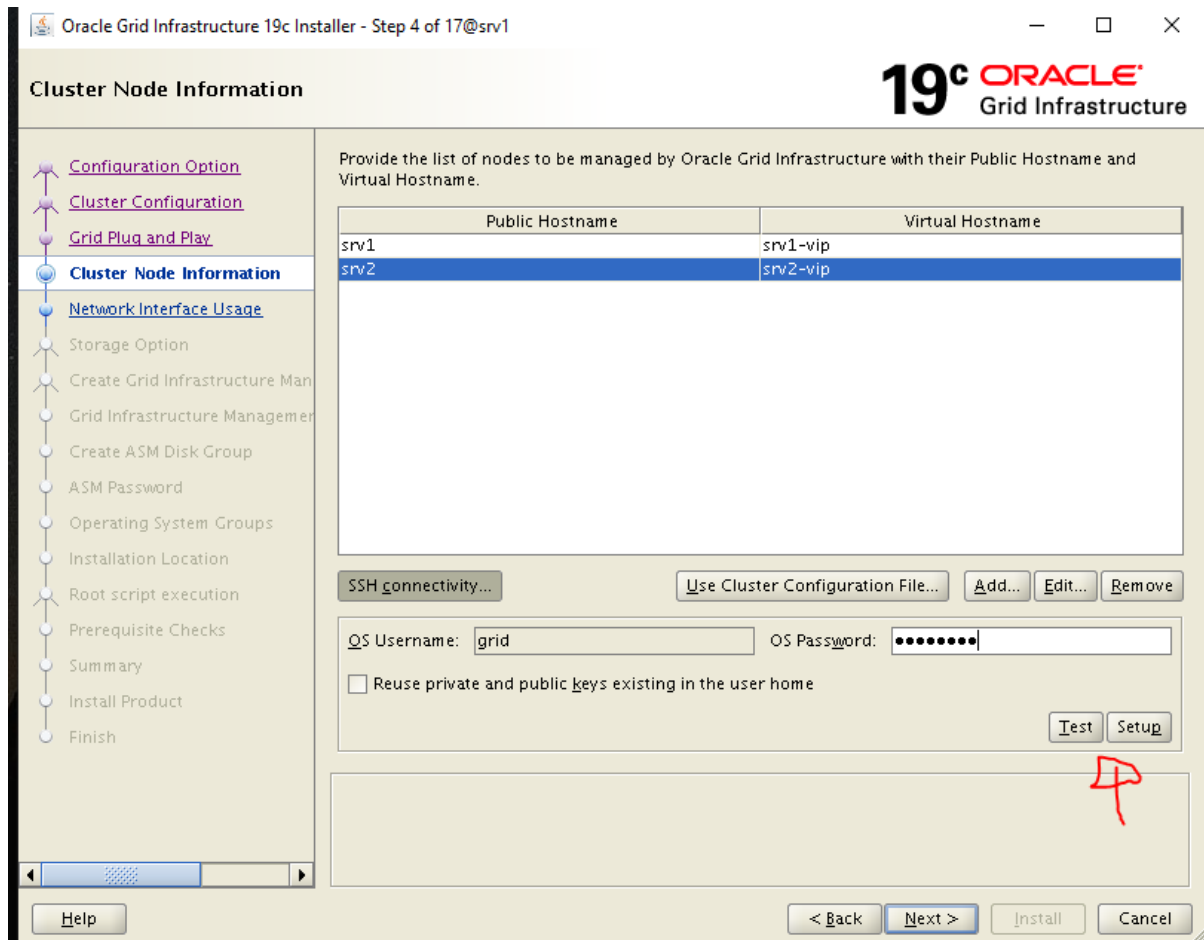
☐ Reuse private and public keys existing in the user home

Test Setup

Help < Back Next > Install Cancel

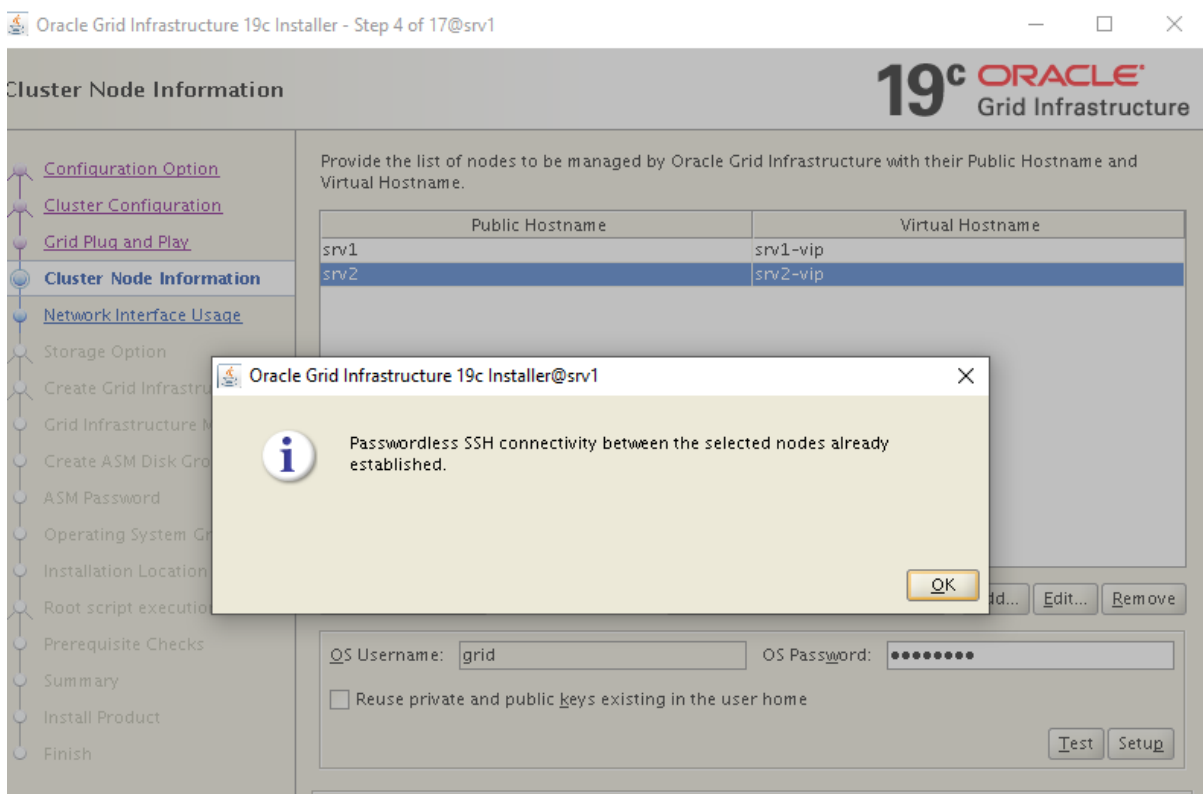
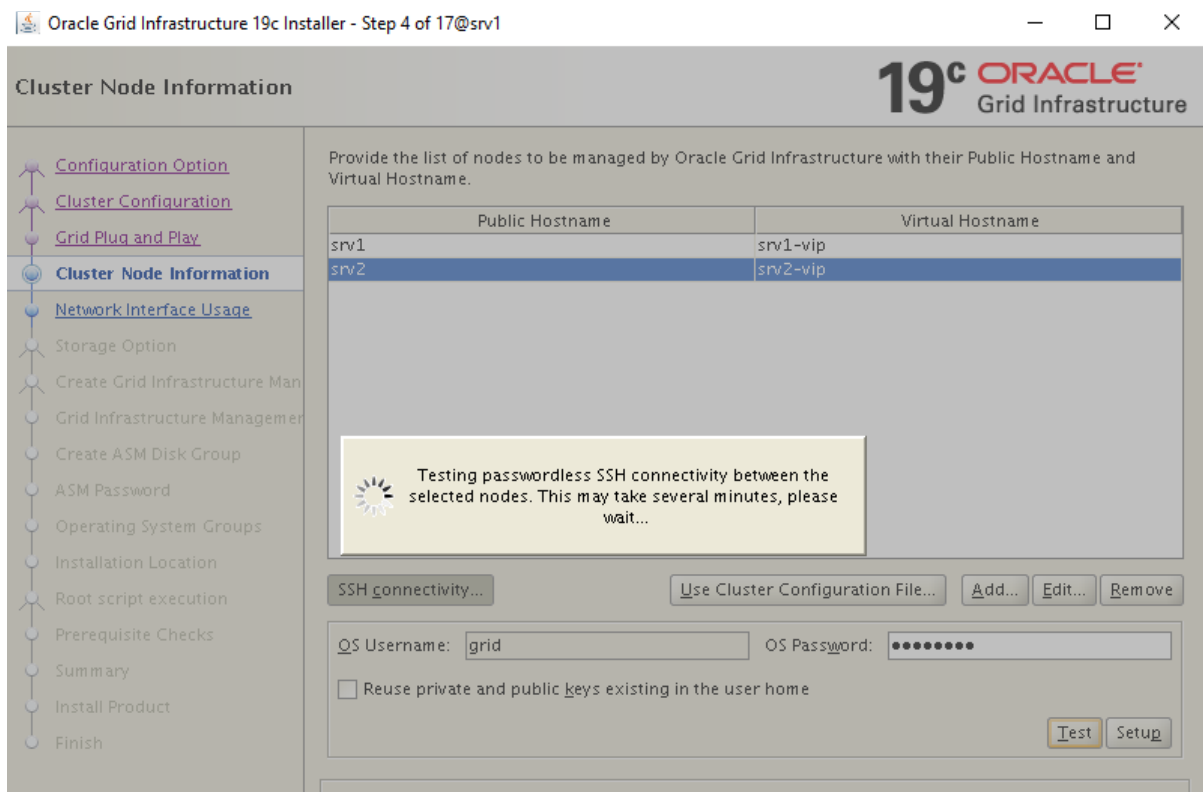
Click on the “Setup” and “Test” buttons

Installation of 19c RAC on Linux7 in Virtual Box VMs

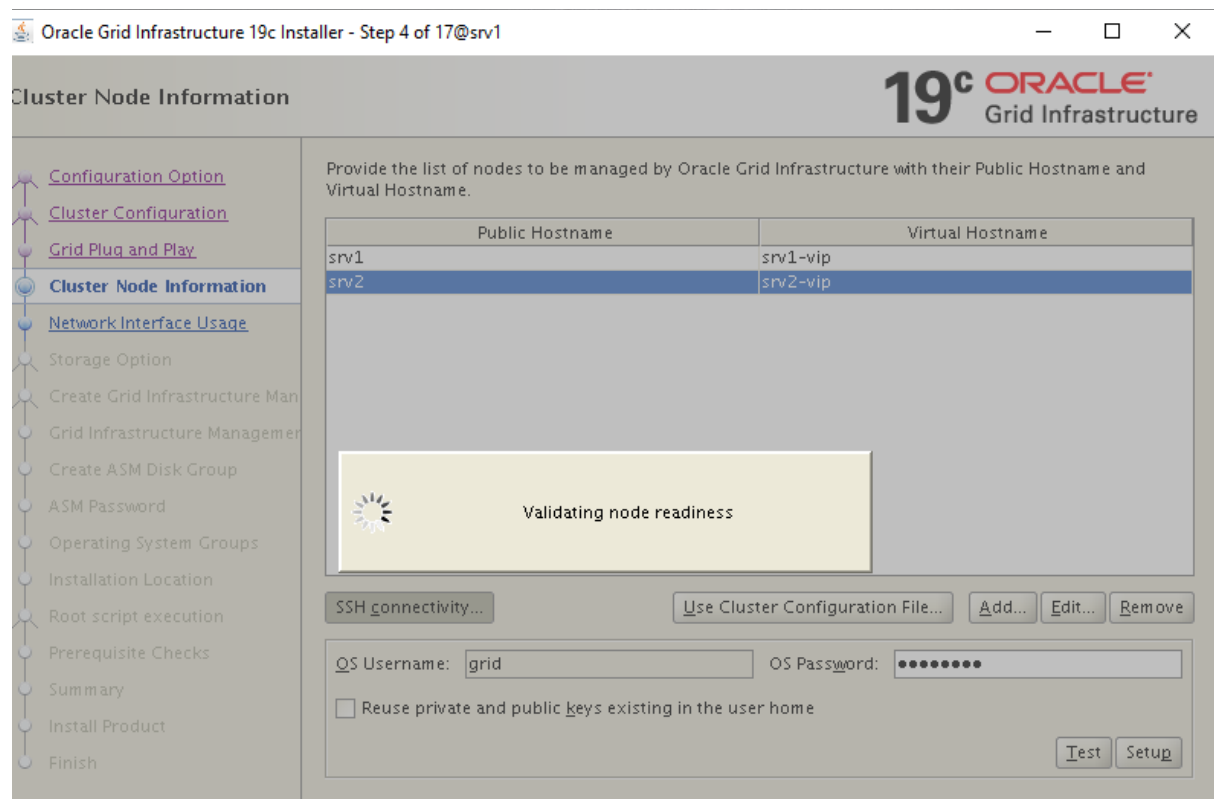


Installation of 19c RAC on Linux7 in Virtual Box VMs

If it is unsuccessful, verify that you have entered the correct password and it is the same on both servers and redo the Setup and Test



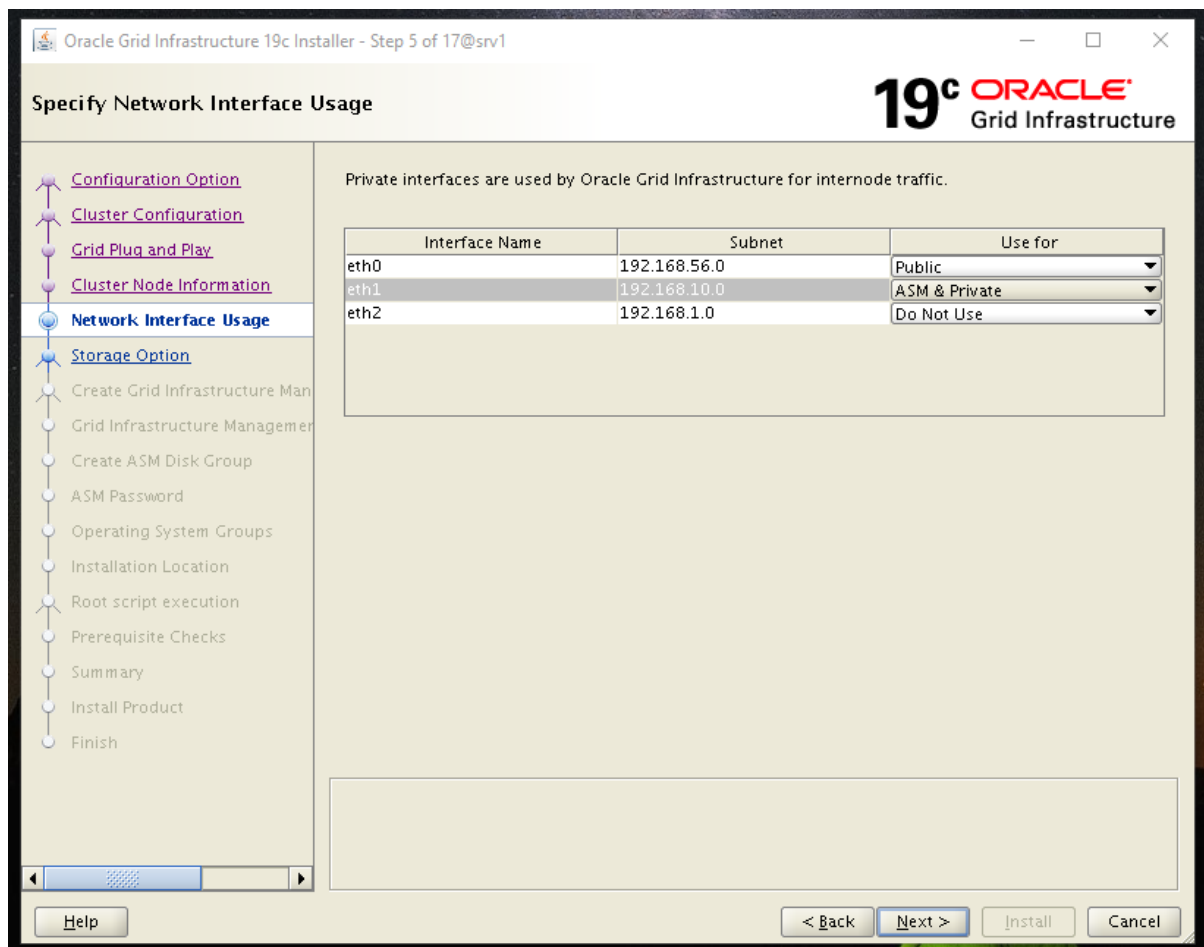
At “Next” it will validate Node Readiness



Then specify the Network Interfaces :

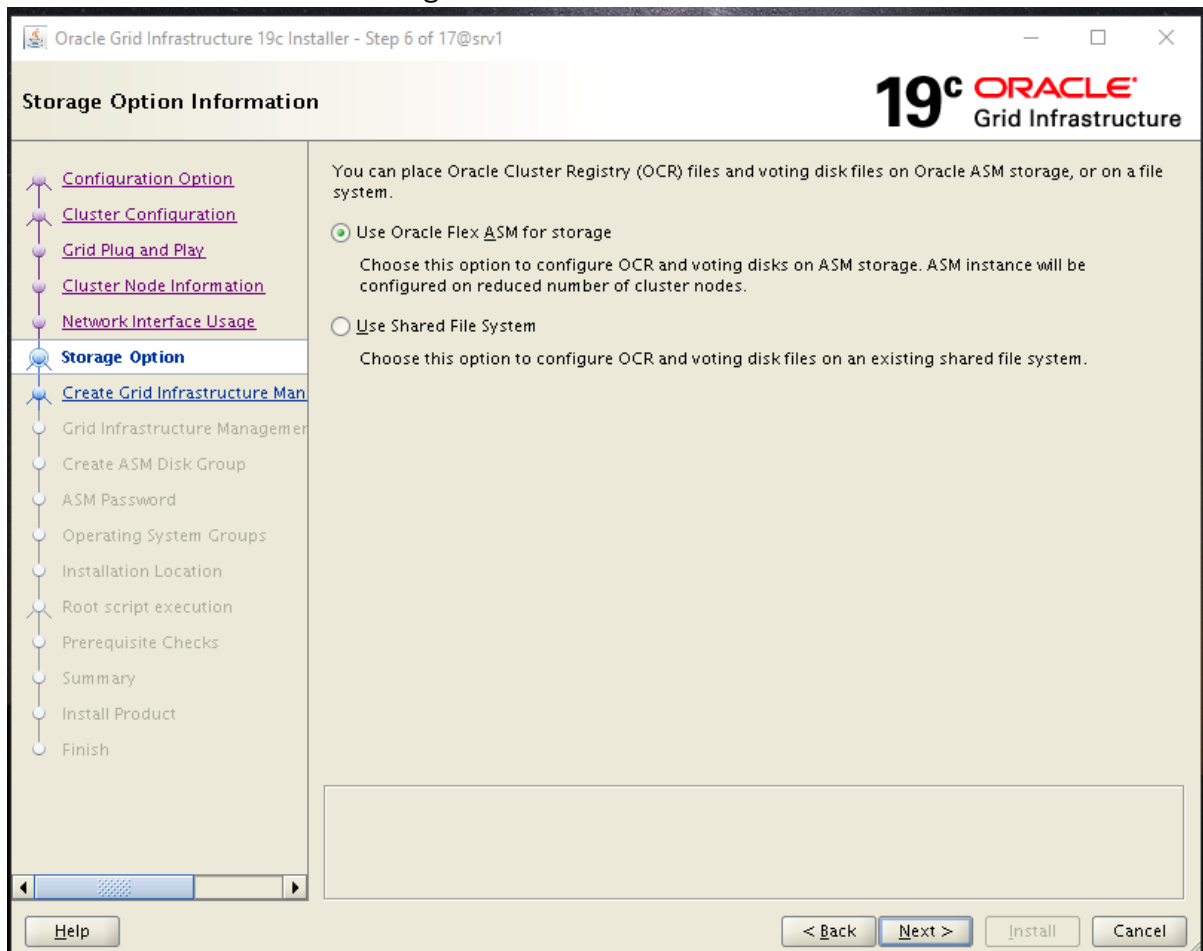
Note : for “eth1” (192.168.10.x) select “ASM & Private” as the “User for” from the drop down list.

“eth2” should be “Do Not Use” – it is for the nodes to connect outside the cluster (e.g. to the Internet) and should not be used by the Cluster

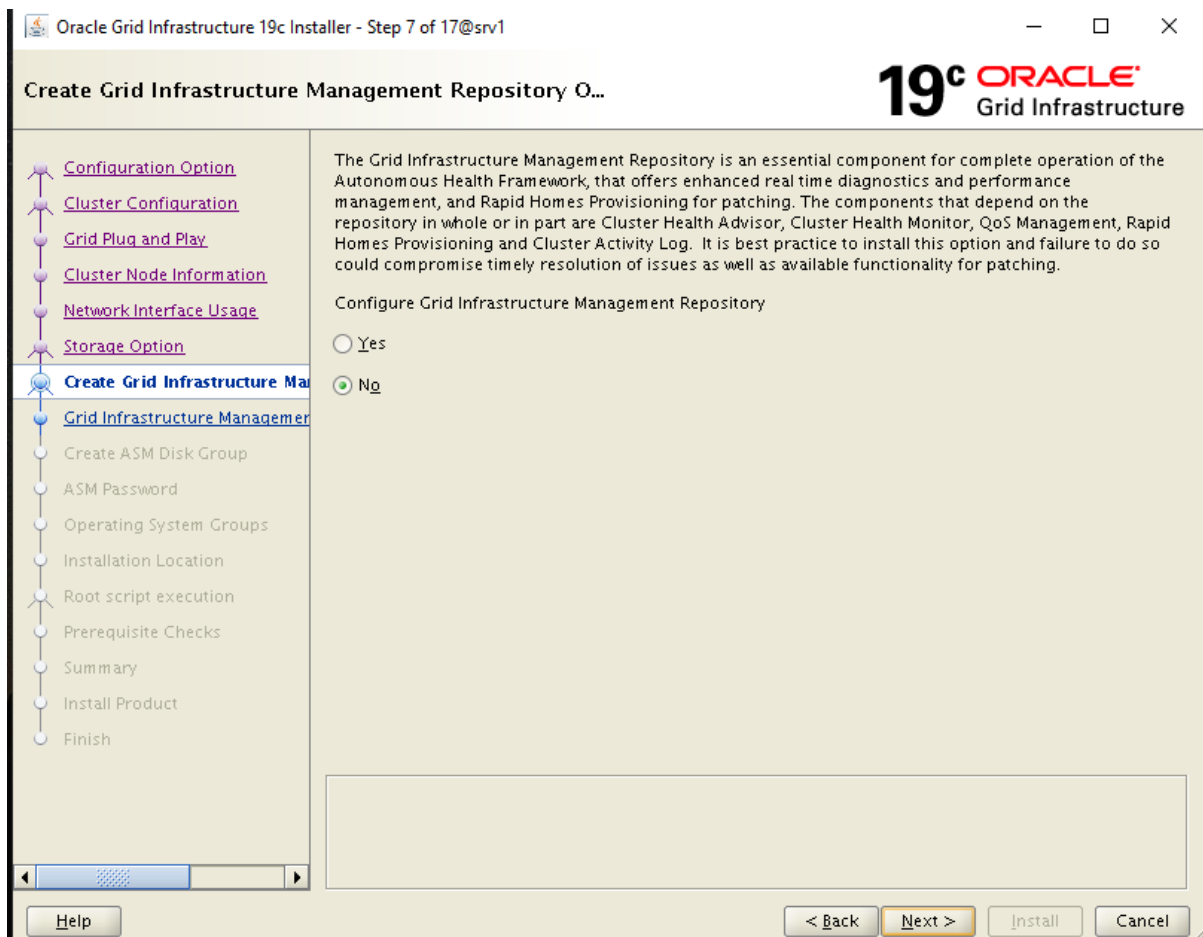


It will then Validate the two selected networks.

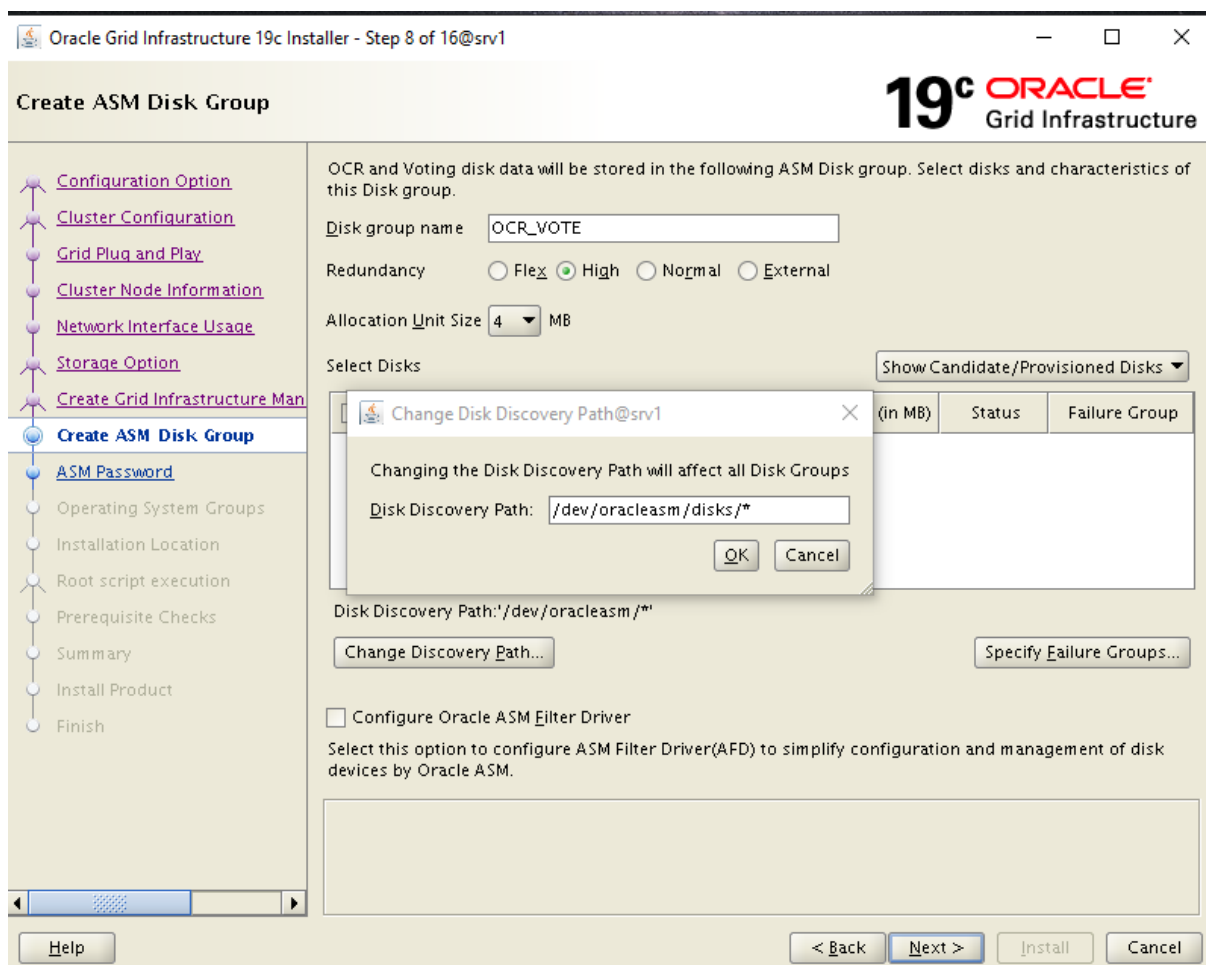
Select “Use Flex ASM for Storage”



Select “No” for the GIMR (GIMR is not mandatory in 19c) If you do want to use GIMR the OCR+VOTE diskgroup must be 100GB (instead of 5 disks of 2GB)



Specify “OCR_VOTE” and “High Redundancy” and use “Change Discovery Path” to “/dev/oracleasm/disks/*” [instead of /dev/sd*]



Then select the 5 OCR_VOTE ASM Disks

Installation of 19c RAC on Linux7 in Virtual Box VMs

Oracle Grid Infrastructure 19c Installer - Step 8 of 16@srv1

19c ORACLE
Grid Infrastructure

Create ASM Disk Group

OCR and Voting disk data will be stored in the following ASM Disk group. Select disks and characteristics of this Disk group.

Disk group name:

Redundancy: ☐ Flex ☒ High ☐ Normal ☐ External

Allocation Unit Size: MB

Select Disks:

	Disk Path	Size (in MB)	Status	Failure Group
☐	/dev/oracleasm/disks/FRA_DISK_2	15359	Provisioned	
☒	/dev/oracleasm/disks/OCR_VOTE_DISK_1	2047	Provisioned	
☒	/dev/oracleasm/disks/OCR_VOTE_DISK_2	2047	Provisioned	
☒	/dev/oracleasm/disks/OCR_VOTE_DISK_3	2047	Provisioned	
☒	/dev/oracleasm/disks/OCR_VOTE_DISK_4	2047	Provisioned	
☒	/dev/oracleasm/disks/OCR_VOTE_DISK_5	2047	Provisioned	

Disk Discovery Path: /dev/oracleasm/disks/**

☐ Configure Oracle ASM Filter Driver

Select this option to configure ASM Filter Driver (AFD) to simplify configuration and management of disk devices by Oracle ASM.

(Configure "Oracle ASM Filter Driver" may fail with this error below so uncheck it)



For simplicity, I am specifying the same password for SYSASM and ASMSNMP

Oracle Grid Infrastructure 19c Installer - Step 9 of 16@srv1

Specify ASM Password

The new Oracle Automatic Storage Management (Oracle ASM) instance requires its own SYS user with SYSASM privileges for administration. Oracle recommends that you create a less privileged ASMSNMP user with SYSDBA privileges to monitor the ASM instance.

Specify the password for these user accounts.

☐ Use different passwords for these accounts

	Password	Confirm Password
SYS		
ASMSNMP		

☒ Use same passwords for these accounts

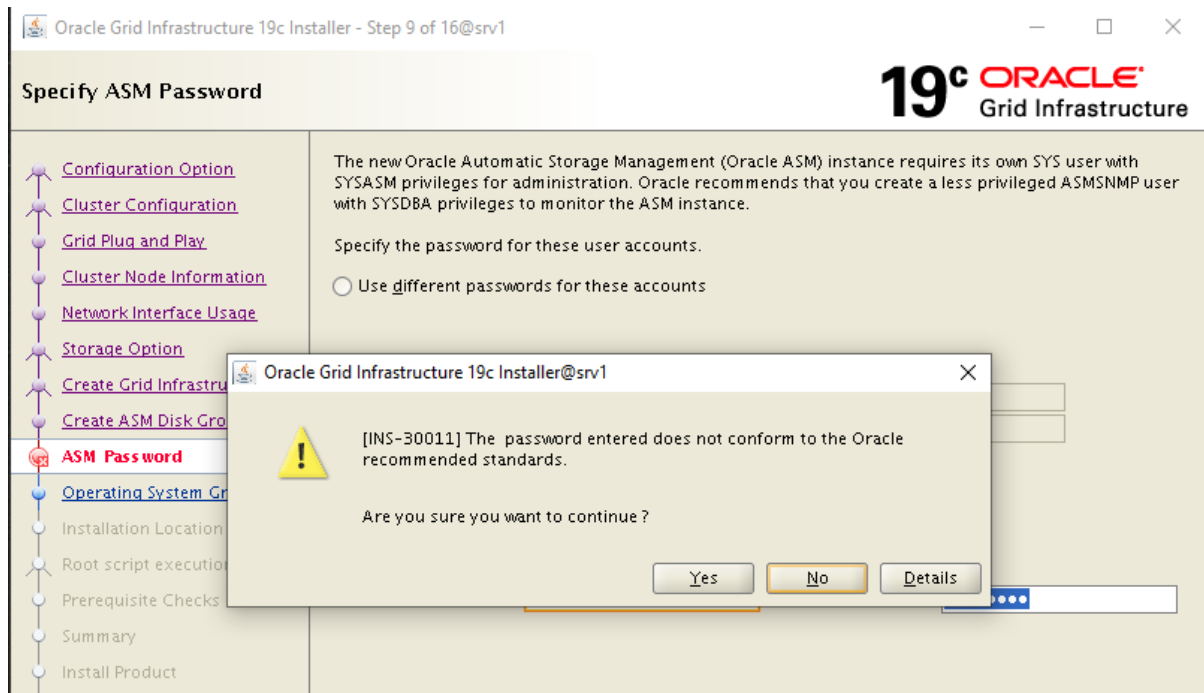
Specify Password: Confirm Password:

Messages:

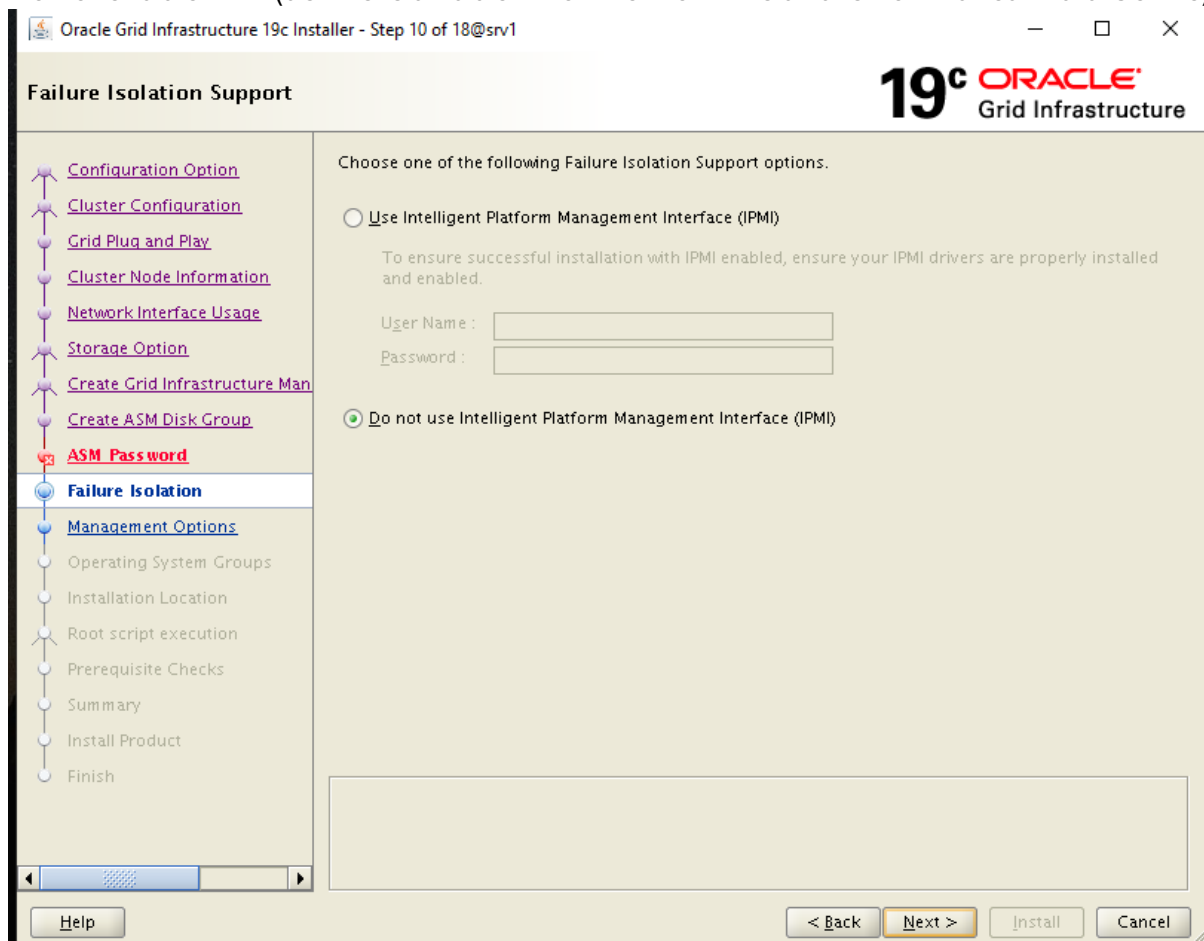
Specify Password:[INS-30011] The password entered does not conform to the Oracle recommended standards.

Help < Back Next > Install Cancel

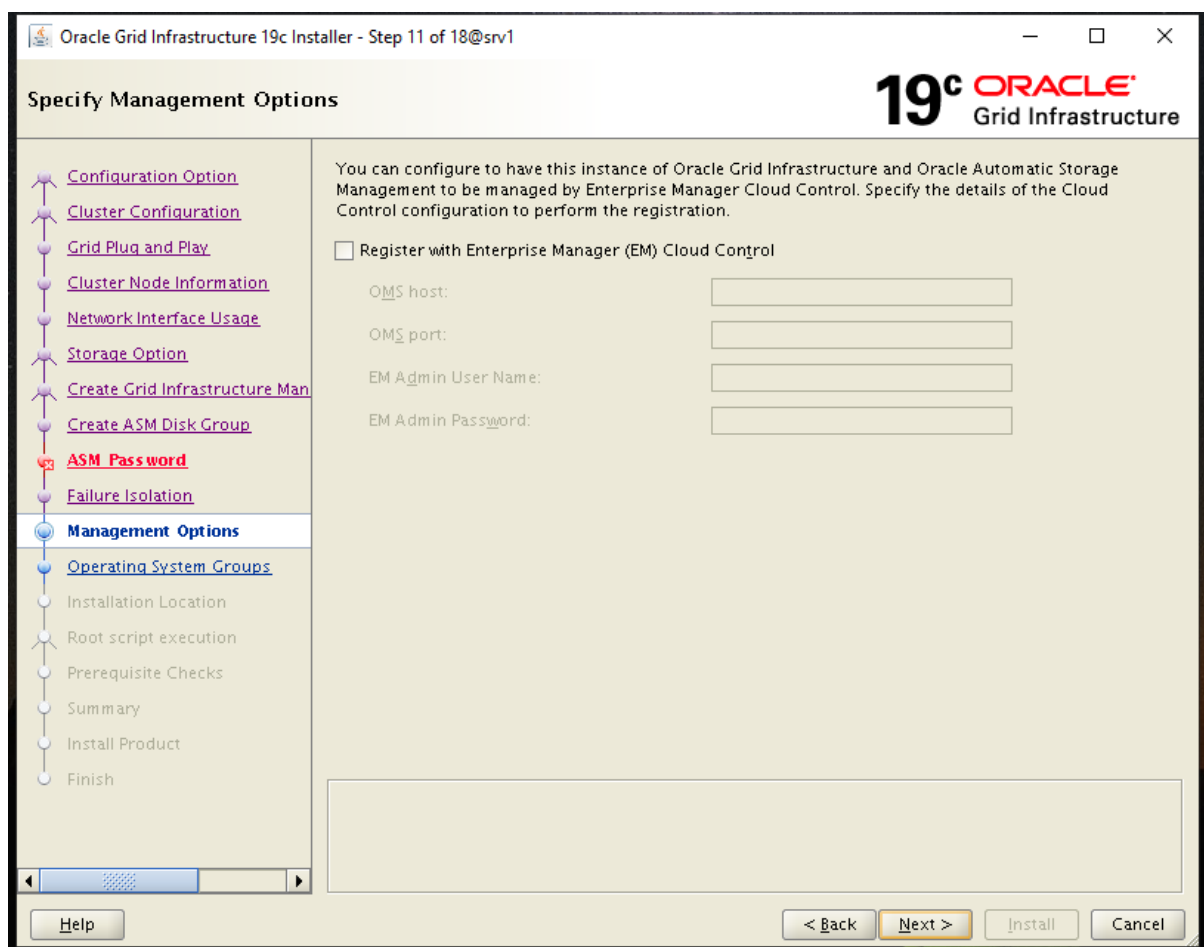
Installation of 19c RAC on Linux7 in Virtual Box VMs



Do not enable IPMI (as this is a Lab environment on VMs and is not in a real Data Centre)



I do not have a separate EM Cloud Control environment, so I am not registering this Cluster in EMCC



Specify the OS Groups

Oracle Grid Infrastructure 19c Installer - Step 12 of 18@srv1

Privileged Operating System Groups

Select the name of the operating system group, that you want to use for operating system authentication to Oracle Automatic Storage Management.

Oracle ASM Administrator (OSASM) Group: oinstall

Oracle ASM DBA (OSDBA for ASM) Group: grid

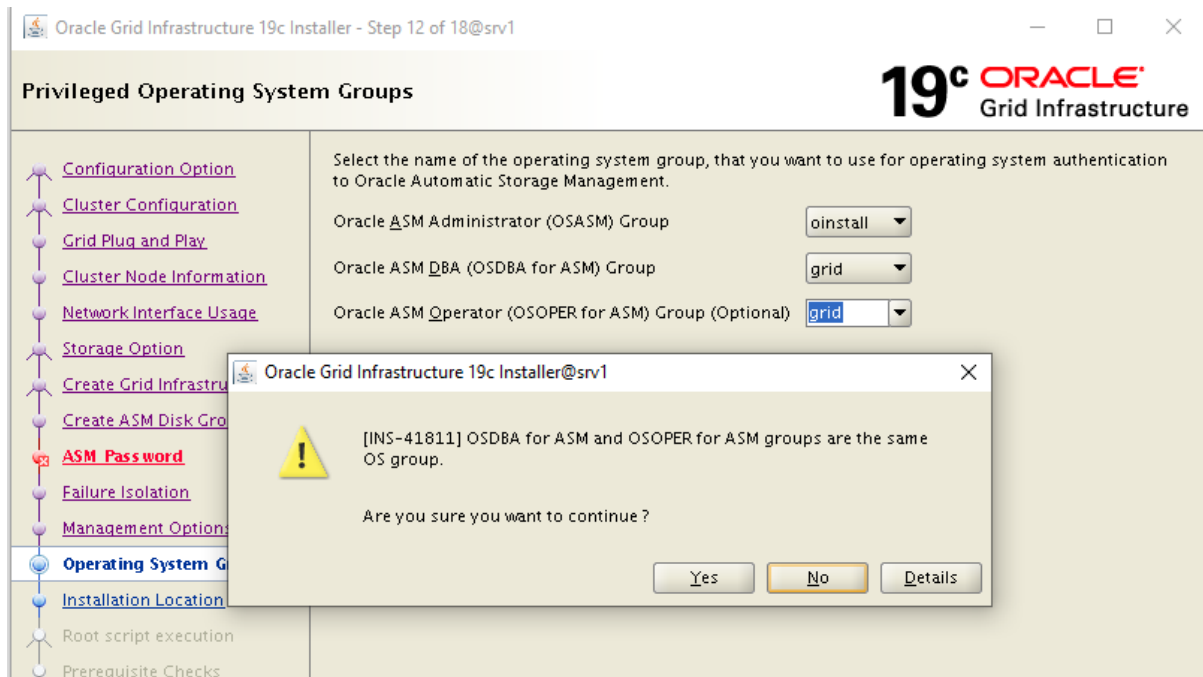
Oracle ASM Operator (OSOPER for ASM) Group (Optional): grid

Navigation pane:

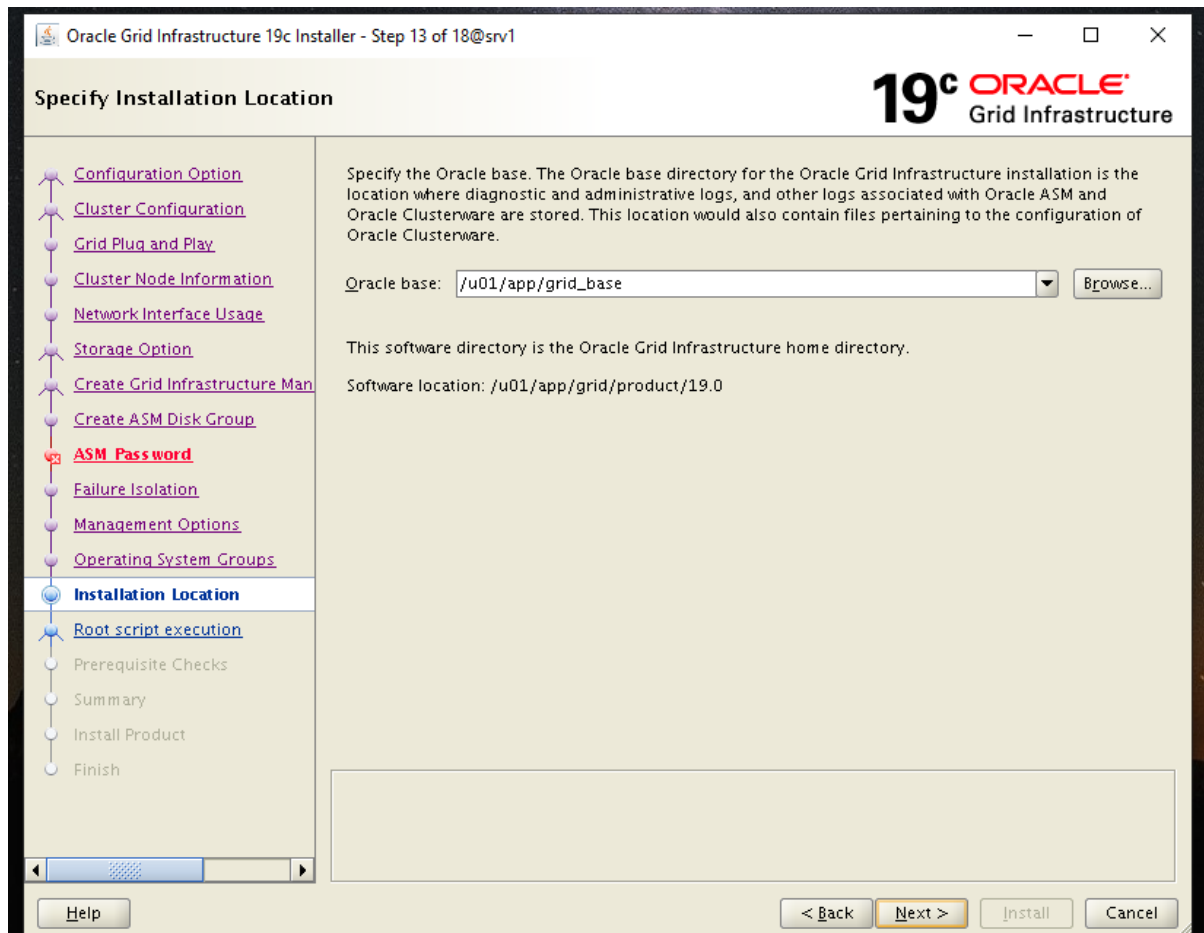
- Configuration Option
- Cluster Configuration
- Grid Plug and Play
- Cluster Node Information
- Network Interface Usage
- Storage Option
- Create Grid Infrastructure Man
- Create ASM Disk Group
- ASM Password
- Failure Isolation
- Management Options
- Operating System Groups
- Installation Location
- Root script execution
- Prerequisite Checks
- Summary
- Install Product
- Finish

Buttons: Help, < Back, Next >, Install, Cancel

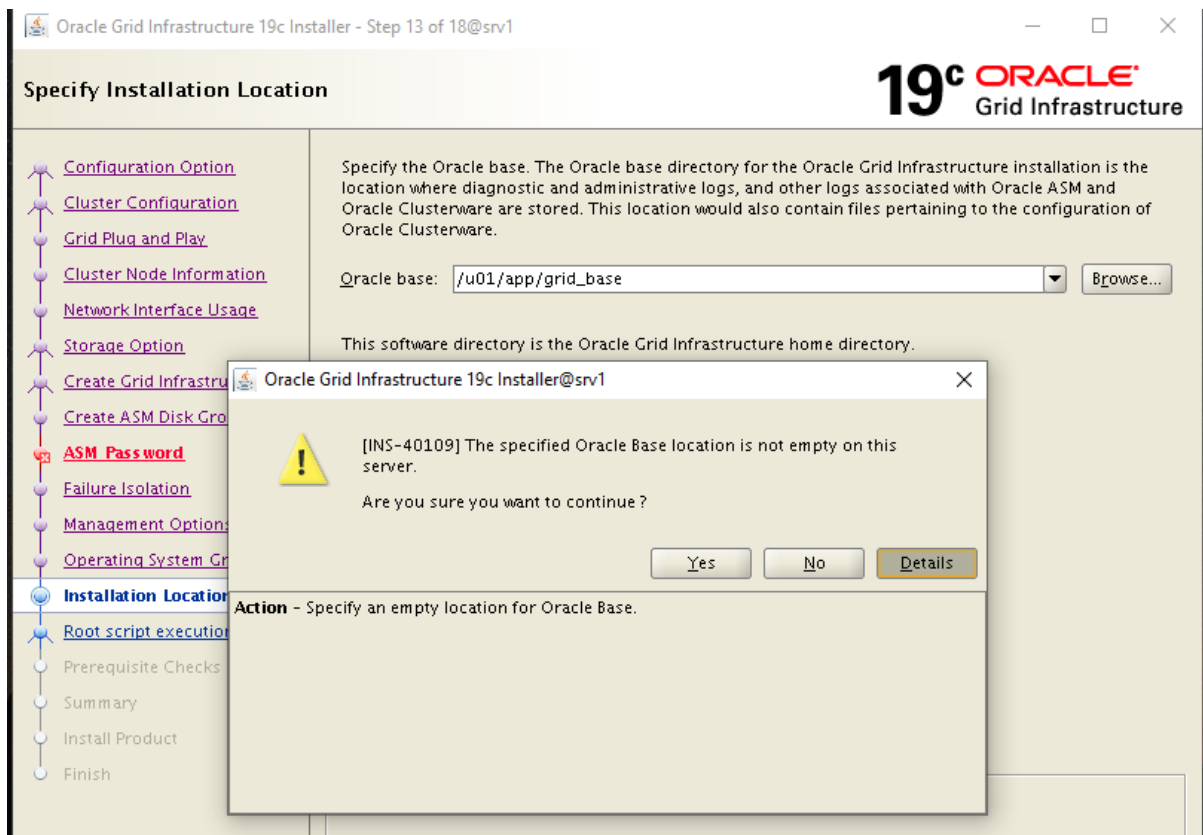
Installation of 19c RAC on Linux7 in Virtual Box VMs



The ORACLE_BASE is read from the environment setup earlier. It must NOT be a parent folder of ORACLE_HOME



You may get a warning about the ORACLE_BASE not being empty if you have run the Installer more than once

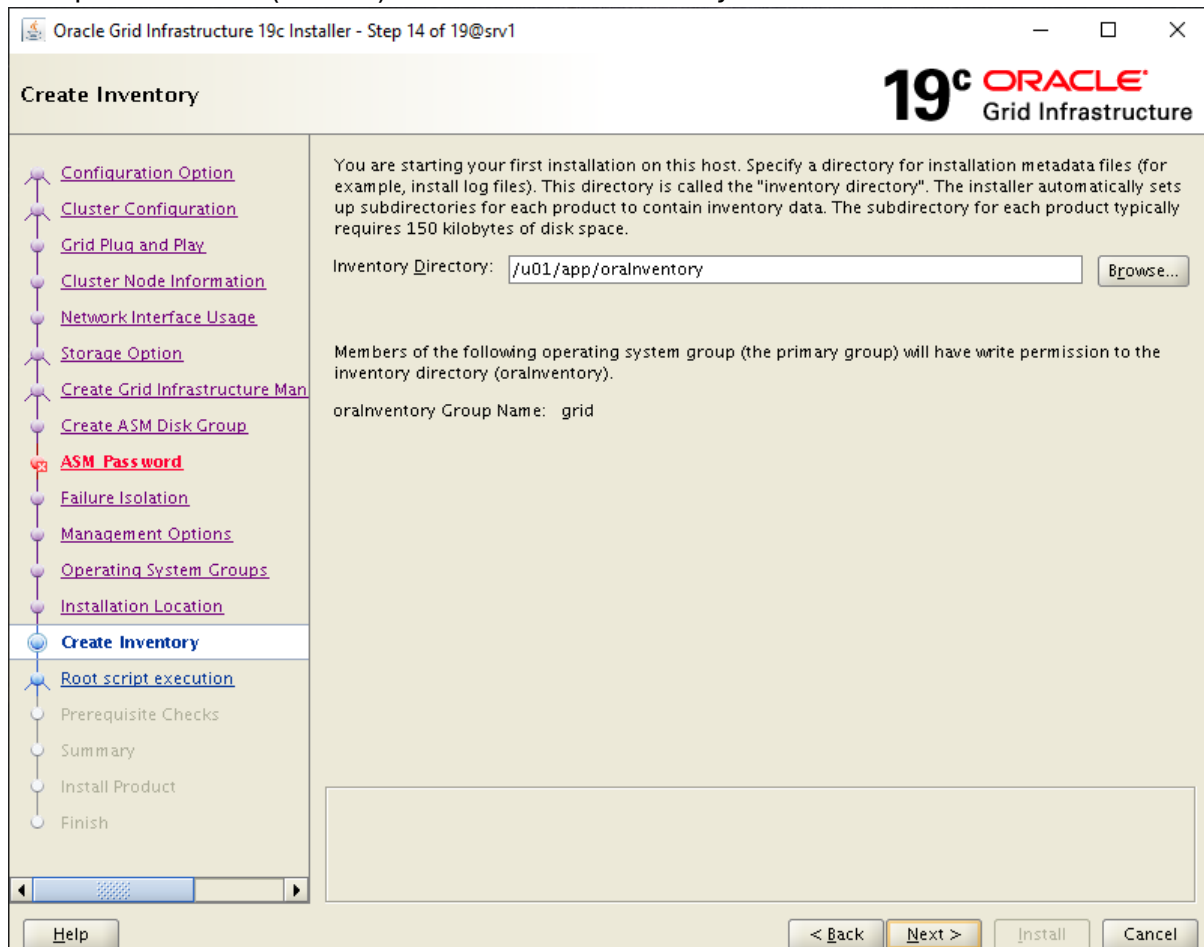


There may be a “diag” folder created by the previous run of the installer. You can remove it

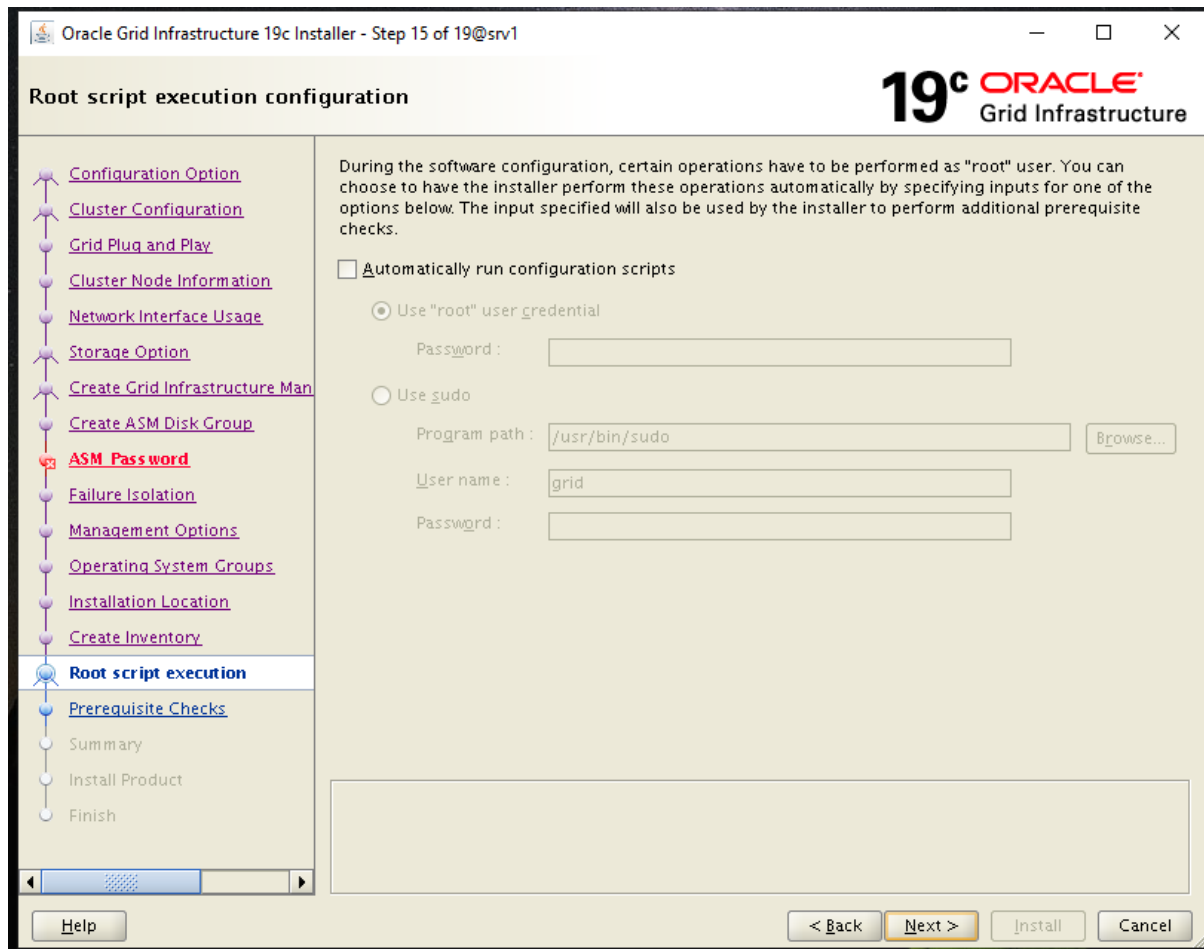
```
[root@srv1 grid_base]# pwd
/u01/app/grid_base
[root@srv1 grid_base]# ls -l
total 0
drwxr-xr-x. 3 root root 20 Mar 17 09:56 diag
[root@srv1 grid_base]#
[root@srv1 grid_base]# rm -rf diag
[root@srv1 grid_base]# pwd
/u01/app/grid_base
[root@srv1 grid_base]# ls -l
total 0
[root@srv1 grid_base]#
```

Select “No” at the “Are you sure you want to continue” . The installer will then re-verify that the ORACLE_BASE folder is empty and resume

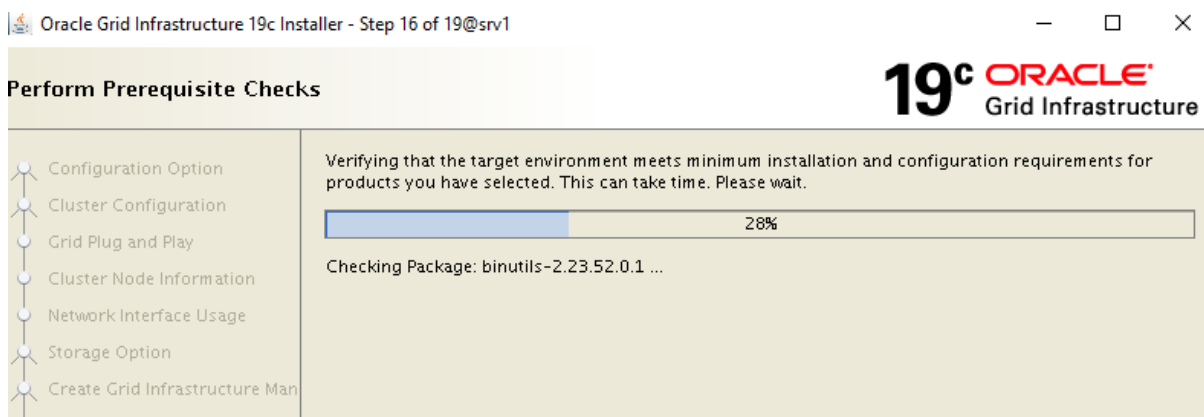
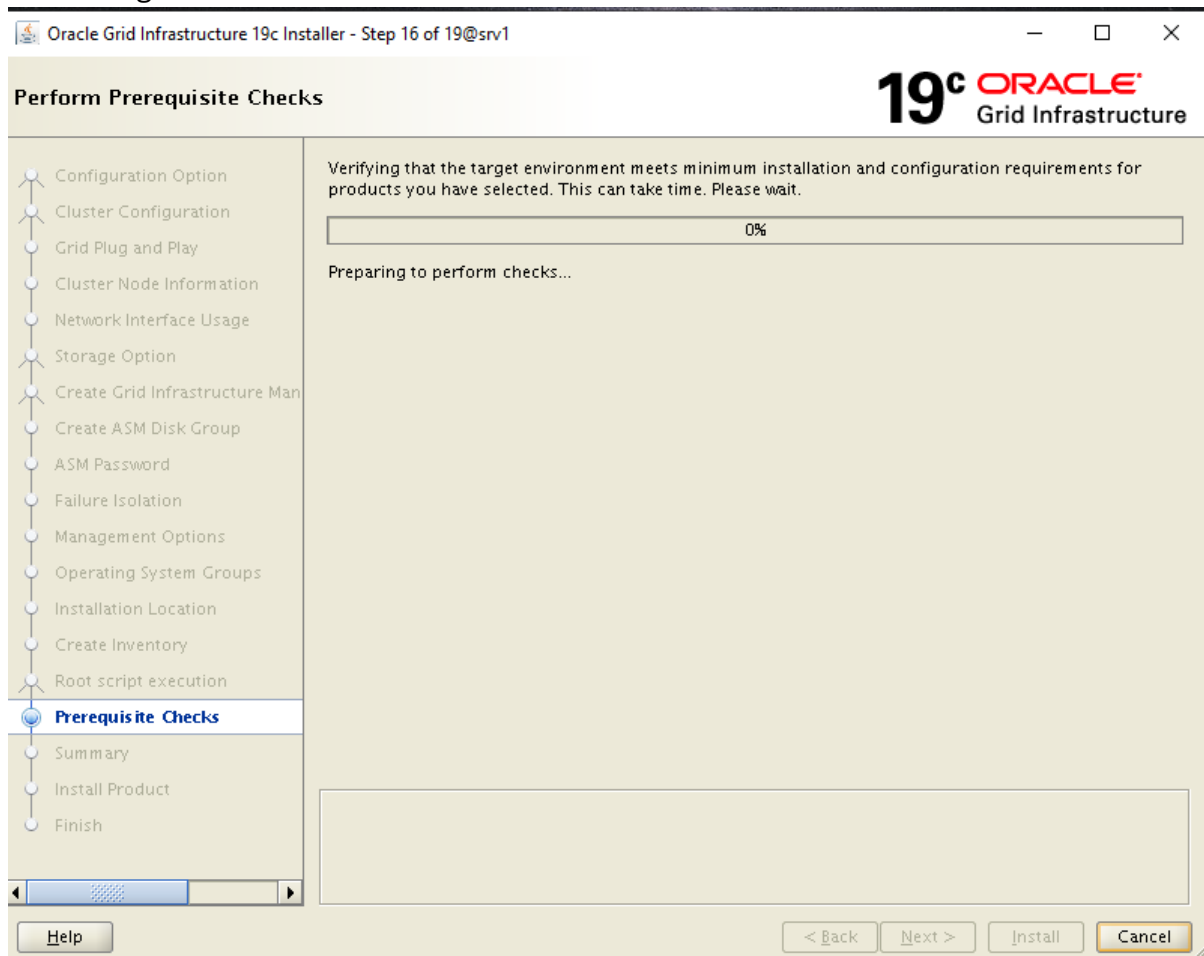
Accept the correct (default) location for oraInventory



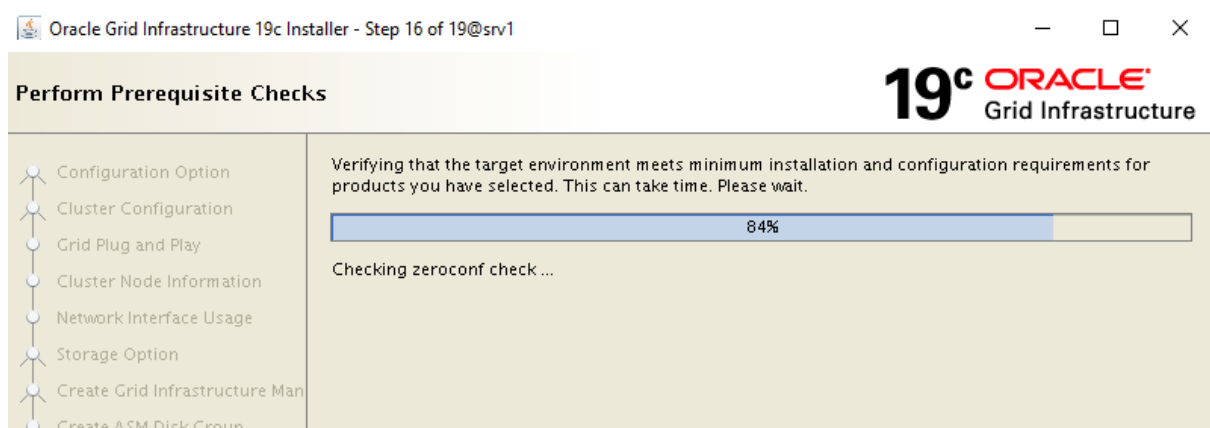
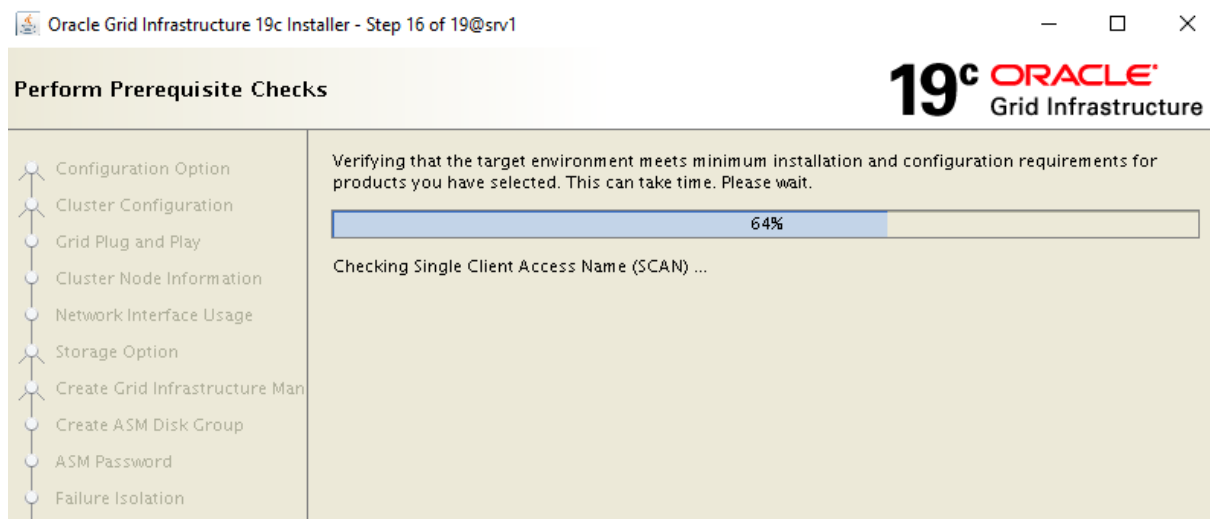
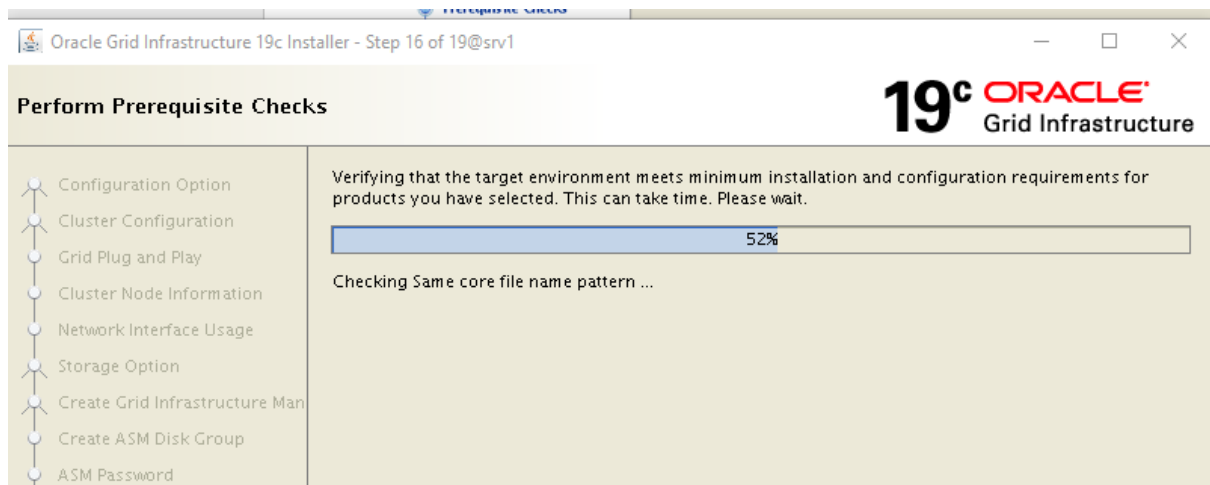
At the next prompt, my preference is to not have the installer run the root script automatically



It then begins to run checks

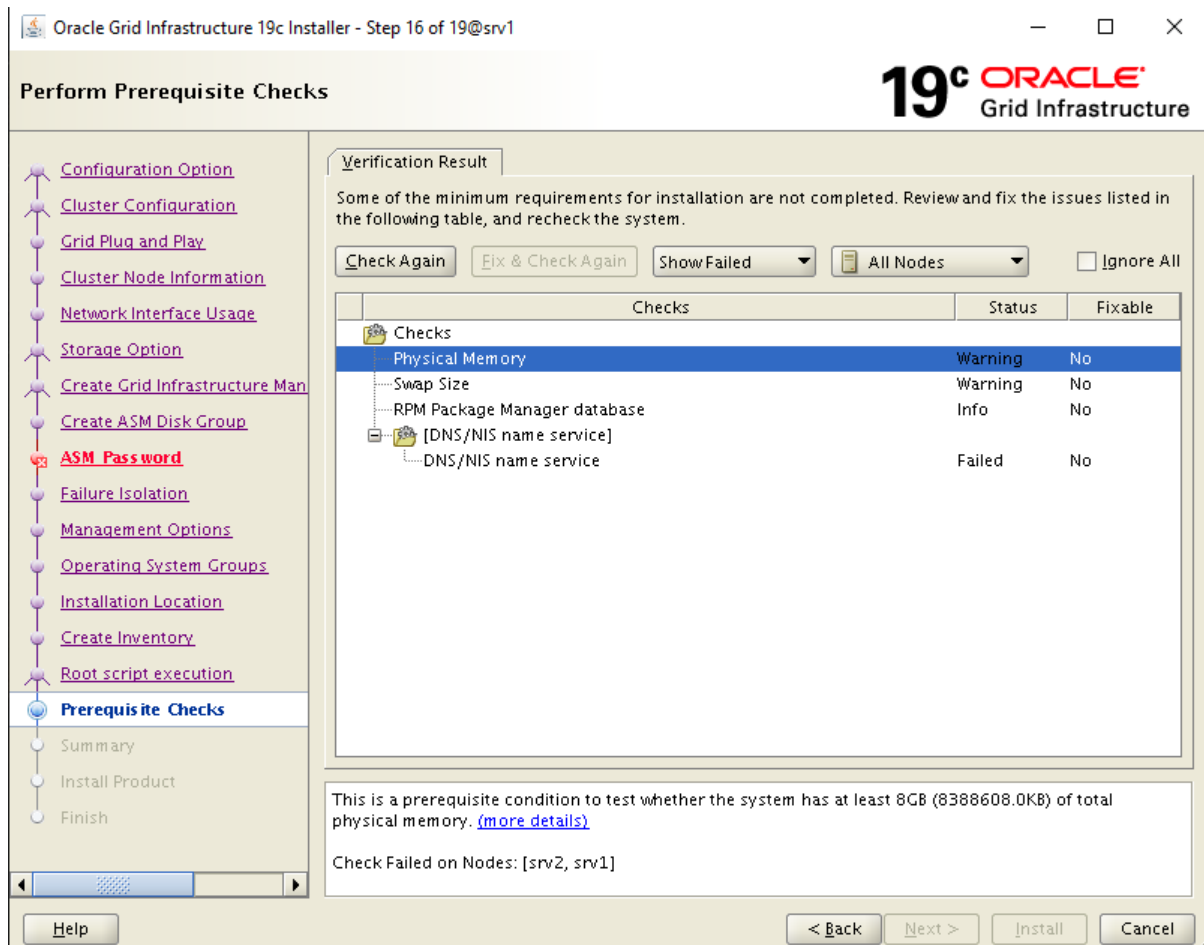


Installation of 19c RAC on Linux7 in Virtual Box VMs



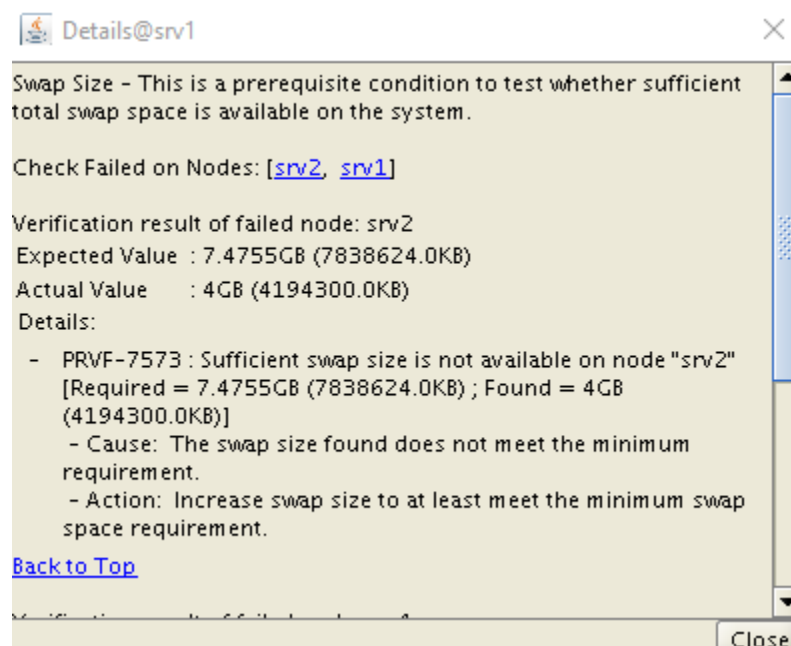
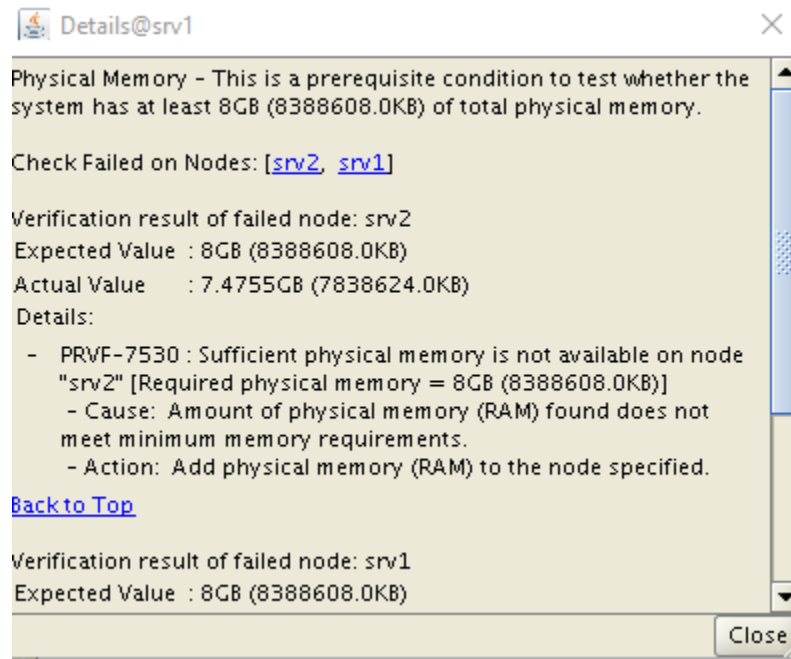
At this page :

Installation of 19c RAC on Linux7 in Virtual Box VMs

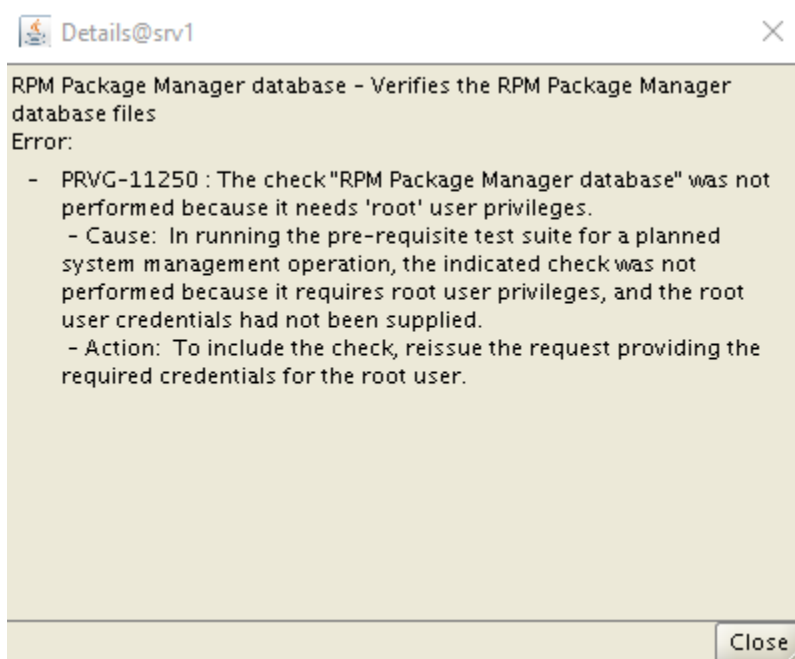


You can check the warnings on Physical RAM and Swap Size by clicking on “more details” in the lower pane

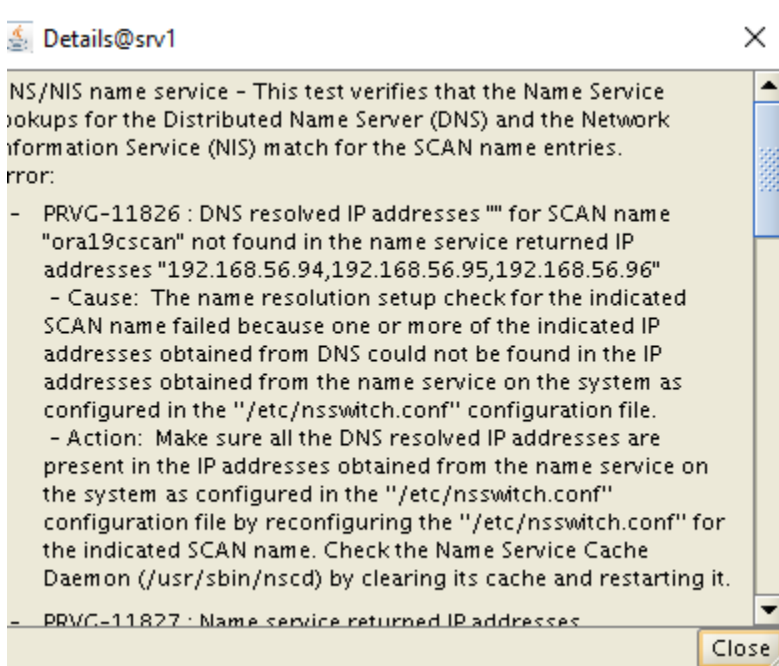
It shows that detected RAM is 7.4755GB instead of 8GB and Swap Size is 4GB. These can be ignored.



The RPM Package Manager message can be ignored

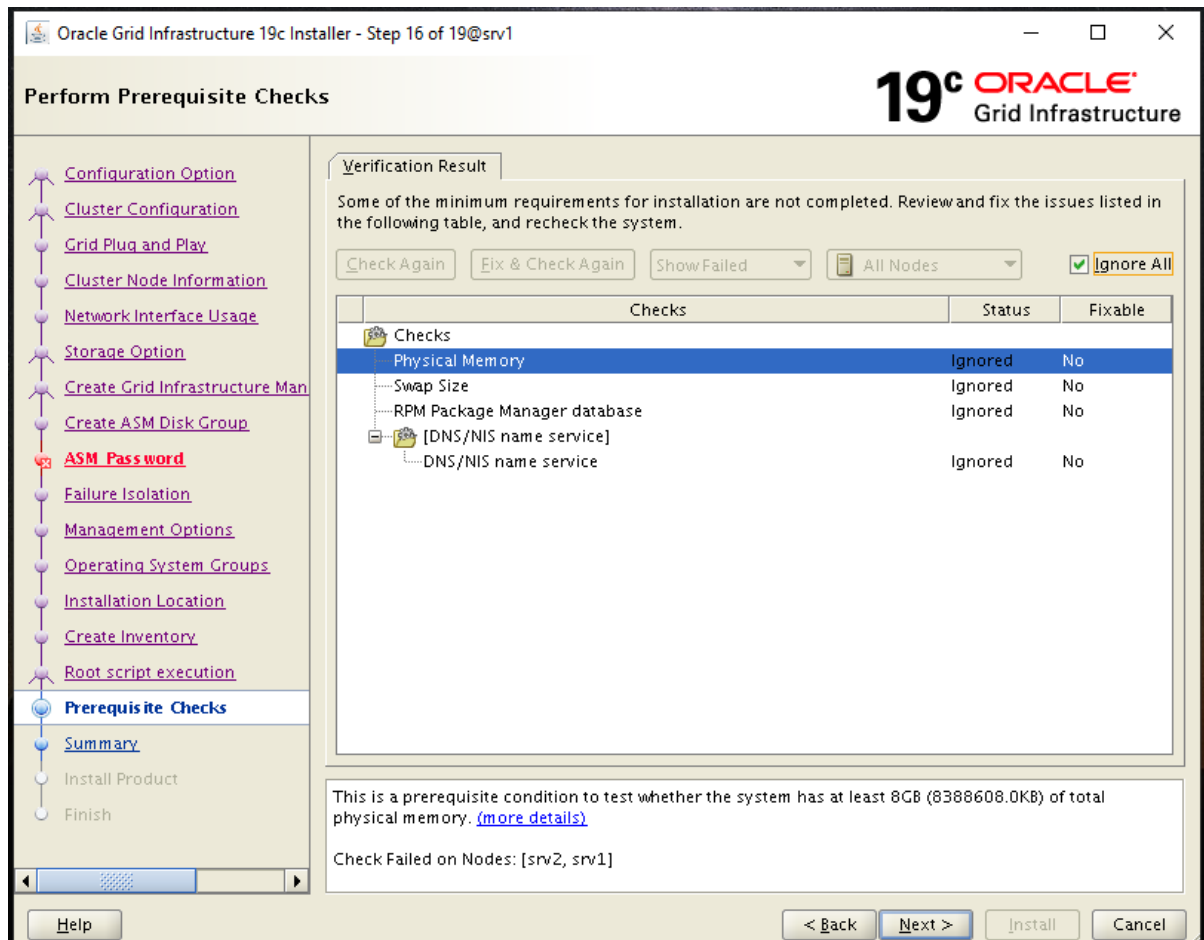


The DNS-NIS error will have to be ignored

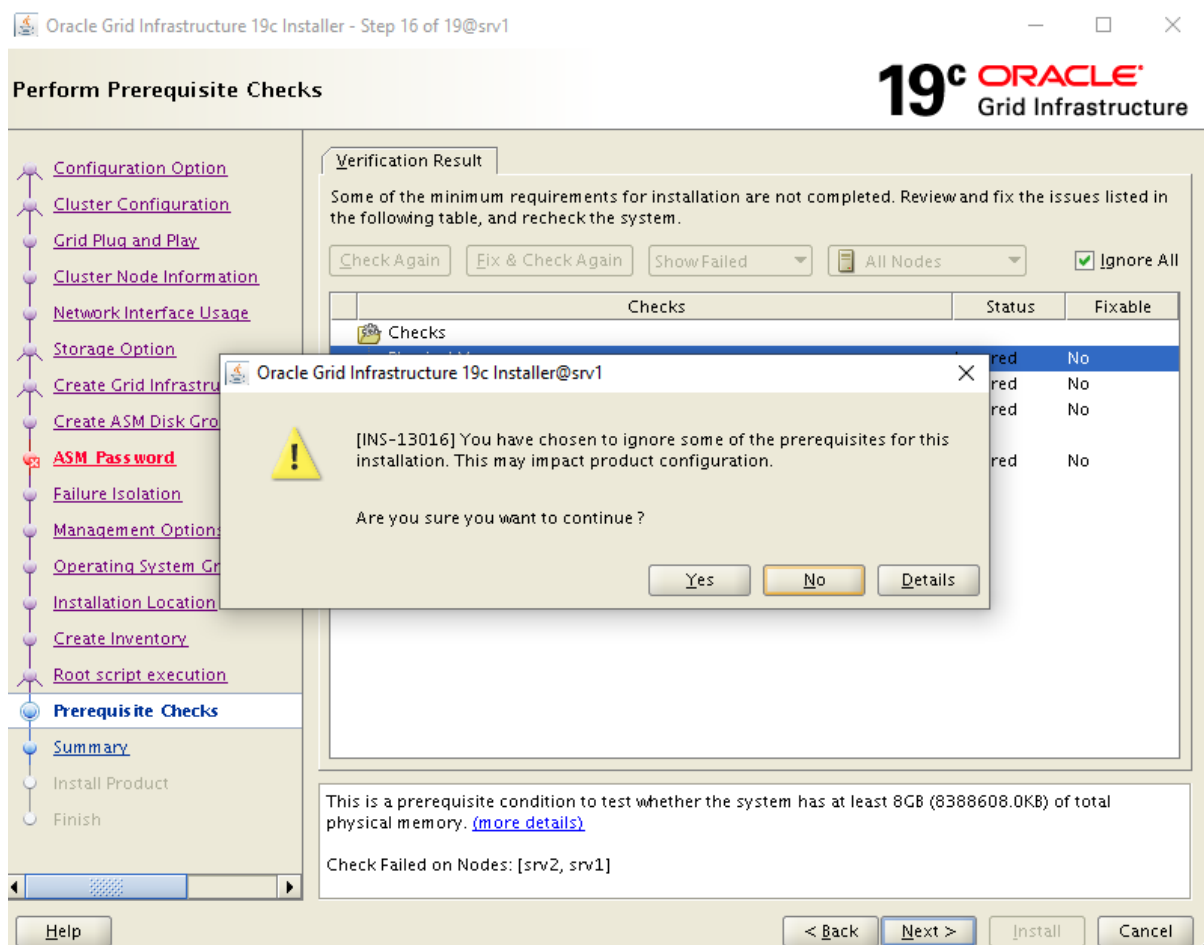


Click on "Ignore All" at the top-right

Installation of 19c RAC on Linux7 in Virtual Box VMs

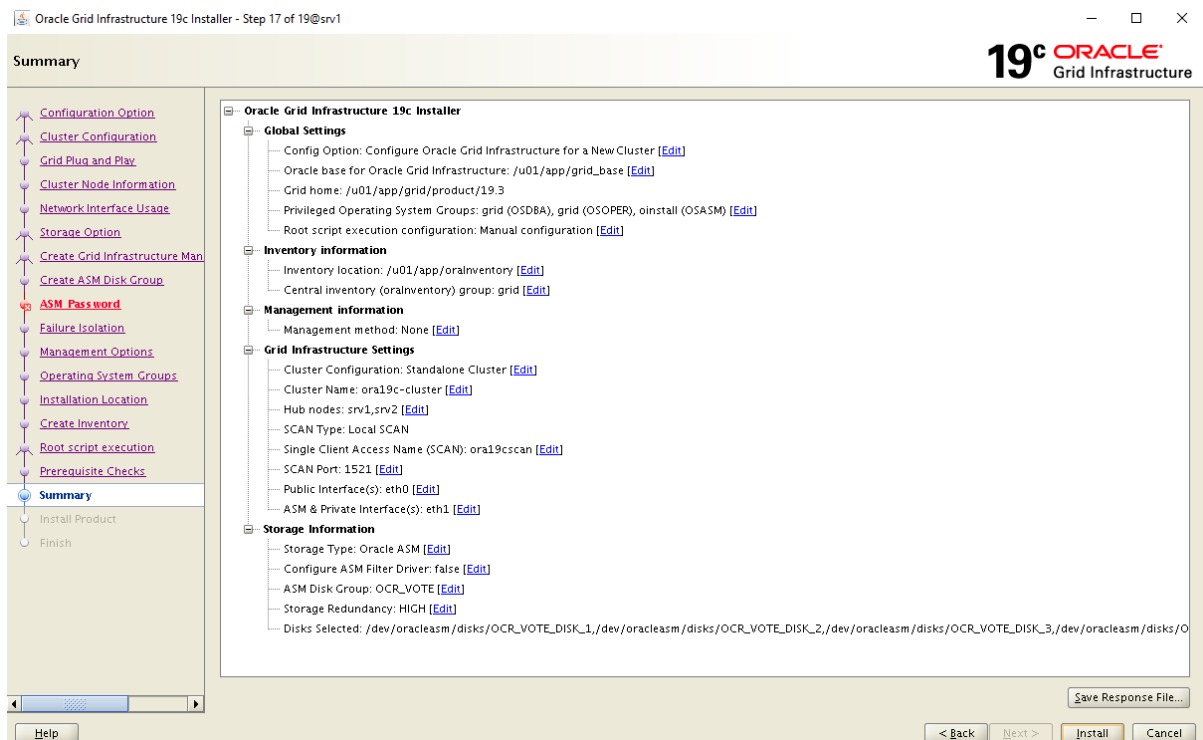


Installation of 19c RAC on Linux7 in Virtual Box VMs

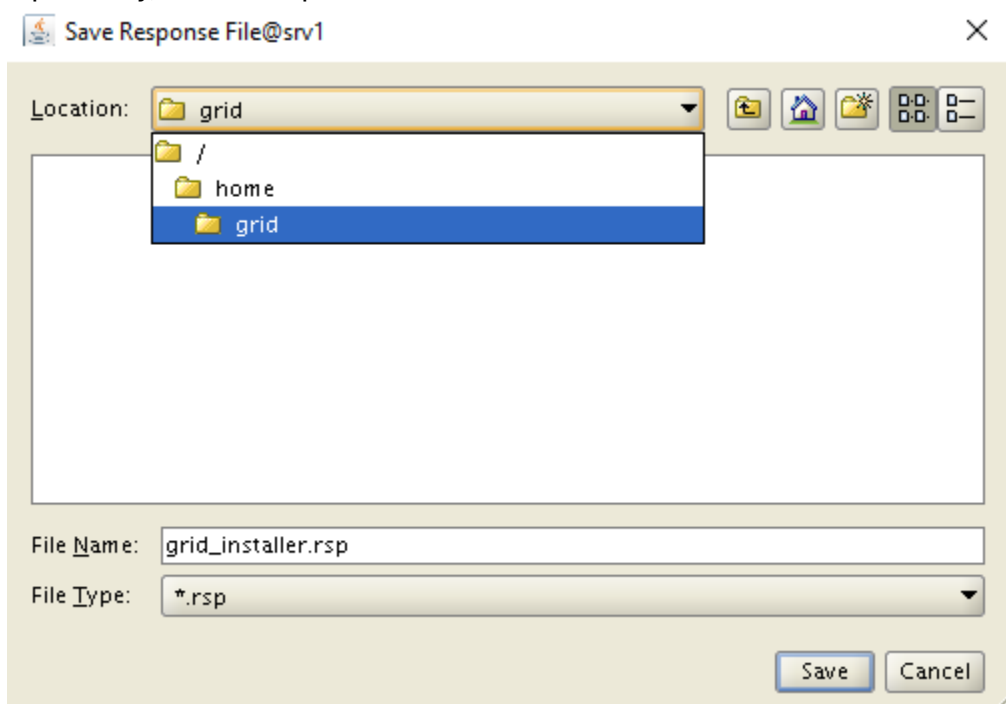


Resize the window that shows all the settings :

Installation of 19c RAC on Linux7 in Virtual Box VMs

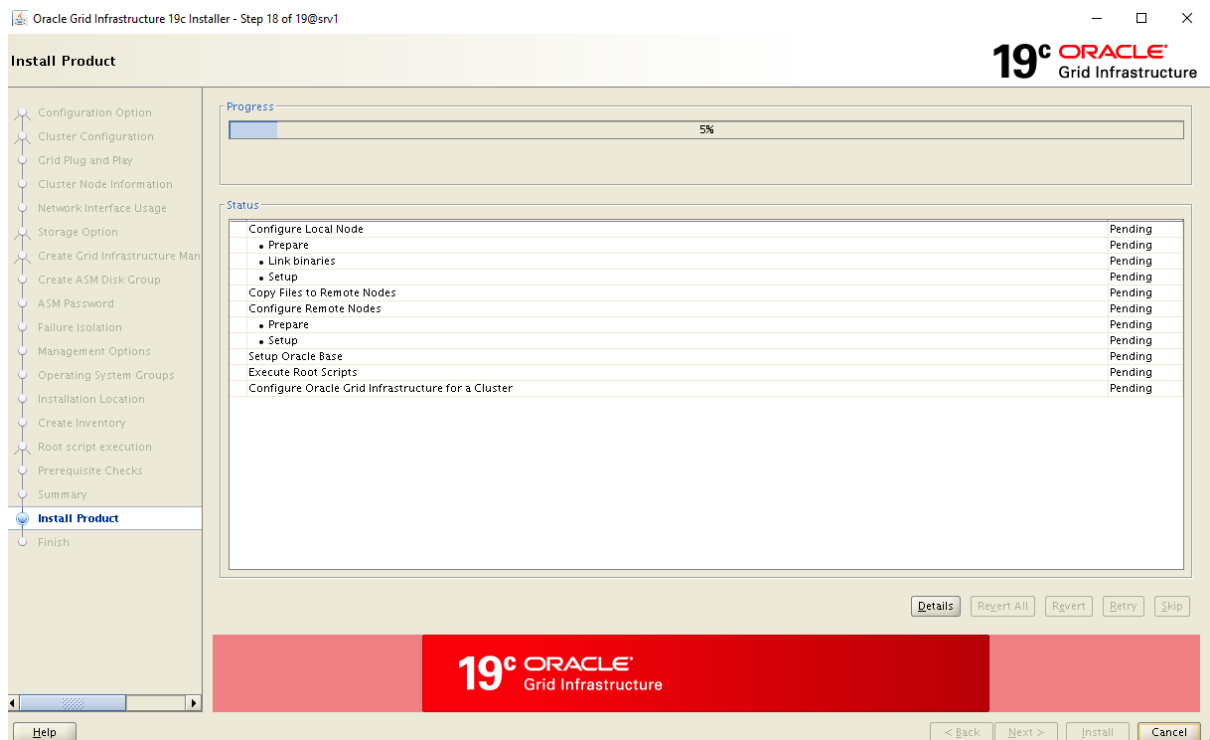


Optionally, “Save Response File”



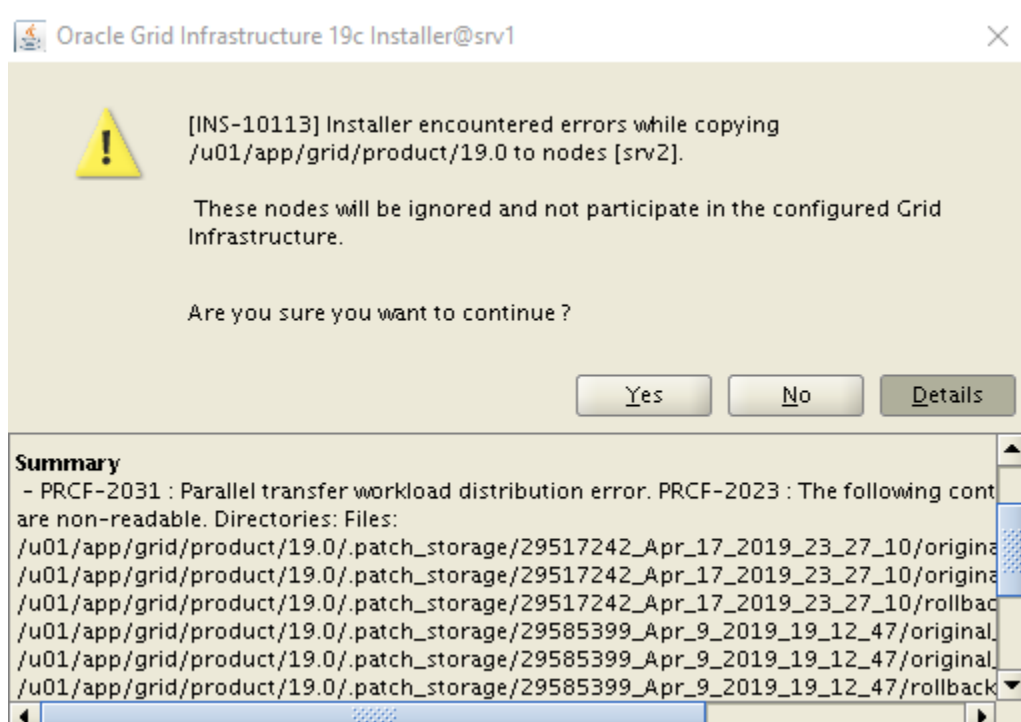
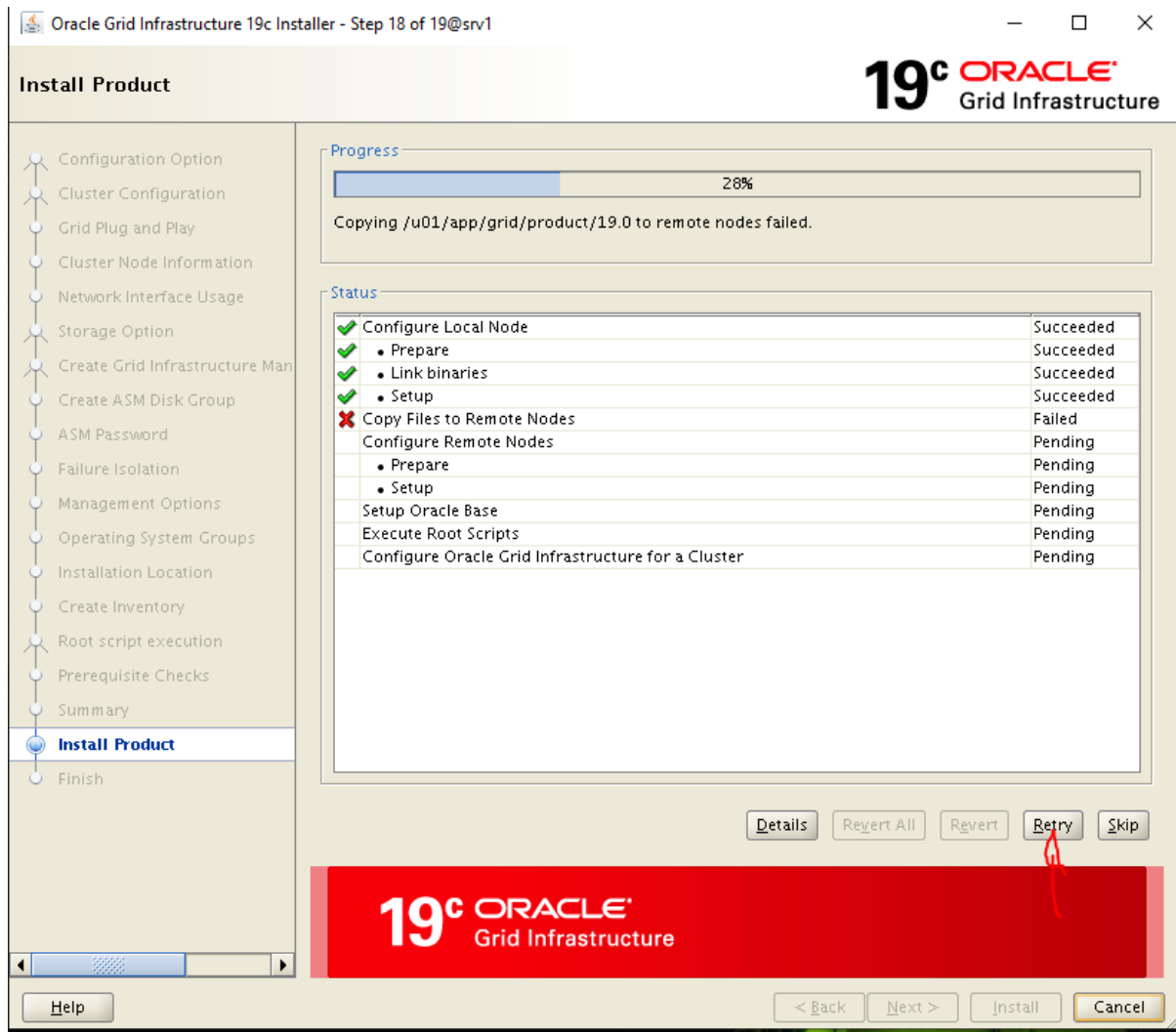
Then proceed to “Install”

Installation of 19c RAC on Linux7 in Virtual Box VMs



This error during installation indicates that extracted files contains a hidden folder “:patch_storage” to which grid does not have permissions. This would happen if you had not changed the ownership and permissions for the folders after unzipping the installer. In this case, change ownership and permissions as below and resume the installation

Installation of 19c RAC on Linux7 in Virtual Box VMs



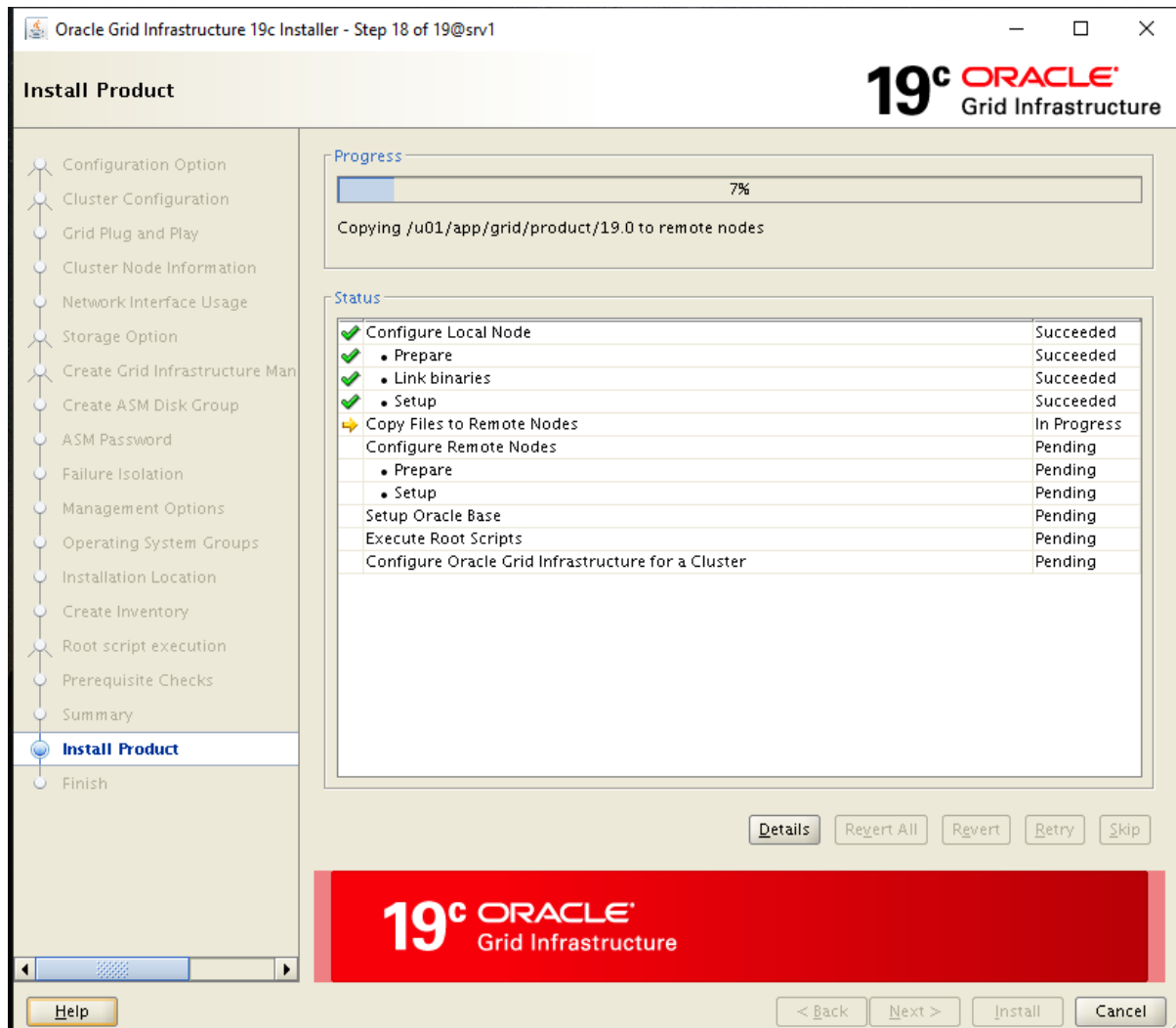
Installation of 19c RAC on Linux7 in Virtual Box VMs

```
[root@srv1 .patch_storage]# pwd
/u01/app/grid/product/19.0/.patch_storage
[root@srv1 .patch_storage]# ls -l
total 24
drwxr-xr-x. 3 root root 74 Apr 18 2019 29401763_Apr_11_2019_22_26_25
drwxr-xr-x. 3 root root 74 Apr 18 2019 29517242_Apr_17_2019_23_27_10
drwxr-xr-x. 3 root root 74 Apr 18 2019 29517247_Apr_1_2019_15_08_20
drwxr-xr-x. 3 root root 74 Apr 18 2019 29585399_Apr_9_2019_19_12_47
-rw-r--r--. 1 root root 6460 Apr 18 2019 interim_inventory.txt
-rw-r--r--. 1 root root 92 Apr 18 2019 LatestOPatchSession.properties
drwxr-xr-x. 10 root root 4096 Apr 18 2019 NApply
-rw-r--r--. 1 root root 5128 Apr 18 2019 record_inventory.txt
[root@srv1 .patch_storage]#
```

If so, change the ownership for this folder as well and retry the step

```
[root@srv1 .patch_storage]# chown -R grid:oinstall *
[root@srv1 .patch_storage]# chown -R grid:oinstall .
[root@srv1 .patch_storage]# pwd
/u01/app/grid/product/19.0/.patch_storage
[root@srv1 .patch_storage]# ls -la
total 32
drwxr-xr-x. 7 grid oinstall 4096 Apr 18 2019 .
drwxr-xr-x. 67 grid oinstall 4096 Mar 19 05:42 ..
drwxr-xr-x. 3 grid oinstall 74 Apr 18 2019
29401763_Apr_11_2019_22_26_25
drwxr-xr-x. 3 grid oinstall 74 Apr 18 2019
29517242_Apr_17_2019_23_27_10
drwxr-xr-x. 3 grid oinstall 74 Apr 18 2019 29517247_Apr_1_2019_15_08_20
drwxr-xr-x. 3 grid oinstall 74 Apr 18 2019 29585399_Apr_9_2019_19_12_47
-rw-r--r--. 1 grid oinstall 6460 Apr 18 2019 interim_inventory.txt
-rw-r--r--. 1 grid oinstall 92 Apr 18 2019
LatestOPatchSession.properties
drwxr-xr-x. 10 grid oinstall 4096 Apr 18 2019 NApply
-rw-r--r--. 1 grid oinstall 5128 Apr 18 2019 record_inventory.txt
[root@srv1 .patch_storage]#
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

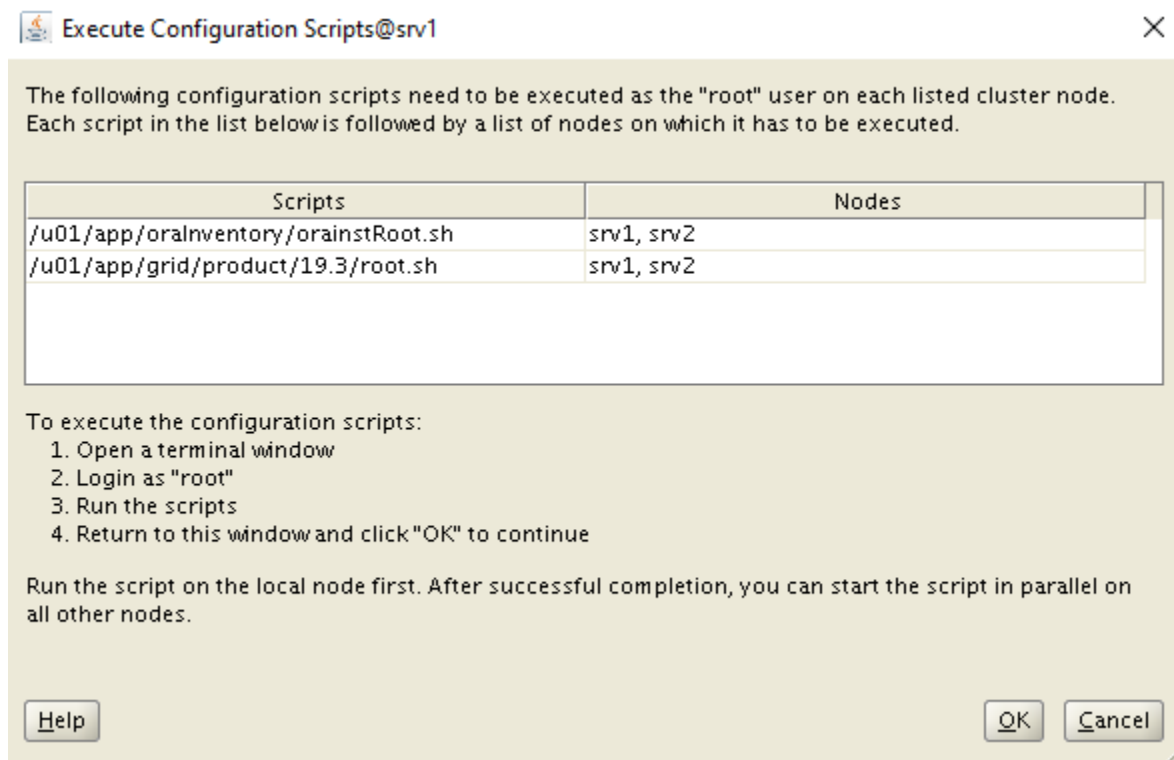
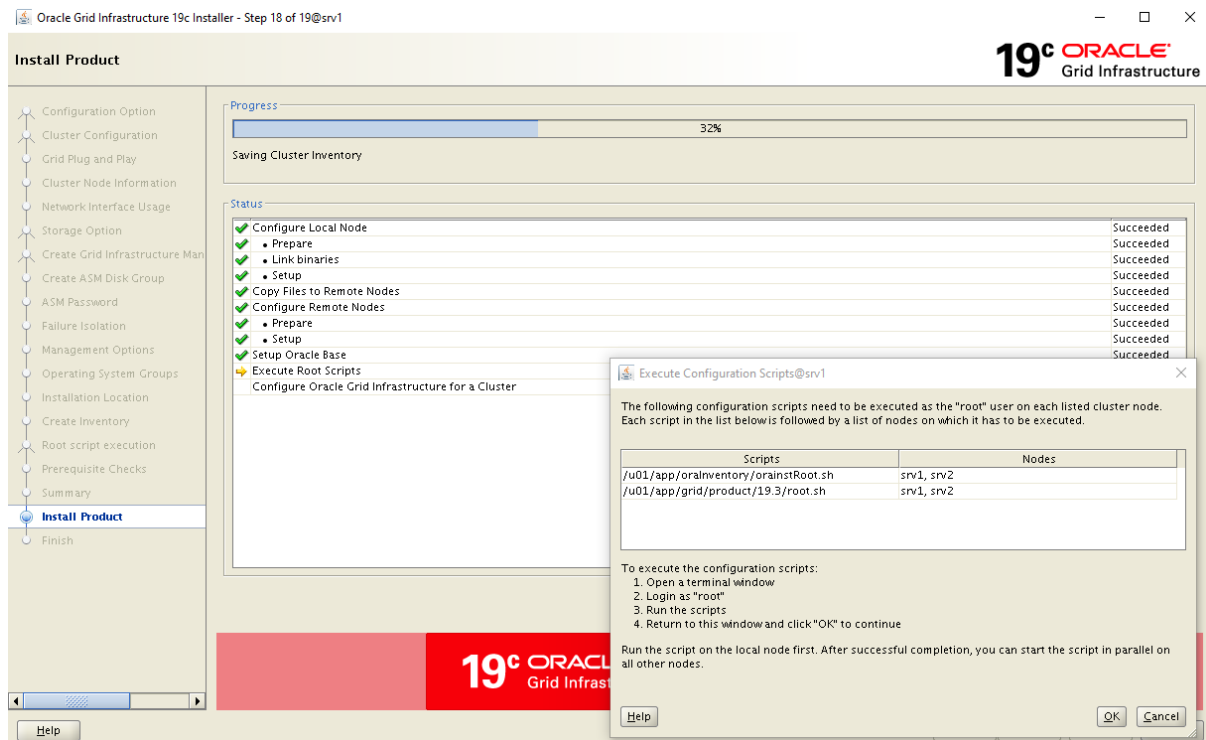


Verify that files are now being copied to srv2 :

```
[root@srv2 19.0]# cd /u01/app/grid
[root@srv2 grid]# du -sh *
4.8G    product
[root@srv2 grid]# du -sh *
5.1G    product
[root@srv2 grid]#
```

The Installer will stop at the “Execute Root Scripts” step :

Installation of 19c RAC on Linux7 in Virtual Box VMs



The two scripts must be run in the same sequence specified on srv1 first. Only on successful completion of the scripts on srv1, should you attempt to run them on srv2 !!

On srv1 :

```
[root@srv1 ~]# cd /u01/app/oraInventory
[root@srv1 oraInventory]# ./orainstRoot.sh
Changing permissions of /u01/app/oraInventory.
Adding read,write permissions for group.
Removing read,write,execute permissions for world.

Changing groupname of /u01/app/oraInventory to grid.
The execution of the script is complete.
[root@srv1 oraInventory]#
```

Note : You can open another ssh session and monitor the `alert.log` file at `/u01/app/grid_base/diag/crs/srv1/crs/trace` to check the CRS startup (which takes some time at Step 17) **[Do NOT edit or write to the alert.log file]** while running `root.sh` below

```
[root@srv1 oraInventory]# cd /u01/app/grid/product/19.3
[root@srv1 19.3]# ls root.sh
root.sh
[root@srv1 19.3]# ./root.sh
Performing root user operation.
```

The following environment variables are set as:

```
ORACLE_OWNER= grid
ORACLE_HOME= /u01/app/grid/product/19.3
```

Enter the full pathname of the local bin directory: [/usr/local/bin]:

```
Copying dbhome to /usr/local/bin ...
Copying oraenv to /usr/local/bin ...
Copying coraenv to /usr/local/bin ...
```

Creating /etc/oratab file...

Entries will be added to the /etc/oratab file as needed by Database Configuration Assistant when a database is created
Finished running generic part of root script.

Now product-specific root actions will be performed.

Relinking oracle with rac_on option

Using configuration parameter file:

`/u01/app/grid/product/19.3/crs/install/crsconfig_params`

The log of current session can be found at:

`/u01/app/grid_base/crsdata/srv1/crsconfig/rootcrs_srv1_2024-03-22_03-44-41AM.log`

2024/03/22 03:44:47 CLSRSC-594: Executing installation step 1 of 19:
'SetupTFA'.

2024/03/22 03:44:47 CLSRSC-594: Executing installation step 2 of 19:
'ValidateEnv'.

2024/03/22 03:44:47 CLSRSC-363: User ignored prerequisites during installation

2024/03/22 03:44:47 CLSRSC-594: Executing installation step 3 of 19:
'CheckFirstNode'.

2024/03/22 03:44:48 CLSRSC-594: Executing installation step 4 of 19:
'GenSiteGUIDs'.

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
2024/03/22 03:44:48 CLSRSC-594: Executing installation step 5 of 19:
'SetupOSD'.
2024/03/22 03:44:48 CLSRSC-594: Executing installation step 6 of 19:
'CheckCRSConfig'.
2024/03/22 03:44:48 CLSRSC-594: Executing installation step 7 of 19:
'SetupLocalGPNP'.
2024/03/22 03:44:57 CLSRSC-594: Executing installation step 8 of 19:
'CreateRootCert'.
2024/03/22 03:44:59 CLSRSC-594: Executing installation step 9 of 19:
'ConfigOLR'.
2024/03/22 03:45:06 CLSRSC-4002: Successfully installed Oracle Trace File
Analyzer (TFA) Collector.
2024/03/22 03:45:09 CLSRSC-594: Executing installation step 10 of 19:
'ConfigCHMOS'.
2024/03/22 03:45:09 CLSRSC-594: Executing installation step 11 of 19:
'CreateOHASD'.
2024/03/22 03:45:12 CLSRSC-594: Executing installation step 12 of 19:
'ConfigOHASD'.
2024/03/22 03:45:12 CLSRSC-330: Adding Clusterware entries to file 'oracle-
ohasd.service'
2024/03/22 03:45:32 CLSRSC-594: Executing installation step 13 of 19:
'InstallAFD'.
2024/03/22 03:45:35 CLSRSC-594: Executing installation step 14 of 19:
'InstallACFS'.
2024/03/22 03:45:38 CLSRSC-594: Executing installation step 15 of 19:
'InstallKA'.
2024/03/22 03:45:41 CLSRSC-594: Executing installation step 16 of 19:
'InitConfig'.
```

ASM has been created and started successfully.

[DBT-30001] Disk groups created successfully. Check
/u01/app/grid_base/cfgtoollogs/asmca/asmca-240322AM034609.log for details.

```
2024/03/22 03:46:59 CLSRSC-482: Running command:
'/u01/app/grid/product/19.3/bin/ocrconfig -upgrade grid grid'
CRS-4256: Updating the profile
Successful addition of voting disk 6c3ea5fbf0254fd5bfd489fc5c674409.
Successful addition of voting disk ff3b9da031064fccbfab4b57933f12e1.
Successful addition of voting disk 44e50015bcf24f7cbfc1b9348fdb568.
Successful addition of voting disk de64da366c164f5cbfba2761df5948d5.
Successful addition of voting disk 4485ff5940384f85bf524a81090c6bd8.
Successfully replaced voting disk group with +OCR_VOTE.
CRS-4256: Updating the profile
CRS-4266: Voting file(s) successfully replaced
##  STATE  File Universal Id  File Name Disk group
--  -
1.  ONLINE  6c3ea5fbf0254fd5bfd489fc5c674409
   (/dev/oracleasm/disks/OCR_VOTE_DISK_1) [OCR_VOTE]
2.  ONLINE  ff3b9da031064fccbfab4b57933f12e1
   (/dev/oracleasm/disks/OCR_VOTE_DISK_2) [OCR_VOTE]
3.  ONLINE  44e50015bcf24f7cbfc1b9348fdb568
   (/dev/oracleasm/disks/OCR_VOTE_DISK_3) [OCR_VOTE]
4.  ONLINE  de64da366c164f5cbfba2761df5948d5
   (/dev/oracleasm/disks/OCR_VOTE_DISK_4) [OCR_VOTE]
5.  ONLINE  4485ff5940384f85bf524a81090c6bd8
   (/dev/oracleasm/disks/OCR_VOTE_DISK_5) [OCR_VOTE]
Located 5 voting disk(s).
2024/03/22 03:48:14 CLSRSC-594: Executing installation step 17 of 19:
'StartCluster'.
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
2024/03/22 03:49:52 CLSRSC-343: Successfully started Oracle Clusterware
stack
2024/03/22 03:49:52 CLSRSC-594: Executing installation step 18 of 19:
'ConfigNode'.
2024/03/22 03:51:06 CLSRSC-594: Executing installation step 19 of 19:
'PostConfig'.
2024/03/22 03:51:24 CLSRSC-325: Configure Oracle Grid Infrastructure for a
Cluster ... succeeded
[root@srv1 19.3]#
```

On completion of the execution at srv1, run this command to check the cluster status :

```
[root@srv1 19.3]# . /home/grid/.bash_profile
[root@srv1 19.3]# crsctl status resource -t
```

```
-----
-----
Name                Target  State          Server          State details
-----
-----
Local Resources
-----
-----
ora.LISTENER.lsnr
                ONLINE  ONLINE        srv1            STABLE
ora.chad
                ONLINE  ONLINE        srv1            STABLE
ora.net1.network
                ONLINE  ONLINE        srv1            STABLE
ora.ons
                ONLINE  ONLINE        srv1            STABLE
-----
-----
Cluster Resources
-----
-----
ora.ASMNET1LSNR_ASM.lsnr(ora.asmgroup)
    1            ONLINE  ONLINE        srv1            STABLE
    2            OFFLINE OFFLINE        srv1            STABLE
    3            OFFLINE OFFLINE        srv1            STABLE
ora.LISTENER_SCAN1.lsnr
    1            ONLINE  ONLINE        srv1            STABLE
ora.LISTENER_SCAN2.lsnr
    1            ONLINE  ONLINE        srv1            STABLE
ora.LISTENER_SCAN3.lsnr
    1            ONLINE  ONLINE        srv1            STABLE
ora.OCR_VOTE.dg(ora.asmgroup)
    1            ONLINE  ONLINE        srv1            STABLE
    2            OFFLINE OFFLINE        srv1            STABLE
    3            OFFLINE OFFLINE        srv1            STABLE
ora.asm(ora.asmgroup)
    1            ONLINE  ONLINE        srv1            Started,STABLE
    2            OFFLINE OFFLINE        srv1            STABLE
    3            OFFLINE OFFLINE        srv1            STABLE
ora.asmnet1.asmnetwork(ora.asmgroup)
    1            ONLINE  ONLINE        srv1            STABLE
    2            OFFLINE OFFLINE        srv1            STABLE
    3            OFFLINE OFFLINE        srv1            STABLE
ora.cvu
    1            ONLINE  ONLINE        srv1            STABLE
ora.qosmserver
    1            ONLINE  ONLINE        srv1            STABLE
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
ora.scan1.vip
  1          ONLINE  ONLINE          srv1          STABLE
ora.scan2.vip
  1          ONLINE  ONLINE          srv1          STABLE
ora.scan3.vip
  1          ONLINE  ONLINE          srv1          STABLE
ora.srv1.vip
  1          ONLINE  ONLINE          srv1          STABLE
-----
-----
[root@srv1 19.3]#
```

Then proceed to execute the two scripts on srv2

```
[root@srv2 ~]# cd /u01/app/oraInventory
[root@srv2 oraInventory]# ./orainstRoot.sh
Changing permissions of /u01/app/oraInventory.
Adding read,write permissions for group.
Removing read,write,execute permissions for world.

Changing groupname of /u01/app/oraInventory to grid.
The execution of the script is complete.
[root@srv2 oraInventory]#

[root@srv2 oraInventory]# cd /u01/app/grid/product/19.3
[root@srv2 19.3]# ls root.sh
root.sh
[root@srv2 19.3]# ./root.sh
Performing root user operation.

The following environment variables are set as:
    ORACLE_OWNER= grid
    ORACLE_HOME=  /u01/app/grid/product/19.3

Enter the full pathname of the local bin directory: [/usr/local/bin]:
    Copying dbhome to /usr/local/bin ...
    Copying oraenv to /usr/local/bin ...
    Copying coraenv to /usr/local/bin ...

Creating /etc/oratab file...
Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
Relinking oracle with rac_on option
Using configuration parameter file:
/u01/app/grid/product/19.3/crs/install/crsconfig_params
The log of current session can be found at:
    /u01/app/grid_base/crsdata/srv2/crsconfig/rootcrs_srv2_2024-03-22_03-56-
03AM.log
2024/03/22 03:56:06 CLSRSC-594: Executing installation step 1 of 19:
'SetupTFA'.
2024/03/22 03:56:06 CLSRSC-594: Executing installation step 2 of 19:
'ValidateEnv'.
2024/03/22 03:56:06 CLSRSC-363: User ignored prerequisites during
installation
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
2024/03/22 03:56:07 CLSRSC-594: Executing installation step 3 of 19:
'CheckFirstNode'.
2024/03/22 03:56:07 CLSRSC-594: Executing installation step 4 of 19:
'GenSiteGUIDs'.
2024/03/22 03:56:07 CLSRSC-594: Executing installation step 5 of 19:
'SetupOSD'.
2024/03/22 03:56:07 CLSRSC-594: Executing installation step 6 of 19:
'CheckCRSConfig'.
2024/03/22 03:56:08 CLSRSC-594: Executing installation step 7 of 19:
'SetupLocalGPNP'.
2024/03/22 03:56:08 CLSRSC-594: Executing installation step 8 of 19:
'CreateRootCert'.
2024/03/22 03:56:08 CLSRSC-594: Executing installation step 9 of 19:
'ConfigOLR'.
2024/03/22 03:56:15 CLSRSC-594: Executing installation step 10 of 19:
'ConfigCHMOS'.
2024/03/22 03:56:15 CLSRSC-594: Executing installation step 11 of 19:
'CreateOHASD'.
2024/03/22 03:56:16 CLSRSC-594: Executing installation step 12 of 19:
'ConfigOHASD'.
2024/03/22 03:56:16 CLSRSC-330: Adding Clusterware entries to file 'oracle-
ohasd.service'
2024/03/22 03:56:26 CLSRSC-4002: Successfully installed Oracle Trace File
Analyzer (TFA) Collector.
2024/03/22 03:56:34 CLSRSC-594: Executing installation step 13 of 19:
'InstallAFD'.
2024/03/22 03:56:35 CLSRSC-594: Executing installation step 14 of 19:
'InstallACFS'.
2024/03/22 03:56:35 CLSRSC-594: Executing installation step 15 of 19:
'InstallKA'.
2024/03/22 03:56:36 CLSRSC-594: Executing installation step 16 of 19:
'InitConfig'.
2024/03/22 03:56:43 CLSRSC-594: Executing installation step 17 of 19:
'StartCluster'.
2024/03/22 03:57:32 CLSRSC-343: Successfully started Oracle Clusterware
stack
2024/03/22 03:57:32 CLSRSC-594: Executing installation step 18 of 19:
'ConfigNode'.
2024/03/22 03:57:42 CLSRSC-594: Executing installation step 19 of 19:
'PostConfig'.
2024/03/22 03:57:47 CLSRSC-325: Configure Oracle Grid Infrastructure for a
Cluster ... succeeded
[root@srv2 19.3]#
```

Rerunning the Cluster Resource Status command should now show both servers (now as nodes of the cluster). This can be checked from both servers

```
[root@srv1 19.3]# crsctl status resource -t
-----
Name                Target    State        Server                State details
-----
Local Resources
-----
ora.LISTENER.lsnr
                   ONLINE    ONLINE      srv1                  STABLE
```

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```

ora.chad          ONLINE  ONLINE          srv2              STABLE
                  ONLINE  ONLINE          srv1              STABLE
                  ONLINE  ONLINE          srv2              STABLE
ora.net1.network  ONLINE  ONLINE          srv1              STABLE
                  ONLINE  ONLINE          srv2              STABLE
ora.ons           ONLINE  ONLINE          srv1              STABLE
                  ONLINE  ONLINE          srv2              STABLE
-----
-----
Cluster Resources
-----
-----
ora.ASMNET1LSNR_ASM.lsnr(ora.asmgroup)
  1      ONLINE  ONLINE          srv1              STABLE
  2      ONLINE  ONLINE          srv2              STABLE
  3      OFFLINE OFFLINE          STABLE
ora.LISTENER_SCAN1.lsnr
  1      ONLINE  ONLINE          srv2              STABLE
ora.LISTENER_SCAN2.lsnr
  1      ONLINE  ONLINE          srv1              STABLE
ora.LISTENER_SCAN3.lsnr
  1      ONLINE  ONLINE          srv1              STABLE
ora.OCR_VOTE.dg(ora.asmgroup)
  1      ONLINE  ONLINE          srv1              STABLE
  2      ONLINE  ONLINE          srv2              STABLE
  3      OFFLINE OFFLINE          STABLE
ora.asm(ora.asmgroup)
  1      ONLINE  ONLINE          srv1              Started,STABLE
  2      ONLINE  ONLINE          srv2              Started,STABLE
  3      OFFLINE OFFLINE          STABLE
ora.asmnet1.asmnetwork(ora.asmgroup)
  1      ONLINE  ONLINE          srv1              STABLE
  2      ONLINE  ONLINE          srv2              STABLE
  3      OFFLINE OFFLINE          STABLE
ora.cvu           1      ONLINE  ONLINE          srv1              STABLE
ora.qosmserver    1      ONLINE  ONLINE          srv1              STABLE
ora.scan1.vip     1      ONLINE  ONLINE          srv2              STABLE
ora.scan2.vip     1      ONLINE  ONLINE          srv1              STABLE
ora.scan3.vip     1      ONLINE  ONLINE          srv1              STABLE
ora.srv1.vip      1      ONLINE  ONLINE          srv1              STABLE
ora.srv2.vip      1      ONLINE  ONLINE          srv2              STABLE
-----
-----
[root@srv1 19.3]#

[root@srv2 19.3]# . /home/grid/.bash_profile
[root@srv2 19.3]# crsctl status resource -t

```

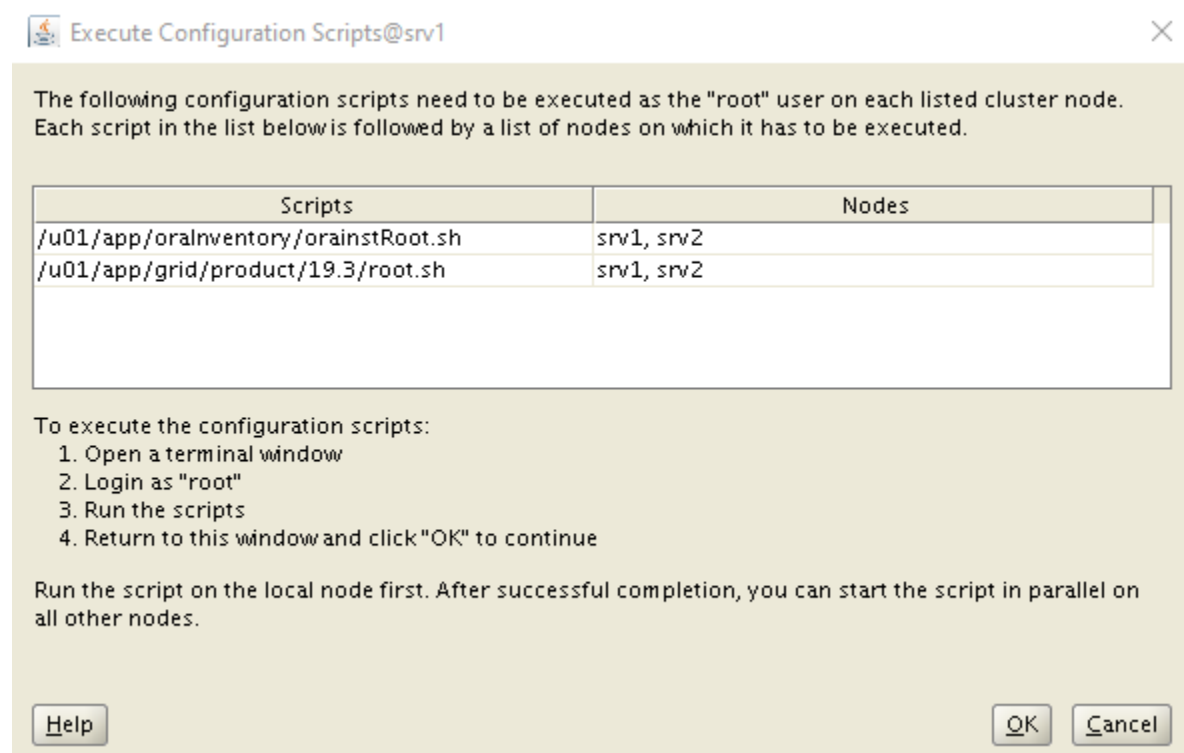
Installation of 19c RAC on Linux7 in Virtual Box VMs

Name	Target	State	Server	State details
Local Resources				
ora.LISTENER.lsnr				
	ONLINE	ONLINE	srv1	STABLE
	ONLINE	ONLINE	srv2	STABLE
ora.chad				
	ONLINE	ONLINE	srv1	STABLE
	ONLINE	ONLINE	srv2	STABLE
ora.net1.network				
	ONLINE	ONLINE	srv1	STABLE
	ONLINE	ONLINE	srv2	STABLE
ora.ons				
	ONLINE	ONLINE	srv1	STABLE
	ONLINE	ONLINE	srv2	STABLE
Cluster Resources				
ora.ASMNET1LSNR_ASM.lsnr(ora.asmgroup)				
1	ONLINE	ONLINE	srv1	STABLE
2	ONLINE	ONLINE	srv2	STABLE
3	OFFLINE	OFFLINE		STABLE
ora.LISTENER_SCAN1.lsnr				
1	ONLINE	ONLINE	srv2	STABLE
ora.LISTENER_SCAN2.lsnr				
1	ONLINE	ONLINE	srv1	STABLE
ora.LISTENER_SCAN3.lsnr				
1	ONLINE	ONLINE	srv1	STABLE
ora.OCR_VOTE.dg(ora.asmgroup)				
1	ONLINE	ONLINE	srv1	STABLE
2	ONLINE	ONLINE	srv2	STABLE
3	OFFLINE	OFFLINE		STABLE
ora.asm(ora.asmgroup)				
1	ONLINE	ONLINE	srv1	Started,STABLE
2	ONLINE	ONLINE	srv2	Started,STABLE
3	OFFLINE	OFFLINE		STABLE
ora.asmnet1.asmnetwork(ora.asmgroup)				
1	ONLINE	ONLINE	srv1	STABLE
2	ONLINE	ONLINE	srv2	STABLE
3	OFFLINE	OFFLINE		STABLE
ora.cvu				
1	ONLINE	ONLINE	srv1	STABLE
ora.qosmserver				
1	ONLINE	ONLINE	srv1	STABLE
ora.scan1.vip				
1	ONLINE	ONLINE	srv2	STABLE
ora.scan2.vip				
1	ONLINE	ONLINE	srv1	STABLE
ora.scan3.vip				
1	ONLINE	ONLINE	srv1	STABLE
ora.srv1.vip				
1	ONLINE	ONLINE	srv1	STABLE
ora.srv2.vip				

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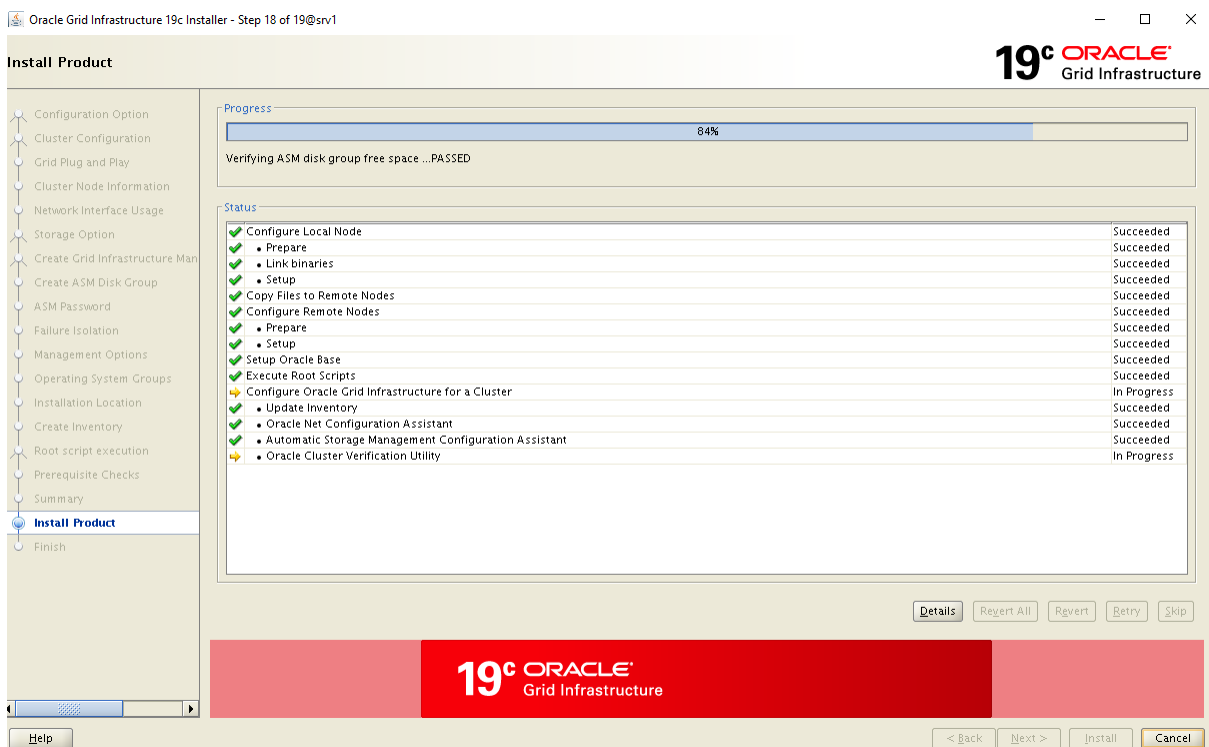
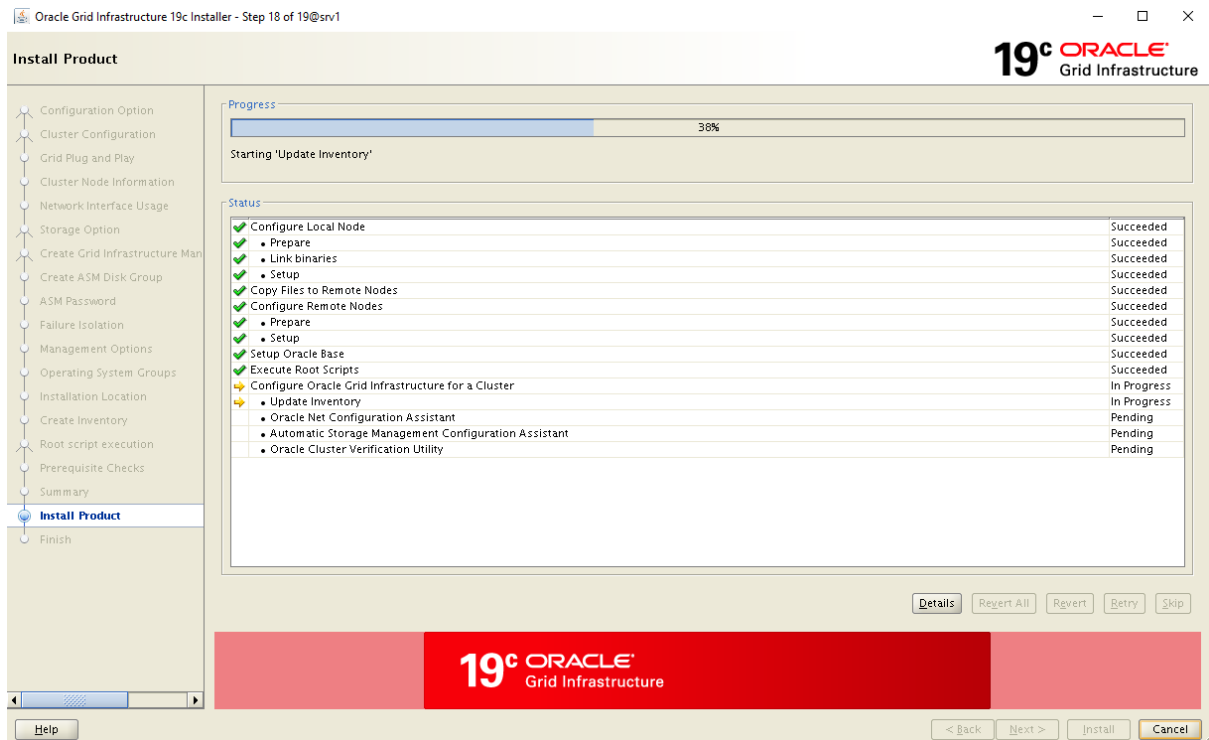
```
1          ONLINE  ONLINE          srv2          STABLE
-----
[root@srv2 19.3]#
```

Click OK on the “Execute Configuration Scripts” window to return to the main Installer window



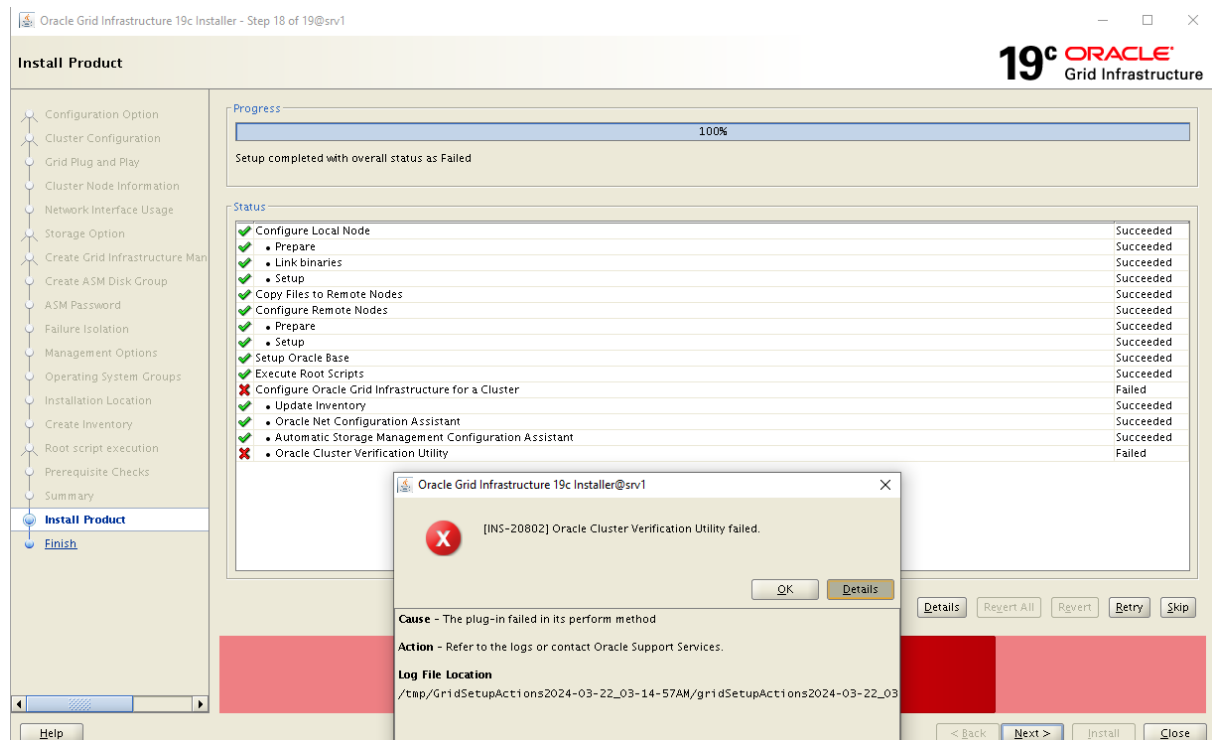
The main Installer will then proceed with the next few steps

Installation of 19c RAC on Linux7 in Virtual Box VMs



If the Sub-Step “Oracle Cluster Verification Utility” of the “Configure Oracle Grid Infrastructure for a Cluster” main step fails, you can view the log file

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```
[root@srv1 GridSetupActions2024-03-22_03-14-57AM]# pwd
/tmp/GridSetupActions2024-03-22_03-14-57AM
[root@srv1 GridSetupActions2024-03-22_03-14-57AM]# ls -ltr
total 16672
-rw-r--r--. 1 grid grid      129 Mar 22 03:14 installerPatchActions_2024-03-22_03-14-57AM.log
-rw-r--r--. 1 grid grid        0 Mar 22 03:15 gridSetupActions2024-03-22_03-14-57AM.log.lck
-rw-r--r--. 1 grid grid        0 Mar 22 03:15 gridSetupActions2024-03-22_03-14-57AM.err
drwxr-xr-x. 4 grid grid      33 Mar 22 03:20 temp_ob
drwxr-xr-x. 3 grid grid    4096 Mar 22 03:33 CVU_19.0.0.0.0_grid
-rw-r-----. 1 grid grid        0 Mar 22 03:39 oraInstall2024-03-22_03-14-57AM.err
-rw-r-----. 1 grid grid     135 Mar 22 03:39 oraInstall2024-03-22_03-14-57AM.out
-rw-r--r--. 1 grid grid 8368540 Mar 22 03:39 gridSetupActions2024-03-22_03-14-57AM.out
-rw-r-----. 1 grid grid     234 Mar 22 03:41 oraInstall2024-03-22_03-14-57AM.out.srv2
-rw-r-----. 1 grid grid        0 Mar 22 03:41 oraInstall2024-03-22_03-14-57AM.err.srv2
-rw-r--r--. 1 grid grid   90884 Mar 22 04:03 time2024-03-22_03-14-57AM.log
-rw-r-----. 1 grid grid 8580515 Mar 22 04:05 gridSetupActions2024-03-22_03-14-57AM.log
[root@srv1 GridSetupActions2024-03-22_03-14-57AM]#
[root@srv1 GridSetupActions2024-03-22_03-14-57AM]# vi gridSetupActions2024-03-22_03-14-57AM.log gridSetupActions2024-03-22_03-14-57AM.out
```

The error text in the log is :

INFO: Checks did not pass for the following nodes:

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
INFO: srv2,srv1
INFO: Failures were encountered during execution of CVU verification
request "stage -post crsinst".
INFO: Verifying Single Client Access Name (SCAN) ...FAILED
INFO: PRVG-11372 : Number of SCAN IP addresses that SCAN "ora19cscan"
resolved to did
INFO: not match the number of SCAN VIP resources
INFO: Verifying DNS/NIS name service 'ora19cscan' ...FAILED
INFO: PRVG-11826 : DNS resolved IP addresses "" for SCAN name "ora19cscan"
not
INFO: found in the name service returned IP addresses
INFO: "192.168.56.94,192.168.56.95,192.168.56.96"
INFO: RVG-11827 : Name service returned IP addresses
INFO: "192.168.56.94,192.168.56.95,192.168.56.96" for SCAN name
"ora19cscan" not
INFO: found in the DNS returned IP addresses ""
INFO: srv2: PRVF-4664 : Found inconsistent name resolution entries for SCAN
name
INFO: "ora19cscan"
INFO: srv1: PRVF-4664 : Found inconsistent name resolution entries for SCAN
name
INFO: "ora19cscan"
INFO: CVU operation performed:      stage -post crsinst
INFO: CVU home:                    /u01/app/grid/product/19.3/
INFO: User:                        grid
INFO: Completed Plugin named: cvu
INFO: Setup completed with overall status as Failed
INFO: ConfigClient.saveSession method called
INFO: Oracle Cluster Verification Utility failed.
INFO: Oracle Cluster Verification Utility failed.
INFO: ConfigClient.executeSelectedToolsInAggregate action performed
INFO: Exiting ConfigClient.executeSelectedToolsInAggregate method
INFO: Adding ExitStatus SUCCESS_MINUS_RECTOOL to the exit status set
INFO: ConfigClient.saveSession method called
INFO: Calling event ConfigSessionEnding
INFO: ConfigClient.endSession method called
INFO: Completed Configuration
INFO: Adding ExitStatus FAILURE to the exit status set
```

This Name Resolution Error can be ignored.

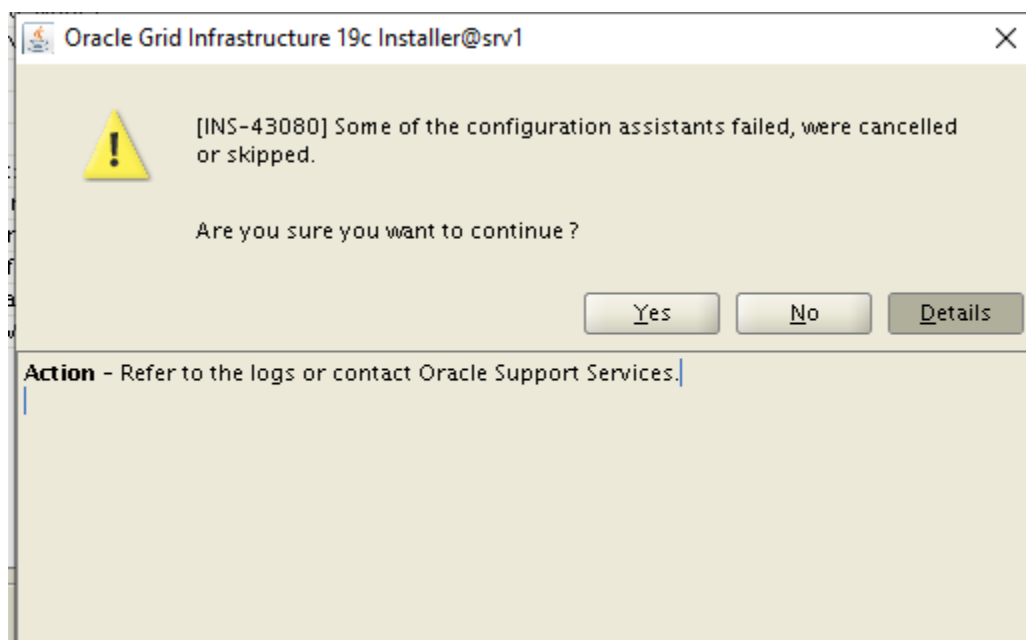
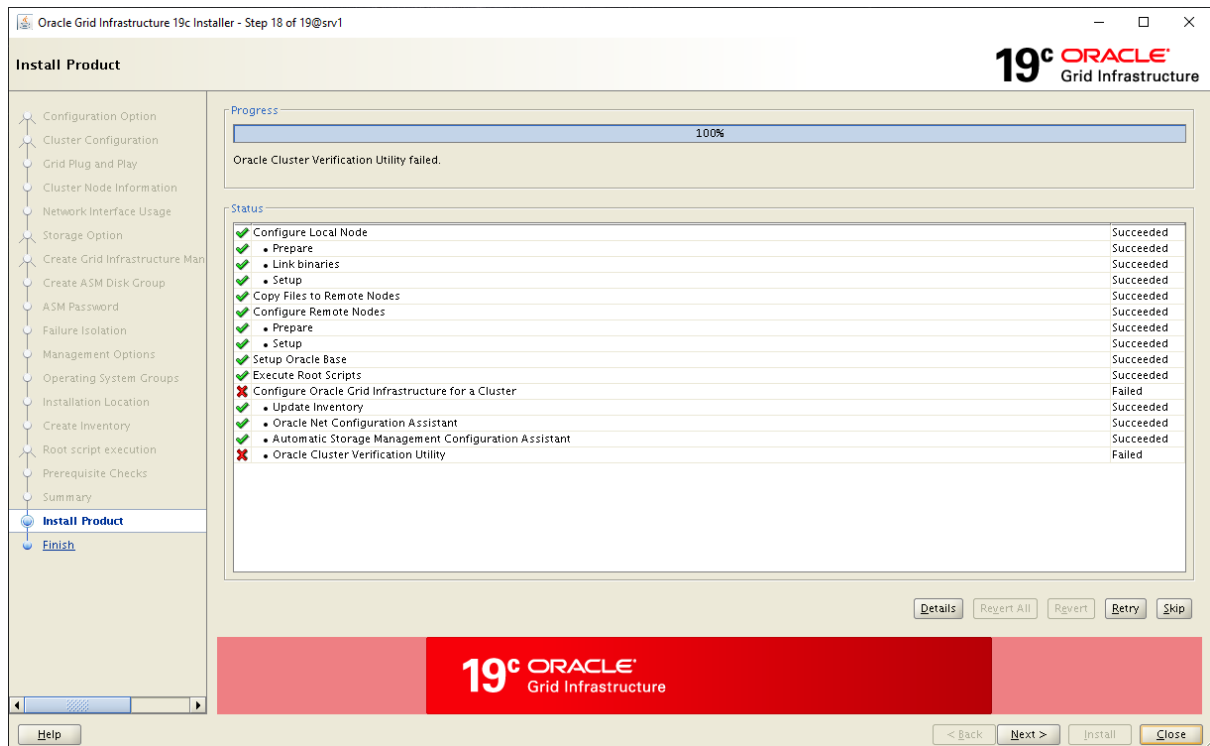
The error is

PRVG-11372 Number of SCAN IP addresses that SCAN "{0}" resolved to did
nnot match the number of SCAN VIP resources
Cause It was found that the SCAN IP addresses that SCAN resolves to were
modified after the SCAN was created.

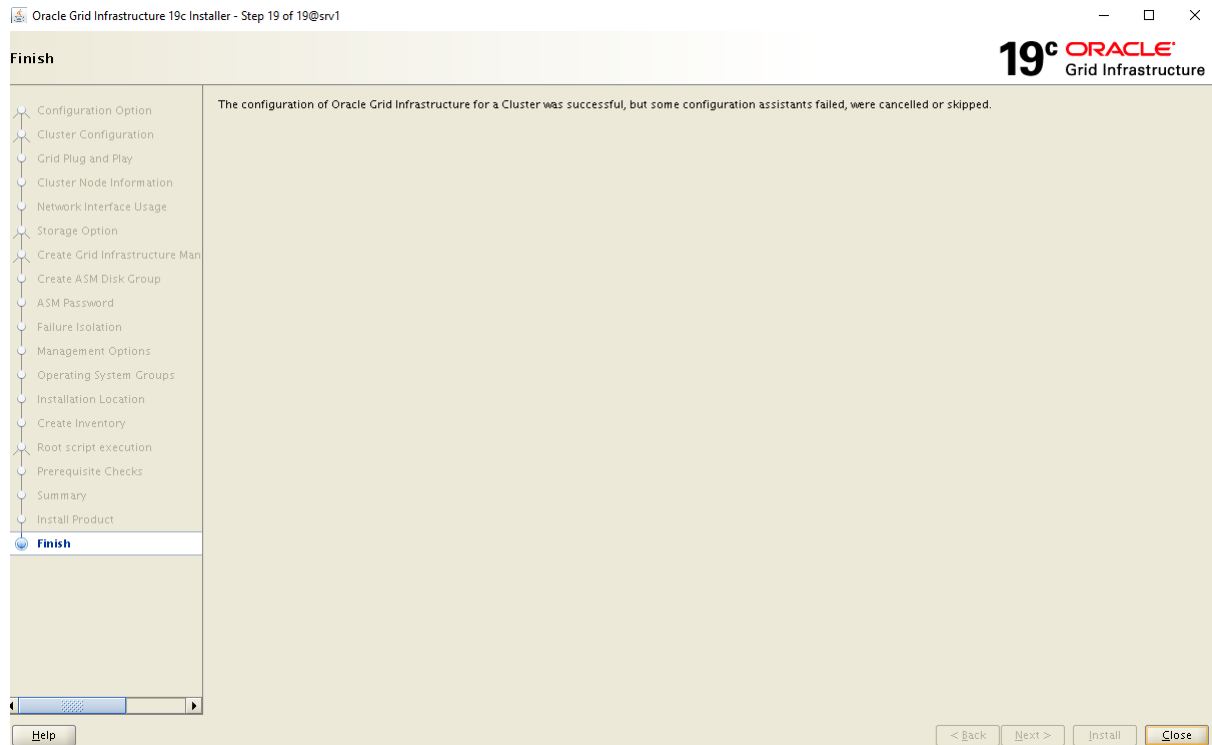
**I have fixed this issue by reconfiguring the SCAN to include the domain name as
ora19scan.localdomain. See [Correct the SCAN to include domain](#)**

Click Next on the Installer Main Screen

Installation of 19c RAC on Linux7 in Virtual Box VMs



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Click Close

The ssh session where you started the gridSetup.sh would return to the command prompt. You can then review the log files

```
[grid@srv1 19.3]$ ./gridSetup.sh
Launching Oracle Grid Infrastructure Setup Wizard...

The response file for this session can be found at:
/u01/app/grid/product/19.3/install/response/grid_2024-03-22_03-14-57AM.rsp

You can find the log of this install session at:
/tmp/GridSetupActions2024-03-22_03-14-57AM/gridSetupActions2024-03-22_03-14-57AM.log
Moved the install session logs to:
/u01/app/oraInventory/logs/GridSetupActions2024-03-22_03-14-57AM
[grid@srv1 19.3]$
```

Then check the Cluster Resources status again from both nodes (remember to invoke :
“`. /home/grid/.bash_profile`” in your root ssh window so that the correct environment parameters are set before running crsctl)

I have highlighted the section where it shows the 3 SCAN VIPs

```
[root@srv1 19.3]# crsctl status resource -t
```

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Name	Target	State	Server	State details
Local Resources				
ora.LISTENER.lsnr				
	ONLINE	ONLINE	srv1	STABLE
	ONLINE	ONLINE	srv2	STABLE
ora.chad				
	ONLINE	ONLINE	srv1	STABLE
	ONLINE	ONLINE	srv2	STABLE
ora.net1.network				
	ONLINE	ONLINE	srv1	STABLE
	ONLINE	ONLINE	srv2	STABLE
ora.ons				
	ONLINE	ONLINE	srv1	STABLE
	ONLINE	ONLINE	srv2	STABLE
Cluster Resources				
ora.ASMNET1LSNR_ASM.lsnr(ora.asmgroup)				
1	ONLINE	ONLINE	srv1	STABLE
2	ONLINE	ONLINE	srv2	STABLE
3	OFFLINE	OFFLINE		STABLE
ora.LISTENER_SCAN1.lsnr				
1	ONLINE	ONLINE	srv2	STABLE
ora.LISTENER_SCAN2.lsnr				
1	ONLINE	ONLINE	srv1	STABLE
ora.LISTENER_SCAN3.lsnr				
1	ONLINE	ONLINE	srv1	STABLE
ora.OCR_VOTE.dg(ora.asmgroup)				
1	ONLINE	ONLINE	srv1	STABLE
2	ONLINE	ONLINE	srv2	STABLE
3	OFFLINE	OFFLINE		STABLE
ora.asm(ora.asmgroup)				
1	ONLINE	ONLINE	srv1	Started,STABLE
2	ONLINE	ONLINE	srv2	Started,STABLE
3	OFFLINE	OFFLINE		STABLE
ora.asmnet1.asmnetwork(ora.asmgroup)				
1	ONLINE	ONLINE	srv1	STABLE
2	ONLINE	ONLINE	srv2	STABLE
3	OFFLINE	OFFLINE		STABLE
ora.cvu				
1	ONLINE	ONLINE	srv1	STABLE
ora.qosmserver				
1	ONLINE	ONLINE	srv1	STABLE
ora.scan1.vip				
1	ONLINE	ONLINE	srv2	STABLE
ora.scan2.vip				
1	ONLINE	ONLINE	srv1	STABLE
ora.scan3.vip				
1	ONLINE	ONLINE	srv1	STABLE
ora.srv1.vip				
1	ONLINE	ONLINE	srv1	STABLE
ora.srv2.vip				

Installation of 19c RAC on Linux7 in Virtual Box VMs

```

1          ONLINE  ONLINE          srv2          STABLE
-----
[root@srv1 19.3]#

[root@srv2 19.3]# crsctl status resource -t
-----
Name          Target  State          Server          State details
-----
Local Resources
-----
ora.LISTENER.lsnr
              ONLINE  ONLINE          srv1          STABLE
              ONLINE  ONLINE          srv2          STABLE
ora.chad
              ONLINE  ONLINE          srv1          STABLE
              ONLINE  ONLINE          srv2          STABLE
ora.net1.network
              ONLINE  ONLINE          srv1          STABLE
              ONLINE  ONLINE          srv2          STABLE
ora.ons
              ONLINE  ONLINE          srv1          STABLE
              ONLINE  ONLINE          srv2          STABLE
-----
Cluster Resources
-----
ora.ASMNET1LSNR_ASM.lsnr(ora.asmgroup)
1          ONLINE  ONLINE          srv1          STABLE
2          ONLINE  ONLINE          srv2          STABLE
3          OFFLINE OFFLINE          STABLE
ora.LISTENER_SCAN1.lsnr
1          ONLINE  ONLINE          srv2          STABLE
ora.LISTENER_SCAN2.lsnr
1          ONLINE  ONLINE          srv1          STABLE
ora.LISTENER_SCAN3.lsnr
1          ONLINE  ONLINE          srv1          STABLE
ora.OCR_VOTE.dg(ora.asmgroup)
1          ONLINE  ONLINE          srv1          STABLE
2          ONLINE  ONLINE          srv2          STABLE
3          OFFLINE OFFLINE          STABLE
ora.asm(ora.asmgroup)
1          ONLINE  ONLINE          srv1          Started,STABLE
2          ONLINE  ONLINE          srv2          Started,STABLE
3          OFFLINE OFFLINE          STABLE
ora.asmnet1.asmnetwork(ora.asmgroup)
1          ONLINE  ONLINE          srv1          STABLE
2          ONLINE  ONLINE          srv2          STABLE
3          OFFLINE OFFLINE          STABLE
ora.cvu
1          ONLINE  ONLINE          srv1          STABLE
ora.qosmserver
1          ONLINE  ONLINE          srv1          STABLE
ora.scan1.vip
1          ONLINE  ONLINE          srv2          STABLE
ora.scan2.vip

```

```

      1      ONLINE  ONLINE      srv1      STABLE
ora.scan3.vip
      1      ONLINE  ONLINE      srv1      STABLE
ora.srv1.vip
      1      ONLINE  ONLINE      srv1      STABLE
ora.srv2.vip
      1      ONLINE  ONLINE      srv2      STABLE
-----
-----
[root@srv2 19.3]#

```

Post-Install Verification

Important Note : After the Grid Infrastructure installation, the startup of ASM and ClusterWare can take a few minutes. Verify that the services are UP before attempting to create ASM DiskGroups or run cluvfy in subsequent steps. Run crsctl status commands to check.

Reboot both nodes and invoke “. /home/grid/.bash_profile” and run “crsctl status resource -t” again from each node.

I have highlighted the section where it shows the 3 SCAN VIPs

```

[root@srv1 ~]# . /home/grid/.bash_profile
[root@srv1 ~]# crsctl check cluster -all
*****
srv1:
CRS-4537: Cluster Ready Services is online
CRS-4529: Cluster Synchronization Services is online
CRS-4533: Event Manager is online
*****
srv2:
CRS-4537: Cluster Ready Services is online
CRS-4529: Cluster Synchronization Services is online
CRS-4533: Event Manager is online
*****
[root@srv1 ~]#
[root@srv1 ~]# crsctl status resource -t
-----
-----
Name                Target  State      Server      State details
-----
-----
Local Resources
-----
-----
ora.LISTENER.lsnr
      ONLINE  ONLINE      srv1      STABLE
      ONLINE  ONLINE      srv2      STABLE
ora.chad
      ONLINE  ONLINE      srv1      STABLE
      ONLINE  ONLINE      srv2      STABLE
ora.net1.network
      ONLINE  ONLINE      srv1      STABLE

```

Installation of 19c RAC on Linux7 in Virtual Box VMs

```

ora.ons
      ONLINE  ONLINE      srv2
                                     STABLE
      ONLINE  ONLINE      srv1
      ONLINE  ONLINE      srv2
                                     STABLE
      STABLE

-----
Cluster Resources
-----
ora.ASMNET1LSNR_ASM.lsnr(ora.asmgroup)
      1      ONLINE  ONLINE      srv1      STABLE
      2      ONLINE  ONLINE      srv2      STABLE
      3      ONLINE  OFFLINE
                                     STABLE
ora.LISTENER_SCAN1.lsnr
      1      ONLINE  ONLINE      srv2      STABLE
ora.LISTENER_SCAN2.lsnr
      1      ONLINE  ONLINE      srv1      STABLE
ora.LISTENER_SCAN3.lsnr
      1      ONLINE  ONLINE      srv1      STABLE
ora.OCR_VOTE.dg(ora.asmgroup)
      1      ONLINE  ONLINE      srv1      STABLE
      2      ONLINE  ONLINE      srv2      STABLE
      3      OFFLINE OFFLINE
                                     STABLE
ora.asm(ora.asmgroup)
      1      ONLINE  ONLINE      srv1      Started,STABLE
      2      ONLINE  ONLINE      srv2      Started,STABLE
      3      OFFLINE OFFLINE
                                     STABLE
ora.asmnet1.asmnetwork(ora.asmgroup)
      1      ONLINE  ONLINE      srv1      STABLE
      2      ONLINE  ONLINE      srv2      STABLE
      3      OFFLINE OFFLINE
                                     STABLE
ora.cvu
      1      ONLINE  ONLINE      srv1      STABLE
ora.qosmserver
      1      ONLINE  ONLINE      srv1      STABLE
ora.scan1.vip
      1      ONLINE  ONLINE      srv2      STABLE
ora.scan2.vip
      1      ONLINE  ONLINE      srv1      STABLE
ora.scan3.vip
      1      ONLINE  ONLINE      srv1      STABLE
ora.srv1.vip
      1      ONLINE  ONLINE      srv1      STABLE
ora.srv2.vip
      1      ONLINE  ONLINE      srv2      STABLE

-----
[root@srv1 ~]#
[root@srv1 ~]# nslookup ora19cscan
Server:      192.168.56.80
Address:     192.168.56.80#53

Name:   ora19cscan
Address: 192.168.56.94
Name:   ora19cscan
Address: 192.168.56.95
Name:   ora19cscan
Address: 192.168.56.96

[root@srv1 ~]# nslookup ora19cscan

```

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
Server:          192.168.56.80
Address:         192.168.56.80#53
```

```
Name:   ora19cscan
Address: 192.168.56.95
Name:   ora19cscan
Address: 192.168.56.96
Name:   ora19cscan
Address: 192.168.56.94
```

```
[root@srv1 ~]# nslookup ora19cscan
Server:          192.168.56.80
Address:         192.168.56.80#53
```

```
Name:   ora19cscan
Address: 192.168.56.96
Name:   ora19cscan
Address: 192.168.56.94
Name:   ora19cscan
Address: 192.168.56.95
```

```
[root@srv1 ~]#
```

```
[root@srv2 ~]# . /home/grid/.bash_profile
```

```
[root@srv2 ~]# crsctl check cluster -all
```

```
*****
```

```
srv1:
```

```
CRS-4537: Cluster Ready Services is online
```

```
CRS-4529: Cluster Synchronization Services is online
```

```
CRS-4533: Event Manager is online
```

```
*****
```

```
srv2:
```

```
CRS-4537: Cluster Ready Services is online
```

```
CRS-4529: Cluster Synchronization Services is online
```

```
CRS-4533: Event Manager is online
```

```
*****
```

```
[root@srv2 ~]#
```

```
[root@srv2 ~]# crsctl status resource -t
```

```
-----
-----
Name                Target   State        Server                State details
-----
-----
Local Resources
-----
-----
ora.LISTENER.lsnr
      ONLINE  ONLINE      srv1                STABLE
      ONLINE  ONLINE      srv2                STABLE
ora.chad
      ONLINE  ONLINE      srv1                STABLE
      ONLINE  ONLINE      srv2                STABLE
ora.net1.network
      ONLINE  ONLINE      srv1                STABLE
      ONLINE  ONLINE      srv2                STABLE
ora.ons
      ONLINE  ONLINE      srv1                STABLE
      ONLINE  ONLINE      srv2                STABLE
-----
-----
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

Cluster Resources

```
-----
-----
ora.ASMNET1LSNR_ASM.lsnr(ora.asmgroup)
  1      ONLINE  ONLINE      srv1      STABLE
  2      ONLINE  ONLINE      srv2      STABLE
  3      ONLINE  OFFLINE     srv2      STABLE
ora.LISTENER_SCAN1.lsnr
  1      ONLINE  ONLINE      srv2      STABLE
ora.LISTENER_SCAN2.lsnr
  1      ONLINE  ONLINE      srv1      STABLE
ora.LISTENER_SCAN3.lsnr
  1      ONLINE  ONLINE      srv1      STABLE
ora.OCR_VOTE.dg(ora.asmgroup)
  1      ONLINE  ONLINE      srv1      STABLE
  2      ONLINE  ONLINE      srv2      STABLE
  3      OFFLINE OFFLINE     srv2      STABLE
ora.asm(ora.asmgroup)
  1      ONLINE  ONLINE      srv1      Started,STABLE
  2      ONLINE  ONLINE      srv2      Started,STABLE
  3      OFFLINE OFFLINE     srv2      STABLE
ora.asmnet1.asmnetwork(ora.asmgroup)
  1      ONLINE  ONLINE      srv1      STABLE
  2      ONLINE  ONLINE      srv2      STABLE
  3      OFFLINE OFFLINE     srv2      STABLE
ora.cvu
  1      ONLINE  ONLINE      srv1      STABLE
ora.qosmserver
  1      ONLINE  ONLINE      srv1      STABLE
ora.scan1.vip
  1      ONLINE  ONLINE      srv2      STABLE
ora.scan2.vip
  1      ONLINE  ONLINE      srv1      STABLE
ora.scan3.vip
  1      ONLINE  ONLINE      srv1      STABLE
ora.srv1.vip
  1      ONLINE  ONLINE      srv1      STABLE
ora.srv2.vip
  1      ONLINE  ONLINE      srv2      STABLE
-----
```

```
-----
[root@srv2 ~]# nslookup ora19cscan
Server:      192.168.56.80
Address:     192.168.56.80#53
```

```
Name:   ora19cscan
Address: 192.168.56.94
Name:   ora19cscan
Address: 192.168.56.95
Name:   ora19cscan
Address: 192.168.56.96
```

```
[root@srv2 ~]# nslookup ora19cscan
Server:      192.168.56.80
Address:     192.168.56.80#53
```

```
Name:   ora19cscan
Address: 192.168.56.95
Name:   ora19cscan
Address: 192.168.56.96
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

Name: ora19cscan
Address: 192.168.56.94

```
[root@srv2 ~]#
```

Create ASM DiskGroups for DATA and FRA

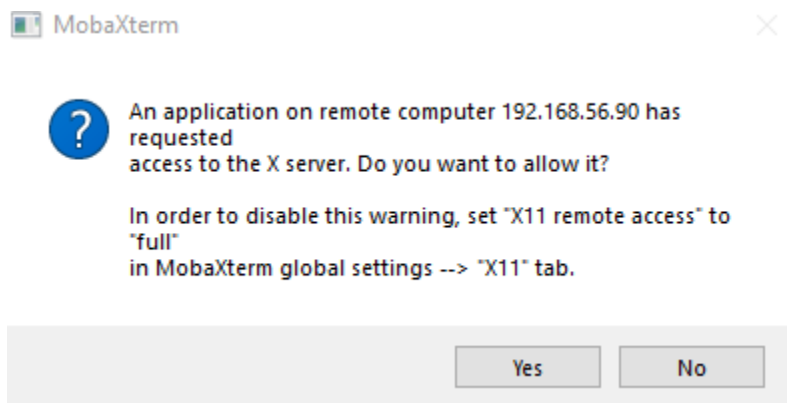
Important Note : After the Grid Infrastructure installation, the startup of ASM and ClusterWare can take a few minutes. Verify that the services are UP before attempting to create ASM DiskGroups or run cluvfy in subsequent steps. Run crsctl status commands to check.

This can be executed from either server (node) of the cluster.

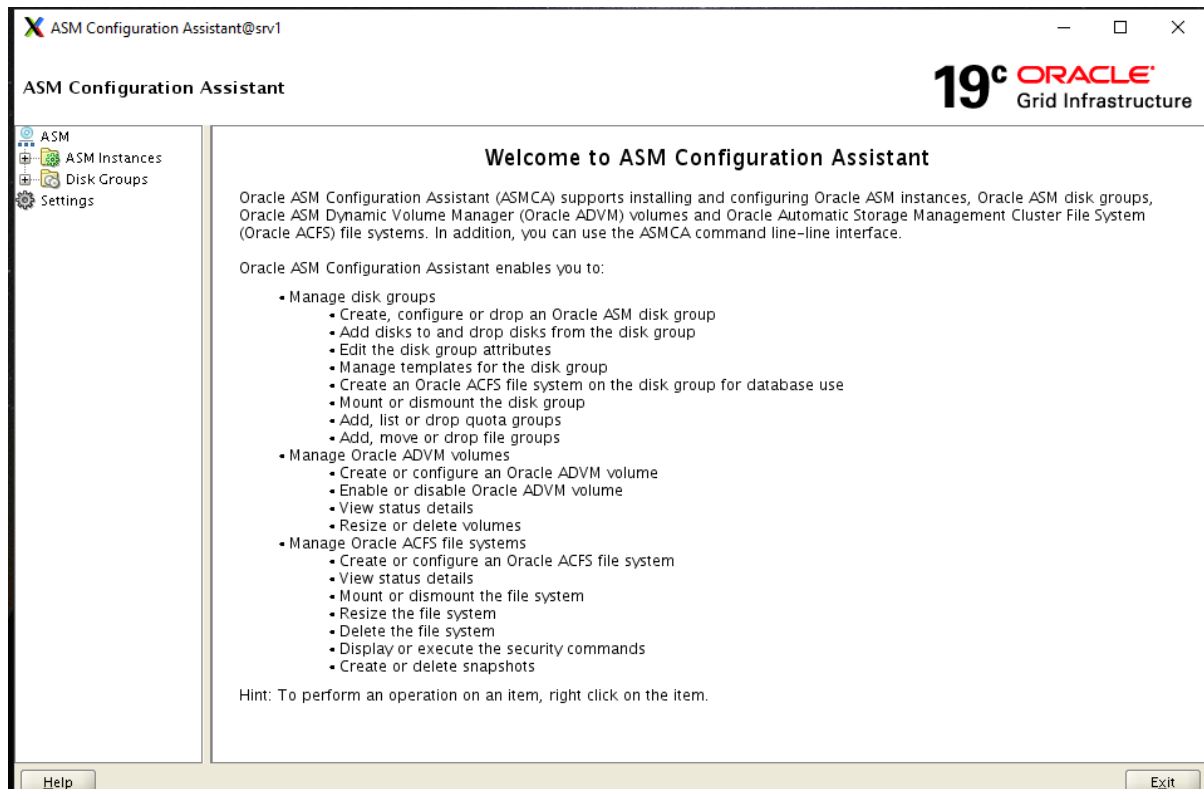
ASMCA

For DATA, I show the method using the ASMCA GUI

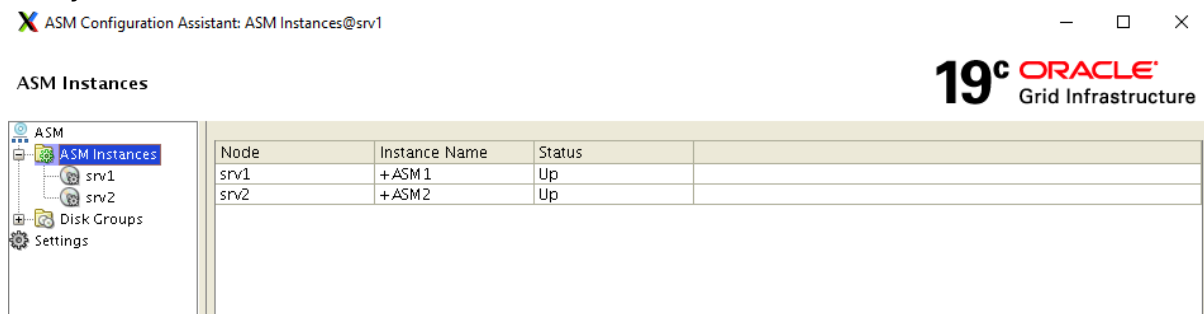
```
[root@srv1 ~]# su - grid  
[grid@srv1 ~]$ env |grep ORA  
ORACLE_SID=+ASM1  
ORACLE_BASE=/u01/app/grid_base  
ORACLE_HOME=/u01/app/grid/product/19.3  
[grid@srv1 ~]$ export DISPLAY=192.168.56.1:0.0  
[grid@srv1 ~]$ asmca
```



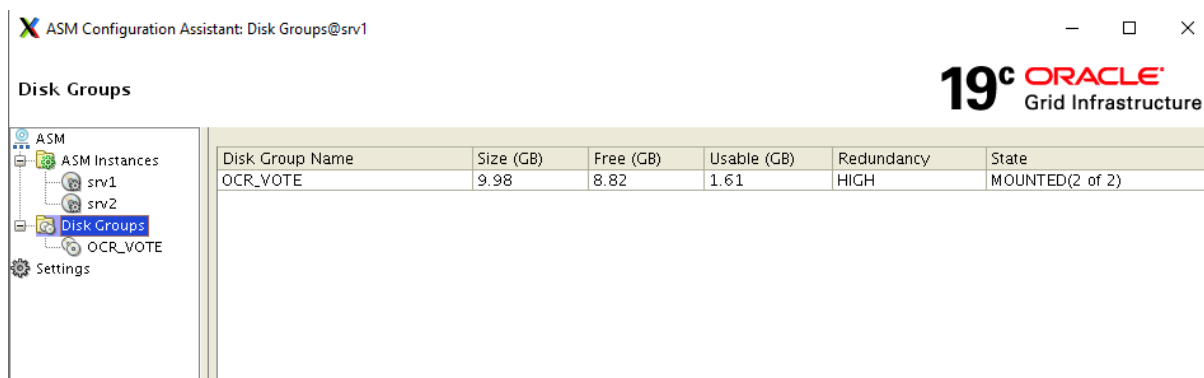
Installation of 19c RAC on Linux7 in Virtual Box VMs



Verify that both ASM Instances are UP

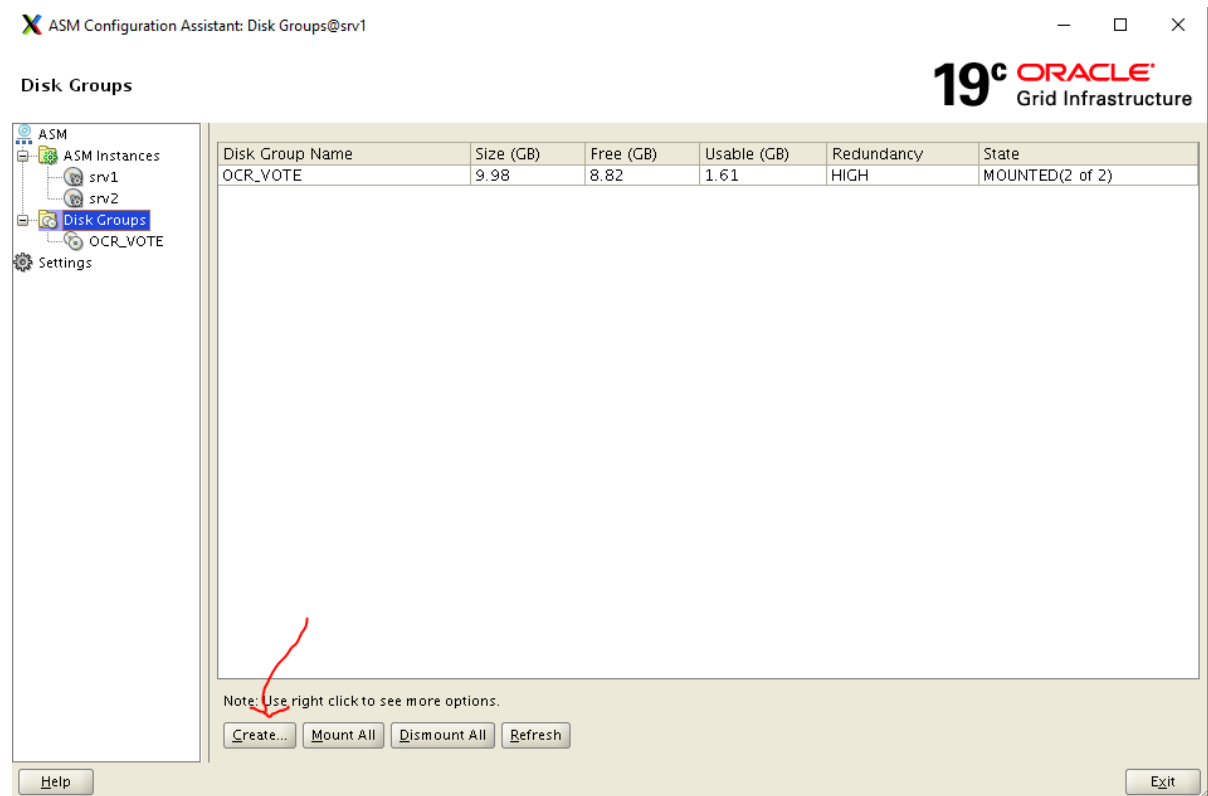


Check the existing OCR_VOTE DiskGroup



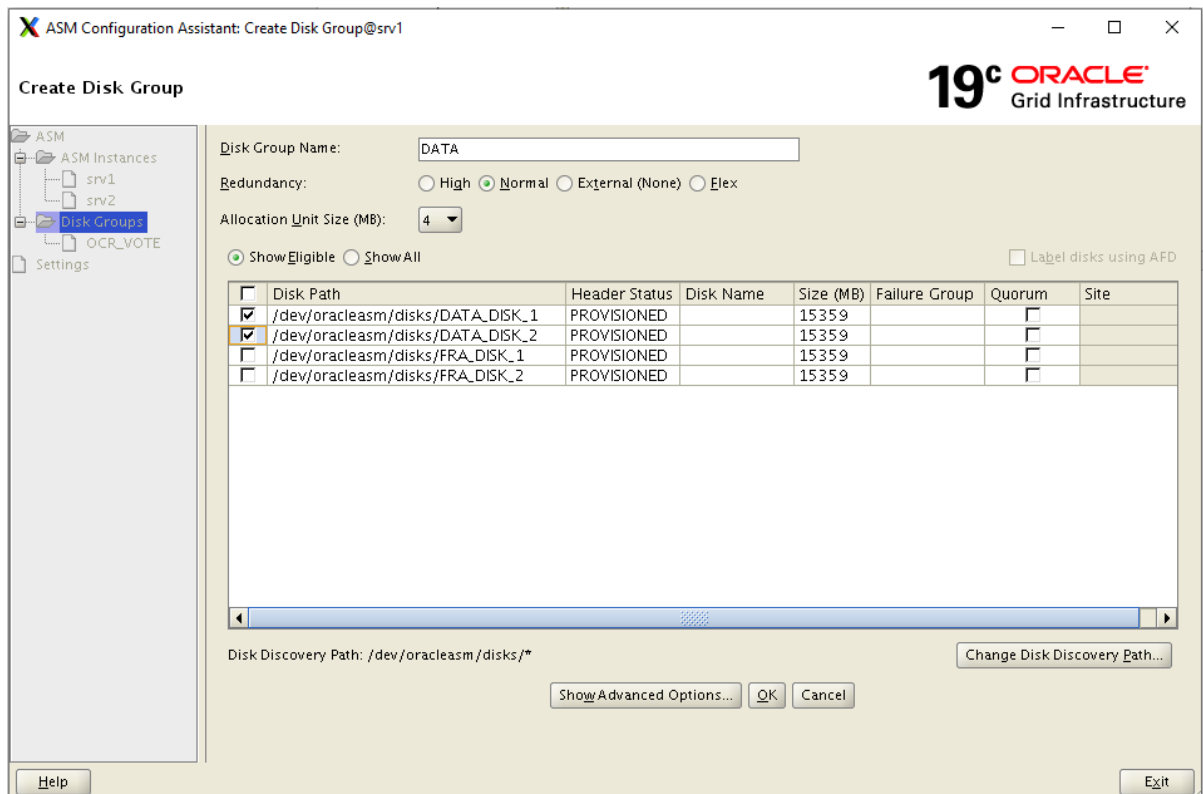
Creating the DATA DiskGroup using ASMCA

Select “CREATE”

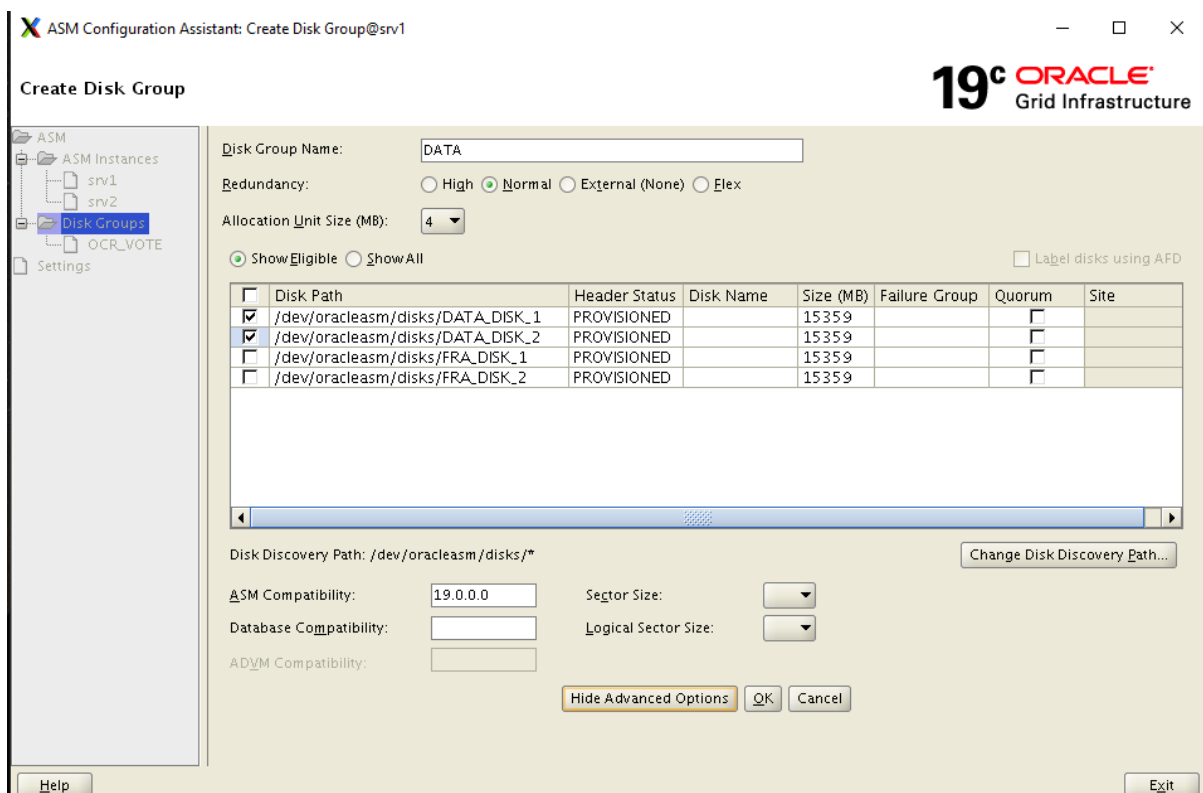


Name the new DiskGroup “DATA”, set it to Normal Redundancy (2 Disks) and select the two DATA Disks

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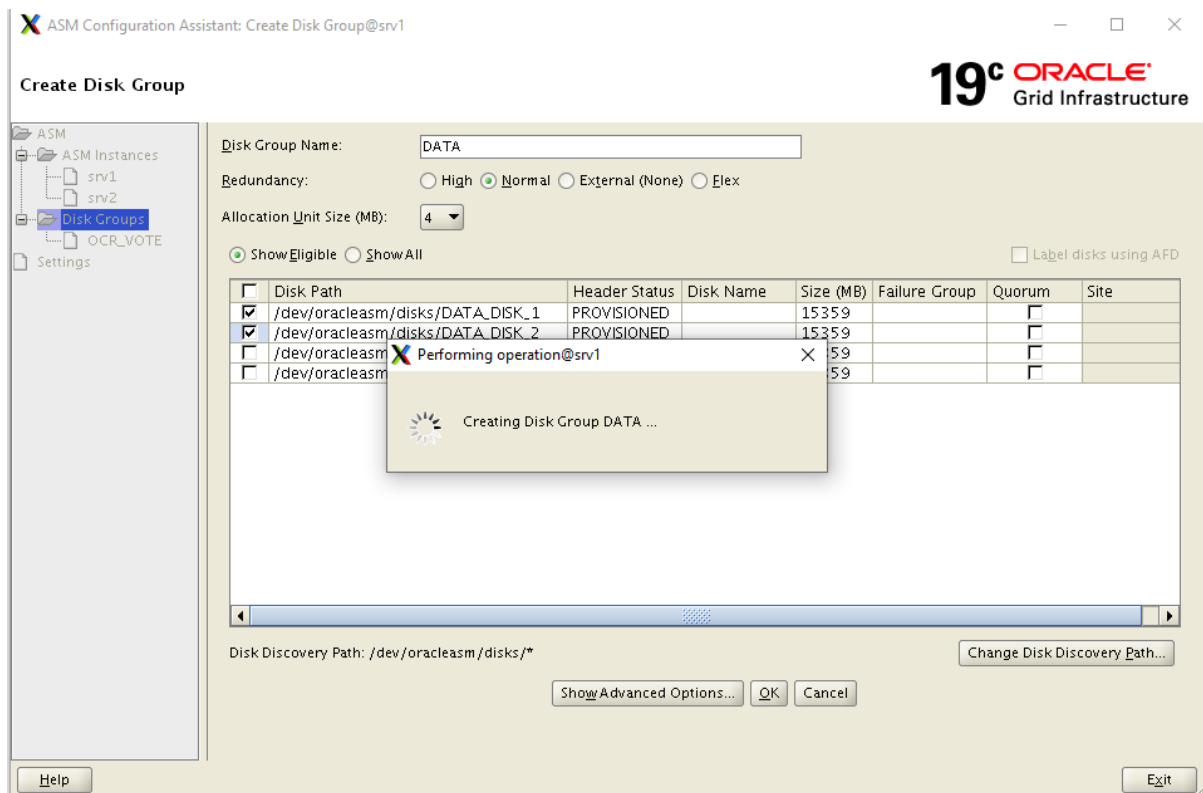
The “Show Advanced Options” allows you to set Compatibility Level, Sector Size etc.



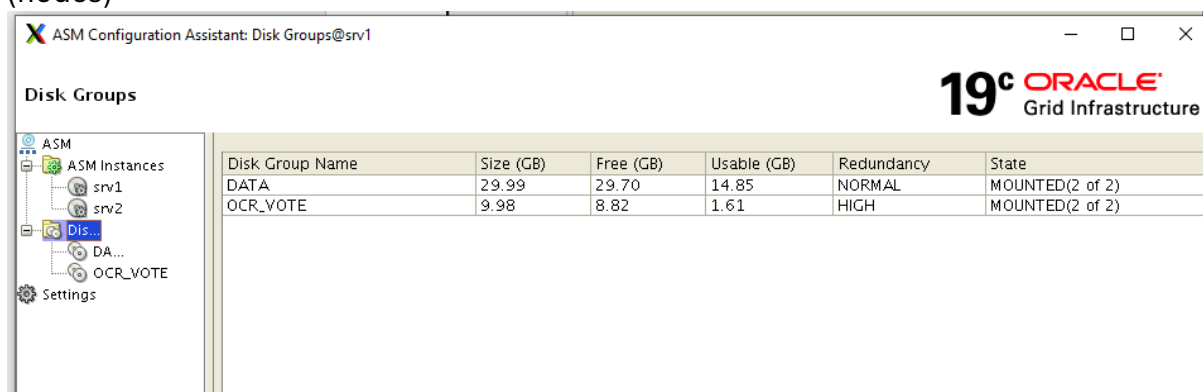
Hide the Advanced Options as we can use the defaults currently

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Click on “OK” and it starts creation of the DiskGroup

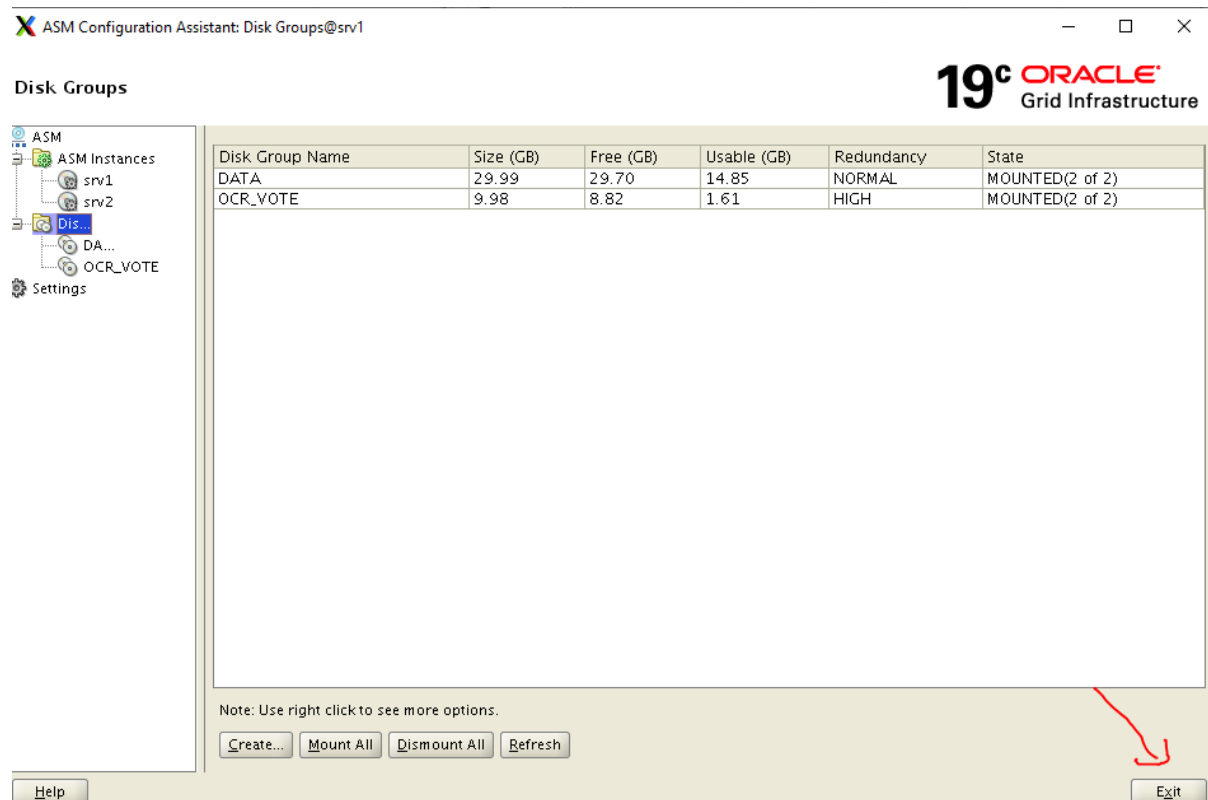


Verify that the DiskGroup is created successfully and also Mounted on both servers (nodes)



Click on Exit at the bottom

Installation of 19c RAC on Linux7 in Virtual Box VMs



Verify the DATA DiskGroup using the SQL Command Line

```
[grid@srv1 ~]$ sqlplus / as sysasm
```

```
SQL*Plus: Release 19.0.0.0.0 - Production on Fri Mar 22 08:50:52 2024  
Version 19.3.0.0.0
```

```
Copyright (c) 1982, 2019, Oracle. All rights reserved.
```

```
Connected to:
```

```
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production  
Version 19.3.0.0.0
```

```
SQL> select name, group_number, state, type, total_mb, free_mb  
2 from v$asm_diskgroup;
```

NAME	GROUP_NUMBER	STATE	TYPE	TOTAL_MB
OCR_VOTE	1	MOUNTED	HIGH	10220
DATA	2	MOUNTED	NORMAL	30712

```
SQL> set linesize 132
```

```
SQL> col name format a12
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
SQL> 1
      1 select name, group_number, state, type, total_mb, free_mb
      2* from v$asm_diskgroup
SQL> /
```

NAME	GROUP_NUMBER	STATE	TYPE	TOTAL_MB	FREE_MB
OCR_VOTE	1	MOUNTED	HIGH	10220	9028
DATA	2	MOUNTED	NORMAL	30712	30416

```
SQL>
SQL> select name, group_number, state, type, total_mb, free_mb,
usable_file_mb
      2 from v$asm_diskgroup
      3 /
```

NAME	GROUP_NUMBER	STATE	TYPE	TOTAL_MB	FREE_MB	USABLE_FILE_MB
OCR_VOTE	1	MOUNTED	HIGH	10220	9028	1646
DATA	2	MOUNTED	NORMAL	30712	30416	15208

```
SQL>
```

Since DATA is created with NORMAL Redundancy with 2 Disks, usable space is half of the total size.

Creating the FRA DiskGroup using SQL

Identify the available disks

```
SQL> set pages60
SQL> col path format a36
SQL> 1
      1 select g.group_number, g.name, d.disk_number, d.label, d.path
      2 from v$asm_diskgroup g right join v$asm_disk d
      3 on g.group_number = d.group_number
      4* order by 1,2
SQL> /
```

GROUP_NUMBER	NAME	DISK_NUMBER	LABEL	PATH
1	OCR_VOTE	1	OCR_VOTE_DISK_2	/dev/oracleasm/disks/OCR_VOTE_DISK_2
1	OCR_VOTE	2	OCR_VOTE_DISK_3	/dev/oracleasm/disks/OCR_VOTE_DISK_3
1	OCR_VOTE	4	OCR_VOTE_DISK_5	/dev/oracleasm/disks/OCR_VOTE_DISK_5
1	OCR_VOTE	0	OCR_VOTE_DISK_1	/dev/oracleasm/disks/OCR_VOTE_DISK_1
1	OCR_VOTE	3	OCR_VOTE_DISK_4	/dev/oracleasm/disks/OCR_VOTE_DISK_4
2	DATA	0	DATA_DISK_1	/dev/oracleasm/disks/DATA_DISK_1

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
2 DATA 1 DATA_DISK_2
/dev/oracleasm/disks/DATA_DISK_2
0
/dev/oracleasm/disks/FRA_DISK_2
1
/dev/oracleasm/disks/FRA_DISK_1

9 rows selected.

SQL>
```

```
SQL> 1
2 select g.group_number, g.name, d.disk_number, d.label, d.path
3 from v$asm_diskgroup g right join v$asm_disk d
4 on g.group_number = d.group_number
5* order by 1,2
SQL> /
```

GROUP_NUMBER	NAME	DISK_NUMBER	LABEL	PATH
1	OCR_VOTE	1	OCR_VOTE_DISK_2	/dev/oracleasm/disks/OCR_VOTE_DISK_2
1	OCR_VOTE	2	OCR_VOTE_DISK_3	/dev/oracleasm/disks/OCR_VOTE_DISK_3
1	OCR_VOTE	4	OCR_VOTE_DISK_5	/dev/oracleasm/disks/OCR_VOTE_DISK_5
1	OCR_VOTE	0	OCR_VOTE_DISK_1	/dev/oracleasm/disks/OCR_VOTE_DISK_1
1	OCR_VOTE	3	OCR_VOTE_DISK_4	/dev/oracleasm/disks/OCR_VOTE_DISK_4
2	DATA	0	DATA_DISK_1	/dev/oracleasm/disks/DATA_DISK_1
2	DATA	1	DATA_DISK_2	/dev/oracleasm/disks/DATA_DISK_2
		0		/dev/oracleasm/disks/FRA_DISK_2
		1		/dev/oracleasm/disks/FRA_DISK_1

```
9 rows selected.

SQL>
```

Thus, we can see that two disks FRA_DISK_1 and FRA_DISK_2 are currently not assigned.

```
SQL> 1
2 create diskgroup FRA normal redundancy
3 disk '/dev/oracleasm/disks/FRA_DISK_1' name FRA_DISK_1 ,
4* '/dev/oracleasm/disks/FRA_DISK_2' name FRA_DISK_2
SQL> /
```

Diskgroup created.

SQL>

Verify both new DiskGroups

Verifying from srv1

```
SQL> 1
2 select g.group_number, g.name, d.disk_number, d.label, d.path
3 from v$asm_diskgroup g right join v$asm_disk d
4 on g.group_number = d.group_number
5* order by 1,2
SQL> /
```

GROUP_NUMBER	NAME	DISK_NUMBER	LABEL	PATH
--------------	------	-------------	-------	------

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
-----  
-----  
1 OCR_VOTE 4 OCR_VOTE_DISK_5  
/dev/oracleasm/disks/OCR_VOTE_DISK_5  
1 OCR_VOTE 3 OCR_VOTE_DISK_4  
/dev/oracleasm/disks/OCR_VOTE_DISK_4  
1 OCR_VOTE 0 OCR_VOTE_DISK_1  
/dev/oracleasm/disks/OCR_VOTE_DISK_1  
1 OCR_VOTE 2 OCR_VOTE_DISK_3  
/dev/oracleasm/disks/OCR_VOTE_DISK_3  
1 OCR_VOTE 1 OCR_VOTE_DISK_2  
/dev/oracleasm/disks/OCR_VOTE_DISK_2  
2 DATA 0 DATA_DISK_1  
/dev/oracleasm/disks/DATA_DISK_1  
2 DATA 1 DATA_DISK_2  
/dev/oracleasm/disks/DATA_DISK_2  
3 FRA 1 FRA_DISK_2  
/dev/oracleasm/disks/FRA_DISK_2  
3 FRA 0 FRA_DISK_1  
/dev/oracleasm/disks/FRA_DISK_1
```

9 rows selected.

SQL>

```
SQL> 1  
2 select g.group_number, g.name, d.disk_number, d.label, d.path  
3 from v$asm_diskgroup g right join v$asm_disk d  
4 on g.group_number = d.group_number  
5* order by 1,2  
SQL> /  
  
GROUP_NUMBER NAME DISK_NUMBER LABEL PATH  
-----  
1 OCR_VOTE 4 OCR_VOTE_DISK_5 /dev/oracleasm/disks/OCR_VOTE_DISK_5  
1 OCR_VOTE 3 OCR_VOTE_DISK_4 /dev/oracleasm/disks/OCR_VOTE_DISK_4  
1 OCR_VOTE 0 OCR_VOTE_DISK_1 /dev/oracleasm/disks/OCR_VOTE_DISK_1  
1 OCR_VOTE 2 OCR_VOTE_DISK_3 /dev/oracleasm/disks/OCR_VOTE_DISK_3  
1 OCR_VOTE 1 OCR_VOTE_DISK_2 /dev/oracleasm/disks/OCR_VOTE_DISK_2  
2 DATA 0 DATA_DISK_1 /dev/oracleasm/disks/DATA_DISK_1  
2 DATA 1 DATA_DISK_2 /dev/oracleasm/disks/DATA_DISK_2  
3 FRA 1 FRA_DISK_2 /dev/oracleasm/disks/FRA_DISK_2  
3 FRA 0 FRA_DISK_1 /dev/oracleasm/disks/FRA_DISK_1  
  
9 rows selected.  
SQL>
```

Verifying from srv2

Note : For DiskGroups manually created using SQL, the other nodes of the Cluster must MOUNT them before they are visible. Thus, FRA has to be mounted on srv2 because it was created using SQL from srv1

```
[grid@srv2 ~]$ sqlplus / as sysasm
```

```
SQL*Plus: Release 19.0.0.0.0 - Production on Fri Mar 22 09:19:58 2024  
Version 19.3.0.0.0
```

```
Copyright (c) 1982, 2019, Oracle. All rights reserved.
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

Connected to:

Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

```
SQL> set pages60
SQL> set linesize 132
SQL> col path format a36
SQL> alter diskgroup FRA mount;
```

Diskgroup altered.

```
SQL> select g.group_number, g.name, d.disk_number, d.label, d.path
  2   from v$asm_diskgroup g right join v$asm_disk d
  3   on g.group_number = d.group_number
  4   order by 1,2
  5   /
```

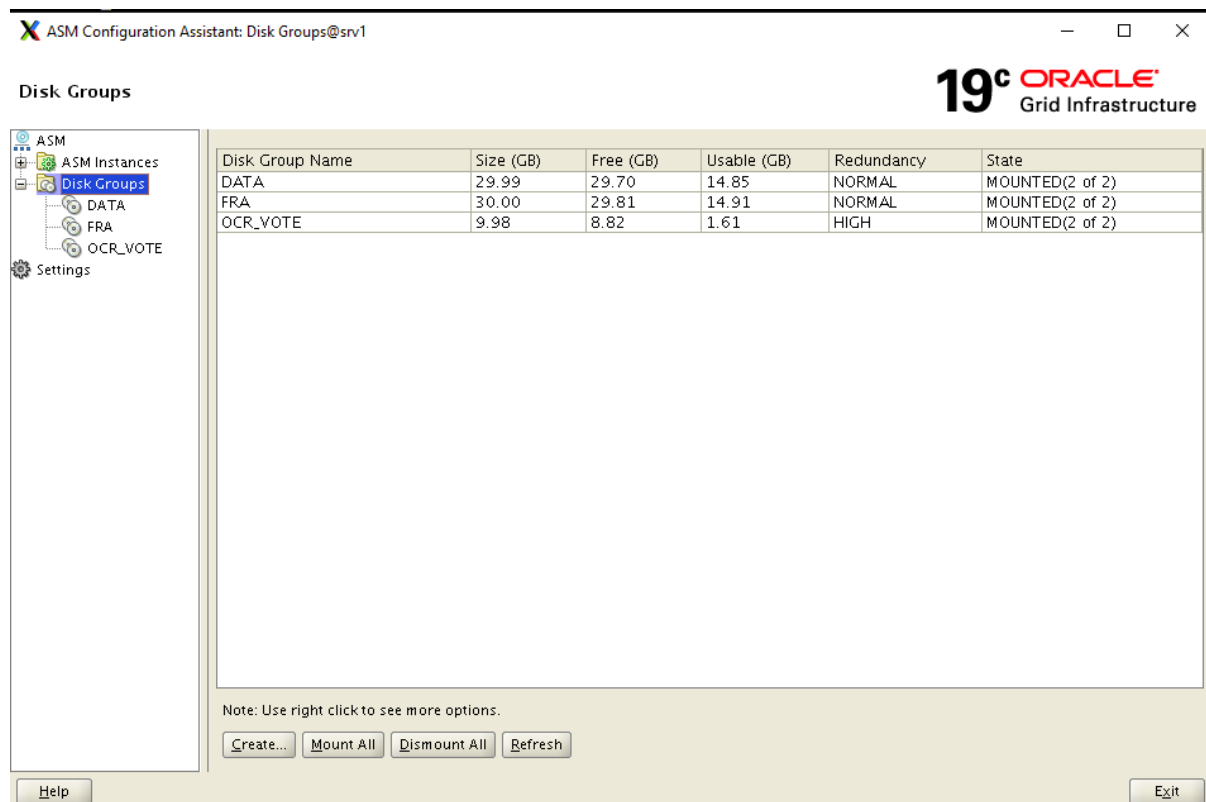
GROUP_NUMBER	NAME	DISK_NUMBER	LABEL
PATH			
1	OCR_VOTE	3	OCR_VOTE_DISK_4
/dev/oracleasm/disks/OCR_VOTE_DISK_4			
1	OCR_VOTE	1	OCR_VOTE_DISK_2
/dev/oracleasm/disks/OCR_VOTE_DISK_2			
1	OCR_VOTE	0	OCR_VOTE_DISK_1
/dev/oracleasm/disks/OCR_VOTE_DISK_1			
1	OCR_VOTE	4	OCR_VOTE_DISK_5
/dev/oracleasm/disks/OCR_VOTE_DISK_5			
1	OCR_VOTE	2	OCR_VOTE_DISK_3
/dev/oracleasm/disks/OCR_VOTE_DISK_3			
2	DATA	0	DATA_DISK_1
/dev/oracleasm/disks/DATA_DISK_1			
2	DATA	1	DATA_DISK_2
/dev/oracleasm/disks/DATA_DISK_2			
3	FRA	1	FRA_DISK_2
/dev/oracleasm/disks/FRA_DISK_2			
3	FRA	0	FRA_DISK_1
/dev/oracleasm/disks/FRA_DISK_1			

9 rows selected.

SQL>

Using ASMCA (from either node of the Cluster)

Installation of 19c RAC on Linux7 in Virtual Box VMs



Prepare for Installation of RDBMS Software for RAC

Run Cluster Verification Utility

Important Note : After the Grid Infrastructure installation, the startup of ASM and ClusterWare can take a few minutes. Verify that the services are UP before attempting to create ASM DiskGroups or run cluvfy in subsequent steps. Run crsctl status commands to check.

Run this from the existing Grid Infrastructure Installation

Note : Most of the parameters and packages would already have been configured by the "oracle-database-preinstall-19c"

```
[grid@srv1 ~]$ cluvfy stage -pre dbinst -n srv1,srv2 -r 19
```

```
Verifying Physical Memory ...PASSED
Verifying Available Physical Memory ...PASSED
Verifying Swap Size ...FAILED (PRVF-7573)
Verifying Free Space: srv2:/tmp ...PASSED
Verifying Free Space: srv1:/tmp ...PASSED
Verifying User Existence: grid ...
  Verifying Users With Same UID: 54322 ...PASSED
Verifying User Existence: grid ...PASSED
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
Verifying Group Existence: dba ...PASSED
Verifying Group Existence: grid ...PASSED
Verifying Group Membership: dba ...FAILED (PRVG-10460)
Verifying Group Membership: grid(Primary) ...PASSED
Verifying Run Level ...PASSED
Verifying Hard Limit: maximum open file descriptors ...PASSED
Verifying Soft Limit: maximum open file descriptors ...PASSED
Verifying Hard Limit: maximum user processes ...PASSED
Verifying Soft Limit: maximum user processes ...PASSED
Verifying Soft Limit: maximum stack size ...PASSED
Verifying Architecture ...PASSED
Verifying OS Kernel Version ...PASSED
Verifying OS Kernel Parameter: semmsl ...PASSED
Verifying OS Kernel Parameter: semmns ...PASSED
Verifying OS Kernel Parameter: semopm ...PASSED
Verifying OS Kernel Parameter: semmni ...PASSED
Verifying OS Kernel Parameter: shmmax ...PASSED
Verifying OS Kernel Parameter: shmmni ...PASSED
Verifying OS Kernel Parameter: shmall ...PASSED
Verifying OS Kernel Parameter: file-max ...PASSED
Verifying OS Kernel Parameter: ip_local_port_range ...PASSED
Verifying OS Kernel Parameter: rmem_default ...PASSED
Verifying OS Kernel Parameter: rmem_max ...PASSED
Verifying OS Kernel Parameter: wmem_default ...PASSED
Verifying OS Kernel Parameter: wmem_max ...PASSED
Verifying OS Kernel Parameter: aio-max-nr ...PASSED
Verifying Package: kmod-20-21 (x86_64) ...PASSED
Verifying Package: kmod-libs-20-21 (x86_64) ...PASSED
Verifying Package: binutils-2.23.52.0.1 ...PASSED
Verifying Package: compat-libcap1-1.10 ...PASSED
Verifying Package: libgcc-4.8.2 (x86_64) ...PASSED
Verifying Package: libstdc++-4.8.2 (x86_64) ...PASSED
Verifying Package: libstdc++-devel-4.8.2 (x86_64) ...PASSED
Verifying Package: sysstat-10.1.5 ...PASSED
Verifying Package: ksh ...PASSED
Verifying Package: make-3.82 ...PASSED
Verifying Package: glibc-2.17 (x86_64) ...PASSED
Verifying Package: glibc-devel-2.17 (x86_64) ...PASSED
Verifying Package: libaio-0.3.109 (x86_64) ...PASSED
Verifying Package: libaio-devel-0.3.109 (x86_64) ...PASSED
Verifying Package: smartmontools-6.2-4 ...PASSED
Verifying Package: net-tools-2.0-0.17 ...PASSED
Verifying Users With Same UID: 0 ...PASSED
Verifying Current Group ID ...PASSED
Verifying Root user consistency ...PASSED
Verifying Host name ...PASSED
Verifying Node Connectivity ...
  Verifying Hosts File ...PASSED
  Verifying Check that maximum (MTU) size packet goes through subnet
...PASSED
  Verifying subnet mask consistency for subnet "192.168.56.0" ...PASSED
  Verifying subnet mask consistency for subnet "192.168.10.0" ...PASSED
Verifying Node Connectivity ...PASSED
Verifying Multicast or broadcast check ...PASSED
Verifying User Mask ...PASSED
Verifying CRS Integrity ...
  Verifying Clusterware Version Consistency ...PASSED
Verifying CRS Integrity ...PASSED
Verifying Cluster Manager Integrity ...PASSED
Verifying Node Application Existence ...PASSED
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

Verifying Clock Synchronization ...

CTSS is in Observer state. Switching over to clock synchronization checks using NTP

```
Verifying Network Time Protocol (NTP) ...
  Verifying '/etc/chrony.conf' ...PASSED
  Verifying Daemon 'chronyd' ...PASSED
  Verifying NTP daemon or service using UDP port 123 ...PASSED
  Verifying chrony daemon is synchronized with at least one external time
source ...PASSED
  Verifying Network Time Protocol (NTP) ...PASSED
Verifying Clock Synchronization ...PASSED
Verifying resolv.conf Integrity ...PASSED
Verifying Time zone consistency ...PASSED
Verifying Single Client Access Name (SCAN) ...
  Verifying DNS/NIS name service 'oral9cscan' ...
    Verifying Name Service Switch Configuration File Integrity ...PASSED
  Verifying DNS/NIS name service 'oral9cscan' ...FAILED (PRVG-11826, PRVG-
11827, PRVF-4664)
Verifying Single Client Access Name (SCAN) ...FAILED (PRVG-11372, PRVG-
11826, PRVG-11827)
Verifying VIP Subnet configuration check ...PASSED
Verifying Database Clusterware Version Compatibility ...PASSED
Verifying ASM storage privileges for the user: grid ...
  Verifying Group Membership: grid ...PASSED
Verifying ASM storage privileges for the user: grid ...PASSED
Verifying Daemon "proxyt" not configured and running ...PASSED
Verifying ACFS device special file ...PASSED
Verifying /dev/shm mounted as temporary file system ...PASSED
Verifying Maximum locked memory check ...PASSED
Verifying Systemd login manager IPC parameter ...PASSED
```

Pre-check for database installation was unsuccessful on all the nodes.

Failures were encountered during execution of CVU verification request "stage -pre dbinst".

Verifying Swap Size ...FAILED

srv2: PRVF-7573 : Sufficient swap size is not available on node "srv2"
[Required = 7.4755GB (7838624.0KB) ; Found = 4GB (4194300.0KB)]

srv1: PRVF-7573 : Sufficient swap size is not available on node "srv1"
[Required = 7.4755GB (7838636.0KB) ; Found = 4GB (4194300.0KB)]

Verifying Group Membership: dba ...FAILED

srv2: PRVG-10460 : User "grid" does not belong to group "dba" selected for
privileges "OSDBA" on node "srv2".

srv1: PRVG-10460 : User "grid" does not belong to group "dba" selected for
privileges "OSDBA" on node "srv1".

Verifying Single Client Access Name (SCAN) ...FAILED

PRVG-11372 : Number of SCAN IP addresses that SCAN "oral9cscan" resolved to
did
not match the number of SCAN VIP resources

```
Verifying DNS/NIS name service 'oral9cscan' ...FAILED
PRVG-11826 : DNS resolved IP addresses "" for SCAN name "oral9cscan" not
found in the name service returned IP addresses
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
"192.168.56.94,192.168.56.95,192.168.56.96"
PRVG-11827 : Name service returned IP addresses
"192.168.56.94,192.168.56.95,192.168.56.96" for SCAN name "ora19cscan"
not
found in the DNS returned IP addresses ""

srv2: PRVF-4664 : Found inconsistent name resolution entries for SCAN
name
      "ora19cscan"

srv1: PRVF-4664 : Found inconsistent name resolution entries for SCAN
name
      "ora19cscan"
```

```
CVU operation performed:      stage -pre dbinst
Date:                        Mar 22, 2024 9:37:34 AM
CVU home:                    /u01/app/grid/product/19.3/
User:                        grid
[grid@srv1 ~]$
```

Correcting Errors/Warnings

We can ignore the Swap Size verification error.

Add grid to the dba group

For the dba group membership error, I add grid to the dba group :

```
[root@srv1 ~]# usermod -G grid,dba,oinstall grid
[root@srv1 ~]#

[root@srv2 ~]# usermod -G grid,dba,oinstall grid
[root@srv2 ~]#
```

Thus, the /etc/group file now has this entry :

```
dba:x:54322:oracle,grid
on both servers (nodes)
```

Correct the SCAN to include domain

Reference : Oracle Support Doc ID 972500.1

On the DNS server, update the entries for the SCAN to include the domain name

```
#SCAN IPs
192.168.56.94 ora19cscan ora19cscan.localdomain
192.168.56.95 ora19cscan ora19cscan.localdomain
192.168.56.96 ora19cscan ora19cscan.localdomain
```

From both nodes (srv1 and srv2) verify nslookup

```
[root@srv1 ~]# nslookup ora19cscan.localdomain
Server:      192.168.56.80
Address:     192.168.56.80#53
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
Name: ora19cscan.localdomain
Address: 192.168.56.96
Name: ora19cscan.localdomain
Address: 192.168.56.94
Name: ora19cscan.localdomain
Address: 192.168.56.95

[root@srv1 ~]# nslookup ora19cscan.localdomain
Server:      192.168.56.80
Address:     192.168.56.80#53

Name: ora19cscan.localdomain
Address: 192.168.56.94
Name: ora19cscan.localdomain
Address: 192.168.56.95
Name: ora19cscan.localdomain
Address: 192.168.56.96

[root@srv1 ~]#

[root@srv2 ~]# nslookup ora19cscan.localdomain
Server:      192.168.56.80
Address:     192.168.56.80#53

Name: ora19cscan.localdomain
Address: 192.168.56.95
Name: ora19cscan.localdomain
Address: 192.168.56.96
Name: ora19cscan.localdomain
Address: 192.168.56.94

[root@srv2 ~]# nslookup ora19cscan.localdomain
Server:      192.168.56.80
Address:     192.168.56.80#53

Name: ora19cscan.localdomain
Address: 192.168.56.96
Name: ora19cscan.localdomain
Address: 192.168.56.94
Name: ora19cscan.localdomain
Address: 192.168.56.95

[root@srv2 ~]#
```

Update SCAN name in the Cluster

Note : The modify command to change the name must be run as “root” as the name is a Network Name

```
[root@srv1 ~]# su - grid
[grid@srv1 ~]$ srvctl status scan_listener
SCAN Listener LISTENER_SCAN1 is enabled
SCAN listener LISTENER_SCAN1 is running on node srv2
SCAN Listener LISTENER_SCAN2 is enabled
SCAN listener LISTENER_SCAN2 is running on node srv1
SCAN Listener LISTENER_SCAN3 is enabled
SCAN listener LISTENER_SCAN3 is running on node srv1
[grid@srv1 ~]$ srvctl stop scan_listener
[grid@srv1 ~]$ srvctl stop scan
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
[grid@srv1 ~]$ logout

[root@srv1 ~]# . /home/grid/.bash_profile
[root@srv1 ~]# srvctl modify scan -n ora19cscan.localdomain
[root@srv1 ~]#

[root@srv1 ~]# su - grid
[grid@srv1 ~]$ srvctl modify scan_listener -u
[grid@srv1 ~]$ srvctl start scan_listener
[grid@srv1 ~]$ srvctl status scan_listener
SCAN Listener LISTENER_SCAN1 is enabled
SCAN listener LISTENER_SCAN1 is running on node srv2
SCAN Listener LISTENER_SCAN2 is enabled
SCAN listener LISTENER_SCAN2 is running on node srv1
SCAN Listener LISTENER_SCAN3 is enabled
SCAN listener LISTENER_SCAN3 is running on node srv1
[grid@srv1 ~]$

[grid@srv1 ~]$ srvctl config scan
SCAN name: ora19cscan.localdomain, Network: 1
Subnet IPv4: 192.168.56.0/255.255.255.0/eth0, static
Subnet IPv6:
SCAN 1 IPv4 VIP: 192.168.56.94
SCAN VIP is enabled.
SCAN 2 IPv4 VIP: 192.168.56.95
SCAN VIP is enabled.
SCAN 3 IPv4 VIP: 192.168.56.96
SCAN VIP is enabled.
[grid@srv1 ~]$
```

Reboot and Rerun cluvfy

Reboot both servers (nodes) and rerun cluvfy

Important Note : After the Grid Infrastructure installation, the startup of ASM and ClusterWare can take a few minutes. Verify that the services are UP before attempting to create ASM DiskGroups or run cluvfy in subsequent steps. Run crsctl status commands to check.

```
[grid@srv1 ~]$ cluvfy stage -pre dbinst -n srv1,srv2 -r 19

Verifying Physical Memory ...PASSED
Verifying Available Physical Memory ...PASSED
Verifying Swap Size ...FAILED (PRVF-7573)
Verifying Free Space: srv2:/tmp ...PASSED
Verifying Free Space: srv1:/tmp ...PASSED
Verifying User Existence: grid ...
  Verifying Users With Same UID: 54322 ...PASSED
Verifying User Existence: grid ...PASSED
Verifying Group Existence: dba ...PASSED
Verifying Group Existence: grid ...PASSED
Verifying Group Membership: dba ...PASSED
Verifying Group Membership: grid(Primary) ...PASSED
Verifying Run Level ...PASSED
Verifying Hard Limit: maximum open file descriptors ...PASSED
Verifying Soft Limit: maximum open file descriptors ...PASSED
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
Verifying Hard Limit: maximum user processes ...PASSED
Verifying Soft Limit: maximum user processes ...PASSED
Verifying Soft Limit: maximum stack size ...PASSED
Verifying Architecture ...PASSED
Verifying OS Kernel Version ...PASSED
Verifying OS Kernel Parameter: semmsl ...PASSED
Verifying OS Kernel Parameter: semmns ...PASSED
Verifying OS Kernel Parameter: semopm ...PASSED
Verifying OS Kernel Parameter: semmni ...PASSED
Verifying OS Kernel Parameter: shmmax ...PASSED
Verifying OS Kernel Parameter: shmmni ...PASSED
Verifying OS Kernel Parameter: shmall ...PASSED
Verifying OS Kernel Parameter: file-max ...PASSED
Verifying OS Kernel Parameter: ip_local_port_range ...PASSED
Verifying OS Kernel Parameter: rmem_default ...PASSED
Verifying OS Kernel Parameter: rmem_max ...PASSED
Verifying OS Kernel Parameter: wmem_default ...PASSED
Verifying OS Kernel Parameter: wmem_max ...PASSED
Verifying OS Kernel Parameter: aio-max-nr ...PASSED
Verifying Package: kmod-20-21 (x86_64) ...PASSED
Verifying Package: kmod-libs-20-21 (x86_64) ...PASSED
Verifying Package: binutils-2.23.52.0.1 ...PASSED
Verifying Package: compat-libcap1-1.10 ...PASSED
Verifying Package: libgcc-4.8.2 (x86_64) ...PASSED
Verifying Package: libstdc++-4.8.2 (x86_64) ...PASSED
Verifying Package: libstdc++-devel-4.8.2 (x86_64) ...PASSED
Verifying Package: sysstat-10.1.5 ...PASSED
Verifying Package: ksh ...PASSED
Verifying Package: make-3.82 ...PASSED
Verifying Package: glibc-2.17 (x86_64) ...PASSED
Verifying Package: glibc-devel-2.17 (x86_64) ...PASSED
Verifying Package: libaio-0.3.109 (x86_64) ...PASSED
Verifying Package: libaio-devel-0.3.109 (x86_64) ...PASSED
Verifying Package: smartmontools-6.2-4 ...PASSED
Verifying Package: net-tools-2.0-0.17 ...PASSED
Verifying Users With Same UID: 0 ...PASSED
Verifying Current Group ID ...PASSED
Verifying Root user consistency ...PASSED
Verifying Host name ...PASSED
Verifying Node Connectivity ...
  Verifying Hosts File ...PASSED
  Verifying Check that maximum (MTU) size packet goes through subnet
...PASSED
  Verifying subnet mask consistency for subnet "192.168.56.0" ...PASSED
  Verifying subnet mask consistency for subnet "192.168.10.0" ...PASSED
Verifying Node Connectivity ...PASSED
Verifying Multicast or broadcast check ...PASSED
Verifying User Mask ...PASSED
Verifying CRS Integrity ...
  Verifying Clusterware Version Consistency ...PASSED
Verifying CRS Integrity ...PASSED
Verifying Cluster Manager Integrity ...PASSED
Verifying Node Application Existence ...PASSED
Verifying Clock Synchronization ...
CTSS is in Observer state. Switching over to clock synchronization checks
using NTP

  Verifying Network Time Protocol (NTP) ...
    Verifying '/etc/chrony.conf' ...PASSED
    Verifying Daemon 'chronyd' ...PASSED
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
Verifying NTP daemon or service using UDP port 123 ...PASSED
Verifying chrony daemon is synchronized with at least one external time
source ...PASSED
Verifying Network Time Protocol (NTP) ...PASSED
Verifying Clock Synchronization ...PASSED
Verifying resolv.conf Integrity ...PASSED
Verifying Time zone consistency ...PASSED
Verifying Single Client Access Name (SCAN) ...
Verifying DNS/NIS name service 'oral9cscan.localdomain' ...
Verifying Name Service Switch Configuration File Integrity ...PASSED
Verifying DNS/NIS name service 'oral9cscan.localdomain' ...PASSED
Verifying Single Client Access Name (SCAN) ...PASSED
Verifying VIP Subnet configuration check ...PASSED
Verifying Database Clusterware Version Compatibility ...PASSED
Verifying ASM storage privileges for the user: grid ...
Verifying Group Membership: grid ...PASSED
Verifying ASM storage privileges for the user: grid ...PASSED
Verifying Daemon "proxyc" not configured and running ...PASSED
Verifying ACFS device special file ...PASSED
Verifying /dev/shm mounted as temporary file system ...PASSED
Verifying Maximum locked memory check ...PASSED
Verifying Systemd login manager IPC parameter ...PASSED
```

Pre-check for database installation was unsuccessful on all the nodes.

Failures were encountered during execution of CVU verification request "stage -pre dbinst".

```
Verifying Swap Size ...FAILED
srv2: PRVF-7573 : Sufficient swap size is not available on node "srv2"
      [Required = 7.4755GB (7838624.0KB) ; Found = 4GB (4194300.0KB)]

srv1: PRVF-7573 : Sufficient swap size is not available on node "srv1"
      [Required = 7.4755GB (7838636.0KB) ; Found = 4GB (4194300.0KB)]
```

```
CVU operation performed:      stage -pre dbinst
Date:                        Mar 25, 2024 2:26:38 AM
CVU home:                    /u01/app/grid/product/19.3/
User:                        grid
[grid@srv1 ~]$
```

As noted earlier, the warning on Swap Size can be ignored.

Extract the RDBMS Installer on srv1 (only)

The unzip command unzips the file from the media folder (i.e. mapped drive from Host) to the target location for the RDBMS installation.

The chown command changes ownership to oracle.

The chown command must also be run for the hidden folder .patch_storage

```
[root@srv1 ~]# ls /media/sf_vagrant/*db*.zip
/media/sf_vagrant/LINUX.X64_193000_db_home.zip
[root@srv1 ~]# cd /u01/app/oracle/product/19.3/dhome_1
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
[root@srv1 dbhome_1]# unzip /media/sf_vagrant/LINUX.X64_193000_db_home.zip
-d .
.
...list of folders and files being extracted appears
...list of folders and files being extracted appears
...listing of symbolic links created by the unzip appears
[root@srv1 dbhome_1]# ls -l |head -5
total 112
drwxr-xr-x.  2 root root   102 Apr 18  2019 addnode
drwxr-xr-x.  6 root root  4096 Apr 17  2019 apex
drwxr-xr-x.  9 root root    93 Apr 17  2019 assistants
drwxr-xr-x.  2 root root  8192 Apr 18  2019 bin
[root@srv1 dbhome_1]# chown -R oracle:oinstall *
[root@srv1 dbhome_1]# cd .patch_storage
[root@srv1 .patch_storage]# chown -R oracle:oinstall *
[root@srv1 .patch_storage]# ls -la
total 32
drwxr-xr-x.  5 root   root    4096 Apr 18  2019 .
drwxr-xr-x. 68 oracle oinstall 4096 Mar 25 03:06 ..
drwxr-xr-x.  3 oracle oinstall   74 Apr 18  2019
29517242_Apr_17_2019_23_27_10
drwxr-xr-x.  3 oracle oinstall   74 Apr 18  2019
29585399_Apr_9_2019_19_12_47
-rw-r--r--.  1 oracle oinstall 6307 Apr 18  2019 interim_inventory.txt
-rw-r--r--.  1 oracle oinstall   92 Apr 18  2019
LatestOPatchSession.properties
drwxr-xr-x.  6 oracle oinstall 4096 Apr 18  2019 NApply
-rw-r--r--.  1 oracle oinstall 4975 Apr 18  2019 record_inventory.txt
[root@srv1 .patch_storage]# chown oracle:oinstall .
[root@srv1 .patch_storage]#
```

Configure the oracle user Environment

Set the environment variables ORACLE_BASE, ORACLE_HOME and ORACLE_SID

```
[root@srv1 ~]# su - oracle
[oracle@srv1 ~]$ cat .bash_profile
# .bash_profile

# Get the aliases and functions
if [ -f ~/.bashrc ]; then
    . ~/.bashrc
fi

# User specific environment and startup programs

PATH=$PATH:$HOME/.local/bin:$HOME/bin

export PATH

ORACLE_BASE=/u01/app/oracle ; export ORACLE_BASE
ORACLE_HOME=/u01/app/oracle/product/19.3/dbhome_1 ; export ORACLE_HOME
ORACLE_SID=RACDB1; export ORACLE_SID
PATH=$PATH:$ORACLE_HOME/bin ; export PATH

umask 0022

[oracle@srv1 ~]$ env |grep ORA
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
ORACLE_SID=RACDB1
ORACLE_BASE=/u01/app/oracle
ORACLE_HOME=/u01/app/oracle/product/19.3/dbhome_1
[oracle@srv1 ~]$
```

On srv2, do the same except that ORACLE_SID would be RACDB2

Install the RDBMS Software

Pre-Checks

Environment Variables

```
[root@srv1 ~]# su - oracle
Last login: Mon Mar 25 03:20:06 UTC 2024 on pts/0
[oracle@srv1 ~]$ env |grep ORA
ORACLE_SID=RACDB1
ORACLE_BASE=/u01/app/oracle
ORACLE_HOME=/u01/app/oracle/product/19.3/dbhome_1
```

nslookup

```
[oracle@srv1 ~]$ nslookup ora19cscan.localdomain
Server:          192.168.56.80
Address:         192.168.56.80#53

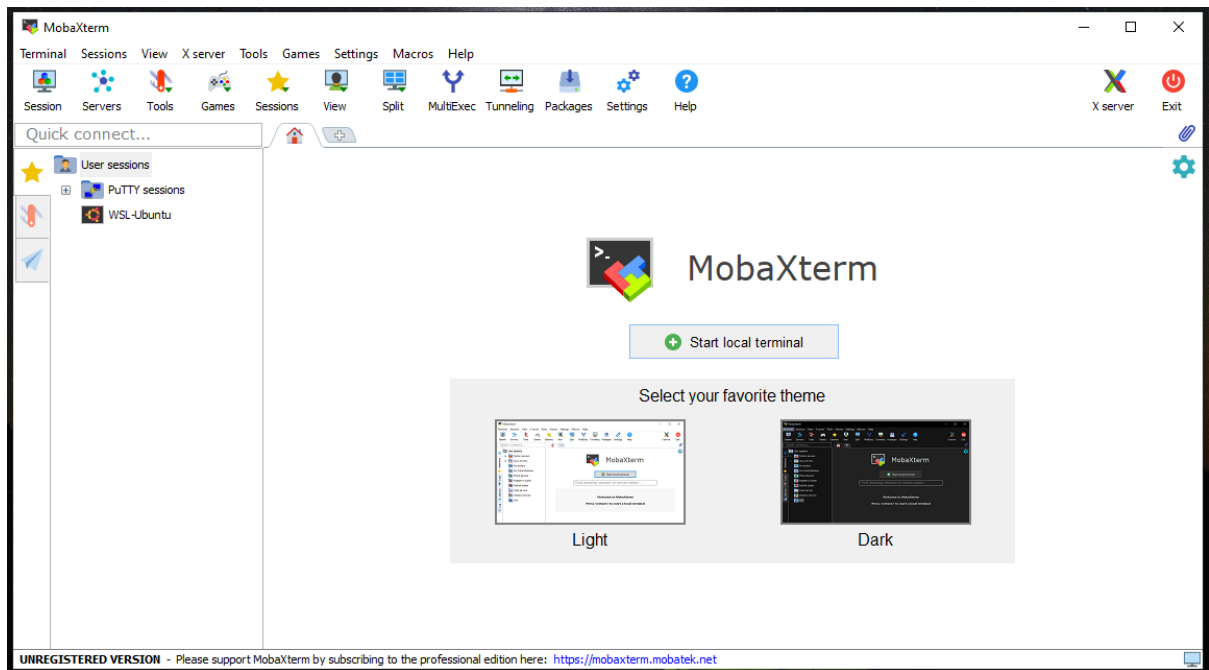
Name:   ora19cscan.localdomain
Address: 192.168.56.94
Name:   ora19cscan.localdomain
Address: 192.168.56.95
Name:   ora19cscan.localdomain
Address: 192.168.56.96

[oracle@srv1 ~]$
```

Start X-Server

Start the installed X-Server (MobaXterm on my Desktop PC)

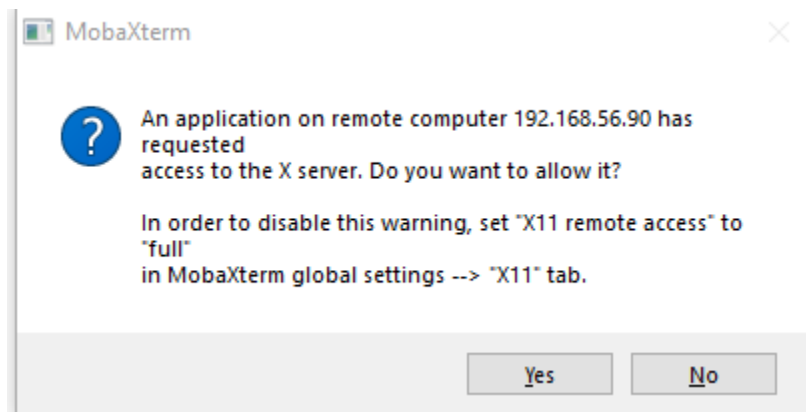
Installation of 19c RAC on Linux7 in Virtual Box VMs



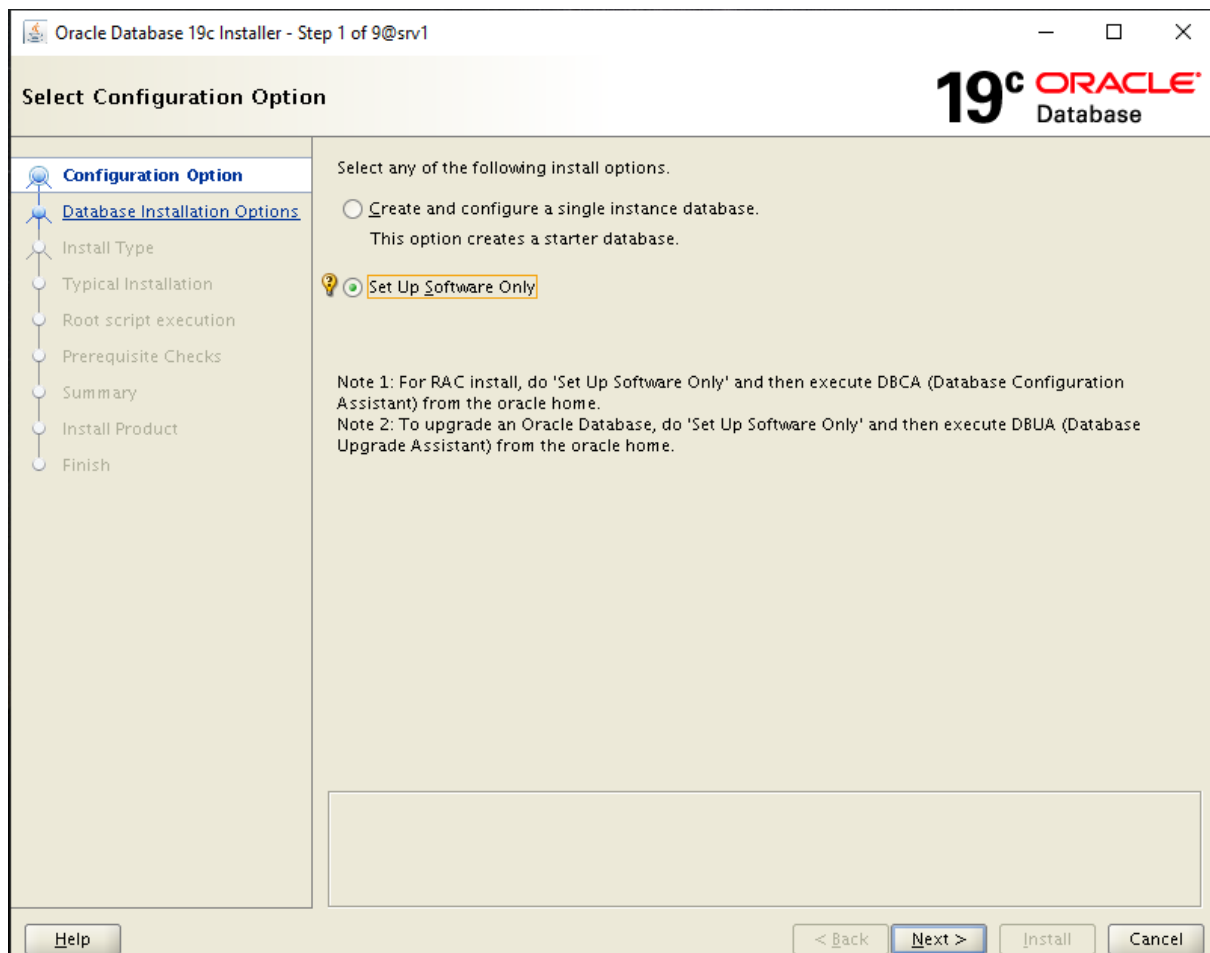
RDBMS Software Installation

Start the Installer from the ORACLE_HOME

```
[oracle@srv1 ~]$ export DISPLAY=192.168.56.1:0.0  
[oracle@srv1 ~]$ cd $ORACLE_HOME  
[oracle@srv1 dbhome_1]$ ./runInstaller
```

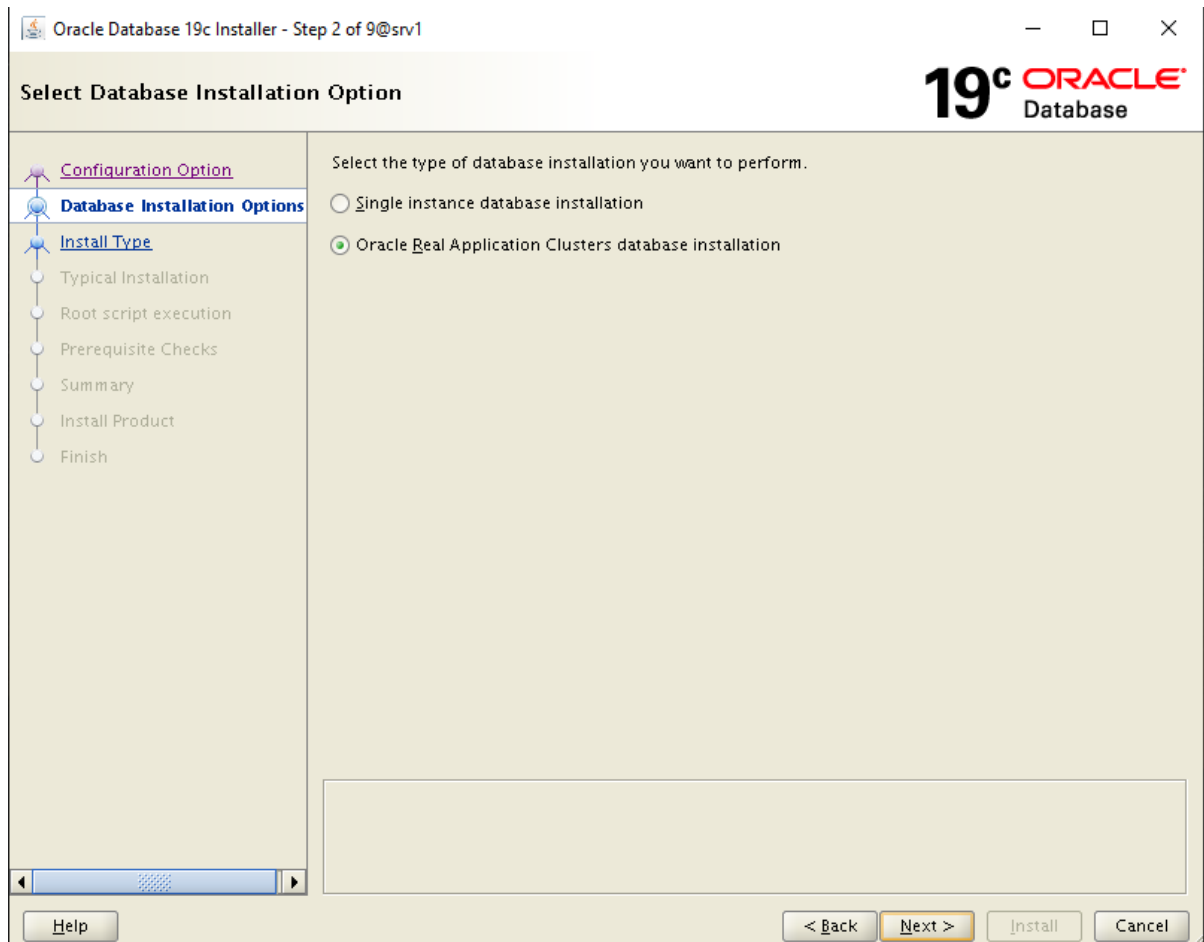


Select "Set Up Software Only"



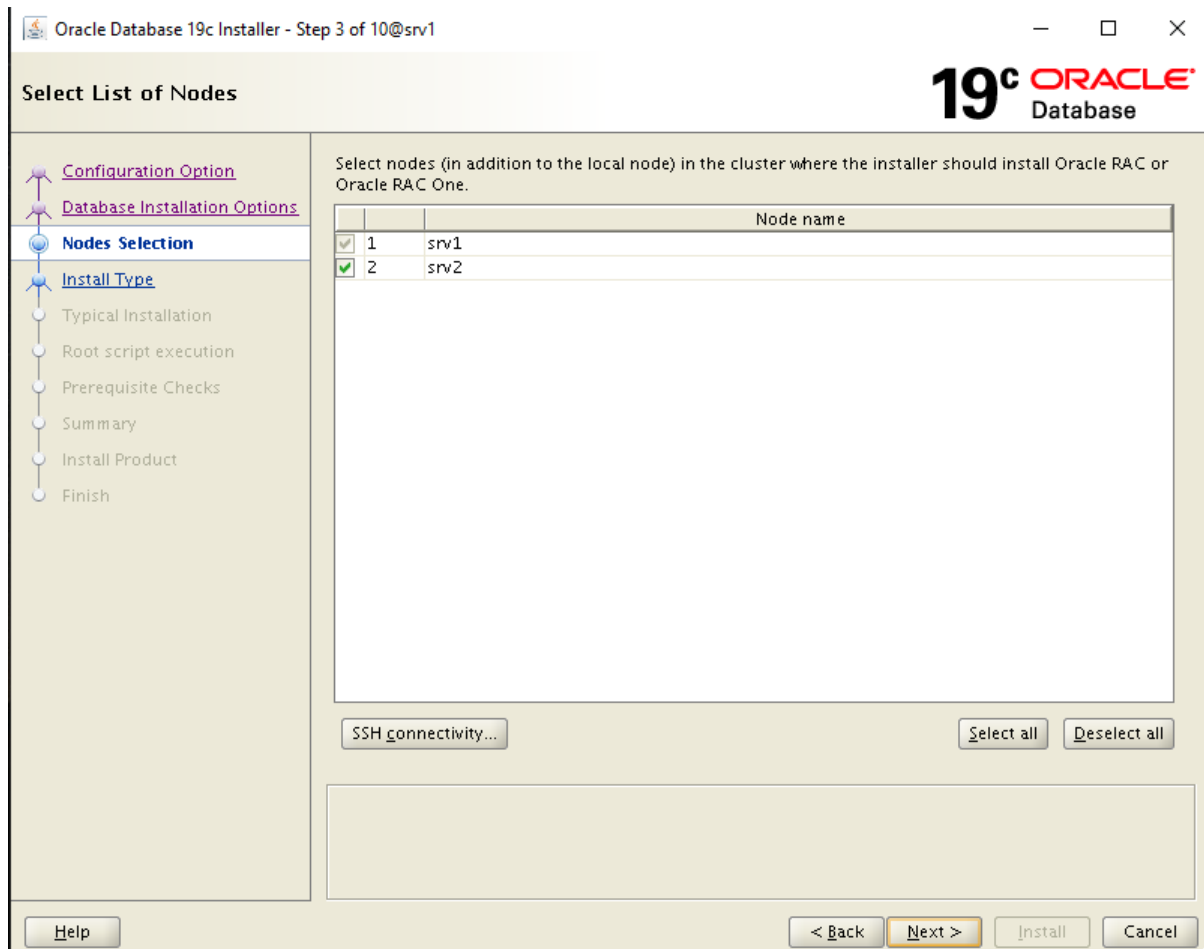
Select "Oracle Real Application Clusters database installation"

Installation of 19c RAC on Linux7 in Virtual Box VMs



Confirm that it identifies both nodes of the Cluster

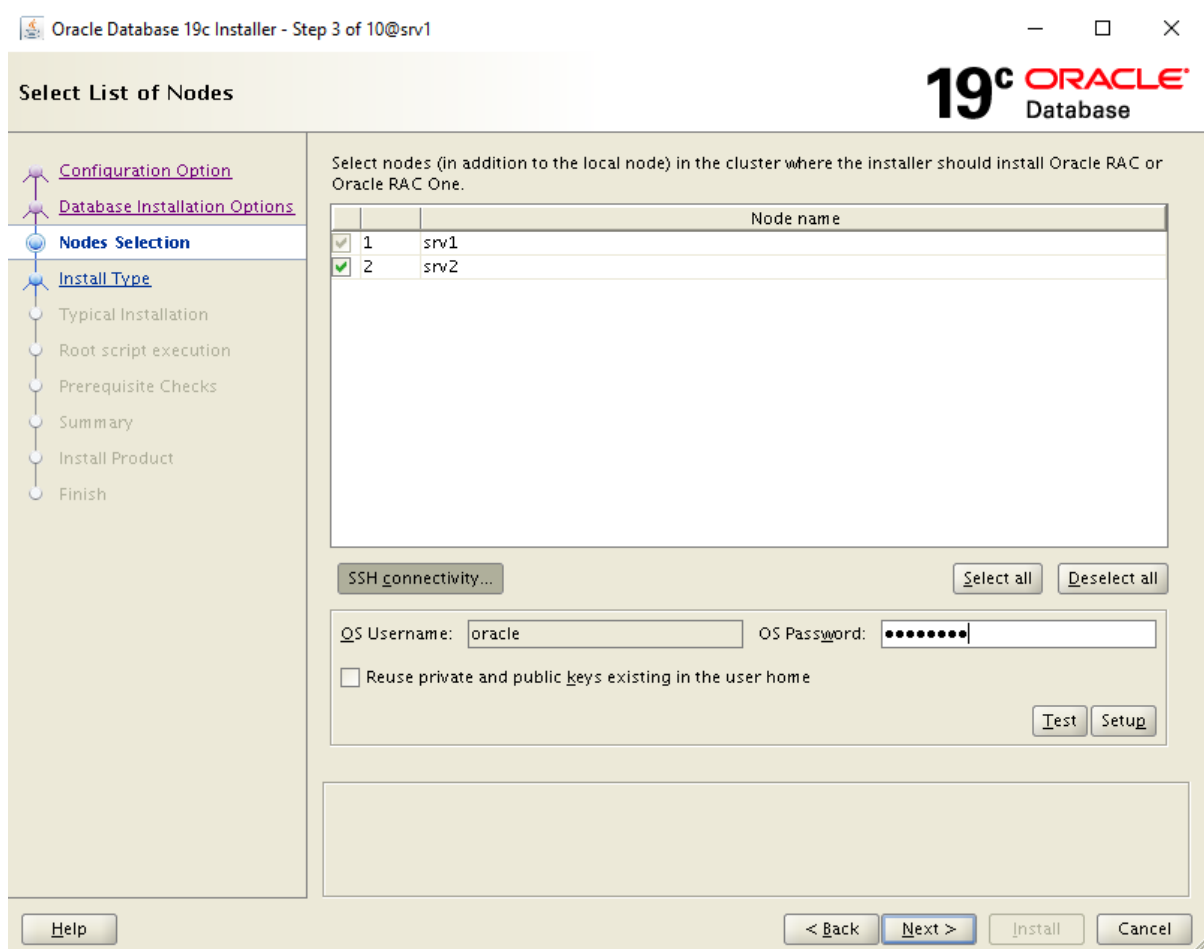
Installation of 19c RAC on Linux7 in Virtual Box VMs



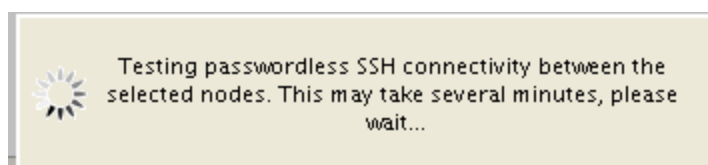
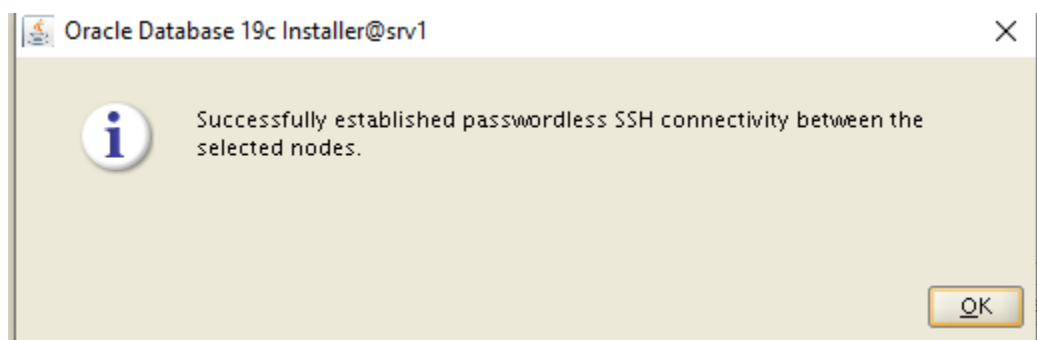
Click on “SSH connectivity”

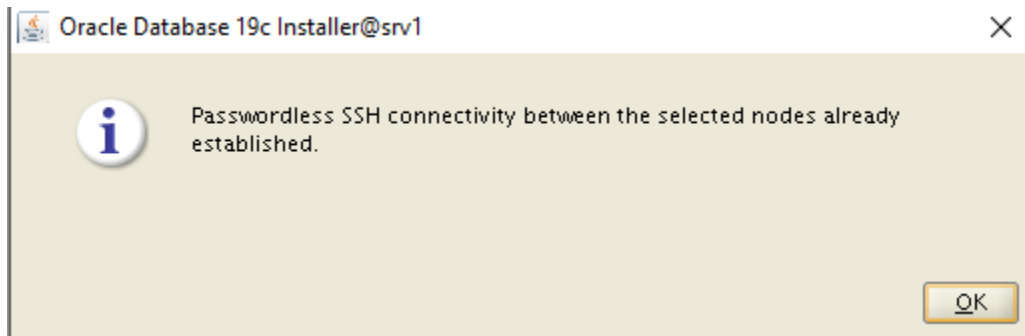
Enter the oracle password (has to be the same on both servers (nodes))

Installation of 19c RAC on Linux7 in Virtual Box VMs

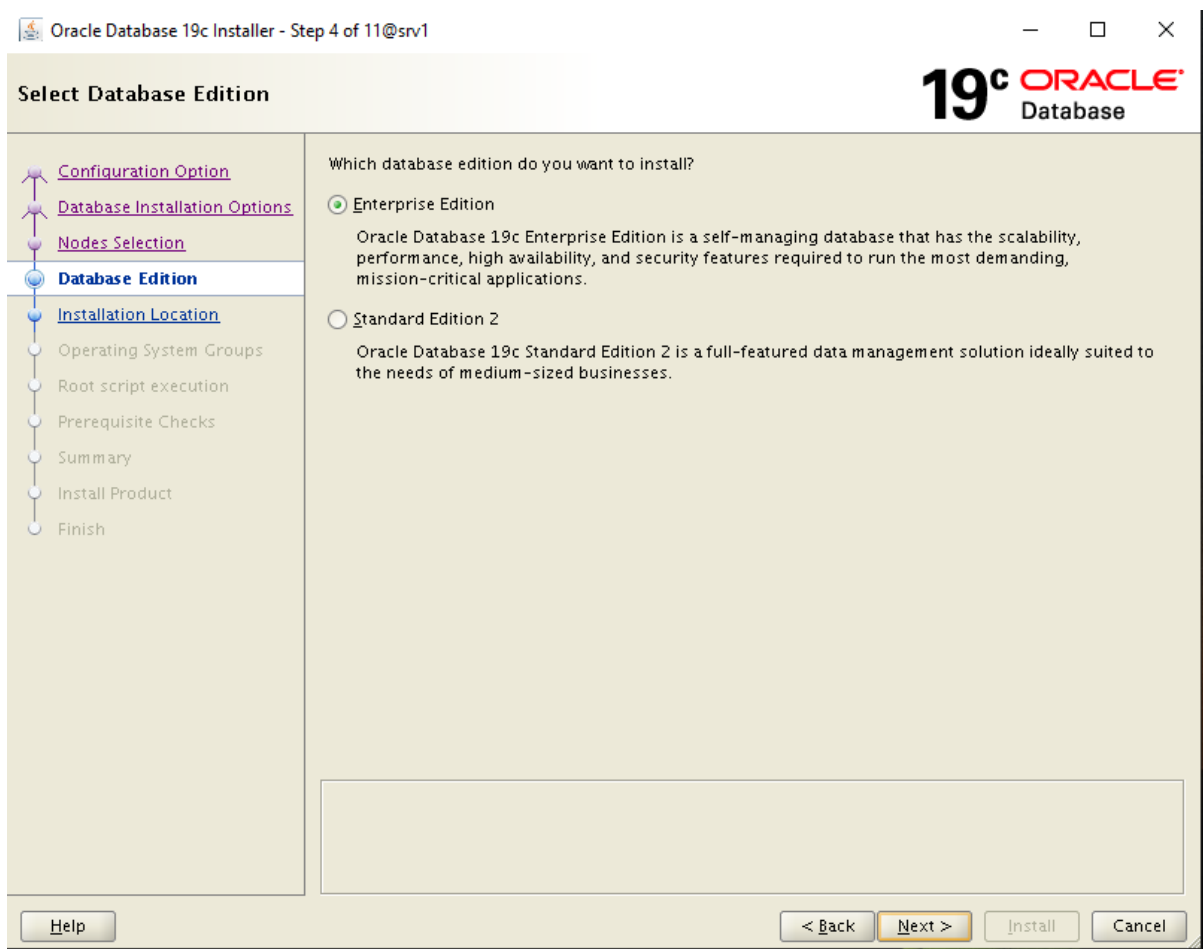


Select Setup and the Test (as you did during the Grid Infrastructure installation)



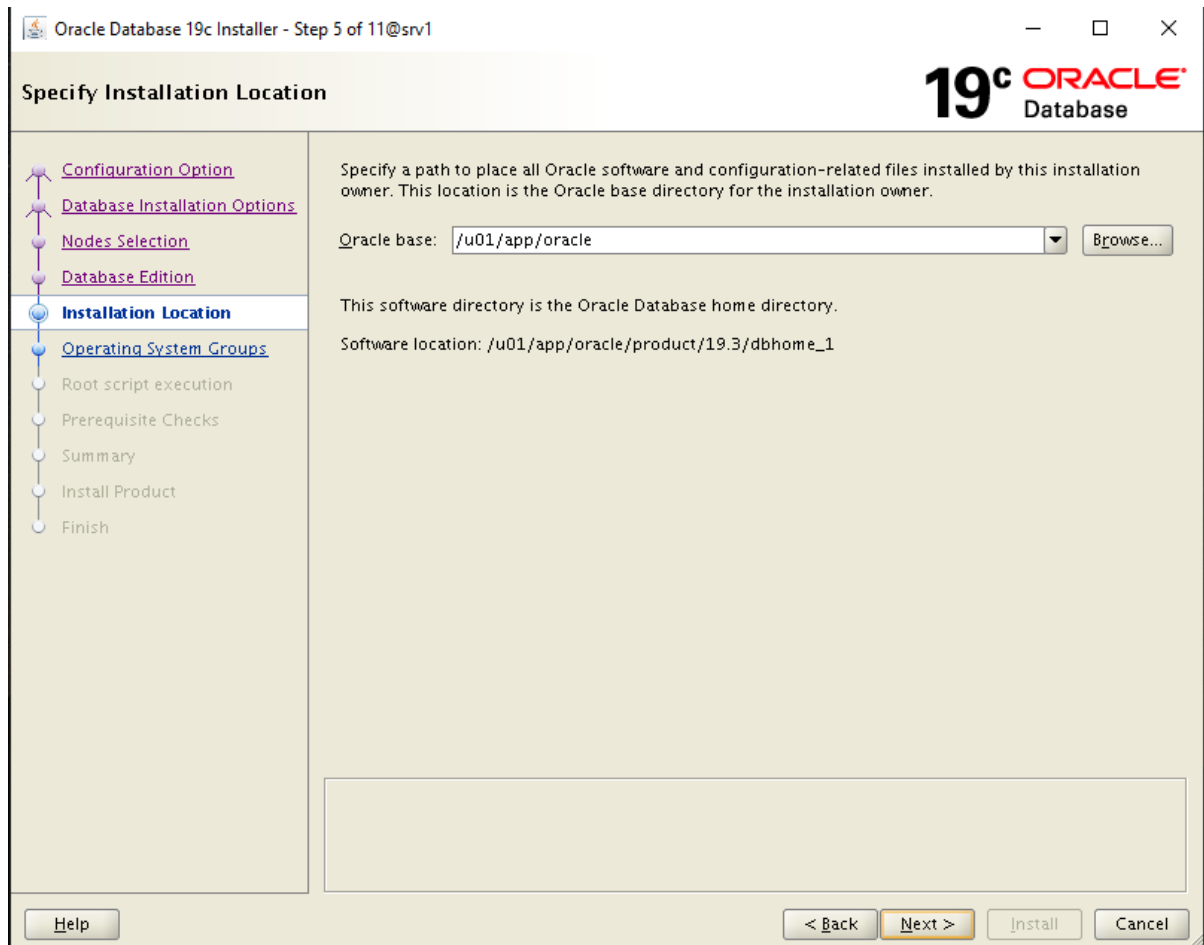


Select "Enterprise Edition" or "Standard Edition 2", depending on the licence purchased. Here I select "Enterprise Edition" so that I can test all the features.



Confirm the ORACLE_BASE (Note : Unlike Grid Infrastructure where the ORACLE_BASE is NOT a parent of the ORACLE_HOME, here the ORACLE_BASE must be a parent folder of the ORACLE_HOME)

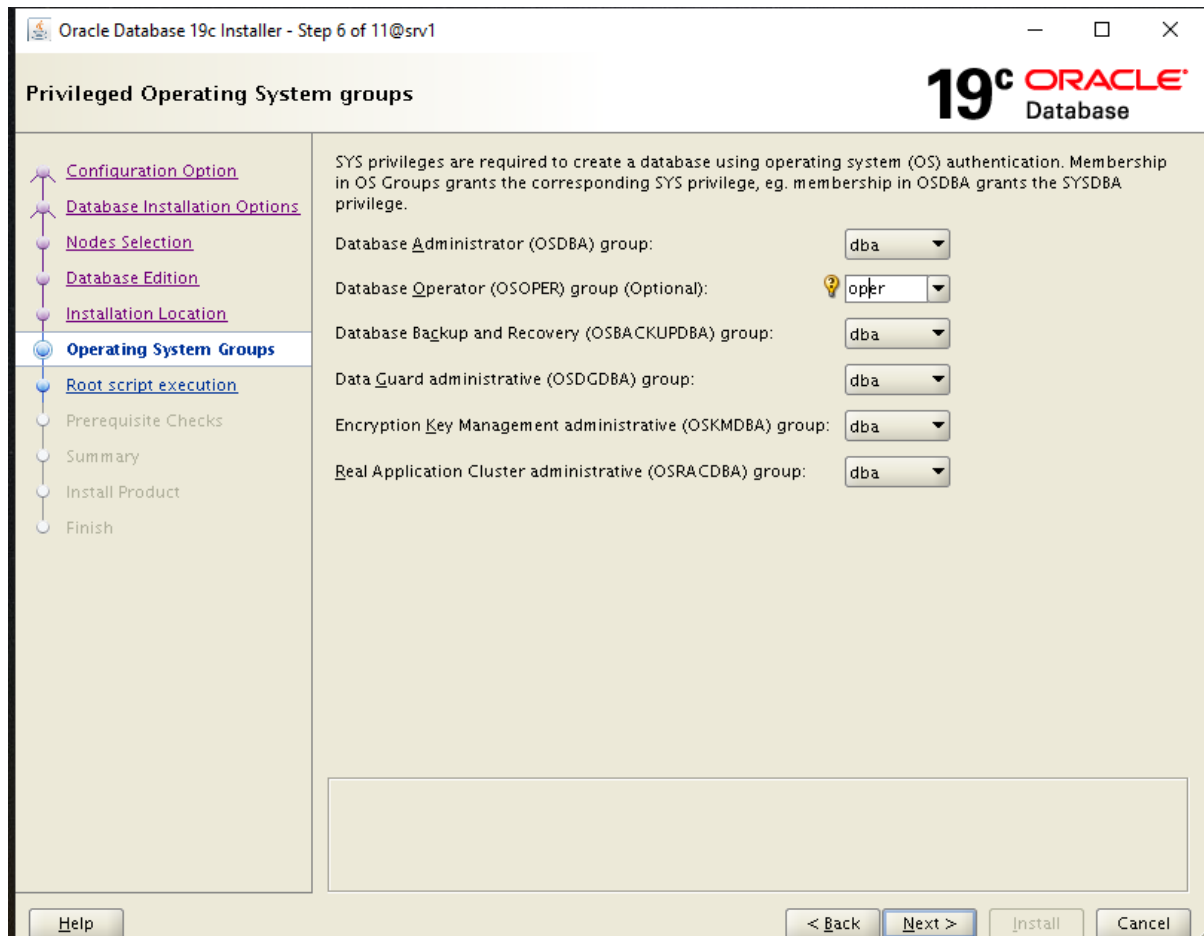
Installation of 19c RAC on Linux7 in Virtual Box VMs



Confirm the OS Groups

Note : Other Groups are setup by the Oracle PreInstall RPM as “backupdba”, “dgdbs”, “kmdbs”, “racdba”. However, I did not run usermod -G to add these groups earlier, so they cannot be assigned now.

Installation of 19c RAC on Linux7 in Virtual Box VMs



I choose to NOT have the installer run configuration scripts automatically. I will run them manually later

Installation of 19c RAC on Linux7 in Virtual Box VMs

Oracle Database 19c Installer - Step 7 of 11@srv1

Root script execution configuration

During the software configuration, certain operations have to be performed as "root" user. You can choose to have the installer perform these operations automatically by specifying inputs for one of the options below. The input specified will also be used by the installer to perform additional prerequisite checks.

☐ Automatically run configuration scripts

☒ Use "root" user credential

Password :

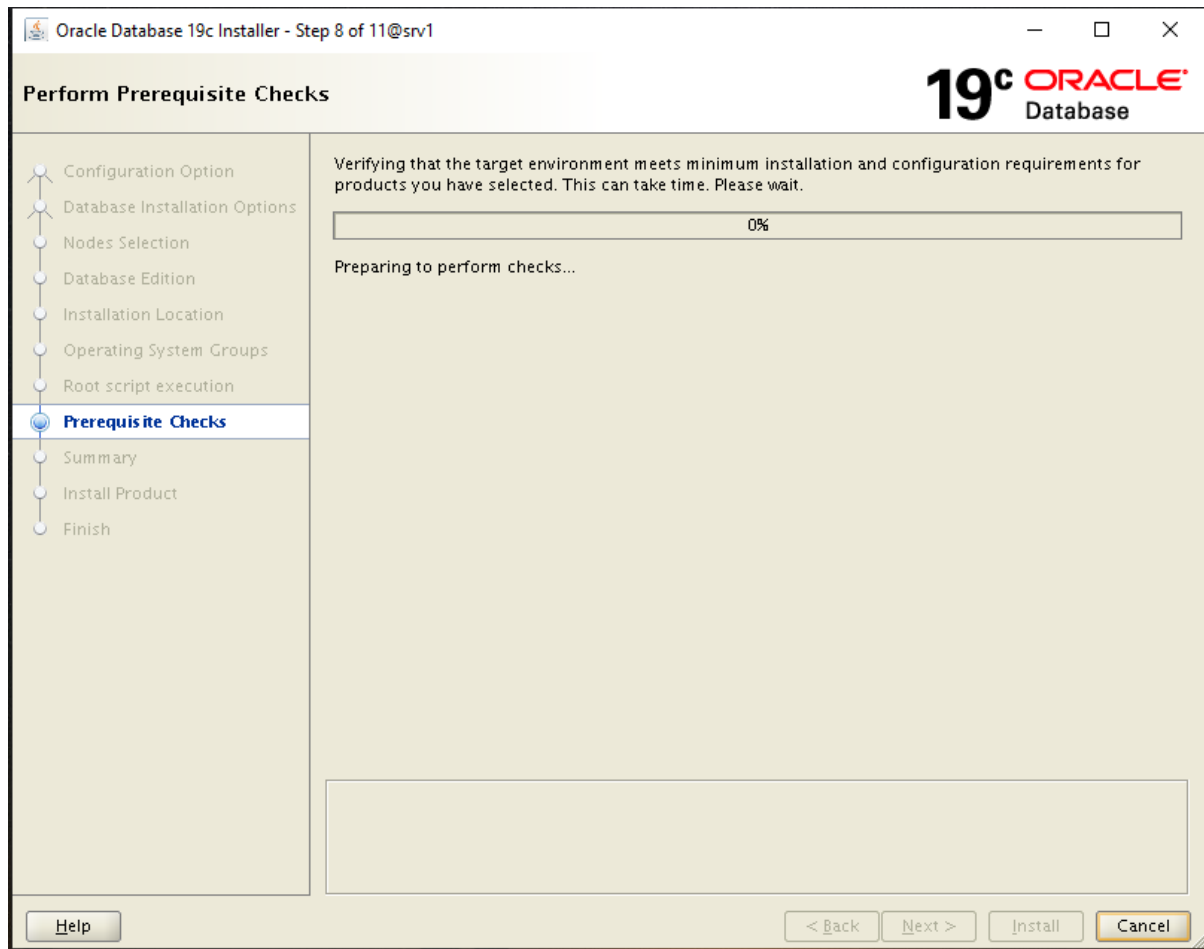
☐ Use sudo

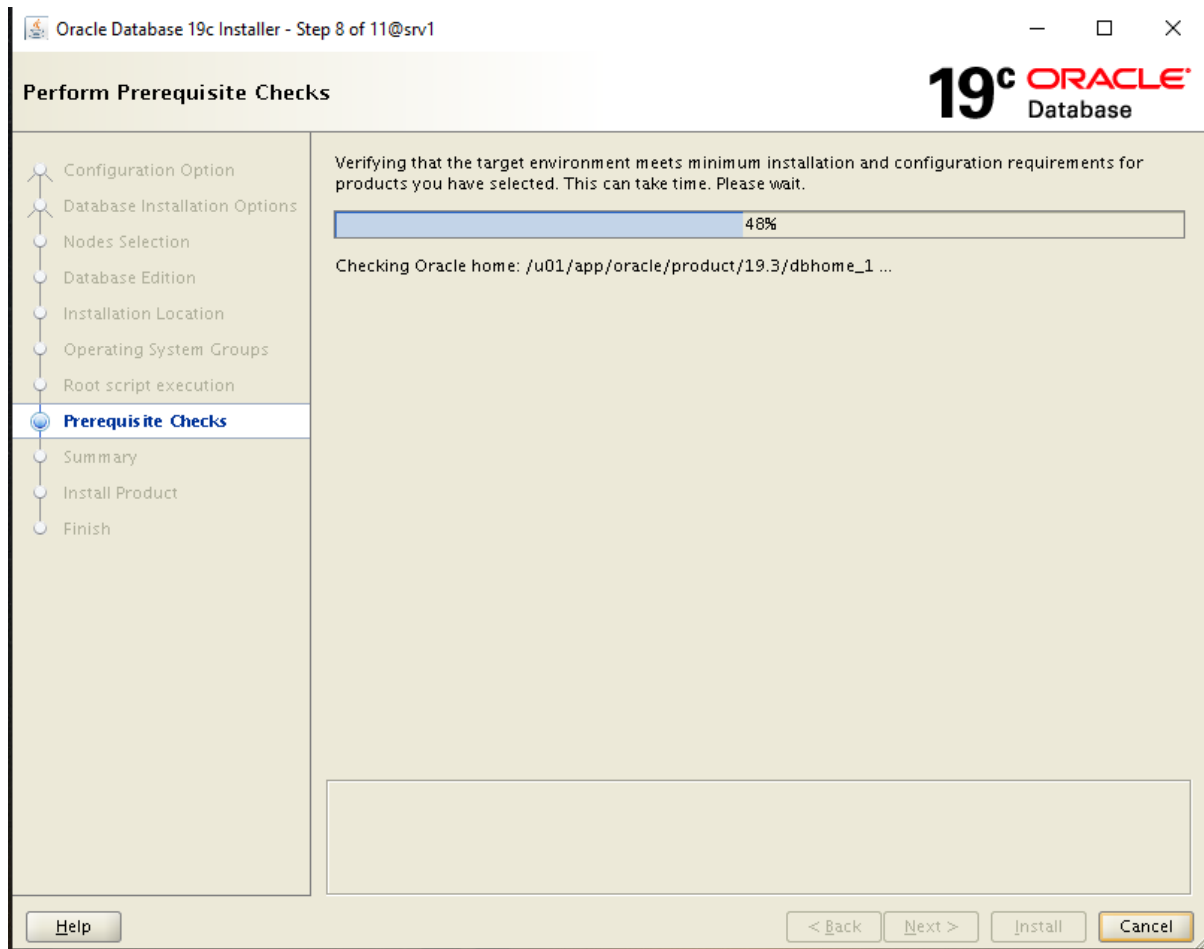
Program path :

User name :

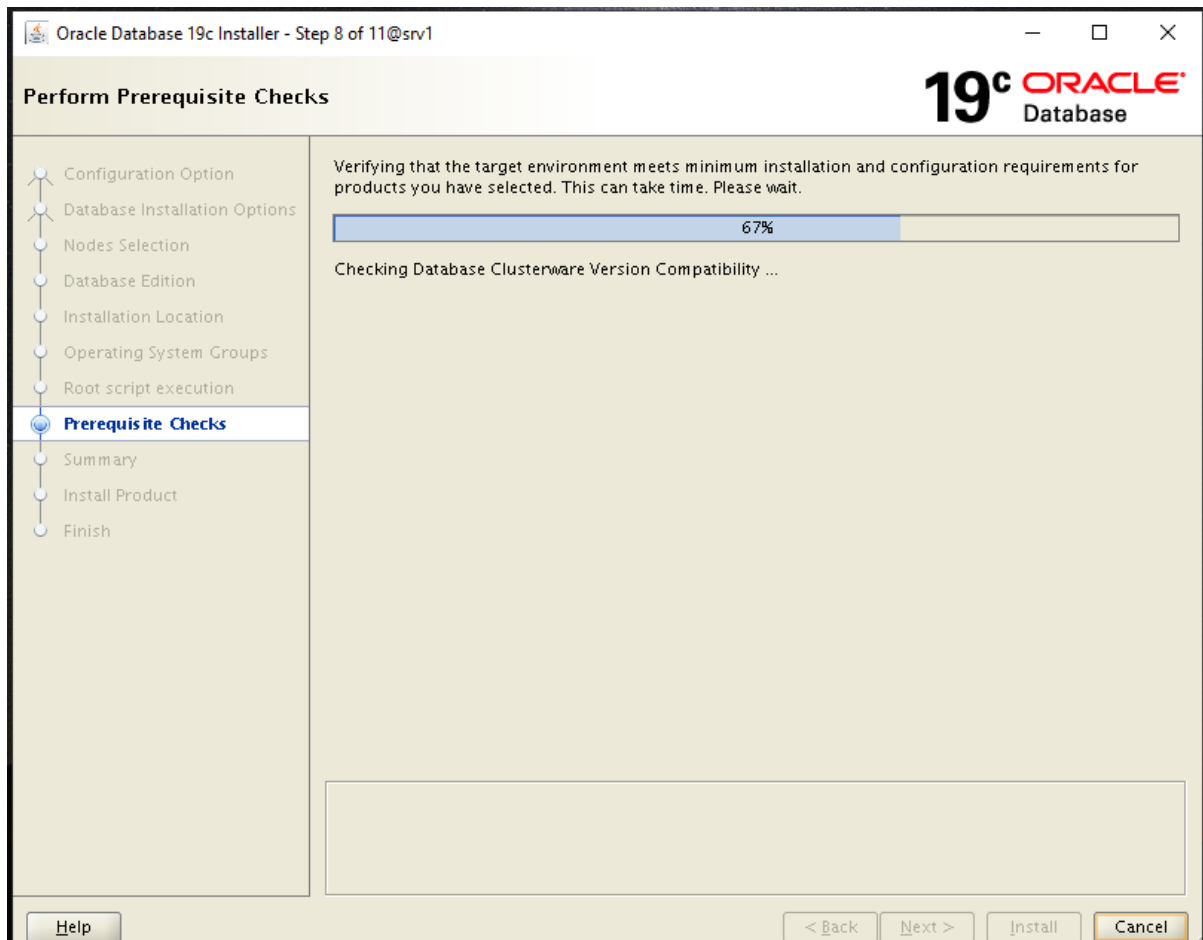
Password :

The installer then runs Pre-Requisite Checks



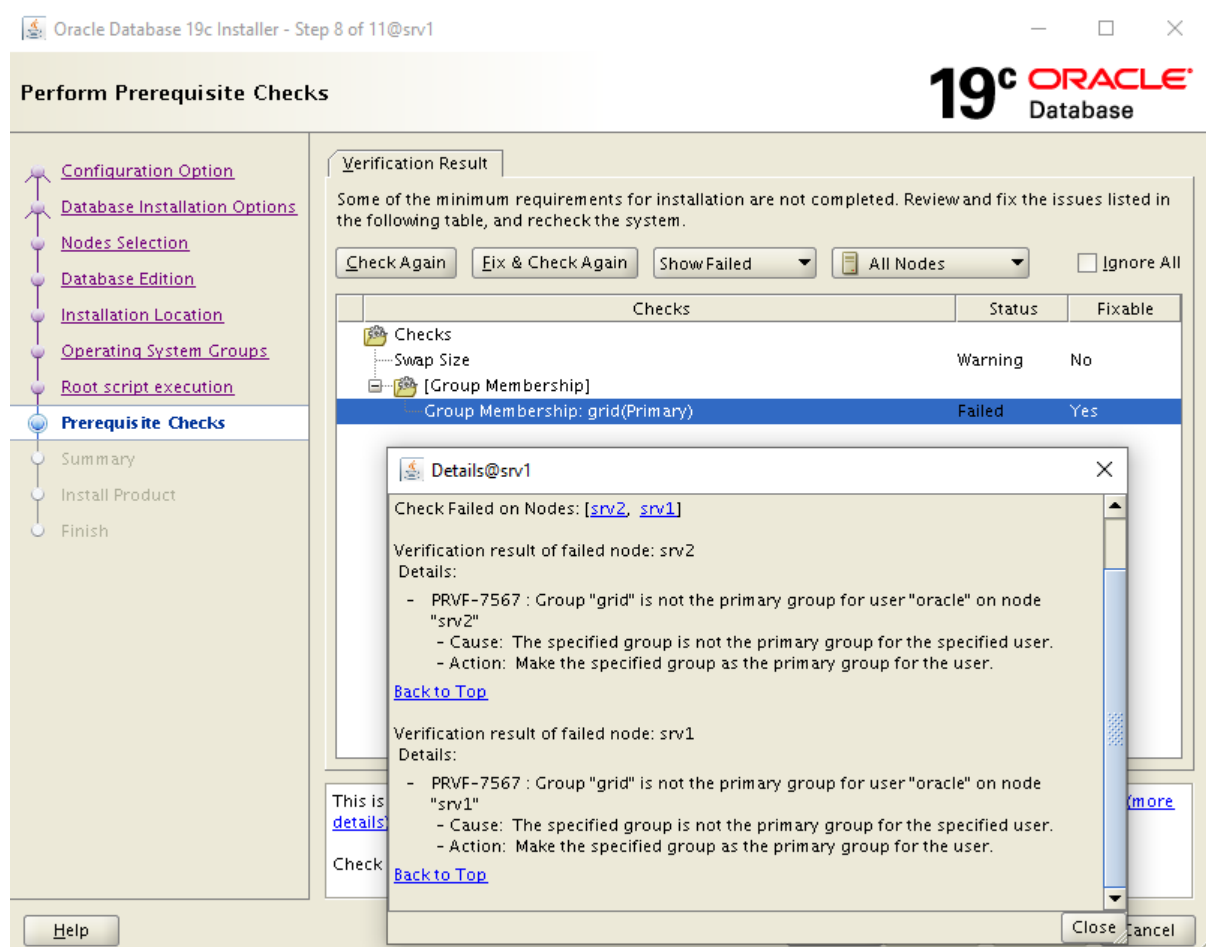


Installation of 19c RAC on Linux7 in Virtual Box VMs



If you get a “Failed” status for the group membership of the grid group

Installation of 19c RAC on Linux7 in Virtual Box VMs

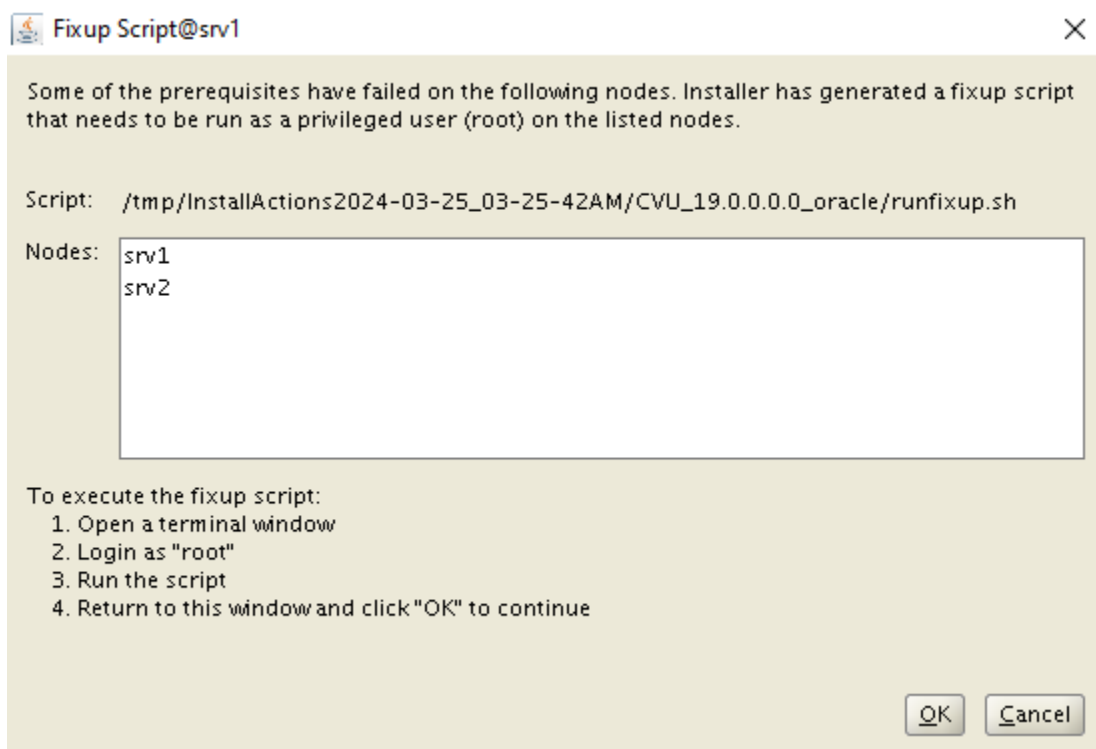


Verify from another terminal on both servers to confirm that oracle is a member of the grid group but not as a primary group :

```
[oracle@srv1 ~]$ id
uid=54321(oracle) gid=54321(oinstall)
groups=54321(oinstall),54322(dba),54323(oper),54331(grid)
context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
[oracle@srv1 ~]$
```

```
[oracle@srv2 ~]$ id
uid=54321(oracle) gid=54321(oinstall)
groups=54321(oinstall),54322(dba),54323(oper),54331(grid)
context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
[oracle@srv2 ~]$
```

Select “Fix and Check Again” so that the Installer generate a Fixup script to execute on both servers (nodes)



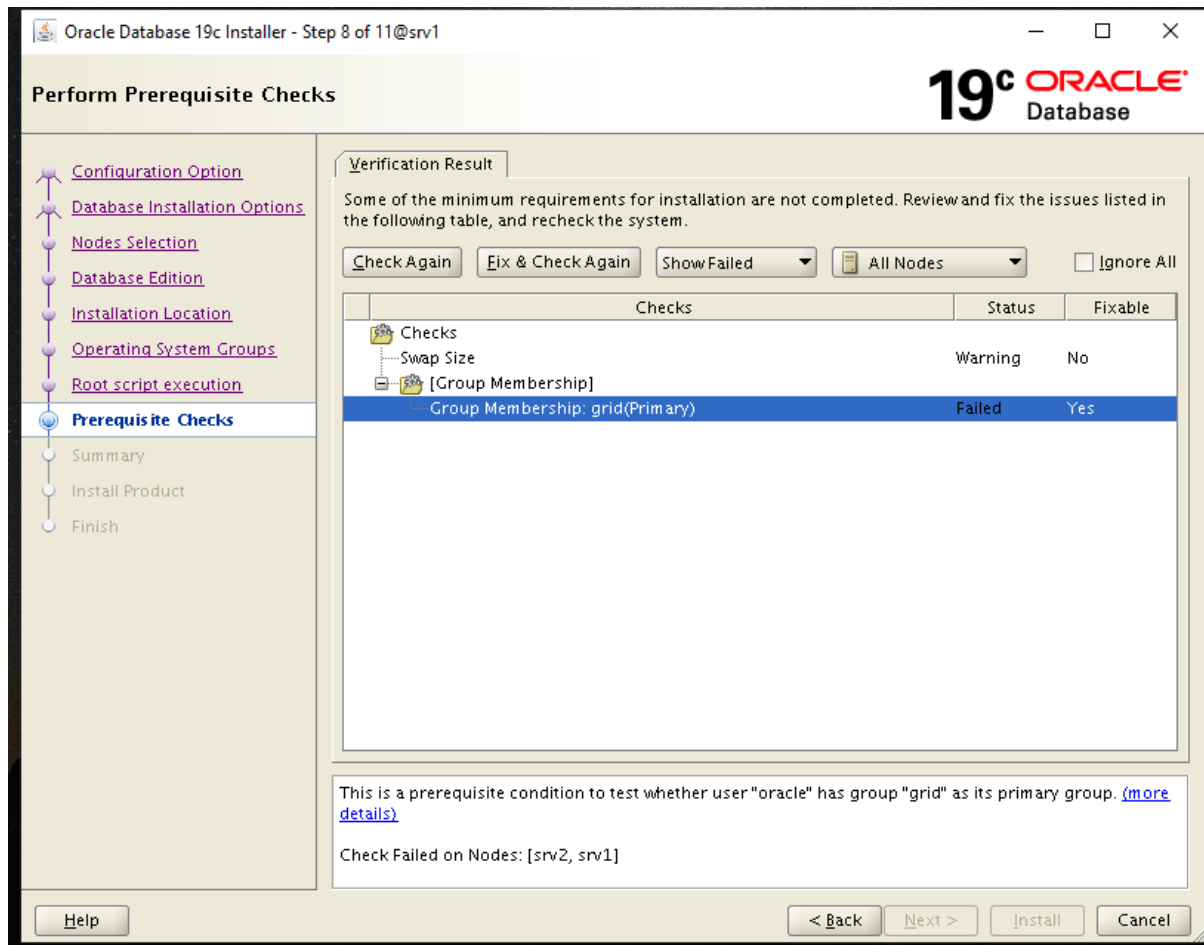
```
[root@srv1 ~]# cd /tmp/InstallActions2024-03-25_03-25-42AM/CVU_19.0.0.0.0_oracle
[root@srv1 CVU_19.0.0.0.0_oracle]# ./runfixup.sh
All Fix-up operations were completed successfully.
[root@srv1 CVU_19.0.0.0.0_oracle]#

[root@srv2 ~]# cd /tmp/InstallActions2024-03-25_03-25-42AM/CVU_19.0.0.0.0_oracle
[root@srv2 CVU_19.0.0.0.0_oracle]# ./runfixup.sh
All Fix-up operations were completed successfully.
[root@srv2 CVU_19.0.0.0.0_oracle]#
```

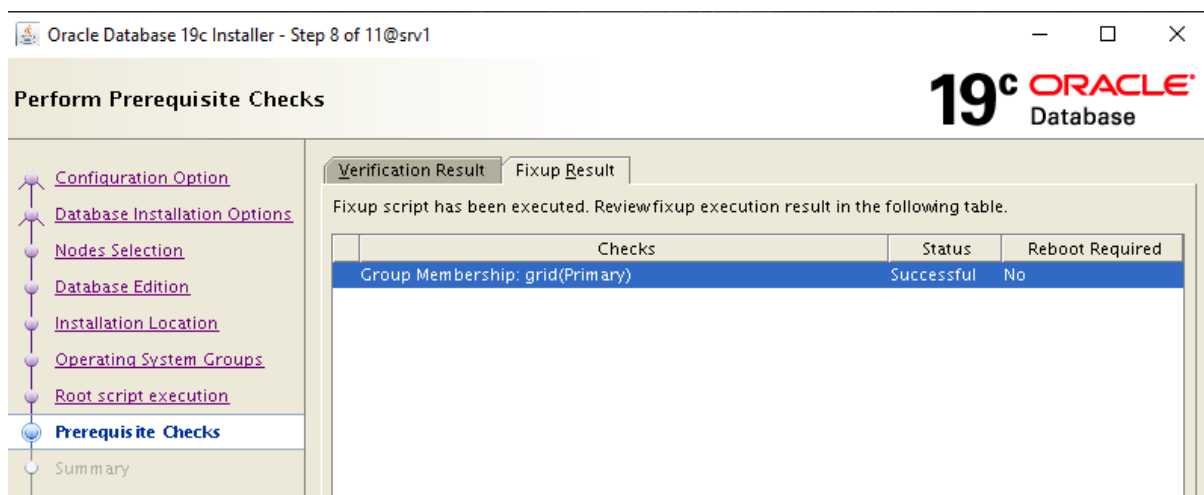
And click on “OK”

Then click on “Check Again”

Installation of 19c RAC on Linux7 in Virtual Box VMs

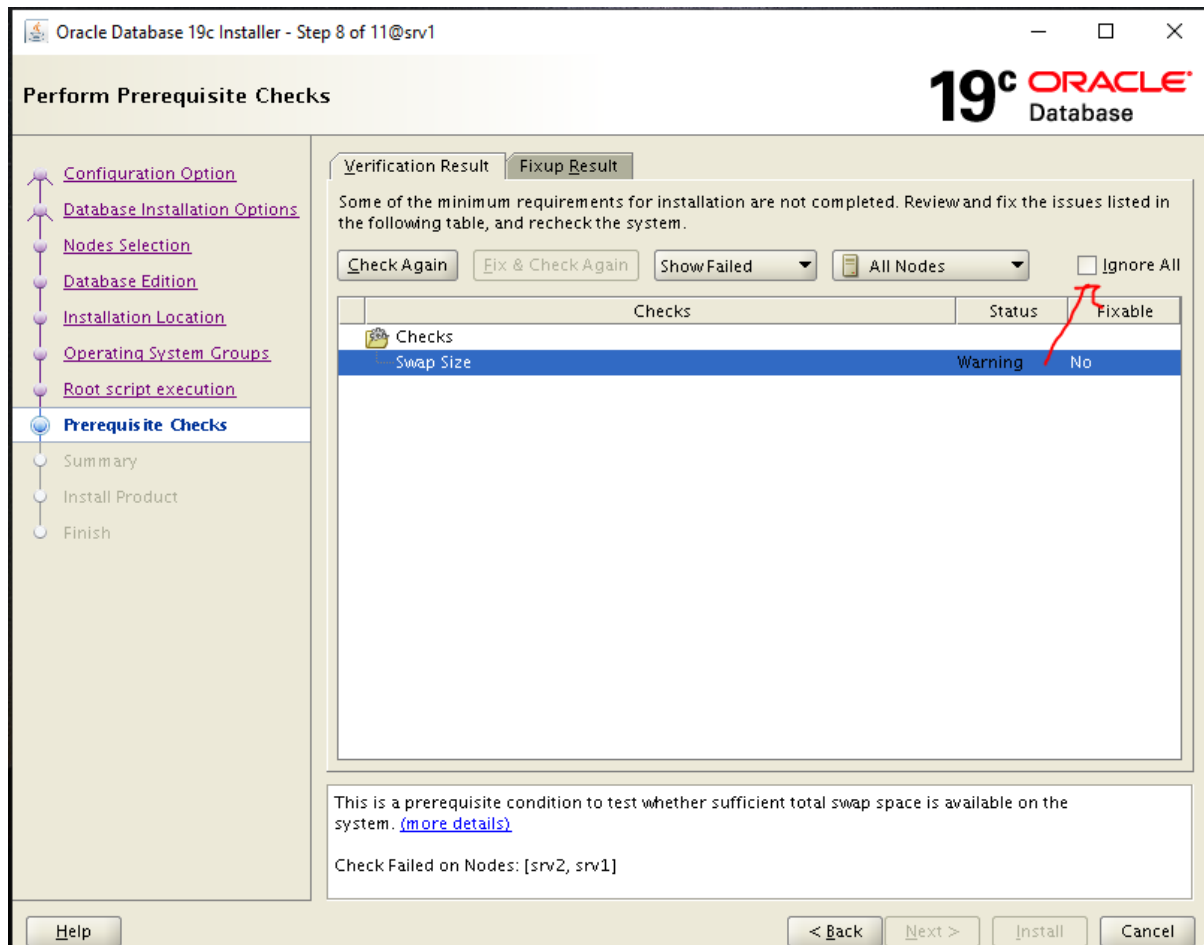


View the Fixup Result

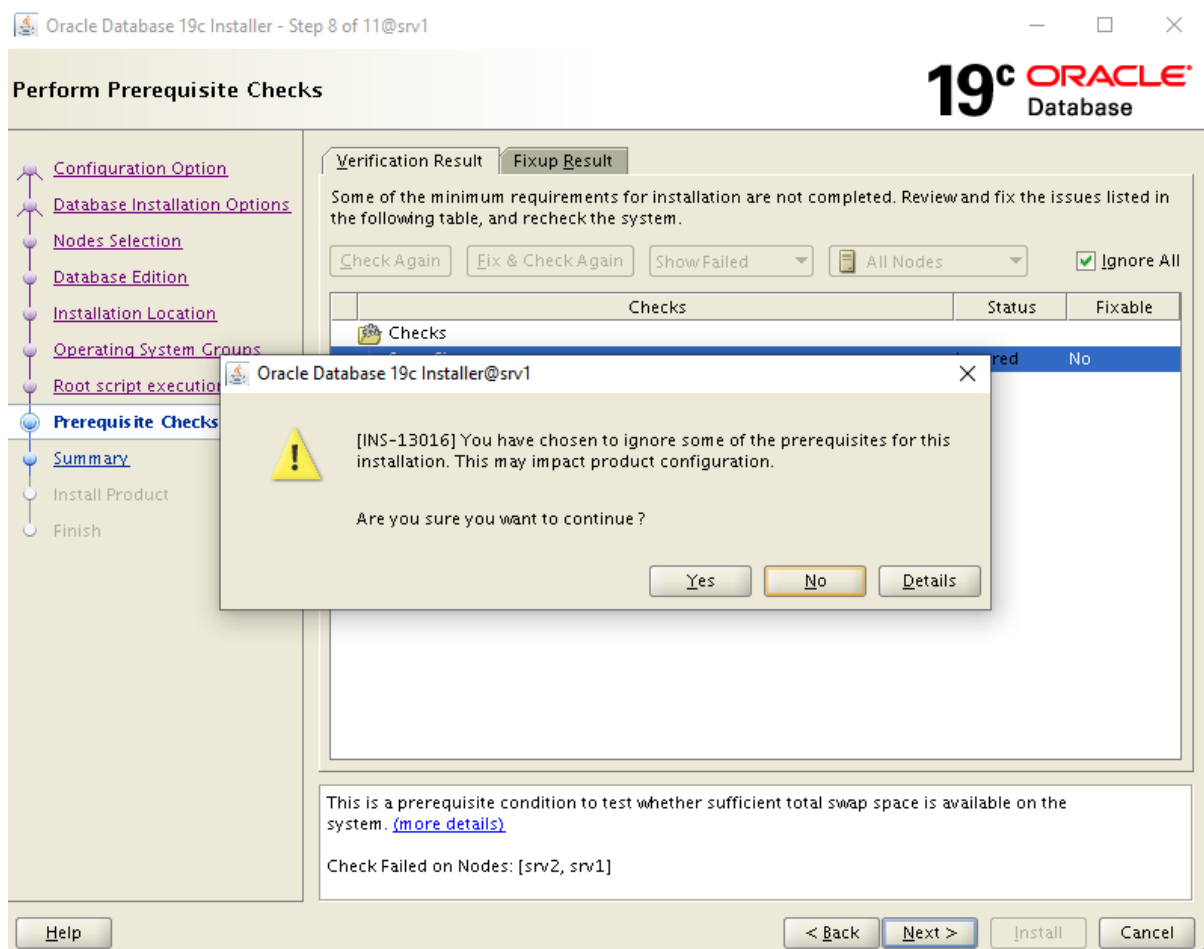


Ignore the Swap Size warning in the Verification Result by selecting the "Ignore All"

Installation of 19c RAC on Linux7 in Virtual Box VMs

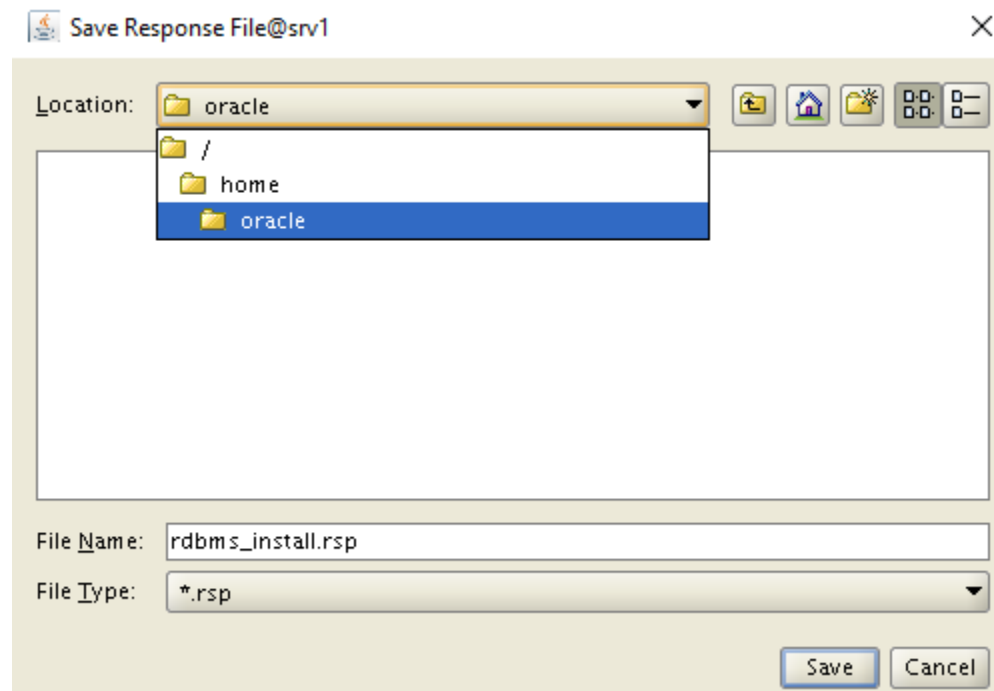
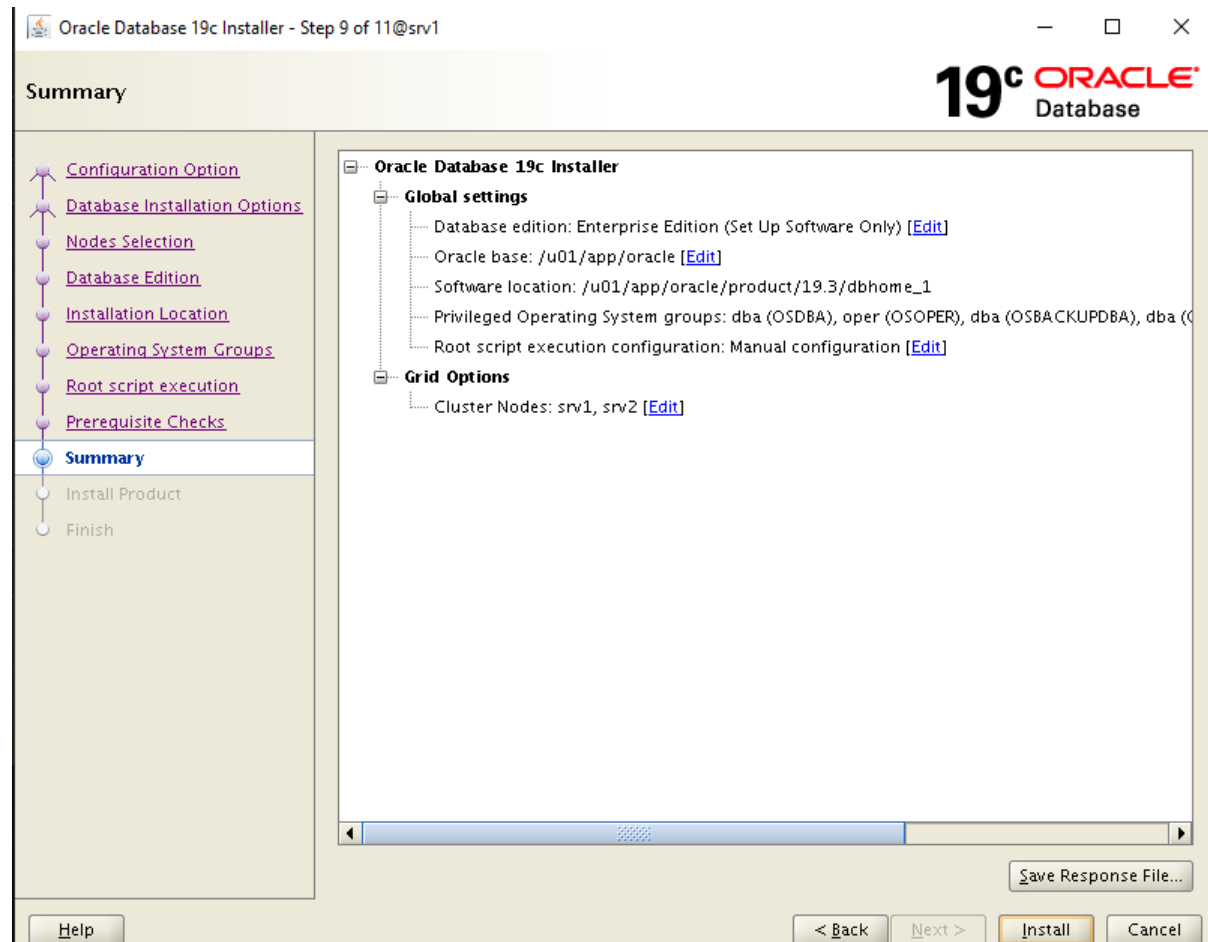


Installation of 19c RAC on Linux7 in Virtual Box VMs



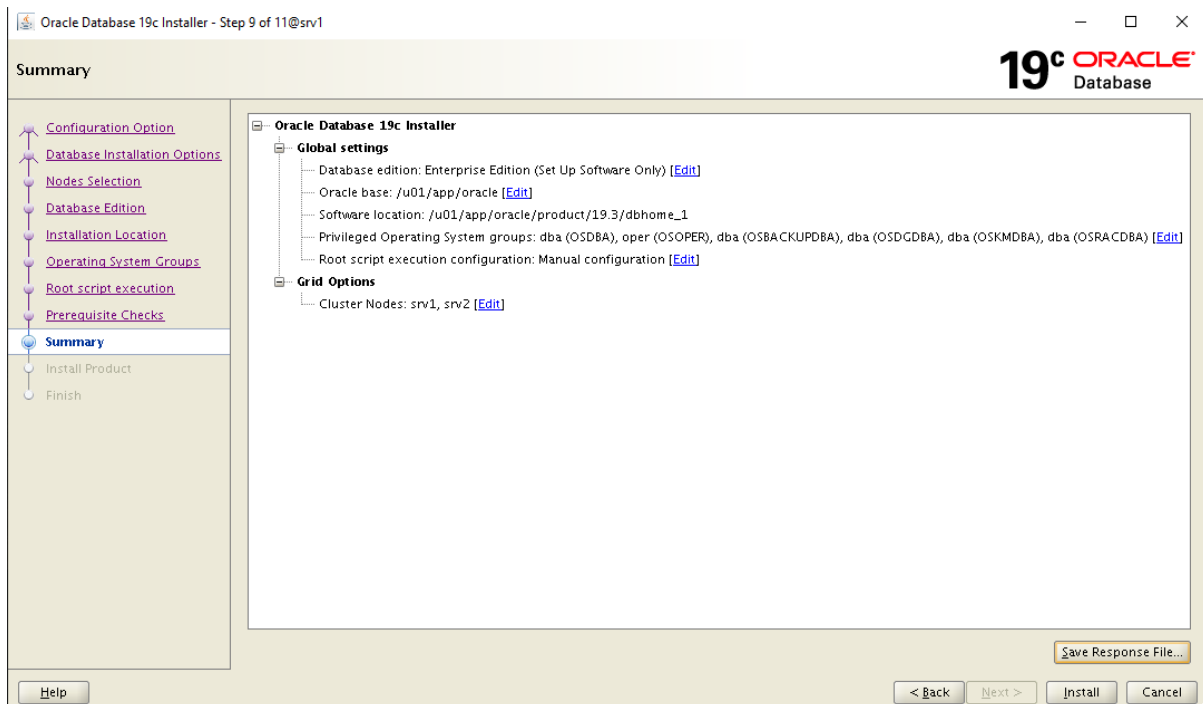
Verify the Summary and use the “Save Response File” button

Installation of 19c RAC on Linux7 in Virtual Box VMs

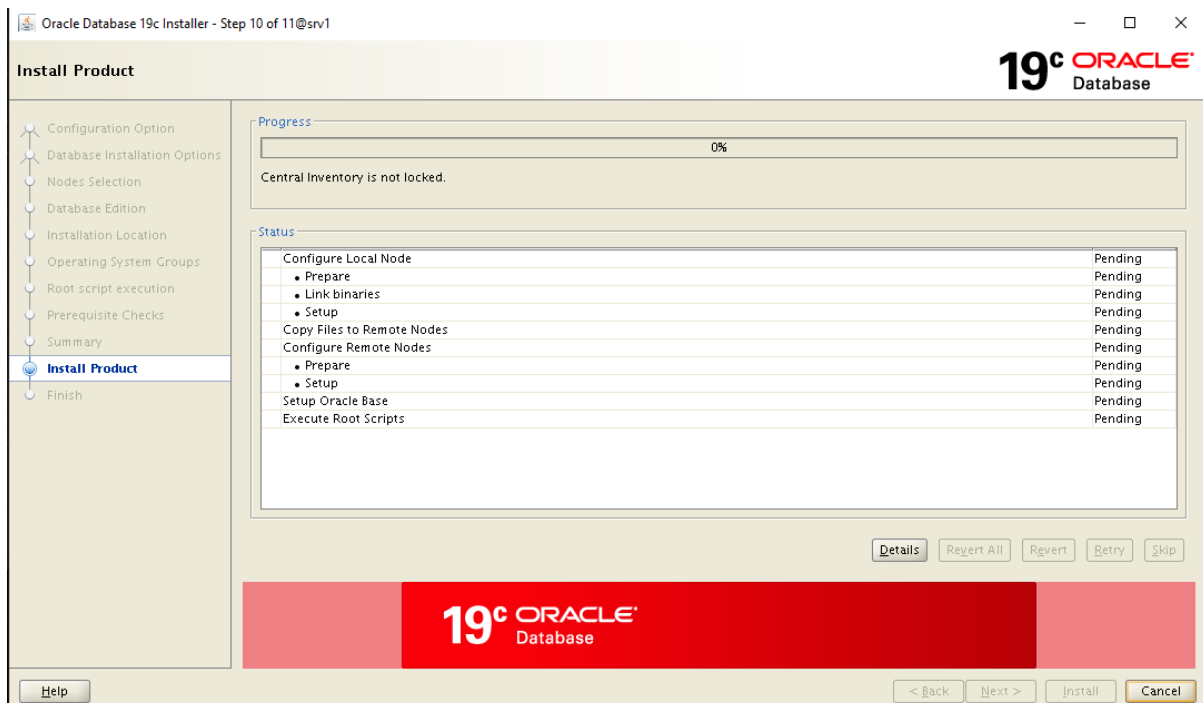


Installation of 19c RAC on Linux7 in Virtual Box VMs

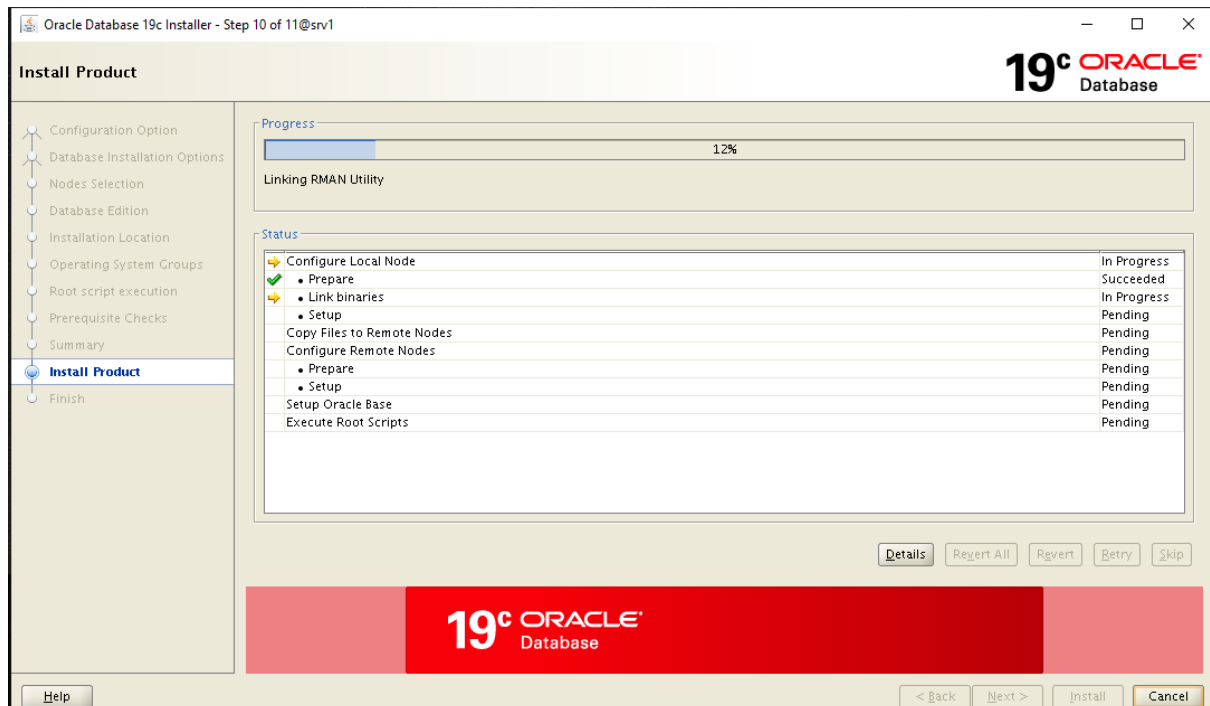
Then proceed with Install :



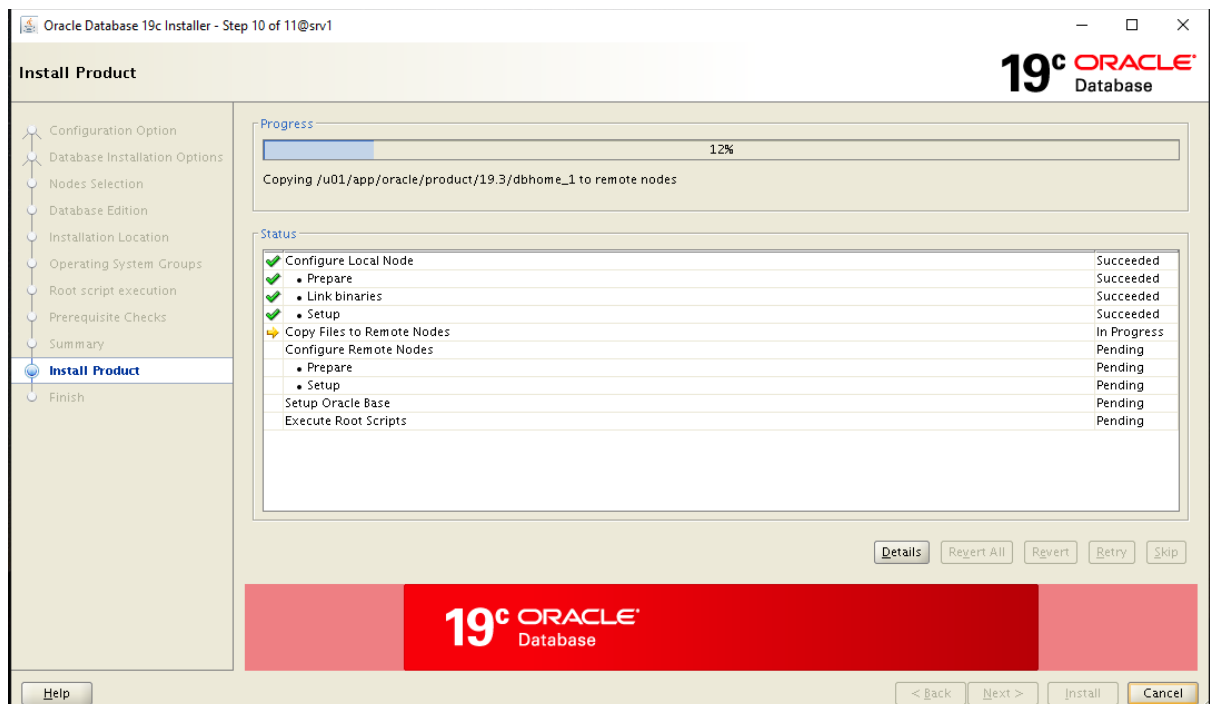
Monitor the progress of the installation



Installation of 19c RAC on Linux7 in Virtual Box VMs



The copy of the ORACLE_HOME to the remote node (srv2) will take some time

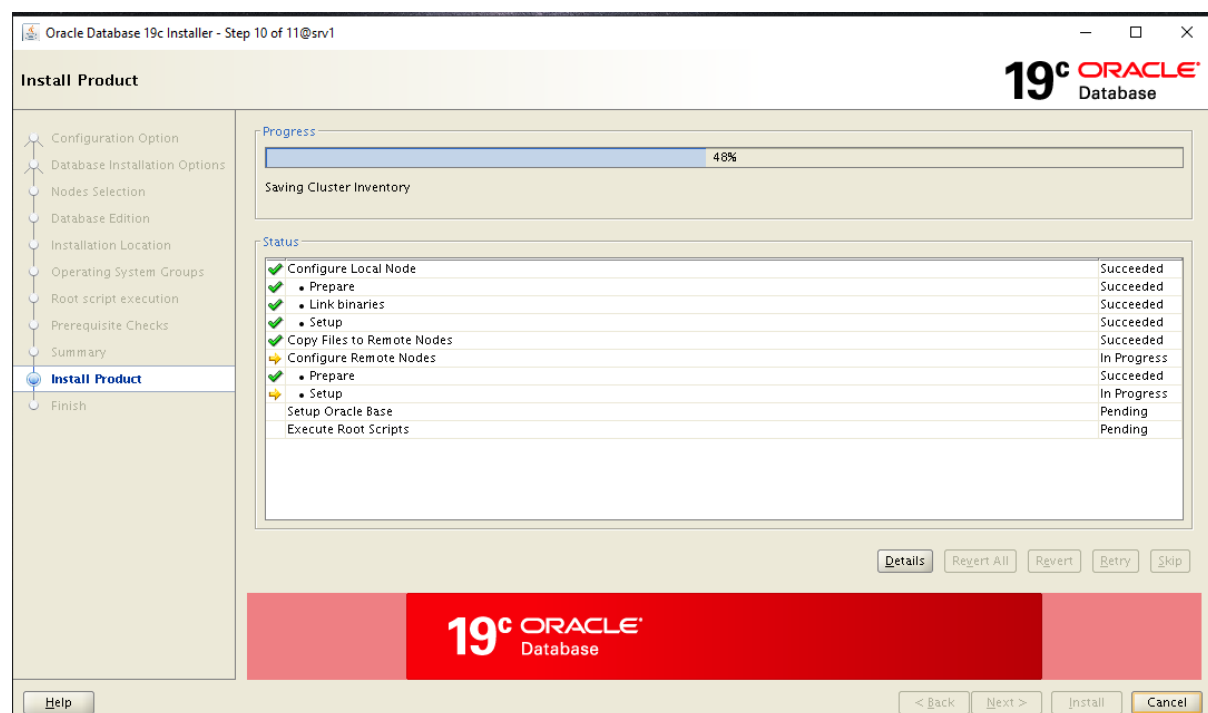


You can monitor it by checking disk space consumption

```
[root@srv1 ~]# cd /u01/app/oracle/product
[root@srv1 product]# du -sh .
7.0G
[root@srv1 product]#
```

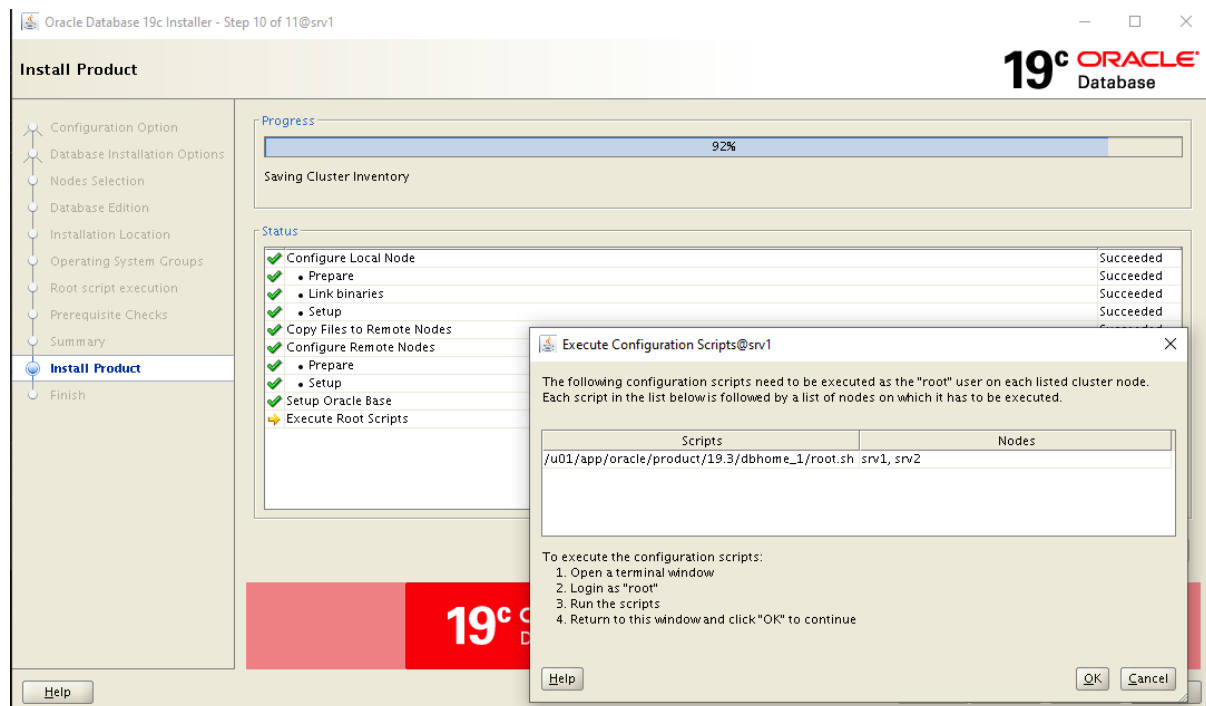
Installation of 19c RAC on Linux7 in Virtual Box VMs

```
[root@srv2 product]# cd
[root@srv2 ~]# cd /u01/app/oracle/product/
[root@srv2 product]# du -sh .
4.8G      .
[root@srv2 product]# du -sh .
5.0G      .
[root@srv2 product]#
[root@srv2 product]# du -sh .
6.2G      .
[root@srv2 product]# du -sh .
6.6G      .
[root@srv2 product]#
```



It will then prompt you with the configuration script to be run by root. As with the Grid Infrastructure installation, ensure that execution on the first server (node) is completed successfully before running it on the second server.

Installation of 19c RAC on Linux7 in Virtual Box VMs



```
[root@srv1 product]# pwd
/u01/app/oracle/product
[root@srv1 product]# cd 19.3
[root@srv1 19.3]# cd dbhome_1
[root@srv1 dbhome_1]# ./root.sh
Performing root user operation.
```

The following environment variables are set as:

```
ORACLE_OWNER= oracle
ORACLE_HOME= /u01/app/oracle/product/19.3/dbhome_1
```

```
Enter the full pathname of the local bin directory: [/usr/local/bin]:
The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.
```

Entries will be added to the /etc/oratab file as needed by Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
[root@srv1 dbhome_1]#

```
[root@srv2 product]# cd 19.3/dbhome_1
[root@srv2 dbhome_1]# pwd
/u01/app/oracle/product/19.3/dbhome_1
[root@srv2 dbhome_1]# ./root.sh
Performing root user operation.
```

The following environment variables are set as:

```
ORACLE_OWNER= oracle
ORACLE_HOME= /u01/app/oracle/product/19.3/dbhome_1
```

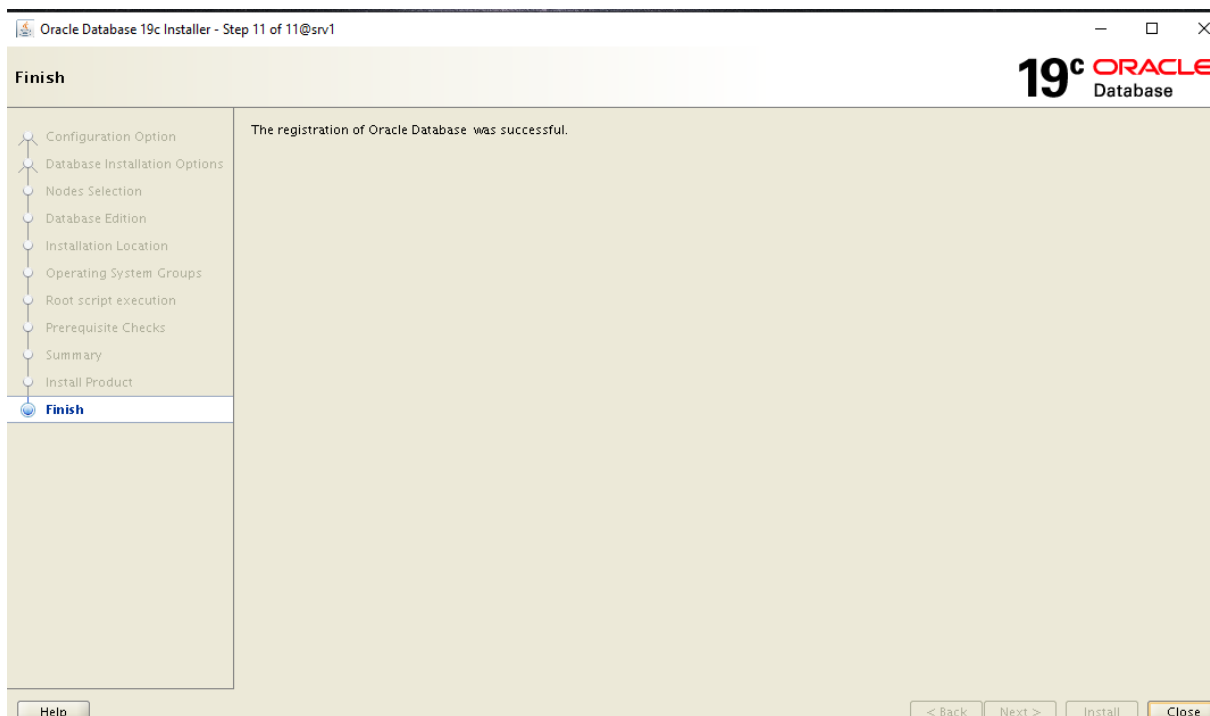
```
Enter the full pathname of the local bin directory: [/usr/local/bin]:
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

The contents of "dbhome" have not changed. No need to overwrite.
The contents of "oraenv" have not changed. No need to overwrite.
The contents of "coraenv" have not changed. No need to overwrite.

Entries will be added to the /etc/oratab file as needed by
Database Configuration Assistant when a database is created
Finished running generic part of root script.
Now product-specific root actions will be performed.
[root@srv2 dbhome_1]#

Then click on OK for the Installer to complete.



After you click on the Close button, the command line returns in the session where you started the installer. You can then review the log file

```
[oracle@srv1 dbhome_1]$ ./runInstaller
Launching Oracle Database Setup Wizard...
```

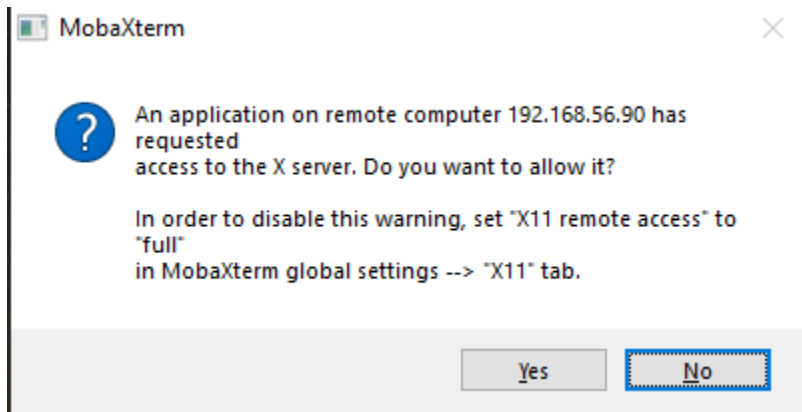
The response file for this session can be found at:
/u01/app/oracle/product/19.3/dbhome_1/install/response/db_2024-03-25_03-25-42AM.rsp

You can find the log of this install session at:
/u01/app/oraInventory/logs/InstallActions2024-03-25_03-25-42AM/installActions2024-03-25_03-25-42AM.log
[oracle@srv1 dbhome_1]\$

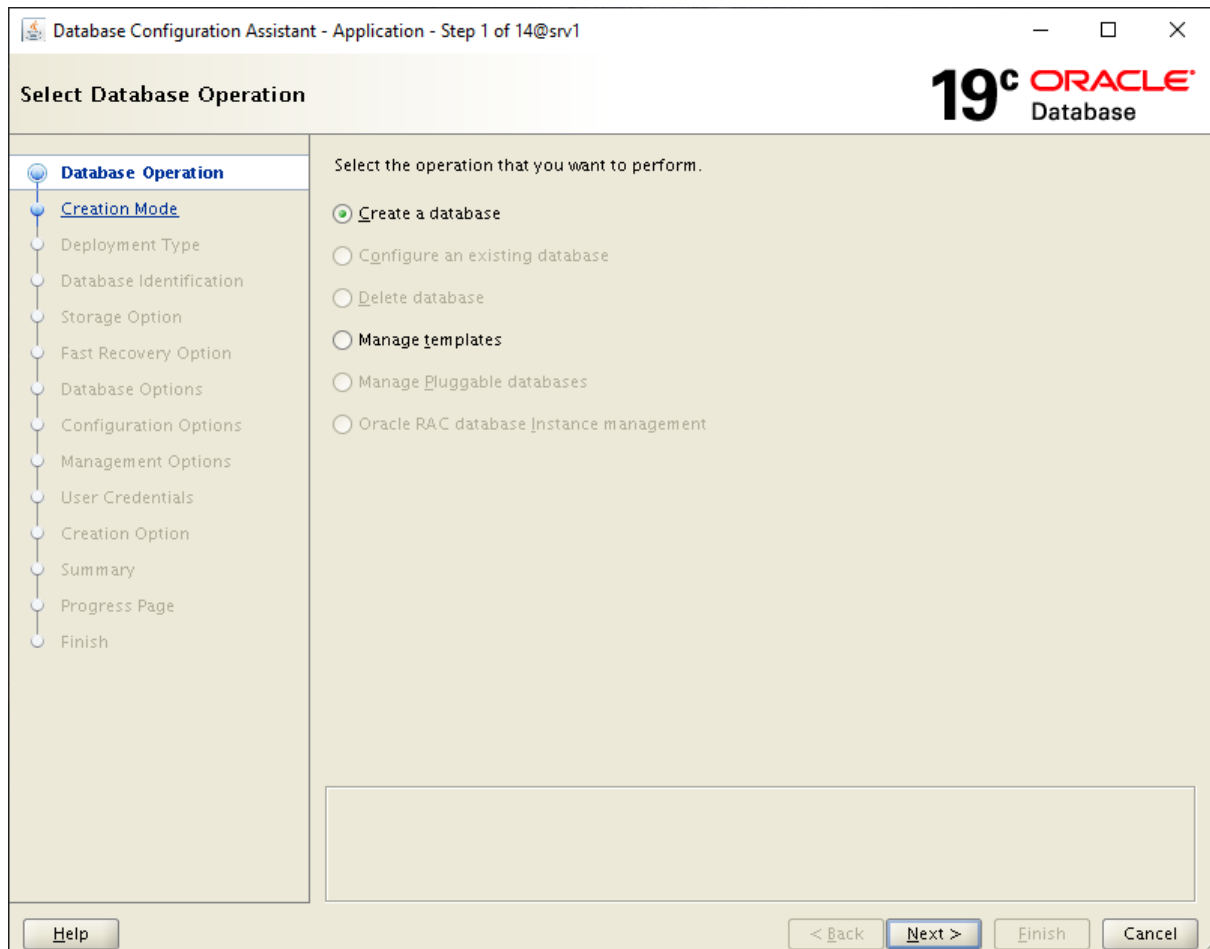
Create the RAC Database

Start the Database Creation Assistant

```
[oracle@srv1 dbhome_1]$ dbca
```



Select "Create a database"



Select “Advanced configuration”

Installation of 19c RAC on Linux7 in Virtual Box VMs

Database Configuration Assistant - Create a database - Step 2 of 14@srv1

19c ORACLE Database

Select Database Creation Mode

Database Operation

Creation Mode

Deployment Type

Database Identification

Storage Option

Fast Recovery Option

Database Options

Configuration Options

Management Options

User Credentials

Creation Option

Summary

Progress Page

Finish

☐ Typical configuration

Global database name: orcl

Storage type: Automatic Storage Management (ASM)

Database files location: +DATA/{DB_UNIQUE_NAME} Browse...

Fast Recovery Area (FRA): +OCR_VOTE Browse...

Database character set: AL32UTF8 - Unicode UTF-8 Universal character set

Administrative password:

Confirm password:

☒ Create as Container database

Pluggable database name:

☒ Advanced configuration

Help < Back Next > Finish Cancel

Select “Oracle Real Application Cluster (RAC) database” as the database type, “Admin Managed” as the Configuration type and “General Purpose or Transaction Processing” as the Template

Database Configuration Assistant - Create a database - Step 3 of 14@srv1

19^c ORACLE Database

Select Database Deployment Type

Select the type of database you want to create.

Database type: Oracle Real Application Cluster (RAC) database

Configuration type: Admin Managed

Select a template for your database.

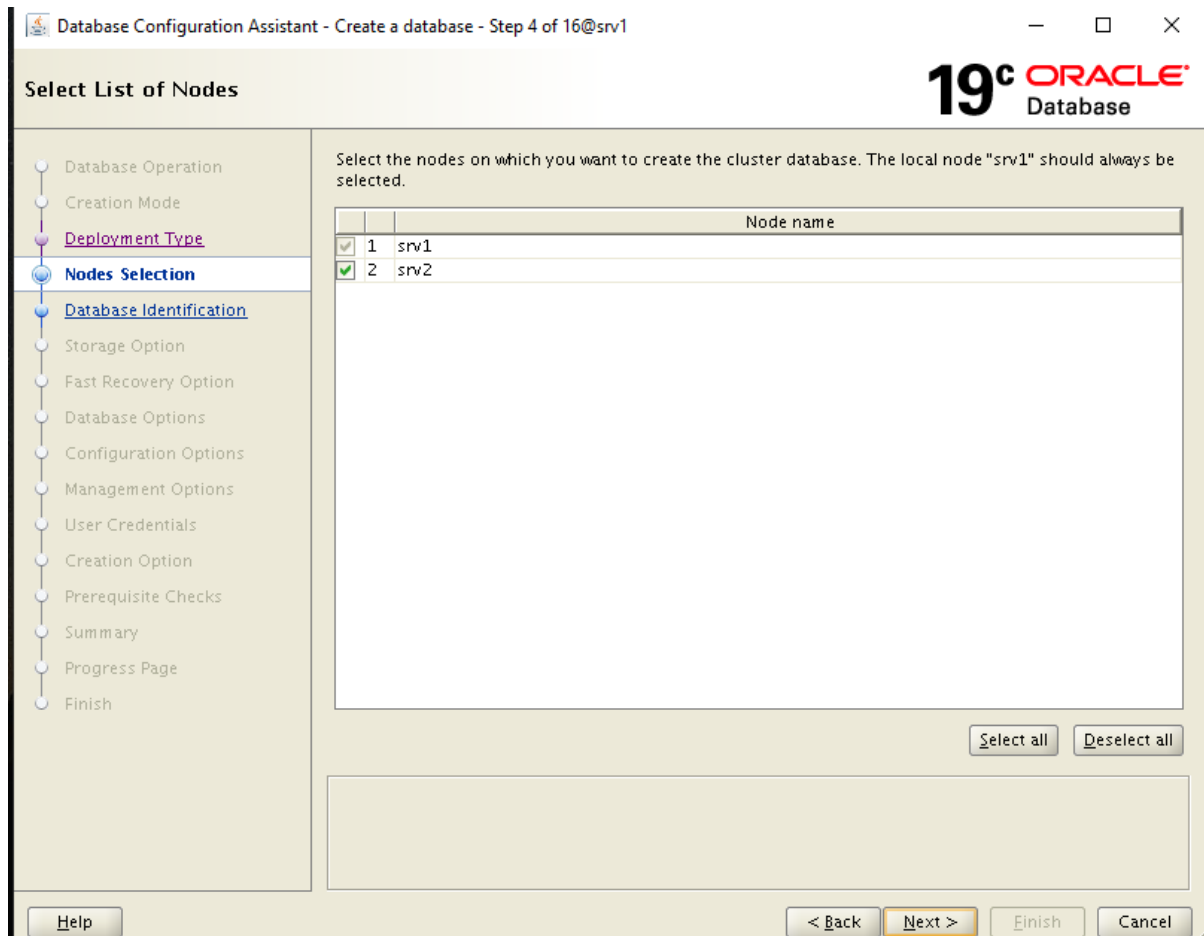
Templates that include datafiles contain pre-created databases. They allow you to create a new database quickly. Use templates without datafiles only when necessary, such as when you need to change attributes like block size that cannot be altered after database creation.

	Template name	Include datafiles	Details
☐	Data Warehouse	Yes	View details
☐	Custom Database	No	View details
☒	General Purpose or Transaction Processing	Yes	View details

Template location: /u01/app/oracle/product/19.3/dbhome_1/assistants/dbca/templates [Change...](#)

[Help](#) [< Back](#) [Next >](#) [Finish](#) [Cancel](#)

Ensure that both servers (nodes) of the Cluster are selected



Specify "RACDB" as the Global Database Name and SID Prefix, ensure that "Create as a Container Database" and "Use Local Undo Tablespace for PDBs" is checked, specify 1 PDB with the "pdb1" as the PDB name

Database Configuration Assistant - Create a database - Step 5 of 16@srv1

Specify Database Identification Details

19^c ORACLE Database

Database Operation
Creation Mode
Deployment Type
Nodes Selection
Database Identification
Storage Option
Fast Recovery Option
Database Options
Configuration Options
Management Options
User Credentials
Creation Option
Prerequisite Checks
Summary
Progress Page
Finish

Provide a unique database identifier information. An Oracle database is uniquely identified by a Global database name, typically of the form "name.domain".

Global database name: RACDB

SID Prefix: RACDB

Service name:

☒ Create as Container database

A Container database can be used for consolidating multiple databases into a single database, and it enables database virtualization. A Container database (CDB) can have zero or more pluggable databases (PDB).

☒ Use Local Undo tablespace for PDBs

☐ Create an empty Container database

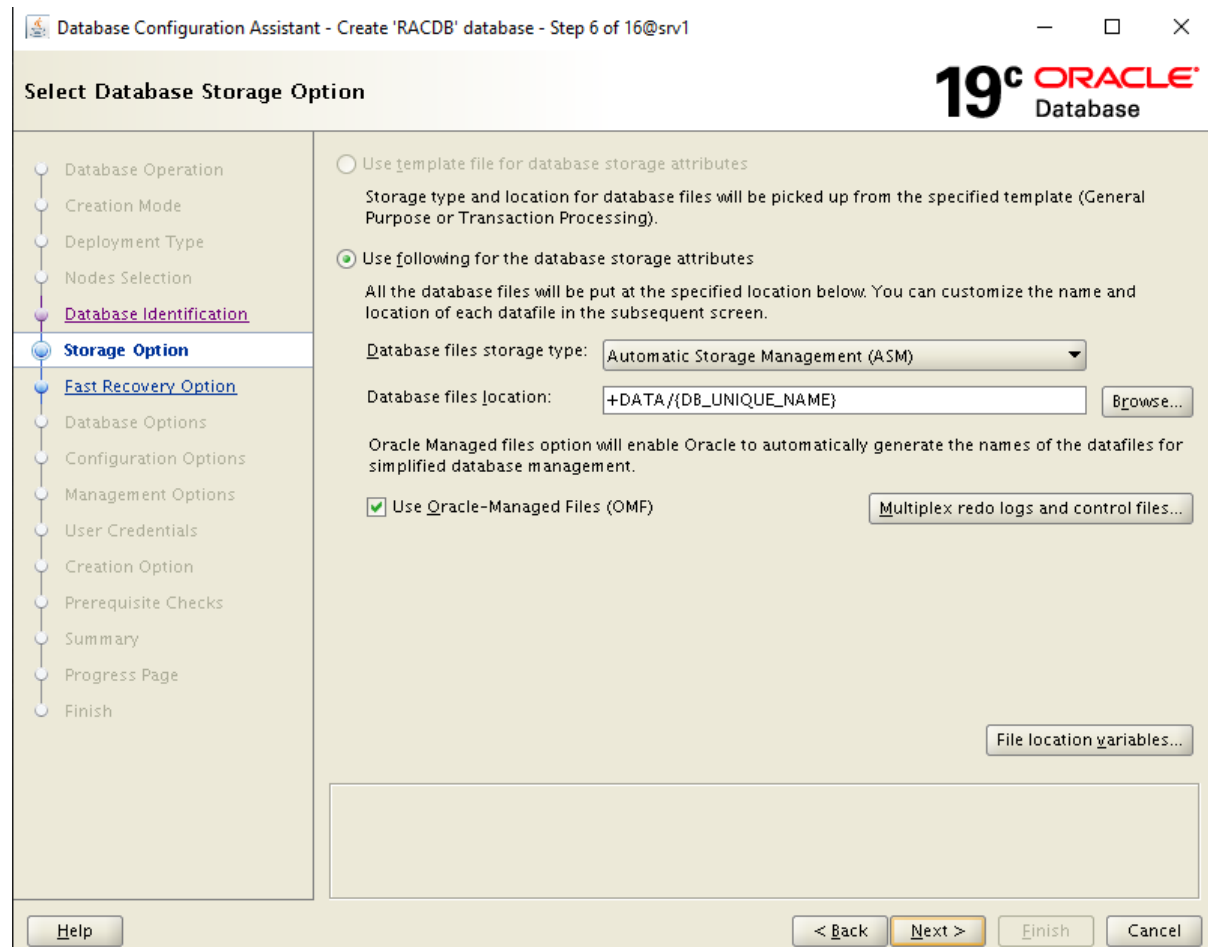
☒ Create a Container database with one or more PDBs

Number of PDBs: 1

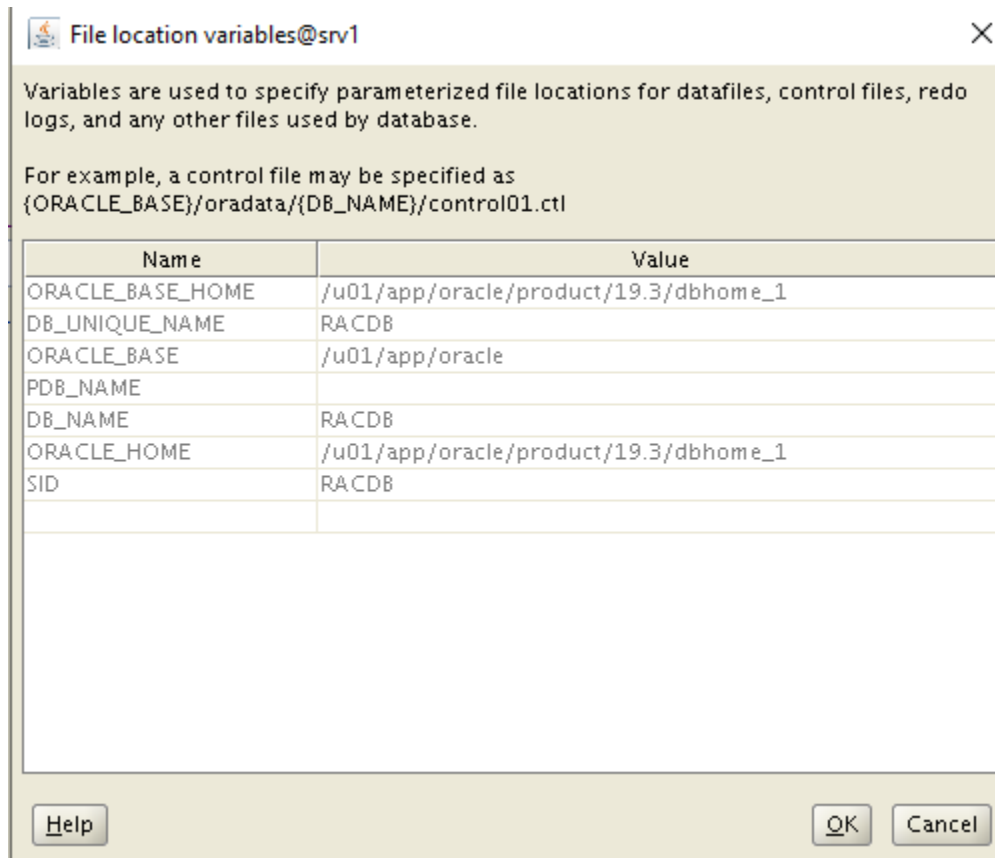
PDB name: PDB1

Help < Back Next > Finish Cancel

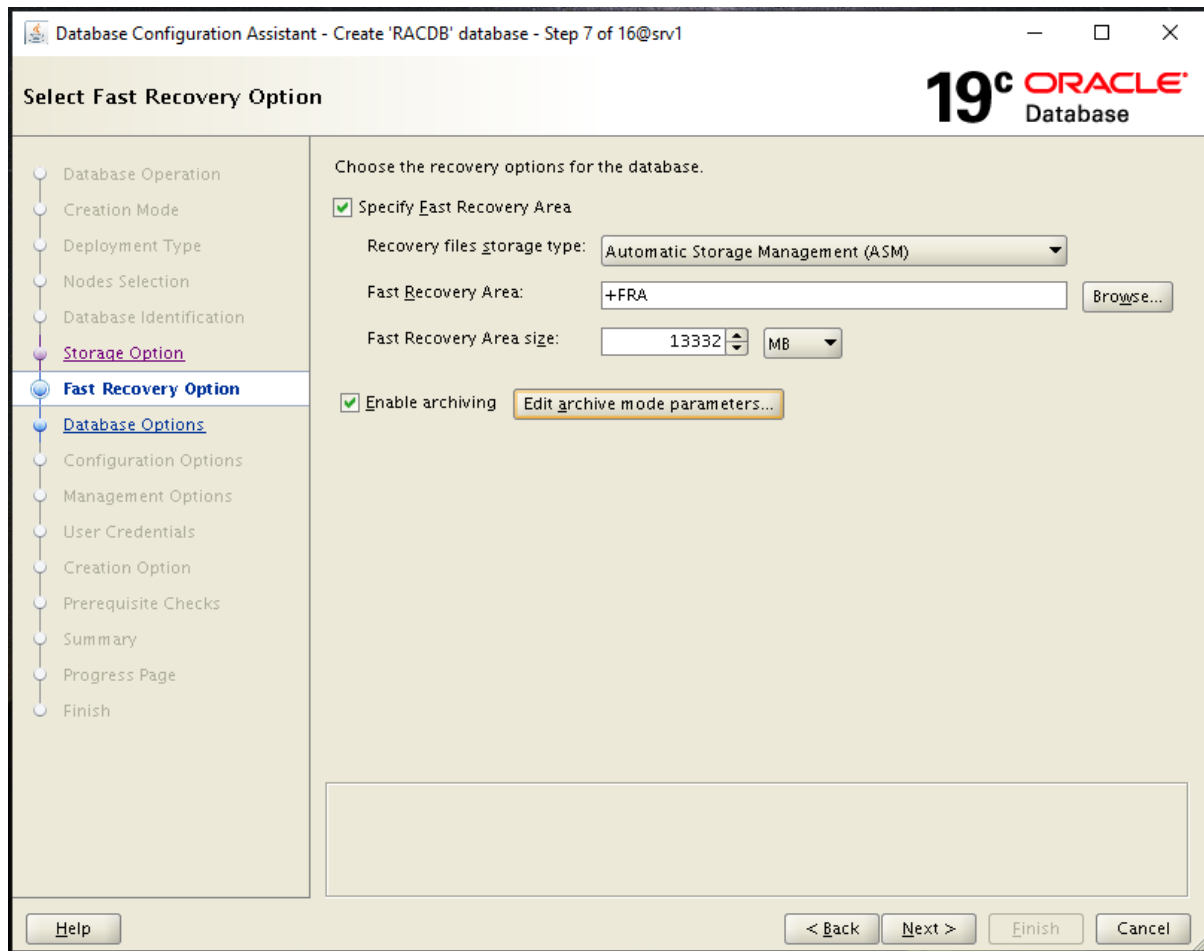
Select "Automatic Storage Management" and "+DATA" (appended by DB_UNIQUE_NAME) as the location of datafiles and select "Use Oracle-Managed Files"



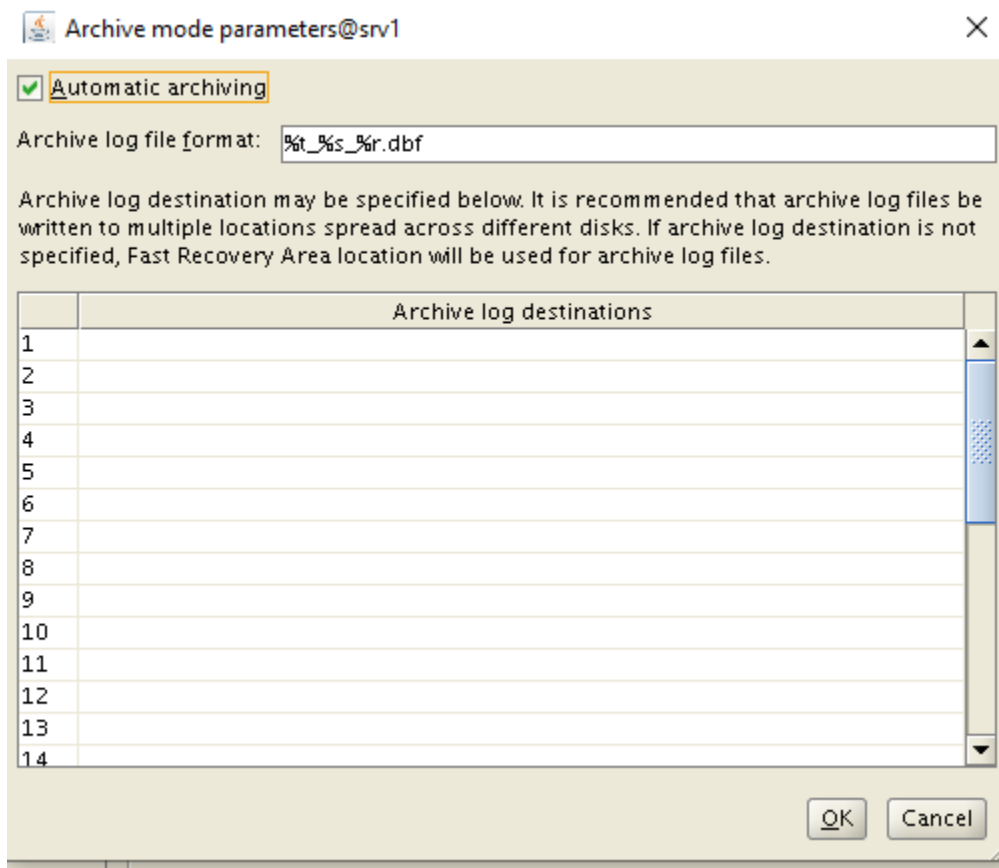
Verify the File Location Variables by clicking on the File location variables button



Specify “+FRA” as the Fast Recovery Area and select “Enable Archiving”



Verify the archive mode parameters
(destination does not need to be specified as the Fast Recovery Area will be used)



Archive mode parameters@srv1

☒ Automatic archiving

Archive log file format:

Archive log destination may be specified below. It is recommended that archive log files be written to multiple locations spread across different disks. If archive log destination is not specified, Fast Recovery Area location will be used for archive log files.

	Archive log destinations
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	

OK Cancel

Do not select Database Vault or Label Security

Database Configuration Assistant - Create 'RACDB' database - Step 8 of 16@srv1

19c ORACLE Database

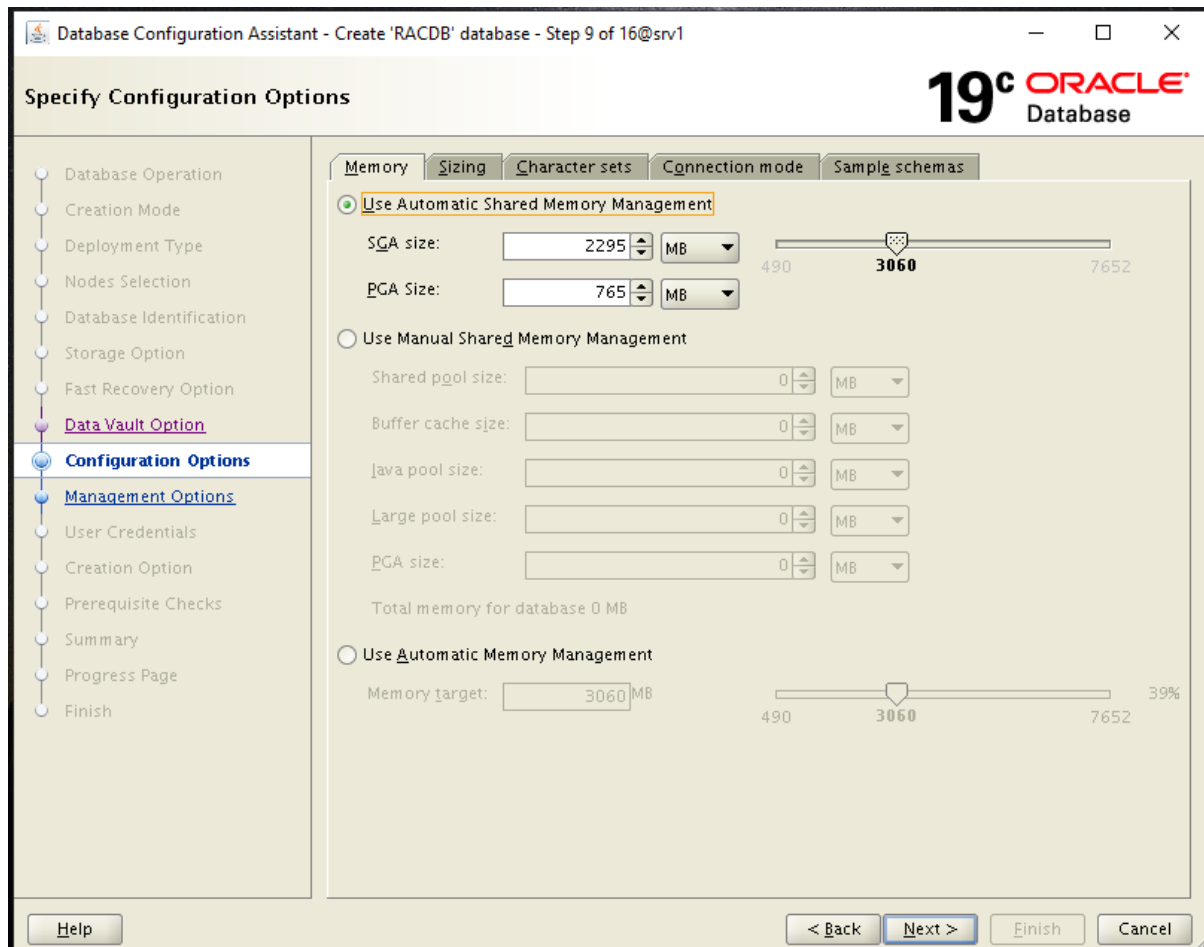
Select Oracle Data Vault Config Option

- ☐ Configure Oracle Database Vault
 - Database Vault owner:
 - Password: Confirm password:
- ☐ Create a separate account manager
 - Account manager:
 - Password: Confirm password:
- ☐ Configure Oracle Label Security
 - ☐ Configure Oracle Label Security with OJD

Help < Back Next > Finish Cancel

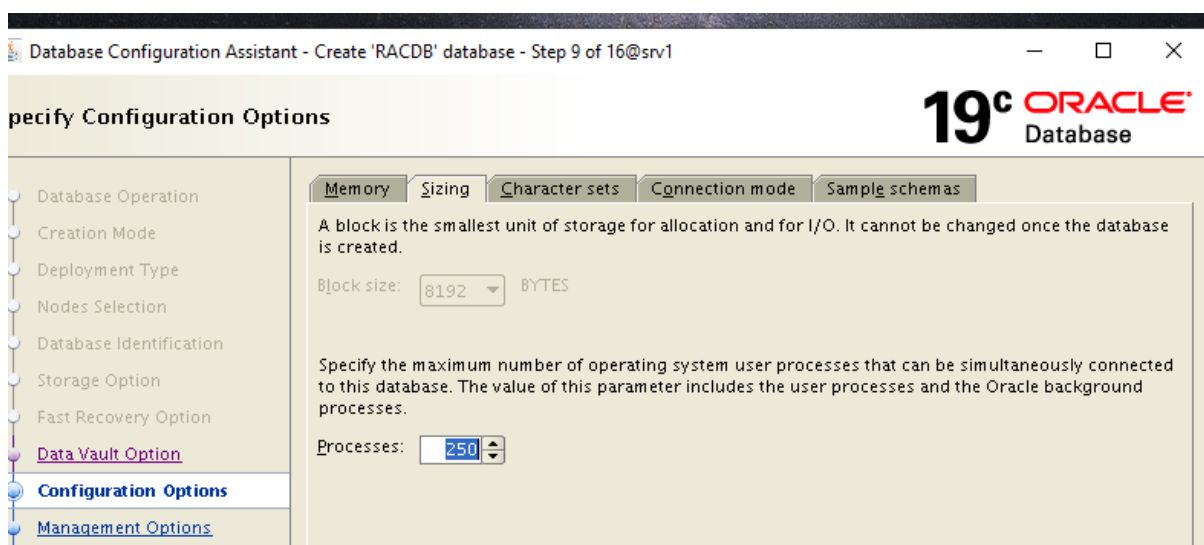
Use the defaults for ASMM (do not select Manual Shared Memory Management OR Automatic Memory Management)

Installation of 19c RAC on Linux7 in Virtual Box VMs



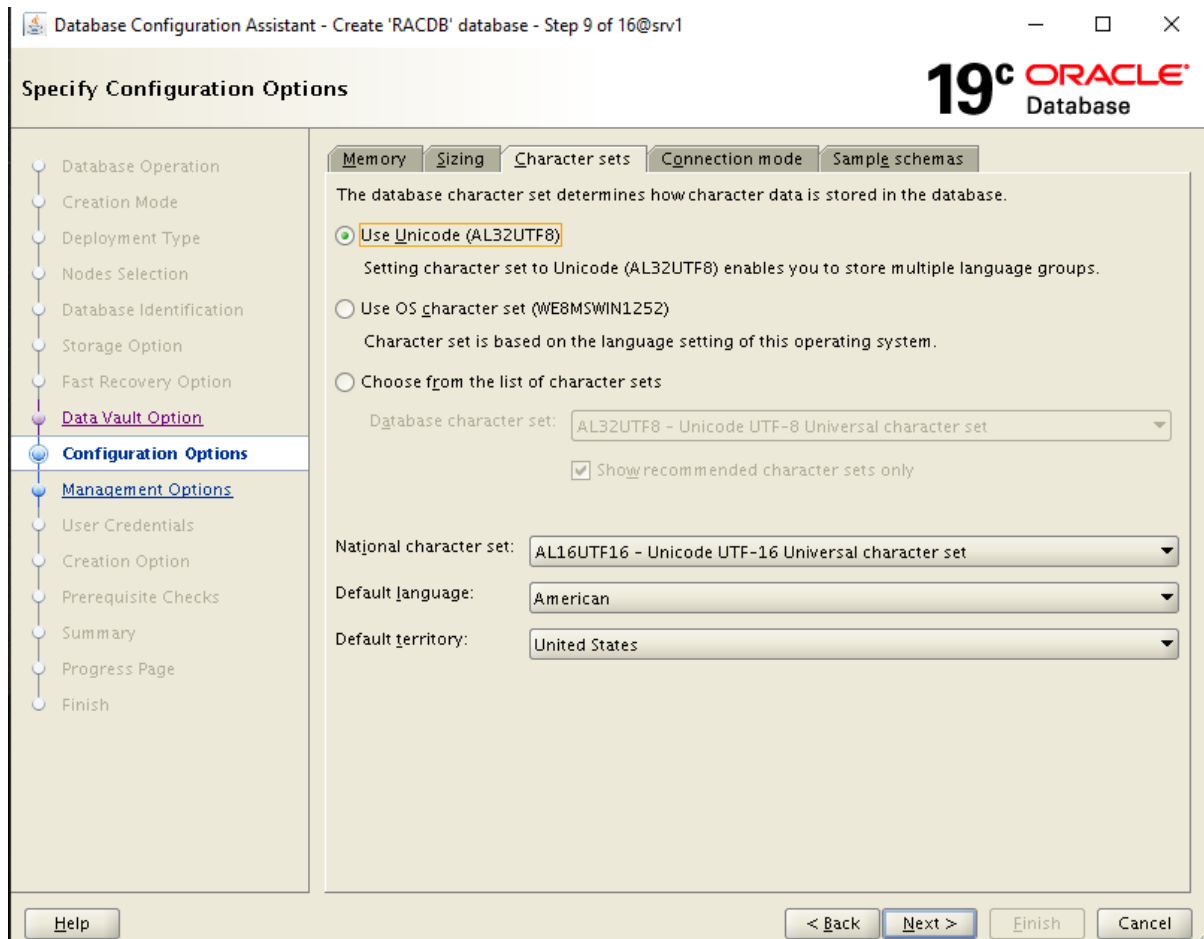
Click on the Sizing tab

Do not attempt to change the default block size (8192). Reduce Processes to 250



Click on the Character sets tab

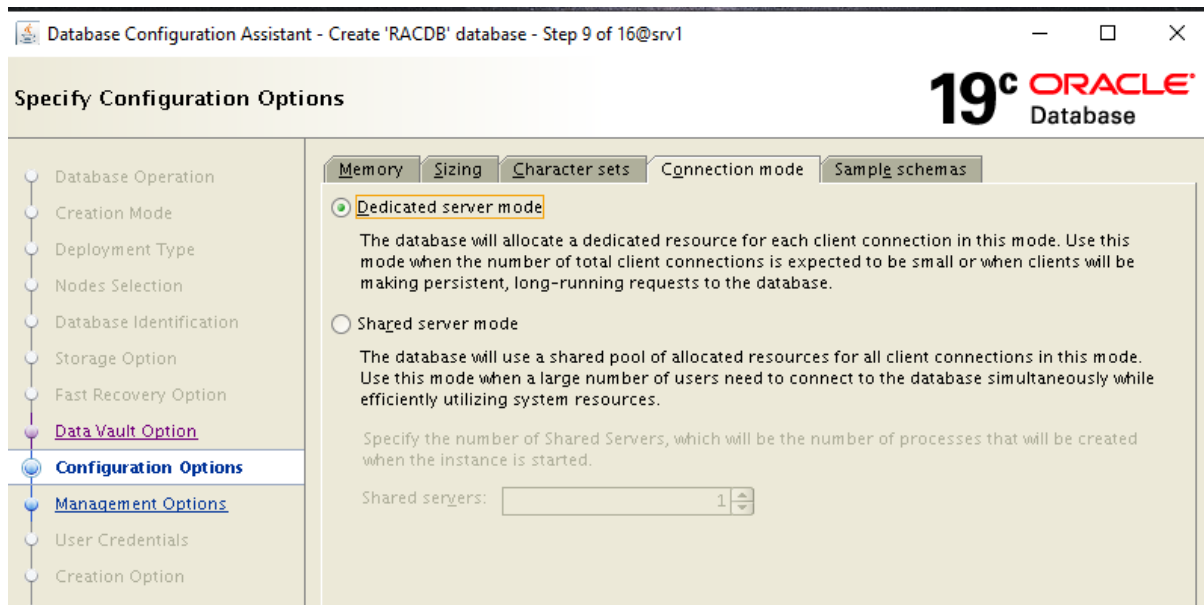
Use the defaults AL32UTF8 and American



Click on the Connection mode tab

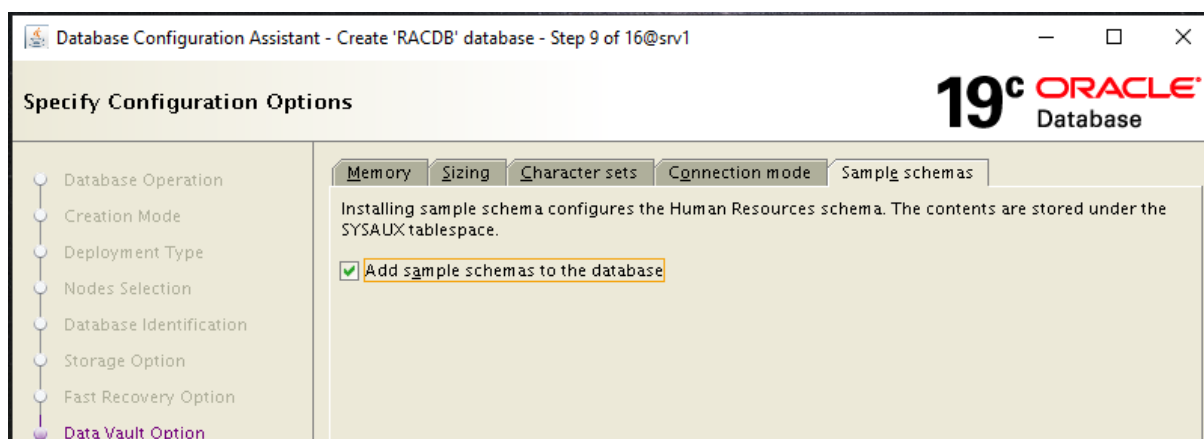
Use the default Dedicated Server mode

Installation of 19c RAC on Linux7 in Virtual Box VMs



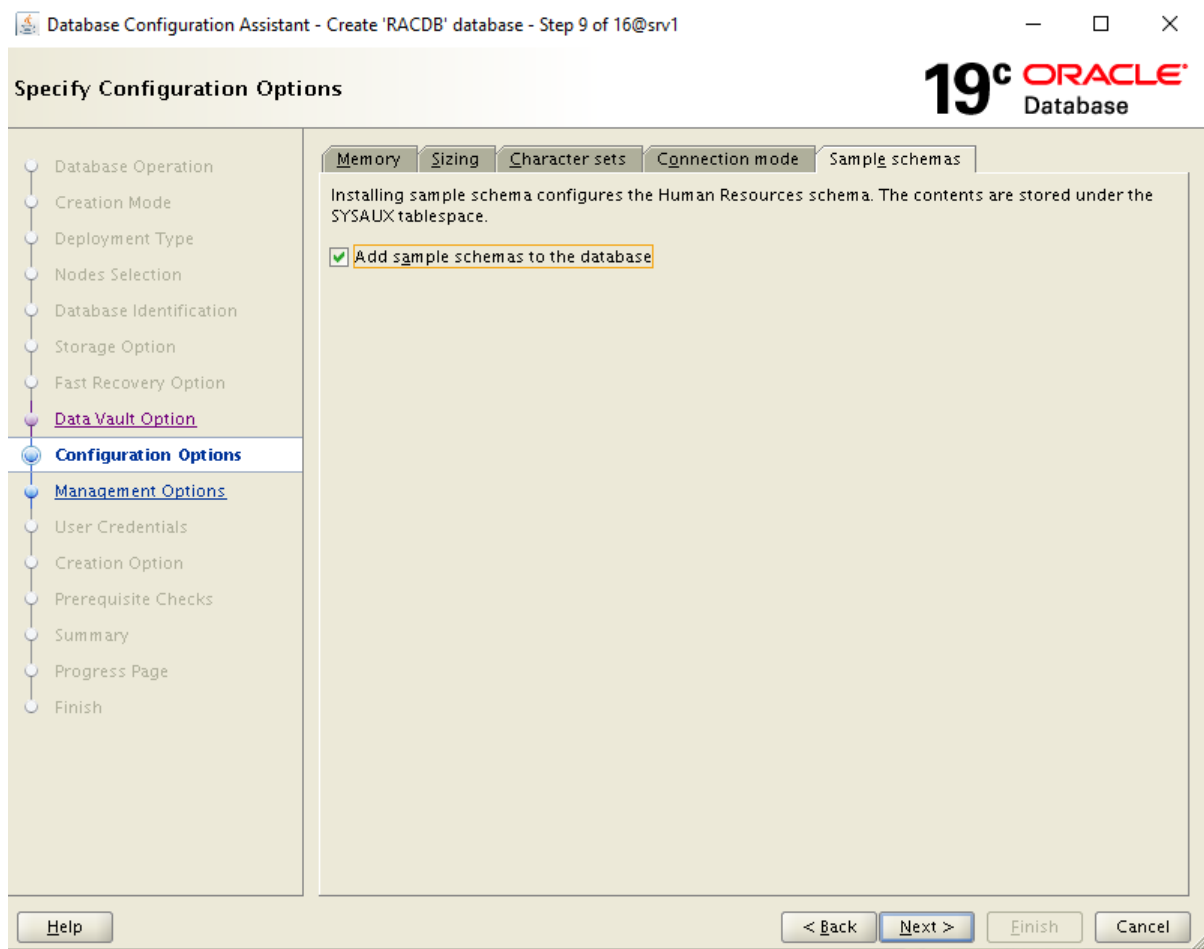
Click on Sample schemas

Select Add Sample Schemas (ie. HR schema)



Click on Next at the bottom of the page

Installation of 19c RAC on Linux7 in Virtual Box VMs



Unselect “Run CVU periodically” , “Configure Enterprise Manager (EM) database express” and “Register with Enterprise Manager cloud control”

The screenshot shows the 'Specify Management Options' window in the Oracle Database Configuration Assistant. The window title is 'Database Configuration Assistant - Create 'RACDB' database - Step 10 of 16@srv1'. The Oracle 19c logo is in the top right. On the left is a navigation pane with steps: Database Operation, Creation Mode, Deployment Type, Nodes Selection, Database Identification, Storage Option, Fast Recovery Option, Data Vault Option, Configuration Options, Management Options (selected), User Credentials, Creation Option, Prerequisite Checks, Summary, Progress Page, and Finish. The main area is titled 'Specify the management options for the database.' and contains several checkboxes and input fields:

- ☐ Run Cluster Verification Utility (CVU) checks periodically
- ☐ Configure Enterprise Manager (EM) database express. Below this is a text field for 'EM database express port:' with the value '5500'.
- ☐ Configure EM database express port as global port
- ☐ Register with Enterprise Manager (EM) cloud control. Below this are four text fields: 'OMS host:', 'OMS port:', 'EM admin username:', and 'EM admin password:'.
- Below the 'Register with Enterprise Manager (EM) cloud control' section is a text field for 'ASMSNMP user password:'.

At the bottom of the window are buttons for 'Help', '< Back', 'Next >', 'Finish', and 'Cancel'.

Specify different passwords for SYS, SYSTEM and PDBADMIN
(here, for simplicity, I have specified the same password for all 3 accounts)

Database Configuration Assistant - Create 'RACDB' database - Step 11 of 16@srv1

19c ORACLE Database

Specify Database User Credentials

You must specify passwords for the following user accounts in the new database for security reasons.

☐ Use different administrative passwords

	Password	Confirm password
SYS		
SYSTEM		
PDBADMIN		

☒ Use the same administrative password for all accounts

Password: Confirm password:

Messages:

Password: [D8T-06208] The 'ADMIN' password entered does not conform to the Oracle recommended standards.

Help < Back Next > Finish Cancel

Database Configuration Assistant@srv1

[D8T-06208] The 'Admin' password entered does not conform to the Oracle recommended standards.

Are you sure you want to continue?

Yes No Details

Select “Create database” and also “Generate database creation scripts” – so that the scripts are available for review and manual execution

Database Configuration Assistant - Create 'RACDB' database - Step 12 of 16@srv1

19c ORACLE Database

Select Database Creation Option

Select the database creation options.

☒ **Create database**

Specify the SQL scripts you want to run after the database is created. The scripts are run in the order listed below.

Post DB creation scripts: [Browse...](#)

☐ Save as a database template

Template name:

Template location: [Browse...](#)

Description:

☒ **Generate database creation scripts**

Destination directory: [Browse...](#)

Following advanced configuration options can be used to configure initialization parameters and customize database storage locations.

[All Initialization Parameters...](#) [Customize Storage Locations...](#)

[Help](#) [< Back](#) [Next >](#) [Finish](#) [Cancel](#)

Review Initialization Parameters (modify any that you need to)

Installation of 19c RAC on Linux7 in Virtual Box VMs

All initialization parameters@srv1

⚠ Update the initialization parameters only when it is required. Refer to the Oracle documentation to learn more about each initialization parameter and its valid set of values.

(Storage related parameter(s) value is shown in MB) ☐ Show advanced parameters

Name	Value	Include in spfile	Category
undo_tablespace (RACDB1)	UNDOTBS1	☒	Cluster Database
undo_tablespace (RACDB2)	UNDOTBS2	☒	Cluster Database
sga_target	2295	☒	SGA Memory
db_block_size (bytes)	8192	☒	Cache and I/O
cluster_database	true	☒	Cluster Database
nls_language	AMERICAN	☒	NLS
remote_login_passwordfile	exclusive	☒	Security and Auditing
db_create_file_dest	+DATA/(DB_UNIQUE_NAME)/	☒	File Configuration
processes	250	☒	Processes and Sessions
pga_aggregate_target	765	☒	Sort, Hash Joins, Bitmap Indexes
nls_territory	AMERICA	☒	NLS
db_recovery_file_dest_size	13332	☒	File Configuration
open_cursors	300	☒	Cursors and Library Cache
compatible	19.0.0	☒	Miscellaneous
db_name	RACDB	☒	Database Identification
instance_number (RACDB1)	1	☒	Cluster Database
instance_number (RACDB2)	2	☒	Cluster Database
db_recovery_file_dest	+FRA	☒	File Configuration
log_archive_dest_1		☐	Archive

Description:

undo_tablespace: Undo tablespaces are used solely for storing undo information. UNDO_TABLESPACE is only allowed in System Managed Undo (SMU) mode. The specified undo tablespace, , will be used by the instance. If the tablespace does not exist, or is not an undo tablespace, or is being used by another instance, the instance STARTUP will fail. Default: Each database contains zero or more undo tablespaces. In the SMU mode, each ORACLE instance is assigned one (and only one) undo tablespace.

[Help](#) [Close](#)

Verify Storage locations

Customize Storage@srv1

Storage

- Control Files
- Datafiles
- Redo Log Groups
 - 1
 - 2
 - 3
 - 4

Edit Control Files

Options

Maximum Datafiles: 100

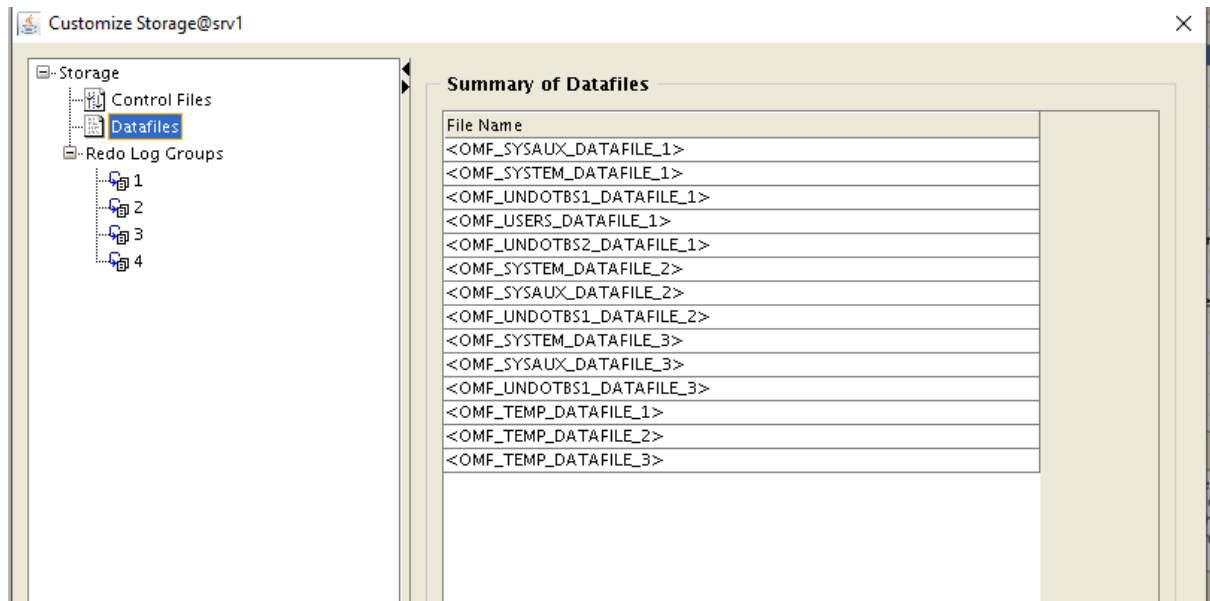
Maximum Instances: 8

Maximum Log History: 1

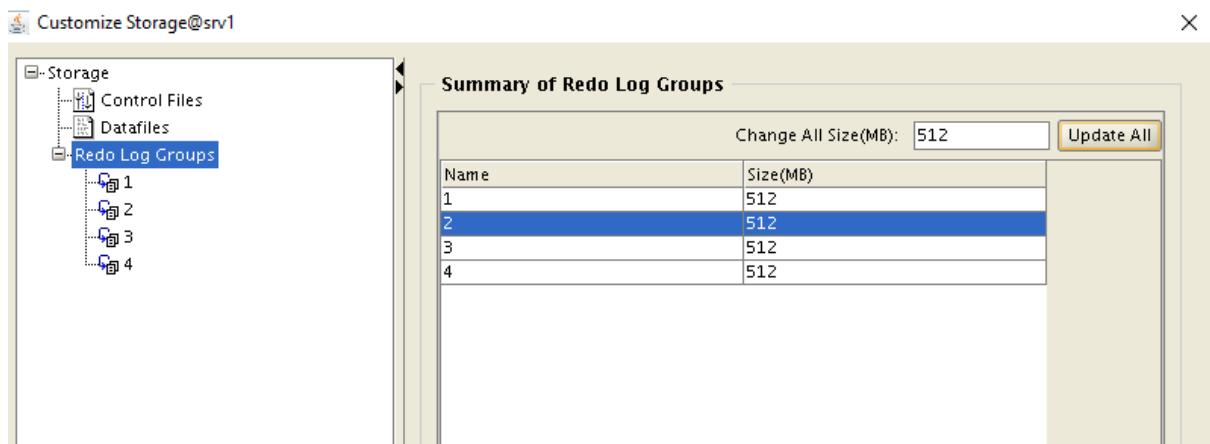
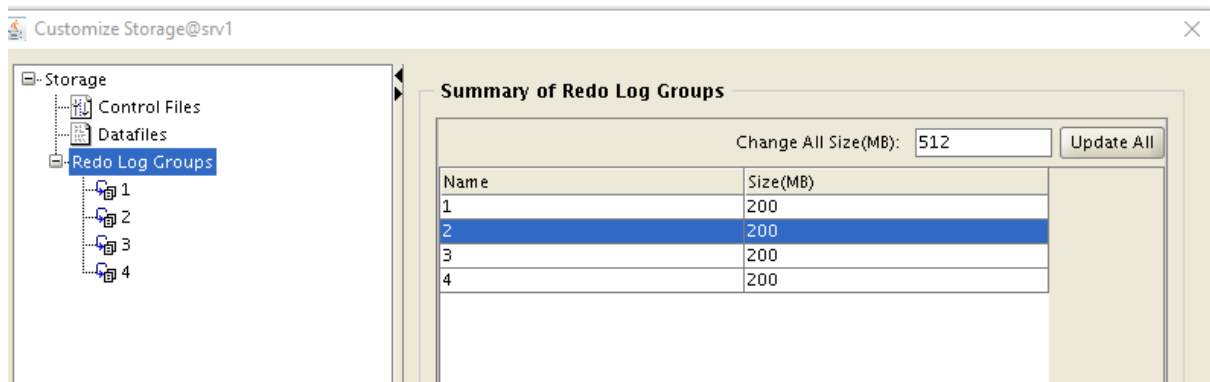
Maximum Redo Log Files: 16

Maximum Log Members: 3

Installation of 19c RAC on Linux7 in Virtual Box VMs

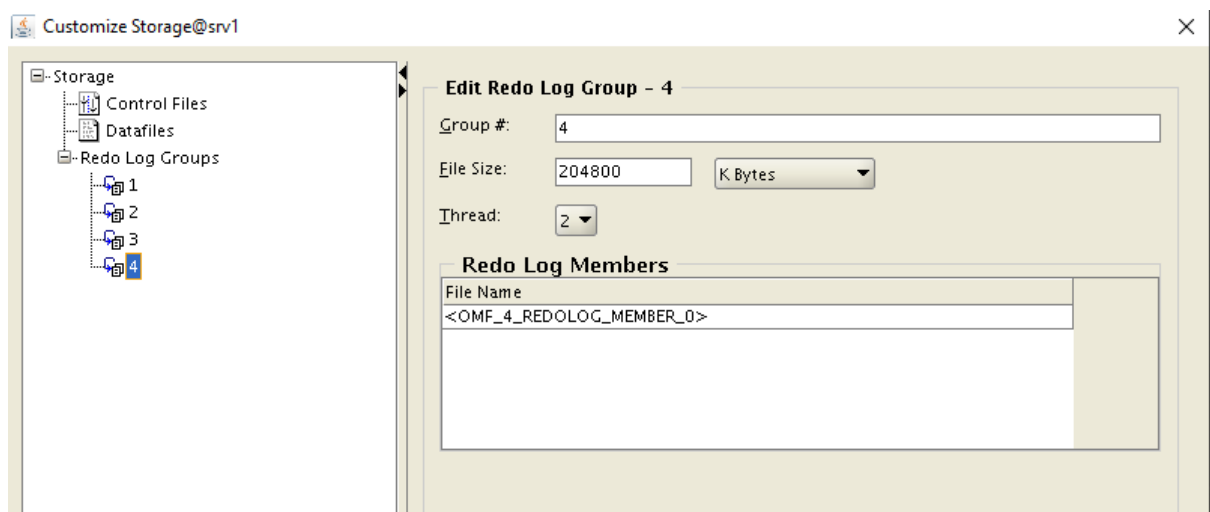
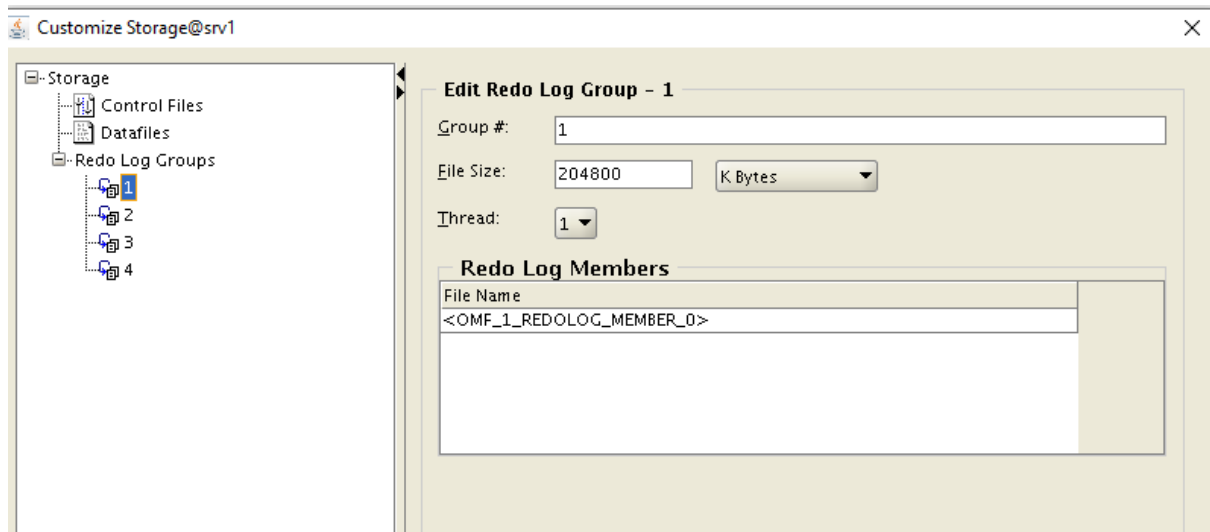


Change Redo Log Group sizes to 512MB and click on Update All

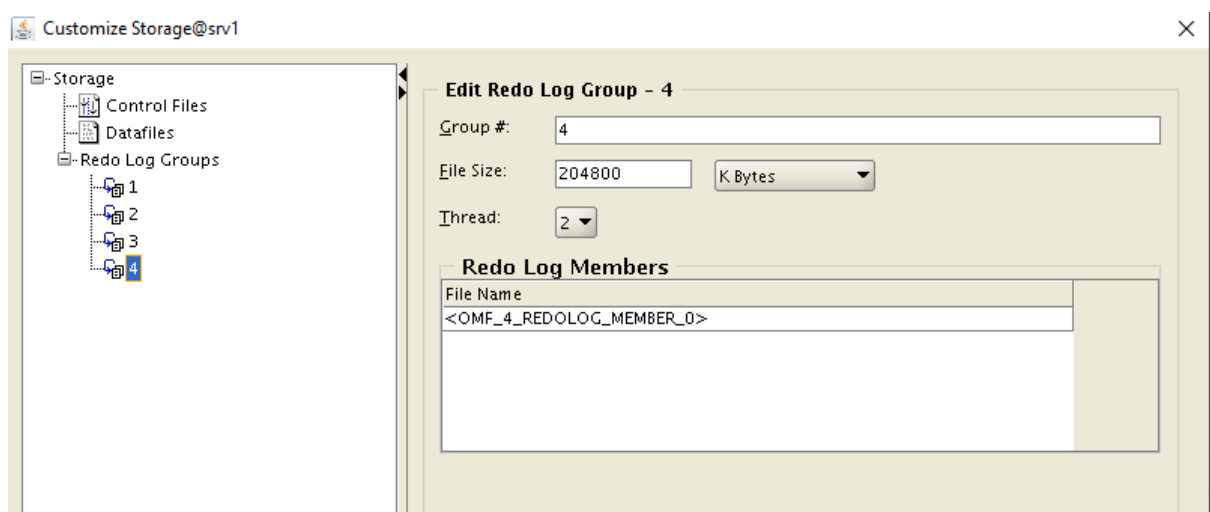


See that the 4 Groups are distributed as 2 Groups in each of the 2 Threads

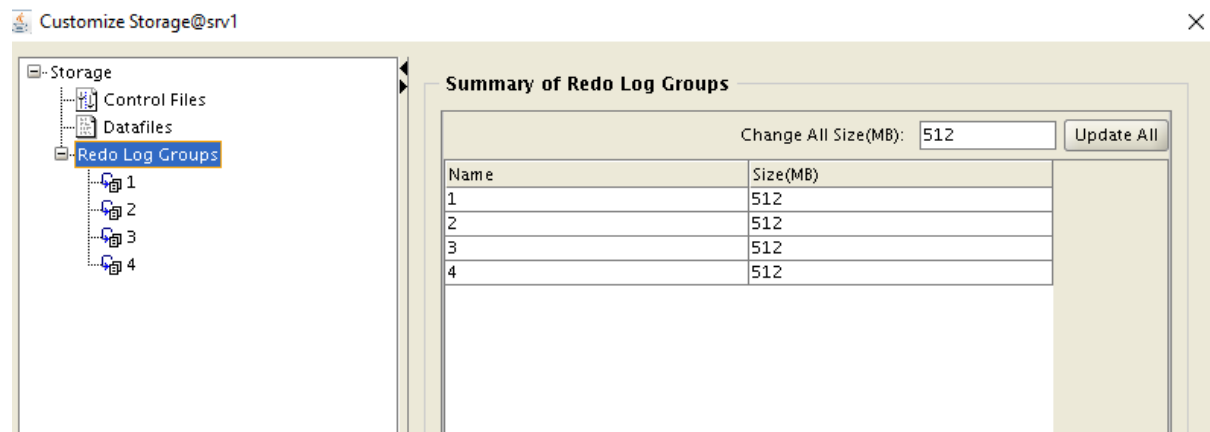
Installation of 19c RAC on Linux7 in Virtual Box VMs



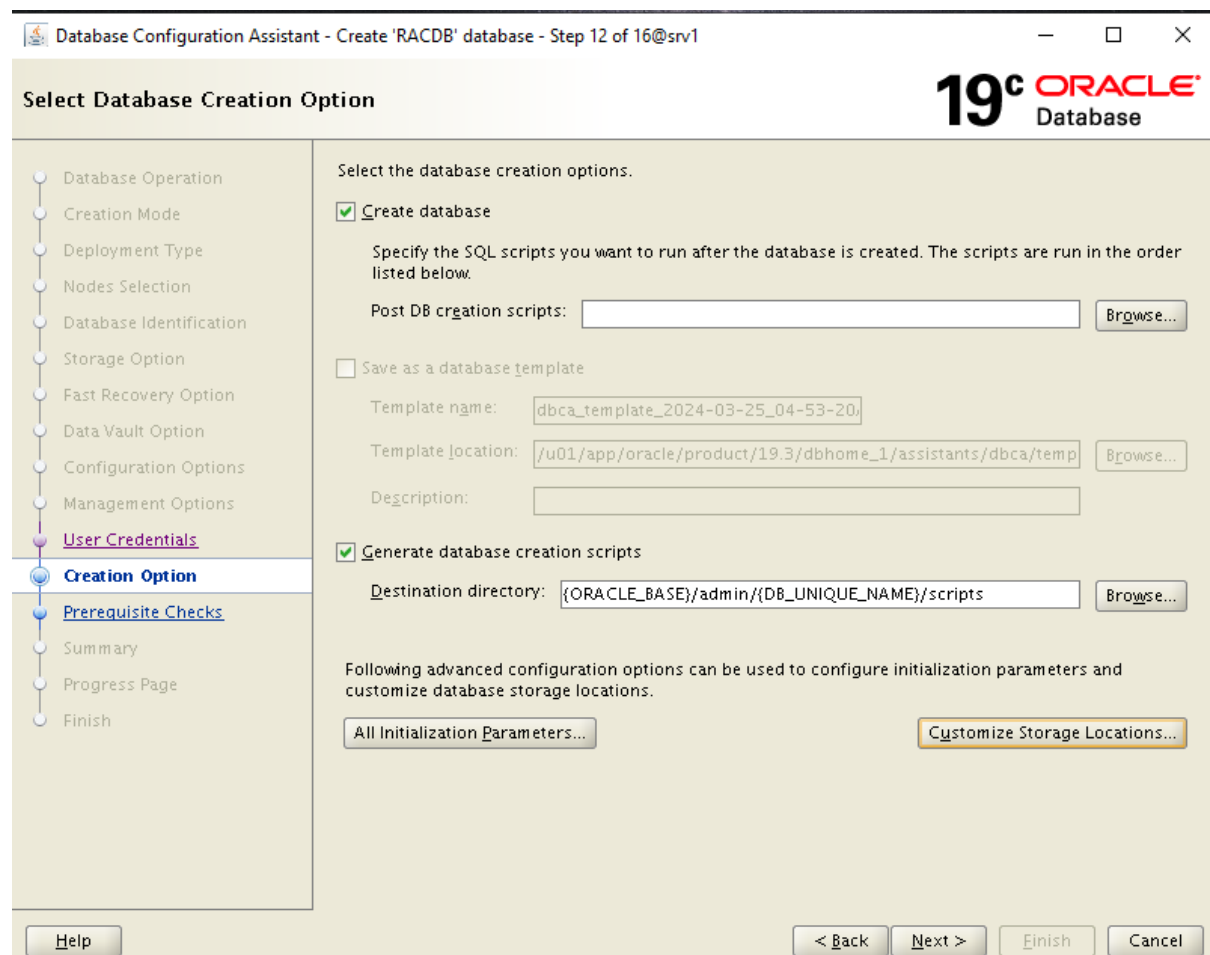
Click on Apply to update the Redo Log File sizes (to 512MB)



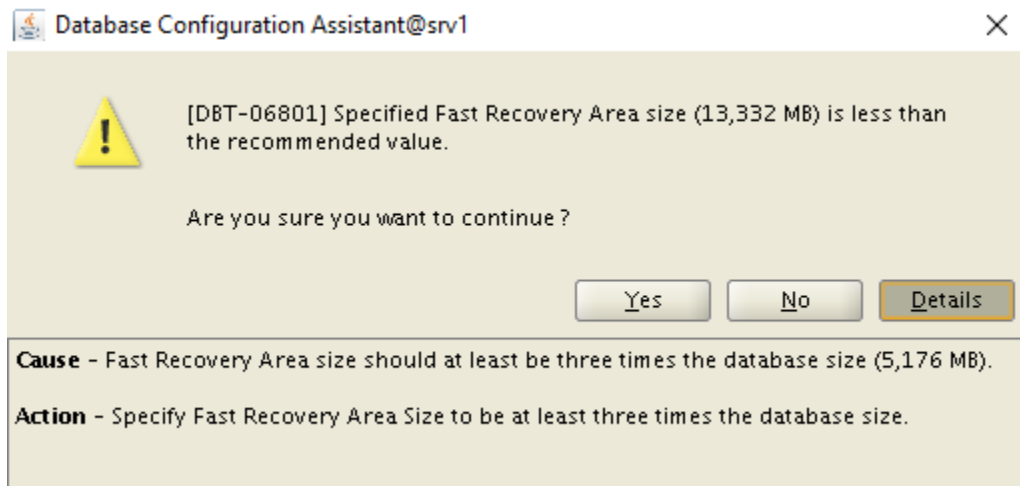
Installation of 19c RAC on Linux7 in Virtual Box VMs



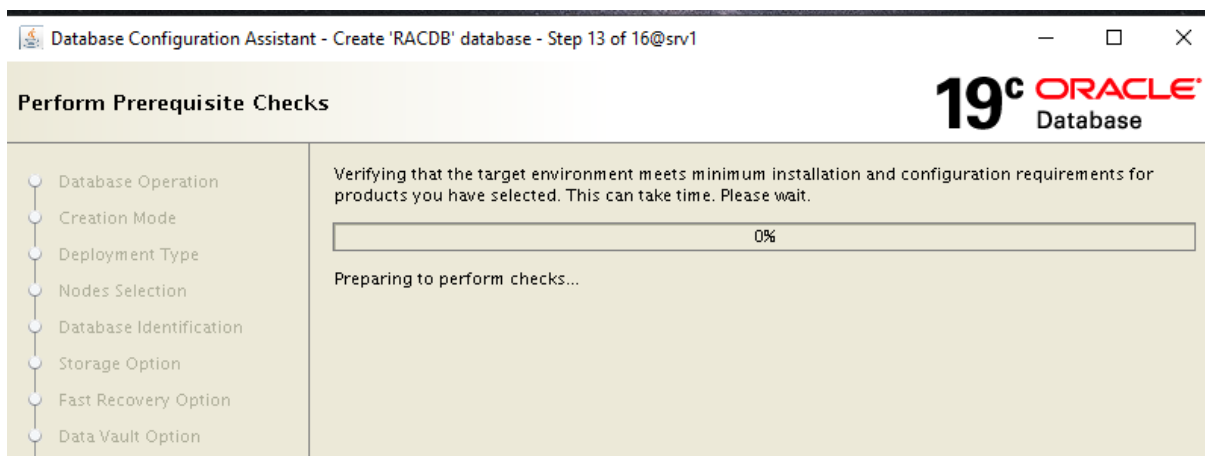
Click on Next



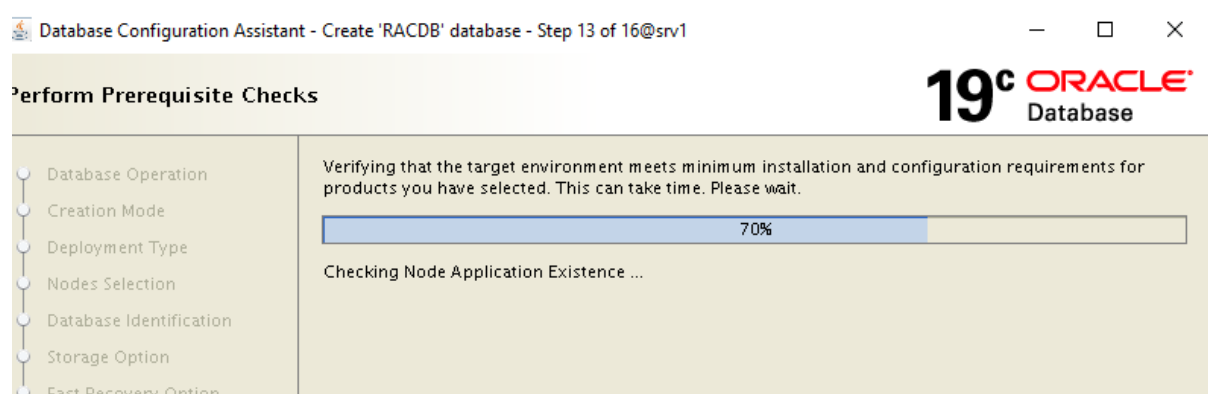
Accept the sizing of the Fast Recovery Area as the FRA created is only 15GB relative to DATA of 15GB (the initial database size may be 5,176MB but it may grow over time !!)



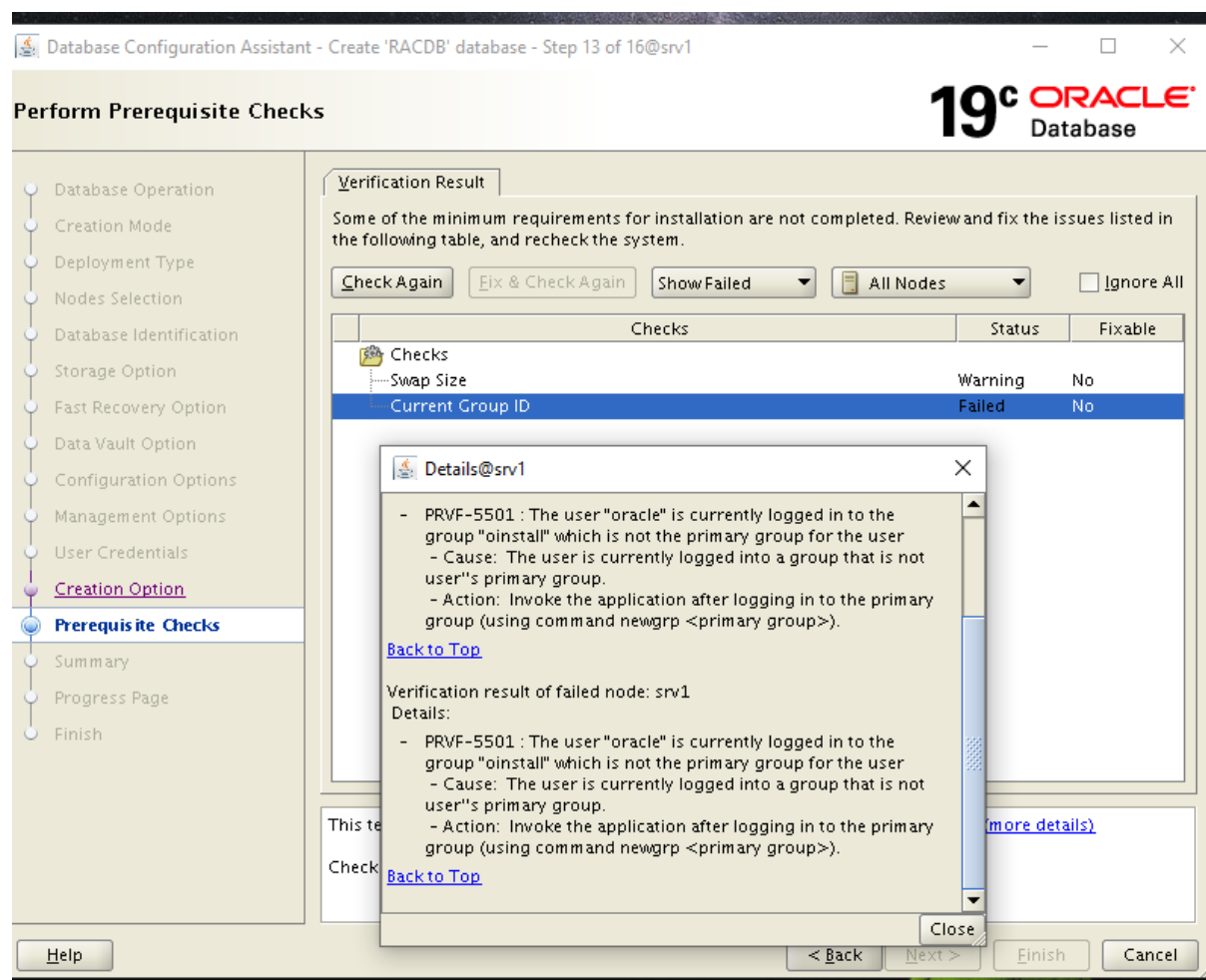
The DBCA will then run multiple checks



Installation of 19c RAC on Linux7 in Virtual Box VMs



If you get a warning on the Current Group versus the Primary Group



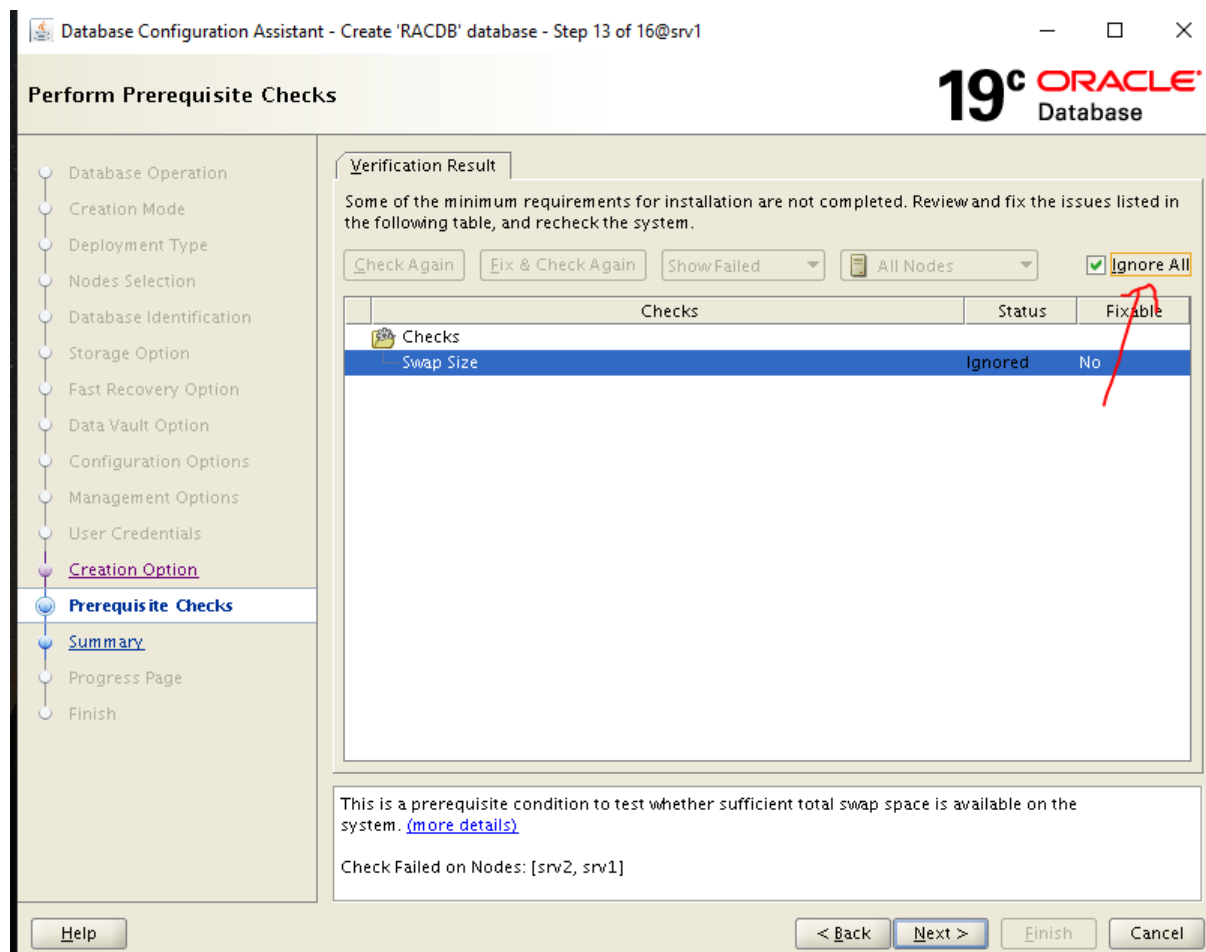
You can exit dbca. Logout of oracle, login again and restart DBCA

```
[oracle@srv1 dbhome_1]$ id
```

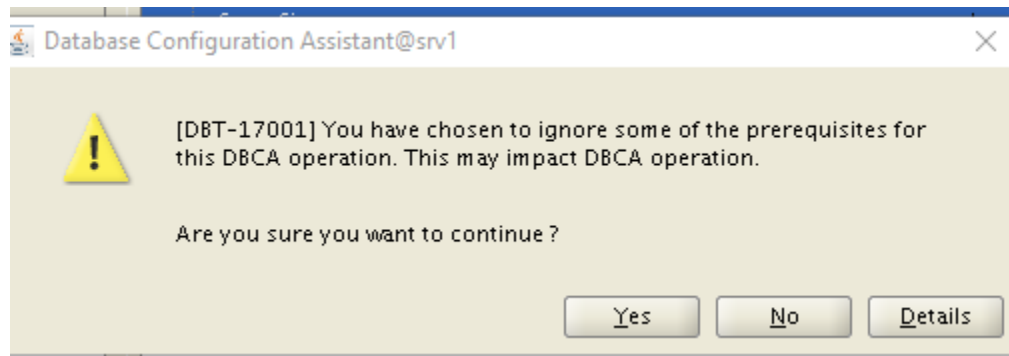
Installation of 19c RAC on Linux7 in Virtual Box VMs

```
uid=54321(oracle) gid=54321(oinstall)
groups=54321(oinstall),54322(dba),54331(grid)
context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
[oracle@srv1 dbhome_1]$ exit
logout
[root@srv1 ~]# su - oracle
Last login: Mon Mar 25 05:24:41 UTC 2024 on pts/6
[oracle@srv1 ~]$ export DISPLAY=192.168.56.1:0.0
[oracle@srv1 ~]$ cd $ORACLE_HOME
[oracle@srv1 dbhome_1]$ id
uid=54321(oracle) gid=54331(grid)
groups=54331(grid),54321(oinstall),54322(dba),54323(oper)
context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
[oracle@srv1 dbhome_1]$ dbca
```

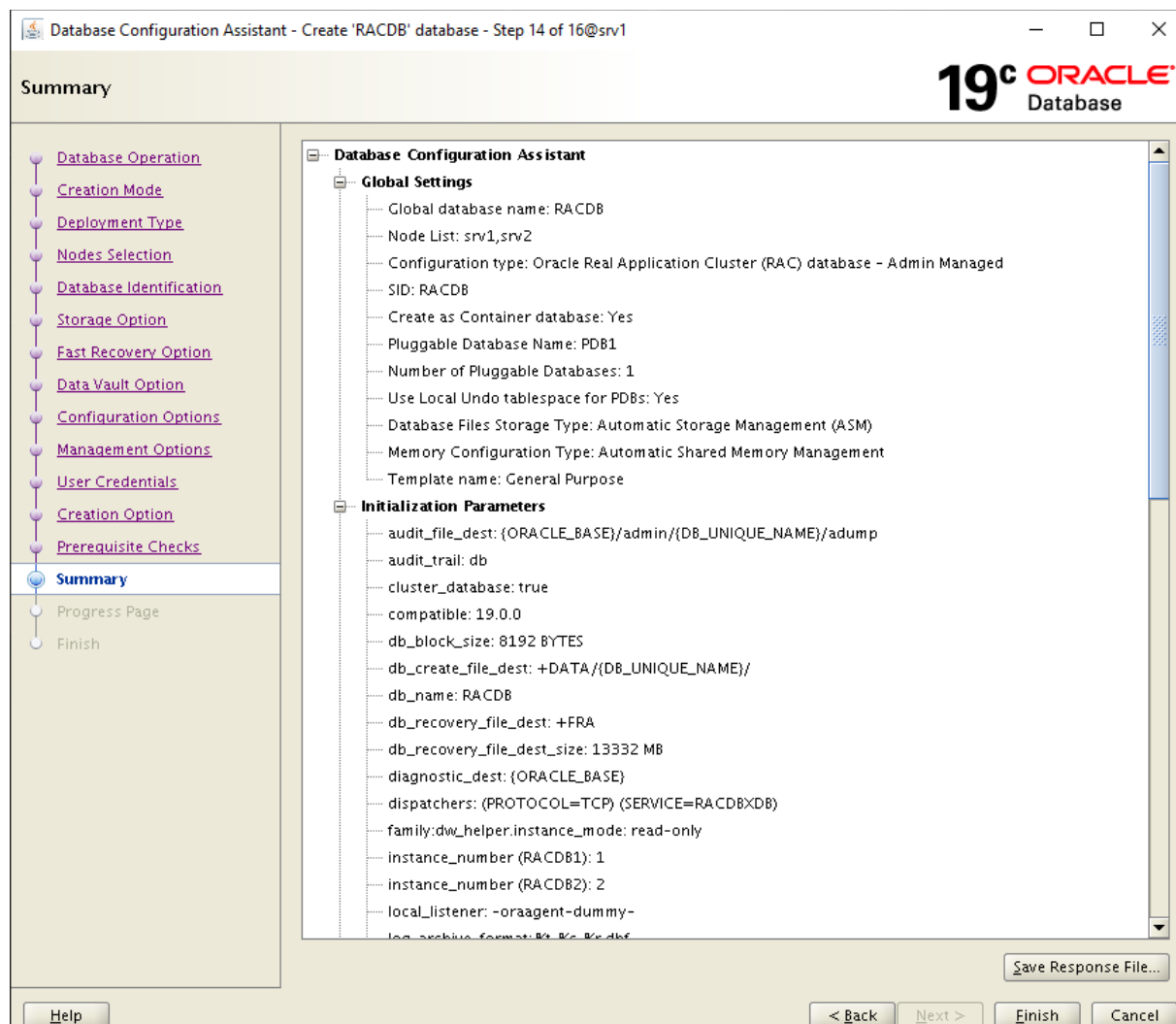
Ignore the Warning on the Swap Space



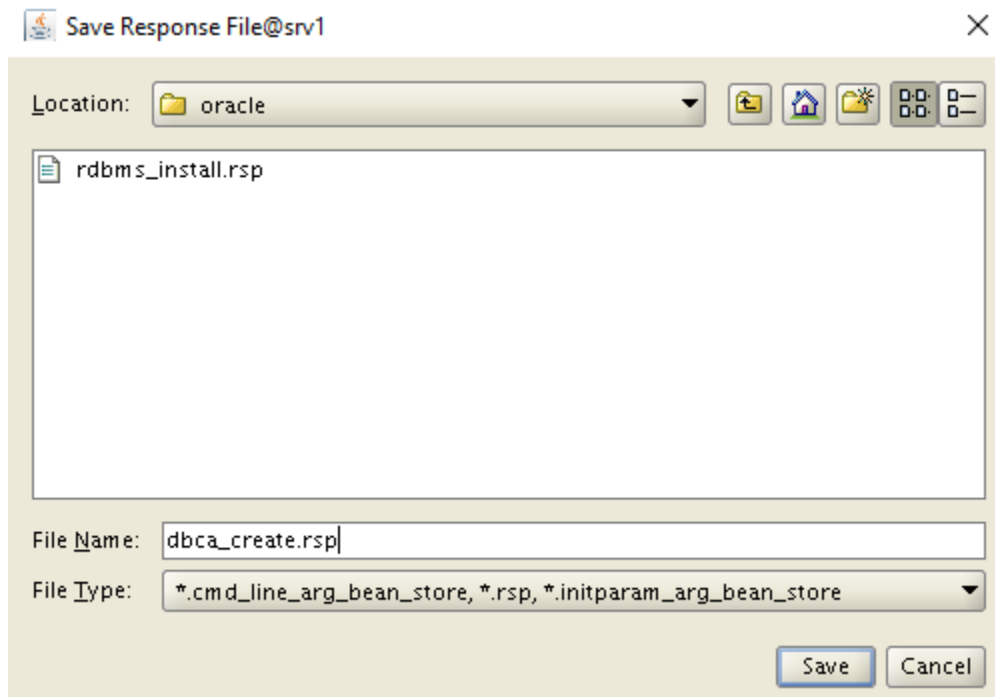
Installation of 19c RAC on Linux7 in Virtual Box VMs



Review the Summary and click on Save Response File as "dbca_create.rsp"

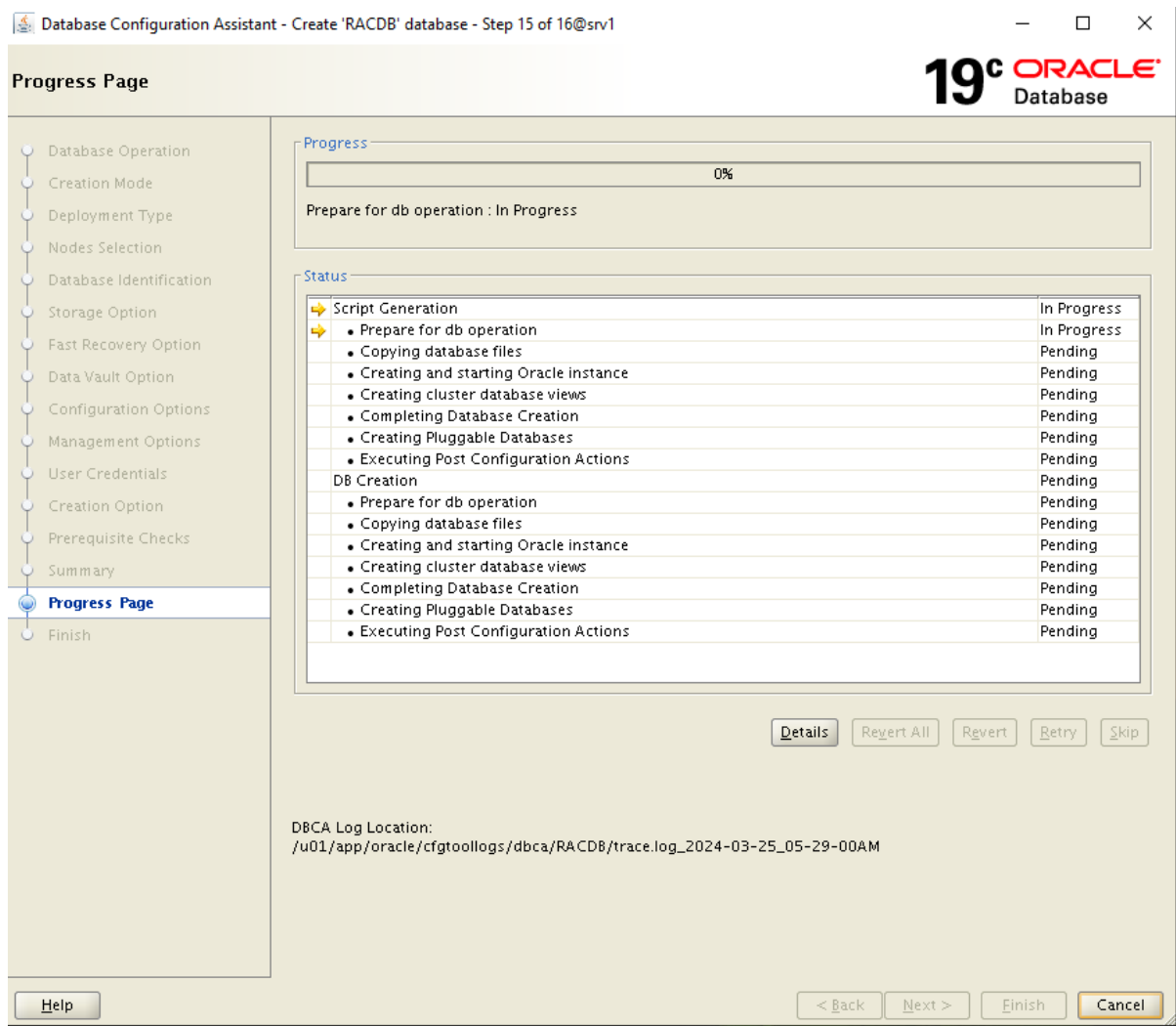


Installation of 19c RAC on Linux7 in Virtual Box VMs



Monitor the progress

Installation of 19c RAC on Linux7 in Virtual Box VMs



Installation of 19c RAC on Linux7 in Virtual Box VMs

Database Configuration Assistant - Create 'RACDB' database - Step 15 of 16@srv1

19c ORACLE Database

Progress Page

Progress

Copying database files : In Progress 22%

Status

✓	Script Generation	Succeeded
✓	• Prepare for db operation	Succeeded
✓	• Copying database files	Succeeded
✓	• Creating and starting Oracle instance	Succeeded
✓	• Creating cluster database views	Succeeded
✓	• Completing Database Creation	Succeeded
✓	• Creating Pluggable Databases	Succeeded
✓	• Executing Post Configuration Actions	Succeeded
➡	DB Creation	In Progress
✓	• Prepare for db operation	Succeeded
➡	• Copying database files	In Progress
	• Creating and starting Oracle instance	Pending
	• Creating cluster database views	Pending
	• Completing Database Creation	Pending
	• Creating Pluggable Databases	Pending
	• Executing Post Configuration Actions	Pending

Details **Revert All** **Revert** **Retry** **Skip**

DBCA Log Location:
/u01/app/oracle/cfgtoollogs/dbca/RACDB/trace.log_2024-03-25_05-29-00AM

Help **< Back** **Next >** **Finish** **Cancel**

Installation of 19c RAC on Linux7 in Virtual Box VMs

Database Configuration Assistant - Create 'RACDB' database - Step 15 of 16@srv1

Progress Page

19c ORACLE Database

Progress

53%

Creating cluster database views : In Progress

Status

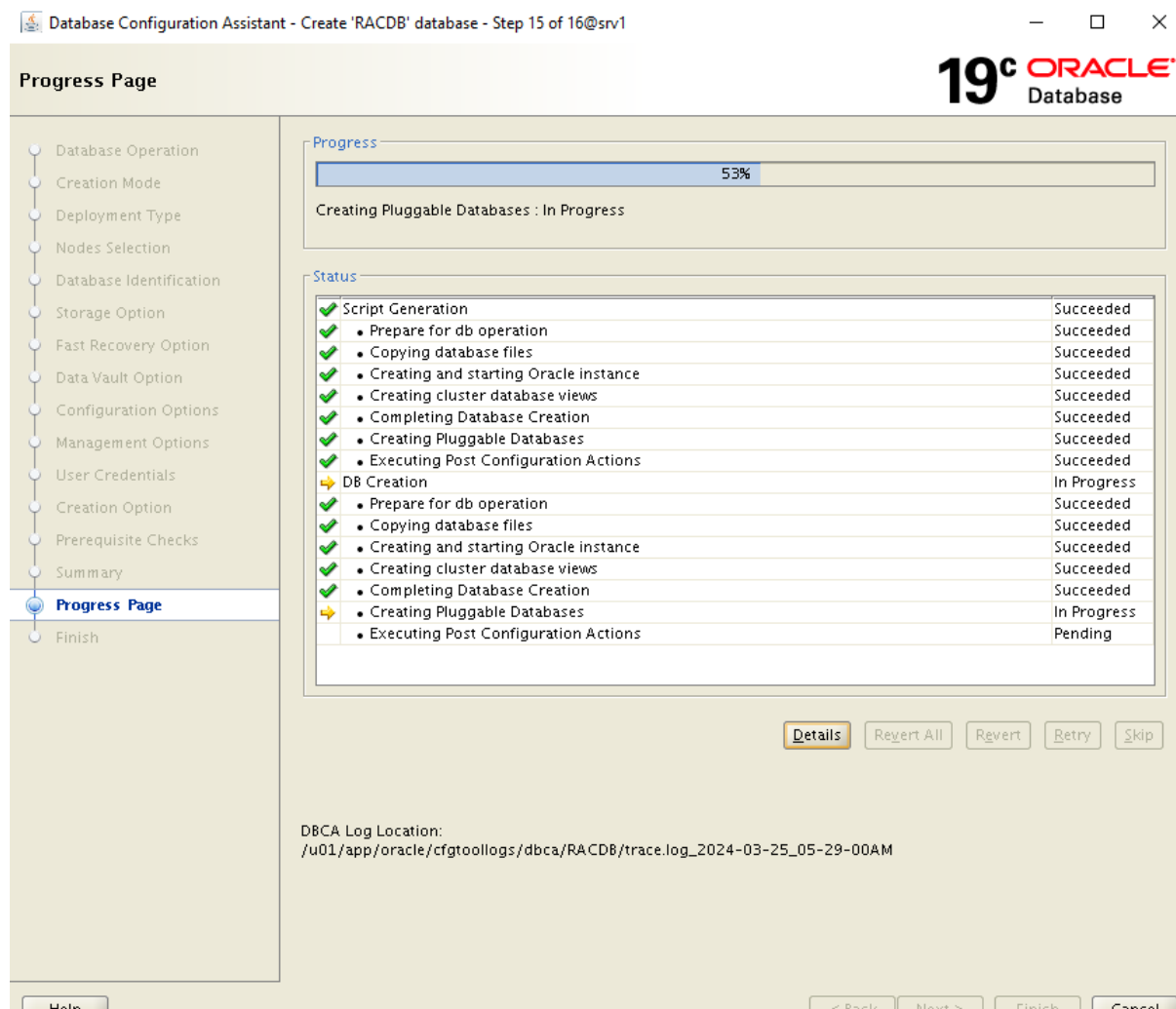
✓	Script Generation	Succeeded
✓	• Prepare for db operation	Succeeded
✓	• Copying database files	Succeeded
✓	• Creating and starting Oracle instance	Succeeded
✓	• Creating cluster database views	Succeeded
✓	• Completing Database Creation	Succeeded
✓	• Creating Pluggable Databases	Succeeded
✓	• Executing Post Configuration Actions	Succeeded
➔	DB Creation	In Progress
✓	• Prepare for db operation	Succeeded
✓	• Copying database files	Succeeded
✓	• Creating and starting Oracle instance	Succeeded
➔	• Creating cluster database views	In Progress
	• Completing Database Creation	Pending
	• Creating Pluggable Databases	Pending
	• Executing Post Configuration Actions	Pending

Details **Revert All** **Revert** **Retry** **Skip**

DBCA Log Location:
/u01/app/oracle/cfgtoollogs/dbca/RACDB/trace.log_2024-03-25_05-29-00AM

Help **< Back** **Next >** **Finish** **Cancel**

Installation of 19c RAC on Linux7 in Virtual Box VMs



If you monitor from a separate terminal you might see that it is also running Patch scripts

```
oracle 23426 20410 0 05:39 ? 00:00:00 oracleRACDB1 (DESCRIPTION=(LOCAL=YES) (ADDRESS=(PROTOCOL=beq)))
oracle 23435 9922 0 05:39 pts/0 00:00:00 /bin/sh /u01/app/oracle/product/19.3/dbhome_1/OPatch/datapatch -skip_upgrade_check -db RACDB1
oracle 23437 23435 0 05:39 pts/0 00:00:00 /bin/sh /u01/app/oracle/product/19.3/dbhome_1/sqlpatch/sqlpatch -skip_upgrade_check -db RACDB1
oracle 23442 23437 1 05:39 pts/0 00:00:00 /u01/app/oracle/product/19.3/dbhome_1/perl/bin/perl -I/u01/app/oracle/product/19.3/dbhome_1/sqlpatch
oracle 23445 23442 1 05:39 ? 00:00:00 oracleRACDB1 (DESCRIPTION=(LOCAL=YES) (ADDRESS=(PROTOCOL=beq)))
oracle 23455 1 0 05:39 ? 00:00:00 ora_w003_RACDB1
oracle 23481 1 0 05:39 ? 00:00:00 ora_w004_RACDB1
oracle 23653 23442 2 05:39 pts/0 00:00:01 /u01/app/oracle/product/19.3/dbhome_1/bin/sqlplus
oracle 23654 23442 0 05:39 pts/0 00:00:00 /u01/app/oracle/product/19.3/dbhome_1/bin/sqlplus
oracle 23655 23442 0 05:39 pts/0 00:00:00 /u01/app/oracle/product/19.3/dbhome_1/bin/sqlplus
oracle 23656 23442 0 05:39 pts/0 00:00:00 /u01/app/oracle/product/19.3/dbhome_1/bin/sqlplus
```

```
oracle 23435 9922 0 05:39 pts/0 00:00:00 /bin/sh
/u01/app/oracle/product/19.3/dbhome_1/OPatch/datapatch -skip_upgrade_check
-db RACDB1
oracle 23437 23435 0 05:39 pts/0 00:00:00 /bin/sh
/u01/app/oracle/product/19.3/dbhome_1/sqlpatch/sqlpatch -skip_upgrade_check
-db RACDB1
oracle 23442 23437 1 05:39 pts/0 00:00:00
/u01/app/oracle/product/19.3/dbhome_1/perl/bin/perl -
I/u01/app/oracle/product/19.3/dbhome_1/sqlpatch
```

You can also monitor the database instance's alert log at this location :

```
[oracle@srv1 trace]$ pwd
/u01/app/oracle/diag/rdbms/racdb/RACDB1/trace
[oracle@srv1 trace]$ ls -l al*
-rw-r----- 1 oracle oinstall 98786 Mar 25 05:44 alert_RACDB1.log
[oracle@srv1 trace]$
```

On completion, the DBCA informs you about locked accounts



Use the Password Management button to unlock accounts and change passwords

Password Management@srv1

Lock / unlock database user accounts and / or change the default passwords:

User Name	Lock Account? ▲	New Password	Confirm Password
SYSTEM			
SYS			
ORDSYS	✓		
SYSDG	✓		
ORACLE_OCM	✓		
ORDDATA	✓		
SYSKM	✓		
ORDPLUGINS	✓		
DIP	✓		
AUDSYS	✓		
SI_INFORMTN_SCHEMA	✓		
GSMROOTUSER	✓		
SYSRAC	✓		
GSMUSER	✓		
REMOTE_SCHEDULER_AGE...	✓		
SYSBACKUP	✓		
MDDATA	✓		
GSMCATUSER	✓		
WM SYS	✓		

OK Cancel Help

Verification of RAC Database

Registration into ClusterWare

```
[root@srv1 dbhome_1]# . /home/grid/.bash_profile
[root@srv1 dbhome_1]# crsctl status resource -t
```

Name	Target	State	Server	State details

Local Resources				

ora.LISTENER.lsnr				
	ONLINE	ONLINE	srv1	STABLE
	ONLINE	ONLINE	srv2	STABLE
ora.chad				
	ONLINE	ONLINE	srv1	STABLE
	ONLINE	ONLINE	srv2	STABLE
ora.net1.network				
	ONLINE	ONLINE	srv1	STABLE
	ONLINE	ONLINE	srv2	STABLE
ora.ons				
	ONLINE	ONLINE	srv1	STABLE
	ONLINE	ONLINE	srv2	STABLE

Cluster Resources				

ora.ASMNET1LSNR_ASM.lsnr(ora.asmgroup)				
1	ONLINE	ONLINE	srv1	STABLE

Installation of 19c RAC on Linux7 in Virtual Box VMs

```

2          ONLINE  ONLINE      srv2          STABLE
3          ONLINE  OFFLINE     STABLE
ora.DATA.dg (ora.asmgroup)
1          ONLINE  ONLINE      srv1          STABLE
2          ONLINE  ONLINE      srv2          STABLE
3          OFFLINE OFFLINE     STABLE
ora.FRA.dg (ora.asmgroup)
1          ONLINE  ONLINE      srv1          STABLE
2          ONLINE  ONLINE      srv2          STABLE
3          OFFLINE OFFLINE     STABLE
ora.LISTENER_SCAN1.lsnr
1          ONLINE  ONLINE      srv2          STABLE
ora.LISTENER_SCAN2.lsnr
1          ONLINE  ONLINE      srv1          STABLE
ora.LISTENER_SCAN3.lsnr
1          ONLINE  ONLINE      srv1          STABLE
ora.OCR_VOTE.dg (ora.asmgroup)
1          ONLINE  ONLINE      srv1          STABLE
2          ONLINE  ONLINE      srv2          STABLE
3          OFFLINE OFFLINE     STABLE
ora.asm (ora.asmgroup)
1          ONLINE  ONLINE      srv1          Started,STABLE
2          ONLINE  ONLINE      srv2          Started,STABLE
3          OFFLINE OFFLINE     STABLE
ora.asmnet1.asmnetwork (ora.asmgroup)
1          ONLINE  ONLINE      srv1          STABLE
2          ONLINE  ONLINE      srv2          STABLE
3          OFFLINE OFFLINE     STABLE
ora.cvu
1          ONLINE  ONLINE      srv1          STABLE
ora.qosmserver
1          ONLINE  ONLINE      srv1          STABLE
ora.racdb.db
1          ONLINE  ONLINE      srv1          Open,HOME=/u01/app/oracle/product/19.3/dbhome_1,STABLE
2          ONLINE  ONLINE      srv2          Open,HOME=/u01/app/oracle/product/19.3/dbhome_1,STABLE
ora.scan1.vip
1          ONLINE  ONLINE      srv2          STABLE
ora.scan2.vip
1          ONLINE  ONLINE      srv1          STABLE
ora.scan3.vip
1          ONLINE  ONLINE      srv1          STABLE
ora.srv1.vip
1          ONLINE  ONLINE      srv1          STABLE
ora.srv2.vip
1          ONLINE  ONLINE      srv2          STABLE
-----
[root@srv1 dbhome_1]#
```

Verify the TNSNAMES.ORA entry

This shows that the created entry (only for the CDB, not for PDB1) uses SCAN.

```
[oracle@srv1 product]$ cd $ORACLE_HOME/network/admin
[oracle@srv1 admin]$ cat tnsnames.ora
# tnsnames.ora Network Configuration File:
/u01/app/oracle/product/19.3/dbhome_1/network/admin/tnsnames.ora
# Generated by Oracle configuration tools.

RACDB =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP) (HOST = ora19cscan.localdomain) (PORT =
1521))
    (CONNECT_DATA =
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
(SERVER = DEDICATED)
(SERVICE_NAME = RACDB)
)
)
```

```
[oracle@srv1 admin]$
```

Add a PDB1 entry to the tnsnames.ora

```
PDB1 =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP) (HOST = ora19cscan.localdomain) (PORT =
1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = PDB1)
    )
  )
)
```

Create a Custom Service for PDB1

```
[oracle@srv1 ~]$ srvctl add service -db RACDB -service mypdb \
> -preferred RACDB1,RACDB2 -tafpolicy BASIC -failovertype SELECT -clbgoal
SHORT -pdb PDB1
```

```
[oracle@srv1 ~]$ srvctl start service -db RACDB -service mypdb
```

```
[oracle@srv1 ~]$ srvctl status service -db RACDB -service mypdb
Service mypdb is running on instance(s) RACDB1,RACDB2
```

```
[oracle@srv1 ~]$ srvctl config service -db RACDB -service mypdb
Service name: mypdb
Server pool:
Cardinality: 2
Service role: PRIMARY
Management policy: AUTOMATIC
DTP transaction: false
AQ HA notifications: false
Global: false
Commit Outcome: false
Failover type: SELECT
Failover method:
Failover retries:
Failover delay:
Failover restore: NONE
Connection Load Balancing Goal: SHORT
Runtime Load Balancing Goal: NONE
TAF policy specification: BASIC
Edition:
Pluggable database name: PDB1
Hub service:
Maximum lag time: ANY
SQL Translation Profile:
Retention: 86400 seconds
Replay Initiation Time: 300 seconds
Drain timeout:
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
Stop option:
Session State Consistency: DYNAMIC
GSM Flags: 0
Service is enabled
Preferred instances: RACDB1,RACDB2
Available instances:
CSS critical: no
Service uses Java: false
[oracle@srv1 ~]$
```

Add a TNSNAMES.ORA entry

```
mypdb =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP) (HOST = ora19cscan.localdomain) (PORT =
1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = mypdb)
    )
  )
```

Client-Side Load Balancing

```
mypdb_clb_scan =
  (DESCRIPTION =
    (ADDRESS_LIST = (LOAD_BALANCE = ON) (FAILOVER = ON)
      (ADDRESS = (PROTOCOL = TCP) (HOST =192.168.56.94) (PORT = 1521)) #scan-1
      (ADDRESS = (PROTOCOL = TCP) (HOST =192.168.56.95) (PORT = 1521)) #scan-2
      (ADDRESS = (PROTOCOL = TCP) (HOST =192.168.56.96) (PORT = 1521)) #scan-3
    )
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = mypdb)
    )
  )
```

```
mypdb_clb_vips =
  (DESCRIPTION =
    (ADDRESS_LIST = (LOAD_BALANCE = ON) (FAILOVER = ON)
      (ADDRESS = (PROTOCOL = TCP) (HOST =192.168.56.91) (PORT = 1521)) #srv1-
vip
      (ADDRESS = (PROTOCOL = TCP) (HOST =192.168.56.93) (PORT = 1521)) #srv2-
vip
    )
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = mypdb)
    )
  )
```

Test Connectivity

```
[oracle@srv1 admin]$ sqlplus / as sysdba
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
SQL*Plus: Release 19.0.0.0.0 - Production on Mon Mar 25 05:59:26 2024  
Version 19.3.0.0.0
```

```
Copyright (c) 1982, 2019, Oracle. All rights reserved.
```

```
Connected to:  
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production  
Version 19.3.0.0.0
```

```
SQL> alter session set container=PDB1;
```

```
Session altered.
```

```
SQL> grant dba to pdbadmin;
```

```
Grant succeeded.
```

```
SQL> quit  
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0  
- Production  
Version 19.3.0.0.0  
[oracle@srv1 admin]$
```

```
[oracle@srv1 admin]$ sqlplus pdbadmin@pdb1
```

```
SQL*Plus: Release 19.0.0.0.0 - Production on Mon Mar 25 06:00:00 2024  
Version 19.3.0.0.0
```

```
Copyright (c) 1982, 2019, Oracle. All rights reserved.
```

```
Enter password:  
Last Successful login time: Mon Mar 25 2024 05:58:26 +00:00
```

```
Connected to:  
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production  
Version 19.3.0.0.0
```

```
SQL> select instance_name, host_name from v$instance;
```

```
INSTANCE_NAME  
-----
```

```
HOST_NAME  
-----
```

```
RACDB1  
srv1
```

```
SQL> quit  
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0  
- Production  
Version 19.3.0.0.0  
[oracle@srv1 admin]$
```

Testing CLB and FAILOVER

```
SQL> connect hemant/hemant@mypdb_clb  
Connected.
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
SQL> select i.instance_name, i.host_name, s.failover_type,
s.failover_method, s.failed_over
  2   from v$instance i, v$session s where s.sid = (select distinct sid from
v$mystat)
  3   /
```

INSTANCE	HOST_NAM	FAILOVER_TYPE	FAILOVER_M	FAI
RACDB2	srv2	SELECT	BASIC	NO

SQL>

AFTER KILLING THE RACDB2 INSTANCE (“kill -9 <smonpid>” at OS level), run the query again

SQL> /

INSTANCE	HOST_NAM	FAILOVER_TYPE	FAILOVER_M	FAI
RACDB1	srv1	SELECT	BASIC	YES

SQL>

The session has failed over to srv1 instance RACDB1

Verify Installed Options

```
[oracle@srv1 admin]$ sqlplus / as sysdba
```

```
SQL*Plus: Release 19.0.0.0.0 - Production on Mon Mar 25 06:05:26 2024
Version 19.3.0.0.0
```

```
Copyright (c) 1982, 2019, Oracle. All rights reserved.
```

Connected to:

```
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
```

```
SQL> archive log list;
```

Database log mode	Archive Mode
Automatic archival	Enabled
Archive destination	USE_DB_RECOVERY_FILE_DEST
Oldest online log sequence	2
Next log sequence to archive	3
Current log sequence	3

SQL>

```
SQL> col parameter format a42
```

```
SQL> col value format a12
```

```
SQL> set pagesize 100
```

```
SQL> select parameter, value from v$option where con_id=0 order by 1;
```

PARAMETER	VALUE
ASM Proxy Instance	FALSE
Active Data Guard	TRUE

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Adaptive Execution Plans	TRUE
Advanced Analytics	TRUE
Advanced Compression	TRUE
Advanced Index Compression	TRUE
Advanced replication	TRUE
Application Role	TRUE
Automatic Data Optimization	TRUE
Automatic Storage Management	TRUE
Backup Encryption	TRUE
Basic Compression	TRUE
Bit-mapped indexes	TRUE
Block Change Tracking	TRUE
Block Media Recovery	TRUE
Cache Fusion Lock Accelerator	TRUE
Centrally Managed User	TRUE
Change Data Capture	TRUE
Coalesce Index	TRUE
Connection multiplexing	TRUE
Connection pooling	TRUE
Cross Transportable Backups	TRUE
DICOM	TRUE
Data Mining	TRUE
Data Redaction	TRUE
Database queuing	TRUE
Database resource manager	TRUE
Deferred Segment Creation	TRUE
Duplexed backups	TRUE
Enterprise User Security	TRUE
Exadata Discovery	TRUE
Export transportable tablespaces	TRUE
Fast-Start Fault Recovery	TRUE
File Mapping	TRUE
Fine-grained Auditing	TRUE
Fine-grained access control	TRUE
Flashback Data Archive	TRUE
Flashback Database	TRUE
Flashback Table	TRUE
Global Data Services	TRUE
Heat Map	TRUE
I/O Server	FALSE
In-Memory Aggregation	TRUE
In-Memory Column Store	TRUE
Incremental backup and recovery	TRUE
Instead-of triggers	TRUE
Java	TRUE
Join index	TRUE
Managed Standby	TRUE
Management Database	FALSE
Materialized view rewrite	TRUE
OLAP	TRUE
OLAP Window Functions	TRUE
Objects	TRUE
Online Index Build	TRUE
Online Redefinition	TRUE
Oracle Data Guard	TRUE
Oracle Database Vault	FALSE
Oracle Label Security	FALSE
Parallel backup and recovery	TRUE
Parallel execution	TRUE
Parallel load	TRUE

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Partitioning	TRUE
Plan Stability	TRUE
Point-in-time tablespace recovery	TRUE
Privilege Analysis	TRUE
Proxy authentication/authorization	TRUE
Real Application Clusters	TRUE
Real Application Security	TRUE
Real Application Testing	TRUE
Result Cache	TRUE
SQL Plan Management	TRUE
Sample Scan	TRUE
SecureFiles Encryption	TRUE
Server Flash Cache	TRUE
Snapshot time recovery	TRUE
Spatial	TRUE
Streams Capture	TRUE
Table Clustering	TRUE
Transparent Application Failover	TRUE
Transparent Data Encryption	TRUE
Transparent Sensitive Data Protection	TRUE
Trial Recovery	TRUE
Unified Auditing	FALSE
Unused Block Compression	TRUE
XStream	TRUE
Zone Maps	TRUE

87 rows selected.

SQL>

(Unfortunately, Unified Auditing still defaults to FALSE)

SQL> col instance_name format a12

SQL> col host_name format a12

SQL> select inst_id, instance_name, host_name from gv\$instance order by 1;

	INST_ID	INSTANCE_NAM	HOST_NAME
	1	RACDB1	srv1
	2	RACDB2	srv2

SQL>

Setting up Single-Node Standby

Add Standby Log File at Primary

```
SQL> select * from v$standby_log;
```

no rows selected

```
SQL> alter database add standby logfile ;
```

Database altered.

```
SQL> alter database add standby logfile ;
```

Database altered.

```
SQL> alter database add standby logfile ;
```

Database altered.

```
SQL> set pages600
```

```
SQL> select * from v$standby_log;
```

GROUP#	DBID				THREAD#	SEQUENCE#
BYTES	BLOCKSIZE	USED	ARC	STATUS	FIRST_CHANGE#	FIRST_TIM
NEXT_CHANGE#	NEXT_TIME	LAST_CHANGE#	LAST_TIME	CON_ID		
5	UNASSIGNED				0	0
104857600	512	0	YES	UNASSIGNED	0	
6	UNASSIGNED				0	0
104857600	512	0	YES	UNASSIGNED	0	
7	UNASSIGNED				0	0
104857600	512	0	YES	UNASSIGNED	0	

```
SQL> alter system set standby_file_management='AUTO' scope=BOTH sid='*';
```

System altered.

```
SQL>
```

```
SQL> alter database force logging;
```

Database altered.

```
SQL>
```

```
SQL> select protection_mode, protection_level from v$database;
```

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```
PROTECTION_MODE      PROTECTION_LEVEL
-----
MAXIMUM PERFORMANCE  MAXIMUM PERFORMANCE

SQL>
```

Disable RAC on Standby Binaries

make -f ins_rdbms.mk rac_off

```
[oracle@stdby lib]$ make -f ins_rdbms.mk rac_off
rm -f /u01/app/oracle/product/19.3/dbhome_1/lib/libskgxp19.so
cp /u01/app/oracle/product/19.3/dbhome_1/lib/libskgxp19.so
/u01/app/oracle/product/19.3/dbhome_1/lib/libskgxp19.so
rm -f /u01/app/oracle/product/19.3/dbhome_1/lib/libskgxn2.so
cp /u01/app/oracle/product/19.3/dbhome_1/lib/libskgxn2.so \
    /u01/app/oracle/product/19.3/dbhome_1/lib/libskgxn2.so
/usr/bin/ar d /u01/app/oracle/product/19.3/dbhome_1/rdbms/lib/libknlopt.a
kcsn.o
/usr/bin/ar cr /u01/app/oracle/product/19.3/dbhome_1/rdbms/lib/libknlopt.a
/u01/app/oracle/product/19.3/dbhome_1/rdbms/lib/ksnkcs.o
[oracle@stdby lib]$ pwd
/u01/app/oracle/product/19.3/dbhome_1/rdbms/lib
```

make -f ins_rdbms.mk ioracle

```
[oracle@stdby lib]$ make -f ins_rdbms.mk ioracle
chmod 755 /u01/app/oracle/product/19.3/dbhome_1/bin

- Linking Oracle
rm -f /u01/app/oracle/product/19.3/dbhome_1/rdbms/lib/oracle
/u01/app/oracle/product/19.3/dbhome_1/bin/orald -o
/u01/app/oracle/product/19.3/dbhome_1/rdbms/lib/oracle -m64 -z noexecstack
-Wl,--disable-new-dtags -L/u01/app/oracle/product/19.3/dbhome_1/rdbms/lib/
-L/u01/app/oracle/product/19.3/dbhome_1/lib/ -
L/u01/app/oracle/product/19.3/dbhome_1/lib/stubs/ -Wl,-E
/u01/app/oracle/product/19.3/dbhome_1/rdbms/lib/opimai.o
/u01/app/oracle/product/19.3/dbhome_1/rdbms/lib/ssoraed.o
/u01/app/oracle/product/19.3/dbhome_1/rdbms/lib/ttcsoi.o -Wl,--whole-
archive -lperfsv19 -Wl,--no-whole-archive
/u01/app/oracle/product/19.3/dbhome_1/lib/nautab.o
/u01/app/oracle/product/19.3/dbhome_1/lib/naeet.o
/u01/app/oracle/product/19.3/dbhome_1/lib/naect.o
/u01/app/oracle/product/19.3/dbhome_1/lib/naedhs.o
/u01/app/oracle/product/19.3/dbhome_1/rdbms/lib/config.o -ldmext -
lserver19 -lodm19 -lofs -lcell19 -lnnet19 -lskgxp19 -lsnls19 -lnls19 -
lcore19 -lsnls19 -lnls19 -lcore19 -lsnls19 -lnls19 -lxml19 -lcore19 -
lunls19 -lsnls19 -lnls19 -lcore19 -lnls19 -lclient19 -lvsnst19 -lcommon19
-lgeneric19 -lknlopt -loraolap19 -lskjcx19 -lslox19 -lpls19 -lrtl -lplp19 -
ldmext -lserver19 -lclient19 -lvsnst19 -lcommon19 -lgeneric19 `if [ -f
/u01/app/oracle/product/19.3/dbhome_1/lib/libavserver19.a ] ; then echo "-
lavserver19" ; else echo "-lavstub19"; fi` `if [ -f
/u01/app/oracle/product/19.3/dbhome_1/lib/libavclient19.a ] ; then echo "-
lavclient19" ; fi` -lknlopt -lslox19 -lpls19 -lrtl -lplp19 -ljavavm19 -
lserver19 -lwwg `cat /u01/app/oracle/product/19.3/dbhome_1/lib/ldflags`
```

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```
-lncrypt19 -lnsgr19 -lnzjs19 -ln19 -lnl19 -lngsmshd19 -lnro19 `cat
/u01/app/oracle/product/19.3/dbhome_1/lib/ldflags` -lncrypt19 -lnsgr19 -
lnzjs19 -ln19 -lnl19 -lngsmshd19 -lnnzst19 -lzt19 -lztkg19 -lmm -lsnls19 -
lnls19 -lcore19 -lsnls19 -lnls19 -lcore19 -lsnls19 -lnls19 -lxml19 -
lcore19 -lunls19 -lsnls19 -lnls19 -lcore19 -lnls19 -lztkg19 `cat
/u01/app/oracle/product/19.3/dbhome_1/lib/ldflags` -lncrypt19 -lnsgr19 -
lnzjs19 -ln19 -lnl19 -lngsmshd19 -lnro19 `cat
/u01/app/oracle/product/19.3/dbhome_1/lib/ldflags` -lncrypt19 -lnsgr19 -
lnzjs19 -ln19 -lnl19 -lngsmshd19 -lnnzst19 -lzt19 -lztkg19 -lsnls19 -
lnls19 -lcore19 -lsnls19 -lnls19 -lcore19 -lsnls19 -lnls19 -lxml19 -
lcore19 -lunls19 -lsnls19 -lnls19 -lcore19 -lnls19 `if /usr/bin/ar tv
/u01/app/oracle/product/19.3/dbhome_1/rdbms/lib/libknlopt.a | grep
"xxmnsd.o" > /dev/null 2>&1 ; then echo " " ; else echo "-lordsdo19 -
lserver19"; fi` -L/u01/app/oracle/product/19.3/dbhome_1/ctx/lib/ -lctxc19 -
lctx19 -lzx19 -lgx19 -lctx19 -lzx19 -lgx19 -lclscest19 -loevm -lclsra19 -
lbcfg19 -lhasgen19 -lskgn2 -lnnzst19 -lzt19 -lxml19 -lgeneric19 -locr19 -
locrb19 -locrut19 -lhasgen19 -lskgn2 -lnnzst19 -lzt19 -lxml19 -lgeneric19
-lgeneric19 -lorazip -loraz -llzopro5 -lorabz2 -lorazstd -loralz4 -lipp_z -
lipp_bz2 -lippdc -lipps -lippcore -lippcp -lsnls19 -lnls19 -lcore19 -
lsnls19 -lnls19 -lcore19 -lsnls19 -lnls19 -lxml19 -lcore19 -lunls19 -
lsnls19 -lnls19 -lcore19 -lnls19 -lsnls19 -lunls19 -lsnls19 -lnls19 -
lcore19 -lsnls19 -lnls19 -lcore19 -lsnls19 -lnls19 -lxml19 -lcore19 -
lunls19 -lsnls19 -lnls19 -lcore19 -lnls19 -lasmcint19 -lcommon19 -lcore19
-ledtn19 -laio -lons -lmql1 -lipcl -lfthread19 `cat
/u01/app/oracle/product/19.3/dbhome_1/lib/sysliblist` -Wl,-
rpath,/u01/app/oracle/product/19.3/dbhome_1/lib -lm `cat
/u01/app/oracle/product/19.3/dbhome_1/lib/sysliblist` -ldl -lm -
L/u01/app/oracle/product/19.3/dbhome_1/lib `test -x /usr/bin/hugeedit -a -r
/usr/lib64/libhugetlbfs.so && test -r
/u01/app/oracle/product/19.3/dbhome_1/rdbms/lib/shugetlbfs.o && echo -Wl,-
zcommon-page-size=2097152 -Wl,-zmax-page-size=2097152 -lhugetlbfs`
rm -f /u01/app/oracle/product/19.3/dbhome_1/bin/oracle
mv /u01/app/oracle/product/19.3/dbhome_1/rdbms/lib/oracle
/u01/app/oracle/product/19.3/dbhome_1/bin/oracle
chmod 6751 /u01/app/oracle/product/19.3/dbhome_1/bin/oracle
(if [ ! -f /u01/app/oracle/product/19.3/dbhome_1/bin/crsd.bin ]; then \

getcrshome="/u01/app/oracle/product/19.3/dbhome_1/srv/admin/getcrshome" ;
\
    if [ -f "$getcrshome" ]; then \
        crshome="$getcrshome"; \
        if [ -n "$crshome" ]; then \
            if [ $crshome != /u01/app/oracle/product/19.3/dbhome_1 ]; then
\
                oracle="/u01/app/oracle/product/19.3/dbhome_1/bin/oracle";
\
                $crshome/bin/setasmgidwrap oracle_binary_path=$oracle; \
            fi \
        fi \
    fi \
fi \
);
[oracle@stdby lib]$ ls -l $ORACLE_HOME/bin/oracle
-rwsr-s--x. 1 oracle oinstall 441253104 May  1 02:47
/u01/app/oracle/product/19.3/dbhome_1/bin/oracle
```

Remove “cluster_database” and all parameters with Instance Names from initSTDBY.ora

Copy Password File (after setting/resetting SYS password)

```
[root@srv1 ~]# su - grid
Last login: Wed May  1 02:49:22 UTC 2024
[grid@srv1 ~]$ asmcmd
ASMCMD> cd data
ASMCMD> ls
RACDB/
snapcf_racdb1.f
ASMCMD> cd RACDB
ASMCMD> ls
14769E258FBB5FD8E0635A38A8C09D43/
1476C653214704CFE0635A38A8C08494/
17153493E0D24767E0635A38A8C0BF79/
86B637B62FE07A65E053F706E80A27CA/
CONTROLFILE/
DATAFILE/
ONLINELOG/
PARAMETERFILE/
PASSWORD/
TEMPFILE/
ASMCMD> cd password
ASMCMD> ls
pwracdb.256.1164519419
ASMCMD> cp pwracdb.256.1164519419 /tmp
copying +data/RACDB/password/pwracdb.256.1164519419 ->
/tmp/pwracdb.256.1164519419
ASMCMD> quit
[grid@srv1 ~]$

[oracle@stdby dbs]$ scp oracle@srv1:/tmp/pwracdb.256.1164519419 orapwSTDBY
pwracdb.256.1164519419
100% 2560    1.5MB/s   00:00
[oracle@stdby dbs]$ ls -l orapw*
-rw-r----- 1 oracle oinstall 2560 May  1 03:04 orapwSTDBY
[oracle@stdby dbs]$
```

Configure initSTDBY.ora file

```
audit_file_dest='/u01/app/oracle/admin/RACDB/adump'
audit_trail='db'
compatible='19.0.0'
control_files='/Standby_DB/oradata/control01.ctl,
/Standby_DB/FRA/control02.ctl'
db_block_size=8192
db_create_file_dest='/Standby_DB/oradata'
db_name='RACDB'
db_unique_name='STDBY'
db_recovery_file_dest='/Standby_DB/FRA'
db_recovery_file_dest_size=13332m
diagnostic_dest='/u01/app/oracle'
dispatchers='(PROTOCOL=TCP) (SERVICE=STDBYXDB)'
enable_pluggable_database=true
```

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```
family:dw_helper.instance_mode='read-only'
log_archive_format='%t_%s_%r.dbf'
nls_language='AMERICAN'
nls_territory='AMERICA'
open_cursors=300
pga_aggregate_target=256M
processes=250
remote_login_passwordfile='exclusive'
sga_target=2048M
standby_file_management='AUTO'
log_archive_dest_2='service=racdb sync affirm
valid_for=(all_logfiles,all_roles) db_unique_name=RACDB';
```

Configure Standby listener.ora file

```
SID_LIST_LISTENER =
  (SID_LIST =
    (SID_DESC =
      (GLOBAL_DBNAME = STDBY)
      (SID_NAME = STDBY)
      (ORACLE_HOME = /u01/app/oracle/product/19.3/dbhome_1)
    )
  )
```

Configure Standby tnsnames.ora

```
RACDB1 =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP) (HOST = srv1.localdomain) (PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = RACDB)
    )
  )

RACDB2 =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP) (HOST = srv2.localdomain) (PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = RACDB)
    )
  )

STDBY =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP) (HOST = stdby.localdomain) (PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = STDBY)
    )
  )
```

Add Corresponding tnsnames.ora Entries at both Primaries

```
STDBY =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP) (HOST = stdby.localdomain) (PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = STDBY)
    )
  )

RACDB1 =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP) (HOST = srv1.localdomain) (PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = RACDB)
    )
  )

RACDB2 =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP) (HOST = srv2.localdomain) (PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = RACDB)
    )
  )
```

Start Standby Listener and Database

```
[oracle@stdby dbs]$ lsnrctl start
```

```
LSNRCTL for Linux: Version 19.0.0.0.0 - Production on 01-MAY-2024 03:48:59
```

```
Copyright (c) 1991, 2019, Oracle. All rights reserved.
```

```
Starting /u01/app/oracle/product/19.3/dbhome_1/bin/tnslsnr: please wait...
```

```
TNSLSNR for Linux: Version 19.0.0.0.0 - Production
```

```
System parameter file is
```

```
/u01/app/oracle/product/19.3/dbhome_1/network/admin/listener.ora
```

```
Log messages written to
```

```
/u01/app/oracle/diag/tnslsnr/stdby/listener/alert/log.xml
```

```
Listening on: (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp) (HOST=stdby) (PORT=1521)))
```

```
Connecting to (ADDRESS=(PROTOCOL=tcp) (HOST=) (PORT=1521))
```

```
STATUS of the LISTENER
```

```
-----
```

Alias	LISTENER
Version	TNSLSNR for Linux: Version 19.0.0.0.0 -
Production	
Start Date	01-MAY-2024 03:48:59
Uptime	0 days 0 hr. 0 min. 0 sec
Trace Level	off
Security	ON: Local OS Authentication

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```
SNMP                                OFF
Listener Parameter File
/u01/app/oracle/product/19.3/dbhome_1/network/admin/listener.ora
Listener Log File
/u01/app/oracle/diag/tnslsnr/stdby/listener/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=stdby)(PORT=1521)))
Services Summary...
Service "STDBY" has 1 instance(s).
  Instance "STDBY", status UNKNOWN, has 1 handler(s) for this service...
The command completed successfully
[oracle@stdby dbs]$
[oracle@stdby dbs]$ sqlplus / as sysdba

SQL*Plus: Release 19.0.0.0.0 - Production on Wed May 1 03:52:11 2024
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to an idle instance.

SQL> startup nomount;
ORACLE instance started.

Total System Global Area 2147482248 bytes
Fixed Size                  9136776 bytes
Variable Size              486539264 bytes
Database Buffers          1644167168 bytes
Redo Buffers                7639040 bytes
SQL> quit
Disconnected from Oracle Database 19c Enterprise Edition Release 19.0.0.0.0
- Production
Version 19.3.0.0.0
[oracle@stdby dbs]$
```

Execute RMAN DUPLICATE DATABASE

```
[oracle@stdby dbs]$ rman

Recovery Manager: Release 19.0.0.0.0 - Production on Wed May 1 04:10:57
2024
Version 19.3.0.0.0

Copyright (c) 1982, 2019, Oracle and/or its affiliates. All rights
reserved.

RMAN> connect target sys/manager@RACDB1

connected to target database: RACDB (DBID=1162136313)

RMAN> connect auxiliary sys/manager@STDBY

connected to auxiliary database: RACDB (not mounted)

RMAN> duplicate target database for standby from active database;

Starting Duplicate Db at 01-MAY-24
```

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using target database control file instead of recovery catalog
configuration for DISK channel 2 is ignored
allocated channel: ORA_AUX_DISK_1
channel ORA_AUX_DISK_1: SID=101 device type=DISK

contents of Memory Script:

```
{
  backup as copy reuse
  passwordfile auxiliary format
  '/u01/app/oracle/product/19.3/dbhome_1/dbs/orapwSTDBY' ;
}
```

executing Memory Script

Starting backup at 01-MAY-24
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=31 instance=RACDB1 device type=DISK
Finished backup at 01-MAY-24
duplicating Online logs to Oracle Managed File (OMF) location
duplicating Datafiles to Oracle Managed File (OMF) location

contents of Memory Script:

```
{
  restore clone from service 'RACDB1' standby controlfile;
}
```

executing Memory Script

Starting restore at 01-MAY-24
using channel ORA_AUX_DISK_1

channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service RACDB1
channel ORA_AUX_DISK_1: restoring control file
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:02
output file name=/Standby_DB/oradata/control01.ctl,
/Standby_DB/FRA/control02.ctl
Finished restore at 01-MAY-24

contents of Memory Script:

```
{
  sql clone 'alter database mount standby database';
}
```

executing Memory Script

sql statement: alter database mount standby database

contents of Memory Script:

```
{
  set newname for clone tempfile 1 to new;
  set newname for clone tempfile 2 to new;
  set newname for clone tempfile 3 to new;
  set newname for clone tempfile 4 to new;
  switch clone tempfile all;
  set newname for clone datafile 1 to new;
  set newname for clone datafile 3 to new;
  set newname for clone datafile 4 to new;
  set newname for clone datafile 5 to new;
  set newname for clone datafile 6 to new;
  set newname for clone datafile 7 to new;
  set newname for clone datafile 8 to new;
  set newname for clone datafile 9 to new;
```

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```
set newname for clone datafile 10 to new;
set newname for clone datafile 11 to new;
set newname for clone datafile 12 to new;
set newname for clone datafile 13 to new;
set newname for clone datafile 14 to new;
set newname for clone datafile 15 to new;
set newname for clone datafile 16 to new;
set newname for clone datafile 17 to new;
set newname for clone datafile 18 to new;
set newname for clone datafile 19 to new;
set newname for clone datafile 20 to new;
set newname for clone datafile 21 to new;
restore
from nonsparse from service
'RACDB1' clone database
;
sql 'alter system archive log current';
}
executing Memory Script

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

renamed tempfile 1 to /Standby_DB/oradata/STDBY/datafile/ol_mf_temp_%u_.tmp
in control file
renamed tempfile 2 to
/Standby_DB/oradata/STDBY/14769E258FBB5FD8E0635A38A8C09D43/datafile/ol_mf_t
emp_%u_.tmp in control file
renamed tempfile 3 to
/Standby_DB/oradata/STDBY/1476C653214704CFE0635A38A8C08494/datafile/ol_mf_t
emp_%u_.tmp in control file
renamed tempfile 4 to
/Standby_DB/oradata/STDBY/17153493E0D24767E0635A38A8C0BF79/datafile/ol_mf_t
emp_%u_.tmp in control file

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

executing command: SET NEWNAME

Starting restore at 01-MAY-24

using channel ORA_AUX_DISK_1

channel ORA_AUX_DISK_1: starting datafile backup set restore

channel ORA_AUX_DISK_1: using network backup set from service RACDB1

channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_AUX_DISK_1: restoring datafile 00001 to

/Standby_DB/oradata/STDBY/datafile/ol_mf_system_%u_.dbf

channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:15

channel ORA_AUX_DISK_1: starting datafile backup set restore

channel ORA_AUX_DISK_1: using network backup set from service RACDB1

channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_AUX_DISK_1: restoring datafile 00003 to

/Standby_DB/oradata/STDBY/datafile/ol_mf_sysaux_%u_.dbf

channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:07

channel ORA_AUX_DISK_1: starting datafile backup set restore

channel ORA_AUX_DISK_1: using network backup set from service RACDB1

channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_AUX_DISK_1: restoring datafile 00004 to

/Standby_DB/oradata/STDBY/datafile/ol_mf_undotbs1_%u_.dbf

channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:04

channel ORA_AUX_DISK_1: starting datafile backup set restore

channel ORA_AUX_DISK_1: using network backup set from service RACDB1

channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_AUX_DISK_1: restoring datafile 00005 to

/Standby_DB/oradata/STDBY/14769E258FBB5FD8E0635A38A8C09D43/datafile/ol_mf_system_%u_.dbf

channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:03

channel ORA_AUX_DISK_1: starting datafile backup set restore

channel ORA_AUX_DISK_1: using network backup set from service RACDB1

channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_AUX_DISK_1: restoring datafile 00006 to

/Standby_DB/oradata/STDBY/14769E258FBB5FD8E0635A38A8C09D43/datafile/ol_mf_sysaux_%u_.dbf

channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:03

channel ORA_AUX_DISK_1: starting datafile backup set restore

channel ORA_AUX_DISK_1: using network backup set from service RACDB1

channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set

channel ORA_AUX_DISK_1: restoring datafile 00007 to

/Standby_DB/oradata/STDBY/datafile/ol_mf_users_%u_.dbf

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:01
channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service RACDB1
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00008 to
/Standby_DB/oradata/STDBY/14769E258FBB5FD8E0635A38A8C09D43/datafile/o1_mf_u
ndotbs1_%u_.dbf
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:01
channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service RACDB1
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00009 to
/Standby_DB/oradata/STDBY/datafile/o1_mf_undotbs2_%u_.dbf
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:01
channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service RACDB1
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00010 to
/Standby_DB/oradata/STDBY/1476C653214704CFE0635A38A8C08494/datafile/o1_mf_s
ystem_%u_.dbf
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:04
channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service RACDB1
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00011 to
/Standby_DB/oradata/STDBY/1476C653214704CFE0635A38A8C08494/datafile/o1_mf_s
ysaux_%u_.dbf
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:03
channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service RACDB1
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00012 to
/Standby_DB/oradata/STDBY/1476C653214704CFE0635A38A8C08494/datafile/o1_mf_u
ndotbs1_%u_.dbf
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:01
channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service RACDB1
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00013 to
/Standby_DB/oradata/STDBY/1476C653214704CFE0635A38A8C08494/datafile/o1_mf_u
ndo_2_%u_.dbf
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:01
channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service RACDB1
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00014 to
/Standby_DB/oradata/STDBY/1476C653214704CFE0635A38A8C08494/datafile/o1_mf_u
sers_%u_.dbf
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:01
channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service RACDB1
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00015 to
/Standby_DB/oradata/STDBY/1476C653214704CFE0635A38A8C08494/datafile/o1_mf_u
sers2_%u_.dbf
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:01
channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service RACDB1
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
```

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```
channel ORA_AUX_DISK_1: restoring datafile 00016 to
/Standby_DB/oradata/STDBY/1476C653214704CFE0635A38A8C08494/datafile/o1_mf_u
sers3_%u_.dbf
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:02
channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service RACDB1
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00017 to
/Standby_DB/oradata/STDBY/17153493E0D24767E0635A38A8C0BF79/datafile/o1_mf_s
ystem_%u_.dbf
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:03
channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service RACDB1
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00018 to
/Standby_DB/oradata/STDBY/17153493E0D24767E0635A38A8C0BF79/datafile/o1_mf_s
ysaux_%u_.dbf
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:03
channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service RACDB1
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00019 to
/Standby_DB/oradata/STDBY/17153493E0D24767E0635A38A8C0BF79/datafile/o1_mf_u
ndotbs1_%u_.dbf
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:01
channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service RACDB1
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00020 to
/Standby_DB/oradata/STDBY/17153493E0D24767E0635A38A8C0BF79/datafile/o1_mf_u
ndo_2_%u_.dbf
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:01
channel ORA_AUX_DISK_1: starting datafile backup set restore
channel ORA_AUX_DISK_1: using network backup set from service RACDB1
channel ORA_AUX_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_AUX_DISK_1: restoring datafile 00021 to
/Standby_DB/oradata/STDBY/17153493E0D24767E0635A38A8C0BF79/datafile/o1_mf_r
mancata_%u_.dbf
channel ORA_AUX_DISK_1: restore complete, elapsed time: 00:00:02
Finished restore at 01-MAY-24
```

sql statement: alter system archive log current

contents of Memory Script:

```
{
    switch clone datafile all;
}
```

executing Memory Script

```
datafile 1 switched to datafile copy
input datafile copy RECID=24 STAMP=1167797609 file
name=/Standby_DB/oradata/STDBY/datafile/o1_mf_system_m33j9fqn_.dbf
datafile 3 switched to datafile copy
input datafile copy RECID=25 STAMP=1167797609 file
name=/Standby_DB/oradata/STDBY/datafile/o1_mf_sysaux_m33j9wyo_.dbf
datafile 4 switched to datafile copy
input datafile copy RECID=26 STAMP=1167797609 file
name=/Standby_DB/oradata/STDBY/datafile/o1_mf_undotbs1_m33jb448_.dbf
datafile 5 switched to datafile copy
```

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```
input datafile copy RECID=27 STAMP=1167797609 file
name=/Standby_DB/oradata/STDBY/14769E258FBB5FD8E0635A38A8C09D43/datafile/o1
_mf_system_m33jb79n_.dbf
datafile 6 switched to datafile copy
input datafile copy RECID=28 STAMP=1167797609 file
name=/Standby_DB/oradata/STDBY/14769E258FBB5FD8E0635A38A8C09D43/datafile/o1
_mf_sysaux_m33jbbgz_.dbf
datafile 7 switched to datafile copy
input datafile copy RECID=29 STAMP=1167797609 file
name=/Standby_DB/oradata/STDBY/datafile/o1_mf_users_m33jbmf_.dbf
datafile 8 switched to datafile copy
input datafile copy RECID=30 STAMP=1167797609 file
name=/Standby_DB/oradata/STDBY/14769E258FBB5FD8E0635A38A8C09D43/datafile/o1
_mf_undotbs1_m33jbgrs_.dbf
datafile 9 switched to datafile copy
input datafile copy RECID=31 STAMP=1167797609 file
name=/Standby_DB/oradata/STDBY/datafile/o1_mf_undotbs2_m33jbhyv_.dbf
datafile 10 switched to datafile copy
input datafile copy RECID=32 STAMP=1167797609 file
name=/Standby_DB/oradata/STDBY/1476C653214704CFE0635A38A8C08494/datafile/o1
_mf_system_m33jbk47_.dbf
datafile 11 switched to datafile copy
input datafile copy RECID=33 STAMP=1167797609 file
name=/Standby_DB/oradata/STDBY/1476C653214704CFE0635A38A8C08494/datafile/o1
_mf_sysaux_m33jbn9f_.dbf
datafile 12 switched to datafile copy
input datafile copy RECID=34 STAMP=1167797609 file
name=/Standby_DB/oradata/STDBY/1476C653214704CFE0635A38A8C08494/datafile/o1
_mf_undotbs1_m33jbqh3_.dbf
datafile 13 switched to datafile copy
input datafile copy RECID=35 STAMP=1167797609 file
name=/Standby_DB/oradata/STDBY/1476C653214704CFE0635A38A8C08494/datafile/o1
_mf_undo_2_m33jbro3_.dbf
datafile 14 switched to datafile copy
input datafile copy RECID=36 STAMP=1167797609 file
name=/Standby_DB/oradata/STDBY/1476C653214704CFE0635A38A8C08494/datafile/o1
_mf_users_m33jbss9_.dbf
datafile 15 switched to datafile copy
input datafile copy RECID=37 STAMP=1167797609 file
name=/Standby_DB/oradata/STDBY/1476C653214704CFE0635A38A8C08494/datafile/o1
_mf_users2_m33jbtxy_.dbf
datafile 16 switched to datafile copy
input datafile copy RECID=38 STAMP=1167797609 file
name=/Standby_DB/oradata/STDBY/1476C653214704CFE0635A38A8C08494/datafile/o1
_mf_users3_m33jbw5d_.dbf
datafile 17 switched to datafile copy
input datafile copy RECID=39 STAMP=1167797609 file
name=/Standby_DB/oradata/STDBY/17153493E0D24767E0635A38A8C0BF79/datafile/o1
_mf_system_m33jbx7_.dbf
datafile 18 switched to datafile copy
input datafile copy RECID=40 STAMP=1167797609 file
name=/Standby_DB/oradata/STDBY/17153493E0D24767E0635A38A8C0BF79/datafile/o1
_mf_sysaux_m33jc0kn_.dbf
datafile 19 switched to datafile copy
input datafile copy RECID=41 STAMP=1167797609 file
name=/Standby_DB/oradata/STDBY/17153493E0D24767E0635A38A8C0BF79/datafile/o1
_mf_undotbs1_m33jc3qm_.dbf
datafile 20 switched to datafile copy
```

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```
input datafile copy RECID=42 STAMP=1167797609 file
name=/Standby_DB/oradata/STDBY/17153493E0D24767E0635A38A8C0BF79/datafile/o1
_mf_undo_2_m33jc4xh_.dbf
datafile 21 switched to datafile copy
input datafile copy RECID=43 STAMP=1167797609 file
name=/Standby_DB/oradata/STDBY/17153493E0D24767E0635A38A8C0BF79/datafile/o1
_mf_rmancata_m33jc62q_.dbf
Finished Duplicate Db at 01-MAY-24
```

```
RMAN>
RMAN> quit
```

Recovery Manager complete.

Configure log_archive_dest_2 at both Primaries

```
SQL> alter system set log_archive_dest_2='service=STDBY sync affirm
valid_for=(all_logfiles,all_roles) db_unique_name=STDBY';
```

System altered.

Begin Recovery at Standby

```
[oracle@stdby dbs]$ sqlplus / as sysdba
```

```
SQL*Plus: Release 19.0.0.0.0 - Production on Wed May 1 04:20:39 2024
Version 19.3.0.0.0
```

Copyright (c) 1982, 2019, Oracle. All rights reserved.

```
Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0
```

```
SQL> alter database recover managed standby database using current logfile
disconnect from session;
```

Database altered.

```
SQL>
SQL> set pages600
SQL> set linesize 132
SQL> select * from v$dataguard_stats where name like '%lag%';
```

SOURCE_DBID	SOURCE_DB_UNIQUE_NAME	NAME

--		
VALUE		UNIT
TIME_COMPUTED		

DATUM_TIME		CON_ID

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```
-----
1162136313                                transport lag
+00 00:03:06                                day(2) to
second(0) interval    05/01/2024 04:22:08
05/01/2024 04:21:42                                0
```

```
1162136313                                apply lag
second(0) interval    05/01/2024 04:22:08
                                0
```

```
SQL>
--- Requery to see that the lag has dropped
SQL> /
```

```
SOURCE_DBID SOURCE_DB_UNIQUE_NAME          NAME
-----
--
VALUE                                               UNIT
TIME_COMPUTED
-----
DATUM_TIME                                CON_ID
-----
1162136313                                transport lag
+00 00:00:13                                day(2) to
second(0) interval    05/01/2024 04:22:43
05/01/2024 04:22:42                                0

1162136313                                apply lag
+00 00:00:13                                day(2) to
second(0) interval    05/01/2024 04:22:43
05/01/2024 04:22:42                                0
```

```
SQL>
SQL> select client_process, process, thread#, sequence#, status
2   from v$managed_standby
3  where process='MRP0'
4   /
```

CLIENT_P	PROCESS	THREAD#	SEQUENCE#	STATUS
N/A	MRP0	2	67	WAIT_FOR_LOG

Query status from both Primaries

```
SQL> alter system archive log current;
```

```
System altered.
```

```
SQL> archive log list;
Database log mode          Archive Mode
Automatic archival        Enabled
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
Archive destination          USE_DB_RECOVERY_FILE_DEST
Oldest online log sequence  60
Next log sequence to archive 61
Current log sequence         61
SQL> select scn_to_timestamp(current_scn) - scn_to_timestamp(applied_scn)
as Time_Diff
  2   from v$database d,
  3   (select applied_scn from v$archive_dest a where target = 'STANDBY')
  4   /
```

TIME_DIFF

+0000000000 00:00:57.000000000

```
SQL>
SQL> select scn_to_timestamp(current_scn) - scn_to_timestamp(applied_scn)
as Time_Diff
  2   from v$database d,
  3   (select applied_scn from v$archive_dest a where target = 'STANDBY')
  4   /
```

TIME_DIFF

+0000000000 00:01:54.000000000

```
SQL>
SQL> alter system archive log current;
```

System altered.

```
SQL> select scn_to_timestamp(current_scn) - scn_to_timestamp(applied_scn)
as Time_Diff
  2   from v$database d,
  3   (select applied_scn from v$archive_dest a where target = 'STANDBY')
  4   /
```

TIME_DIFF

+0000000000 00:00:18.000000000

SQL>

Reference

Oracle's Vagrant Projects GitHub Page

<https://github.com/oracle/vagrant-projects>

Server Names

srv1

srv2

dnsssrvr

Virtual Box Configuration

Memory 2GB initially (update to 8GB before installing Oracle ClusterWare (Grid Infrastructure) [The DNS Server can run on 2GB)

Processors : 2 initially (update to 4 before installing Oracle ClusterWare (Grid Infrastructure) [The DNS Server can run with 2 Processors)

IP addresses

– these are based on my Virtual Box Network domain being 192.168.56



Entries in `/etc/hosts` on all 3 servers (SCAN entries on the DB Servers are commented out as the DNS server must return those 3 SCAN Ips)

```
#Host IPs
192.168.56.90  srv1 srv1.localdomain
192.168.56.91  srv1-vip srv1-vip.localdomain
192.168.10.8   srv1-priv srv1-priv

192.168.56.92  srv2 srv2.localdomain
192.168.56.93  srv2-vip srv2-vip.localdomain
192.168.10.10  srv2-priv srv2-priv
```

Installation of 19c RAC on Linux7 in Virtual Box VMs

```
192.168.56.80 dnssrvr dnssrvr.localdomain

#SCAN IPs (DNS Server now configured to dnssrvr
192.168.56.94 ora19cscan ora19cscan.localdomain
192.168.56.95 ora19cscan ora19cscan.localdomain
192.168.56.96 ora19cscan ora19cscan.localdomain

192.168.56.99 stdby stdby.localdomain
```

Also, the initial nameserver entry in /etc/resolv.conf is based on the default gateway on my Host PC's Internet Connection

```
nameserver 192.168.1.254
```

```
IPv4 Address. . . . . : 192.168.1.5
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.254
```